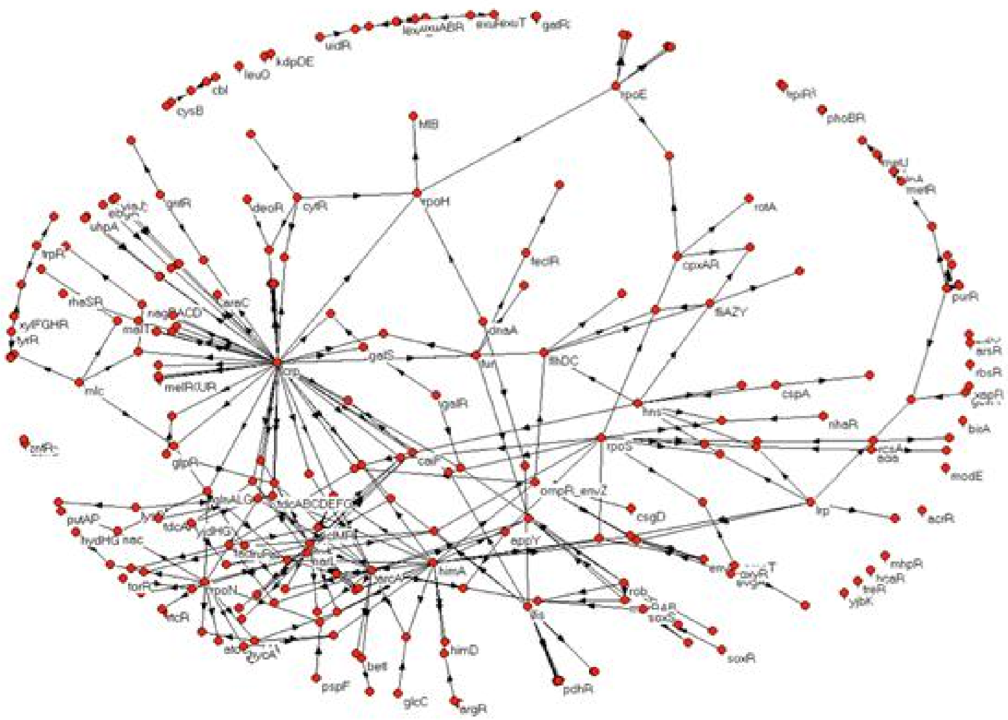
Computational Biology

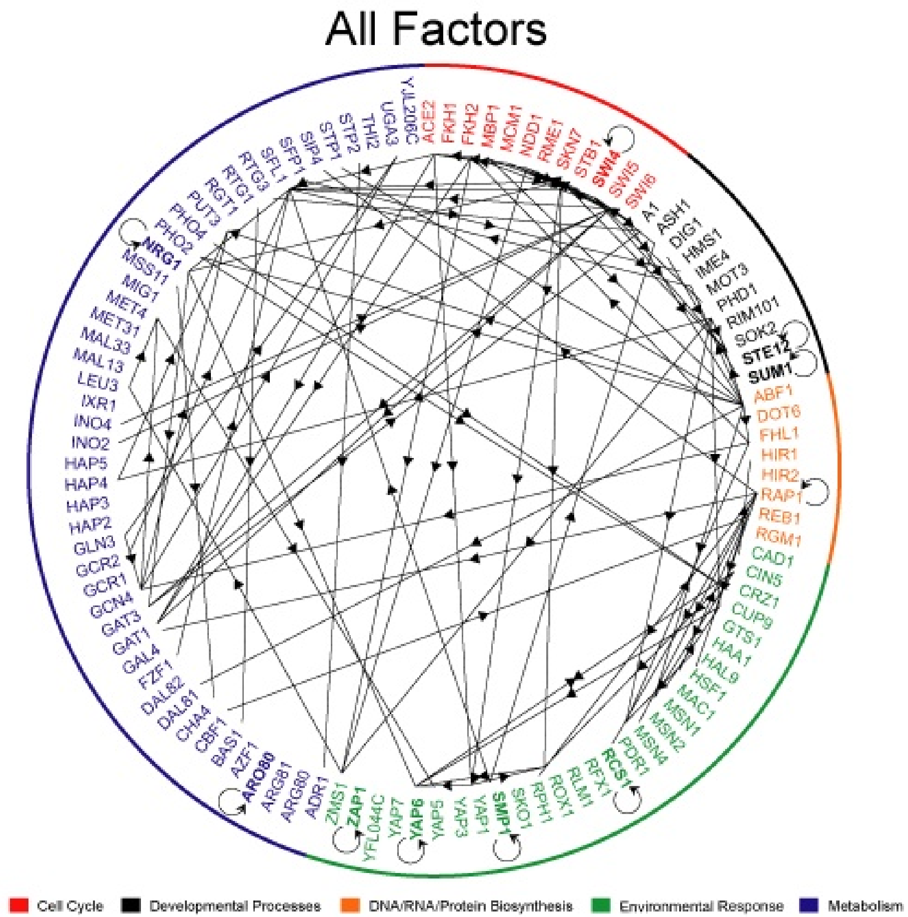
Lecture 3: *Gene Regulation Networks*



Gene regulation, chemical equations, motifs

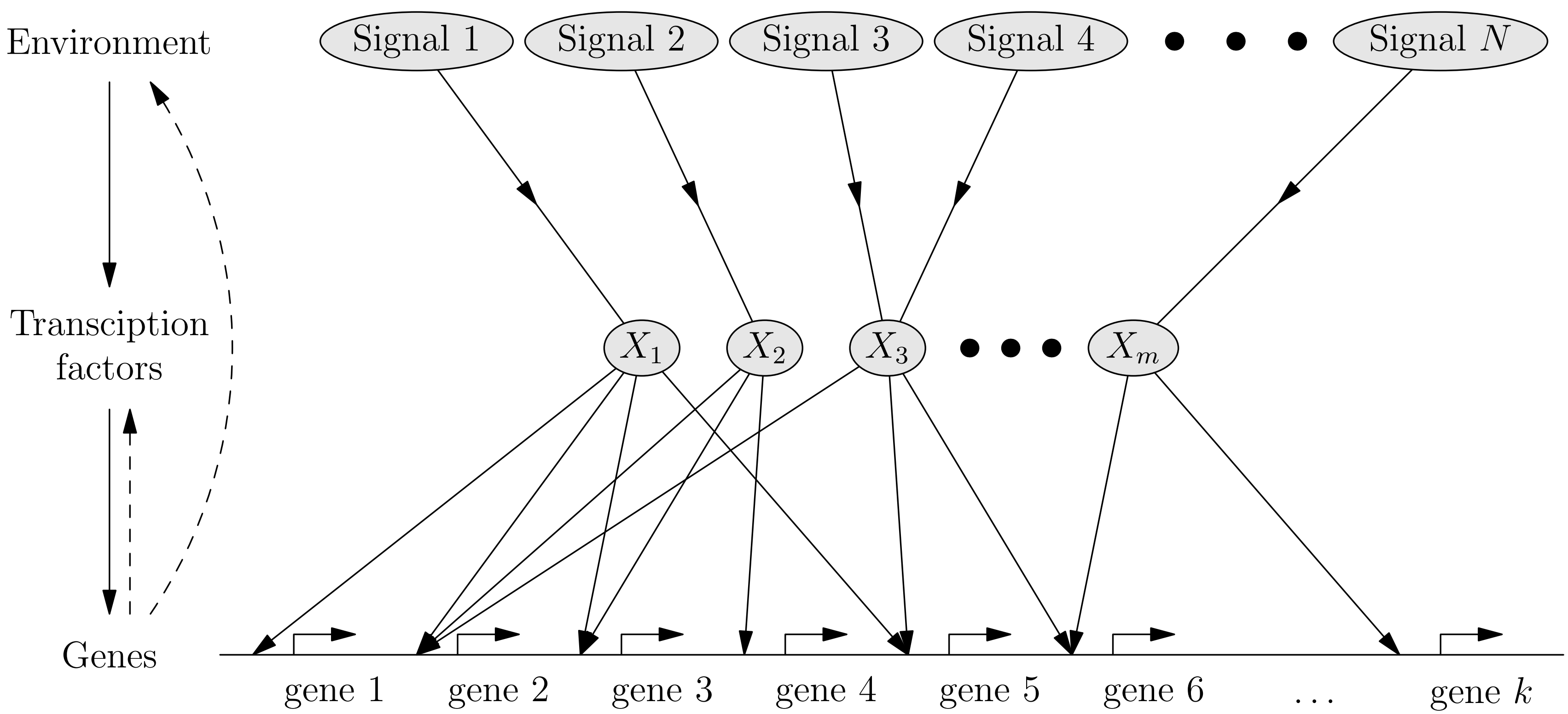
Outline

1. **Gene Regulation**
2. Chemical Equations
3. Motifs
4. Quiz



Transciption Networks

- Signalling molecules combine with transcription factors (proteins) which excite or inhibit the transcription of other proteins



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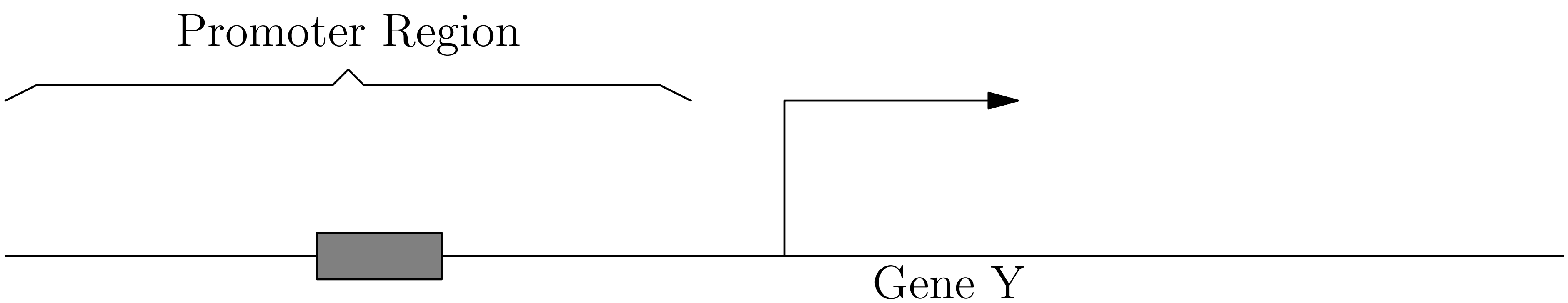

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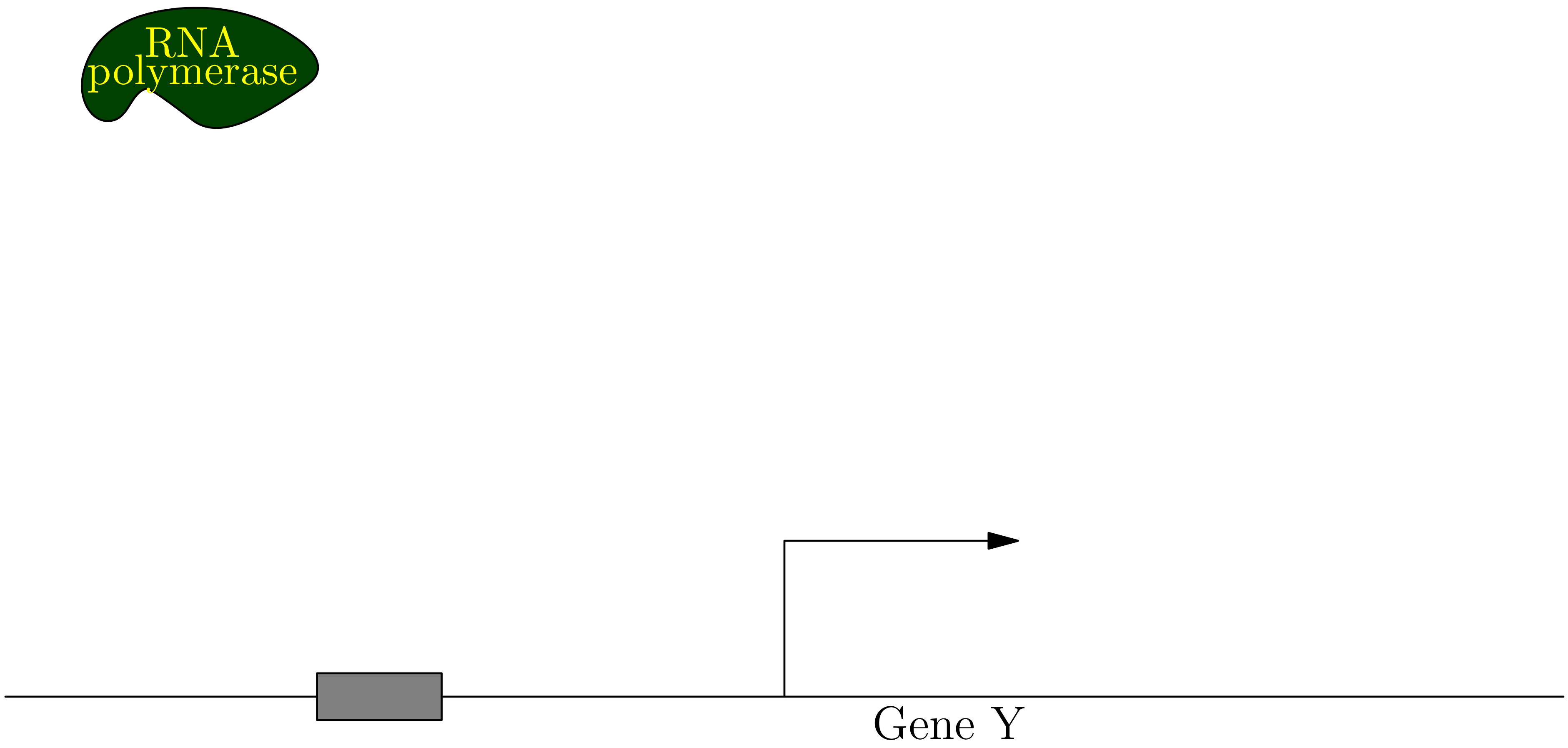

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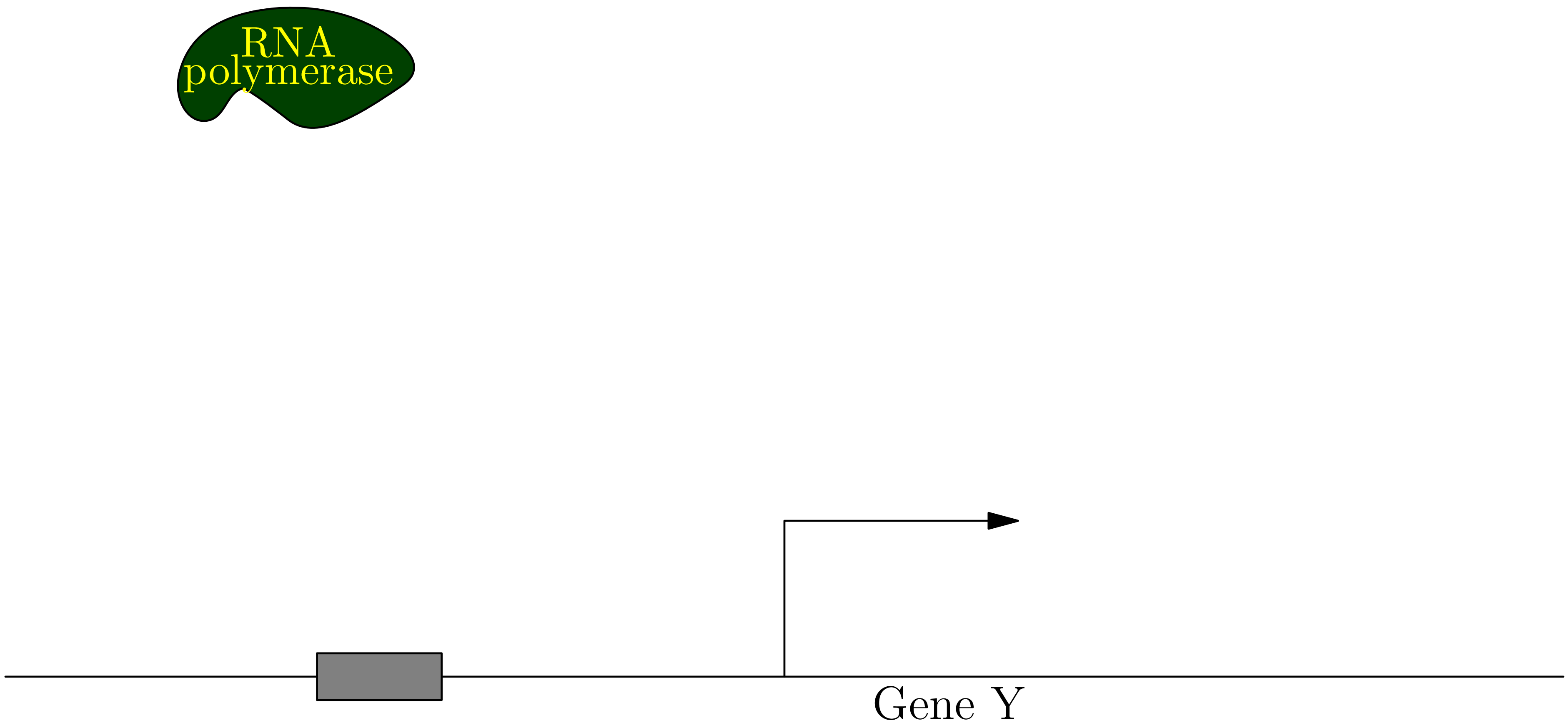

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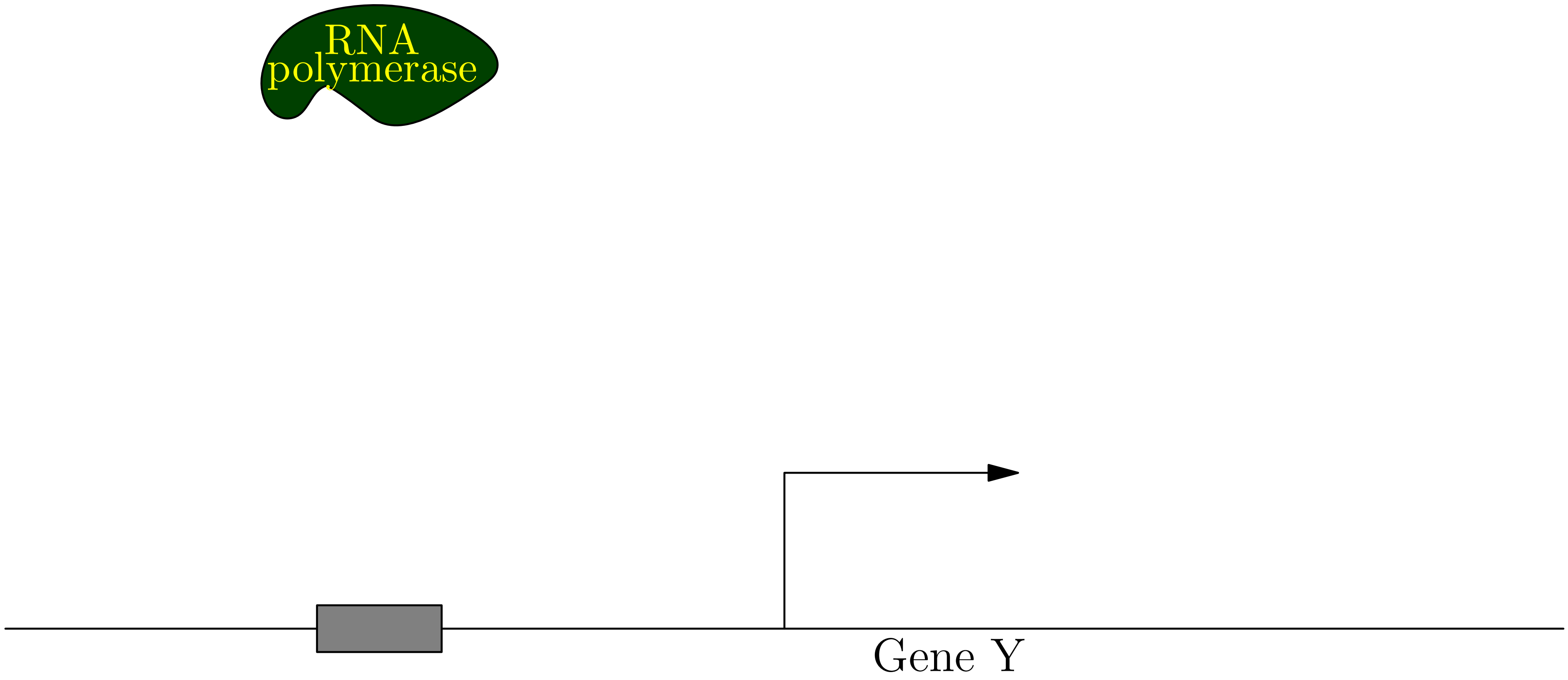

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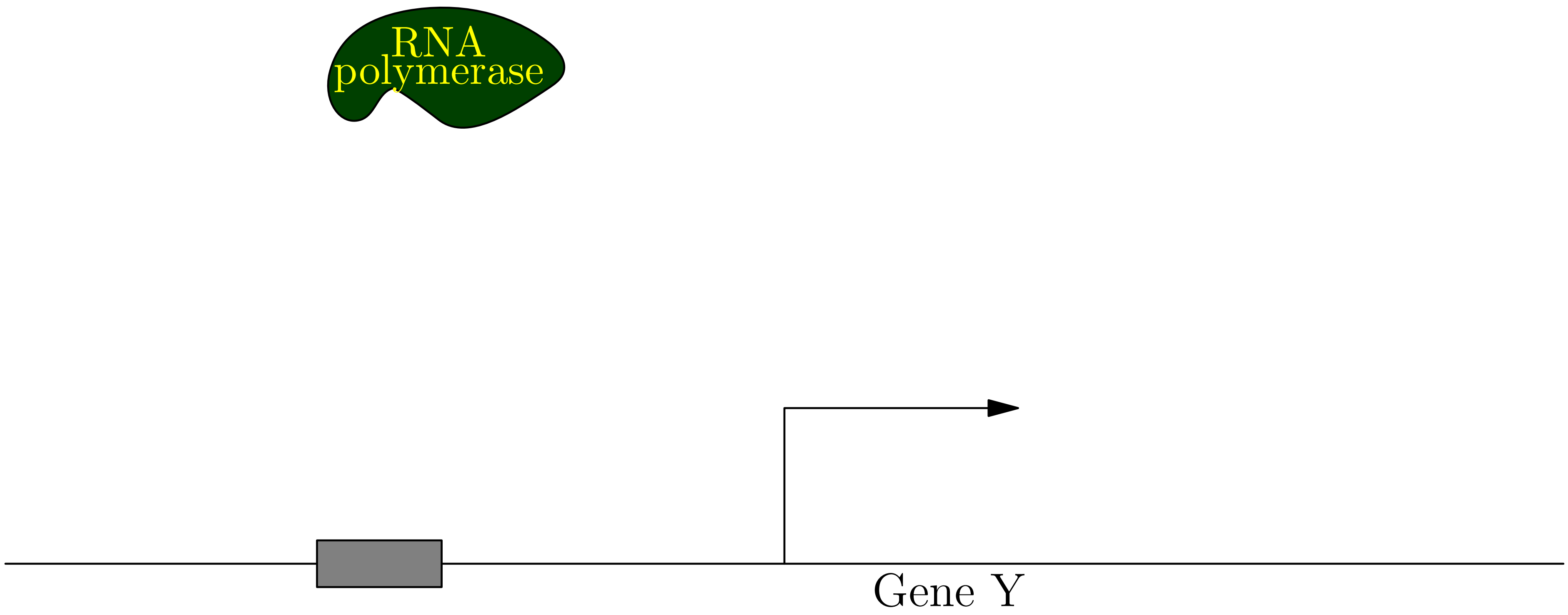

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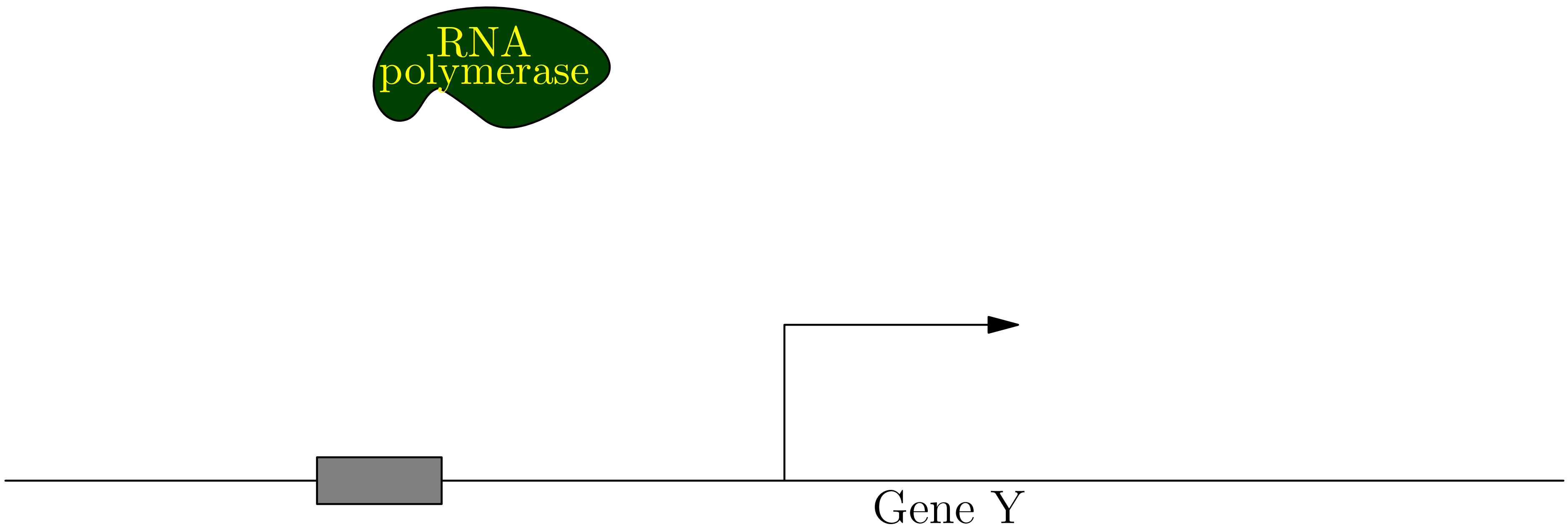

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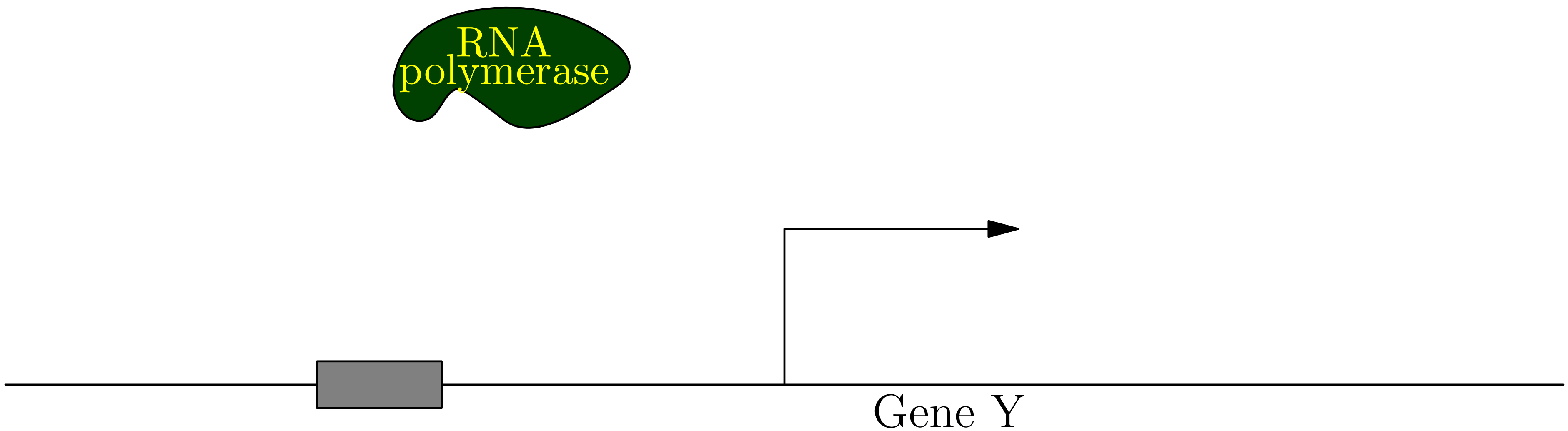

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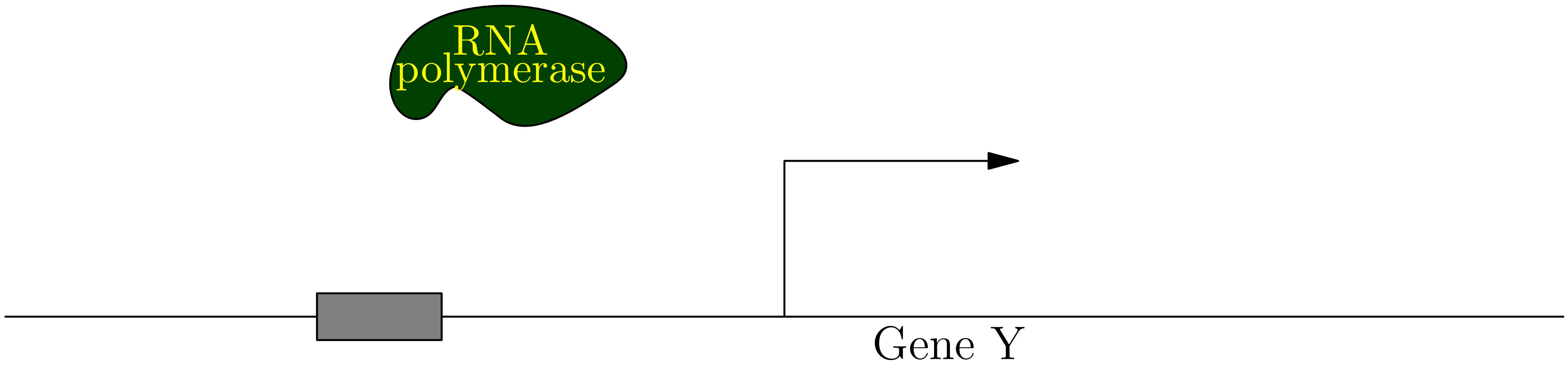

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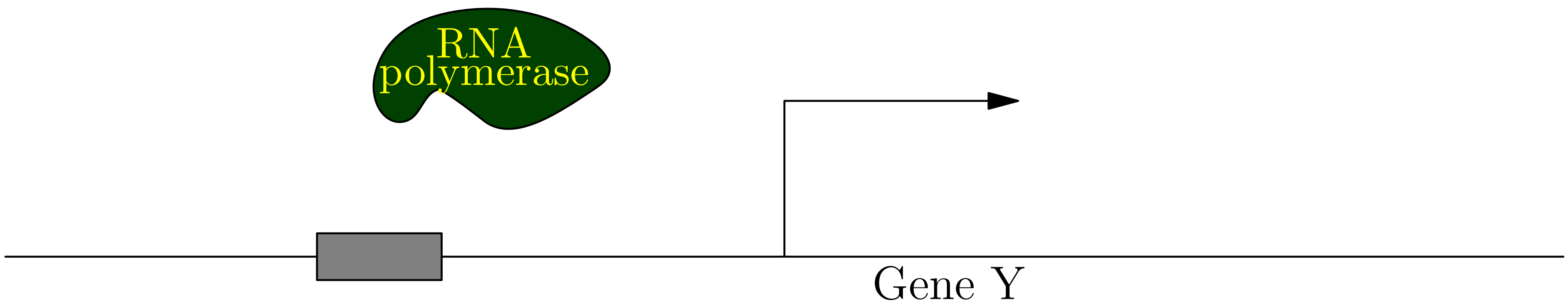

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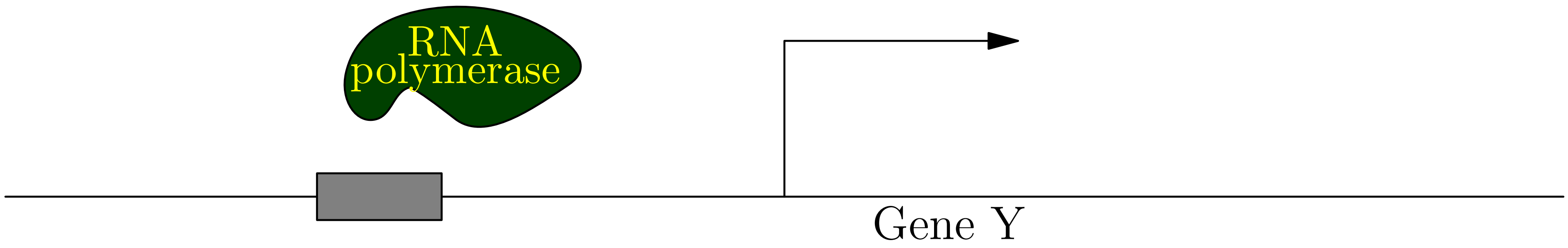

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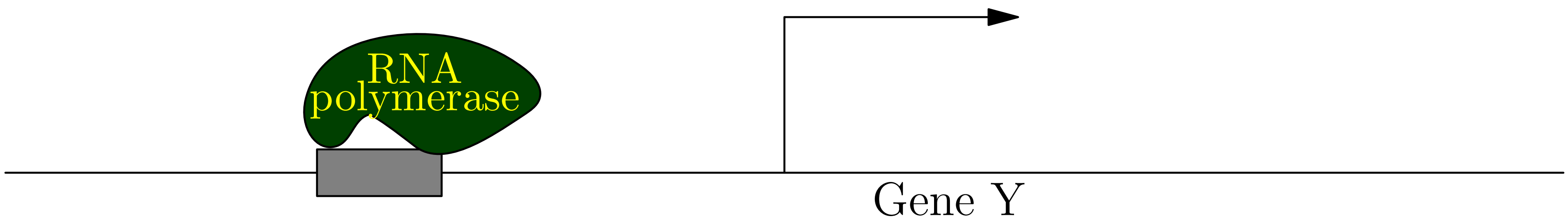

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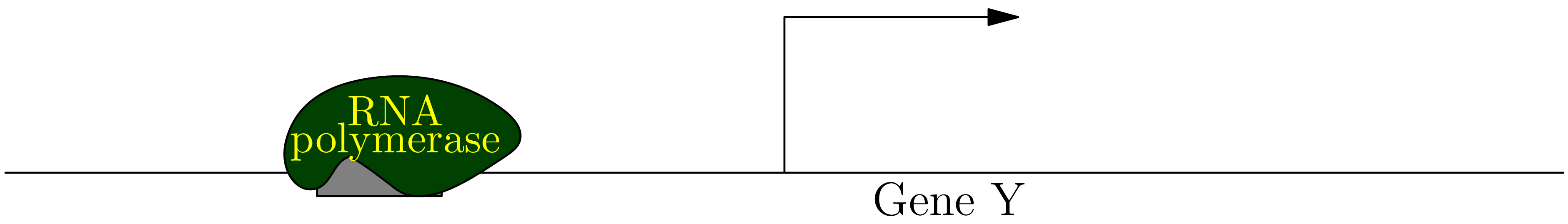

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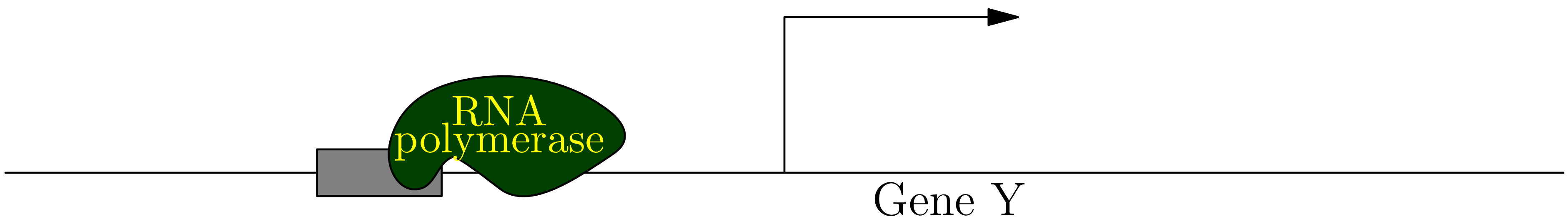

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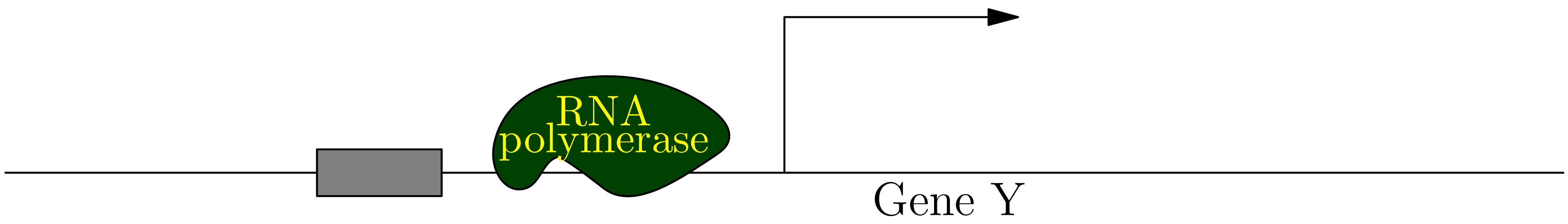

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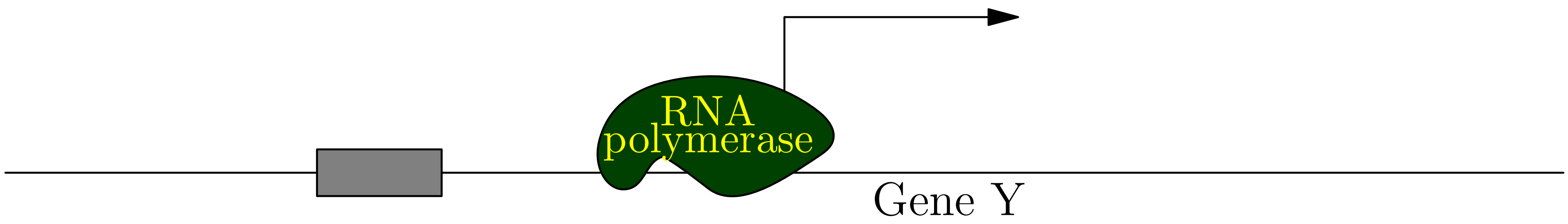

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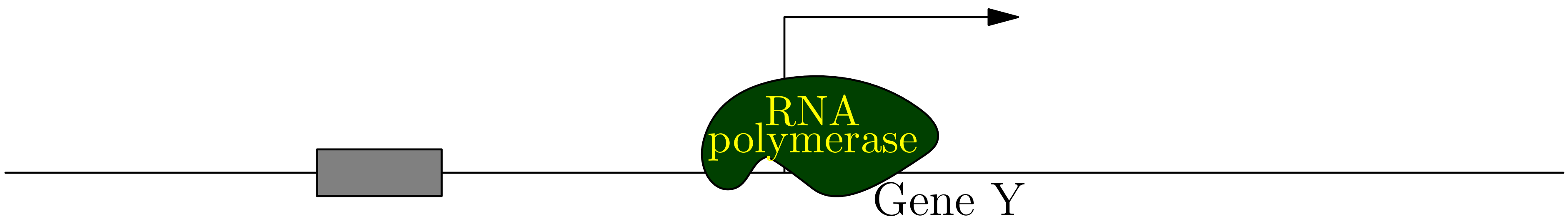

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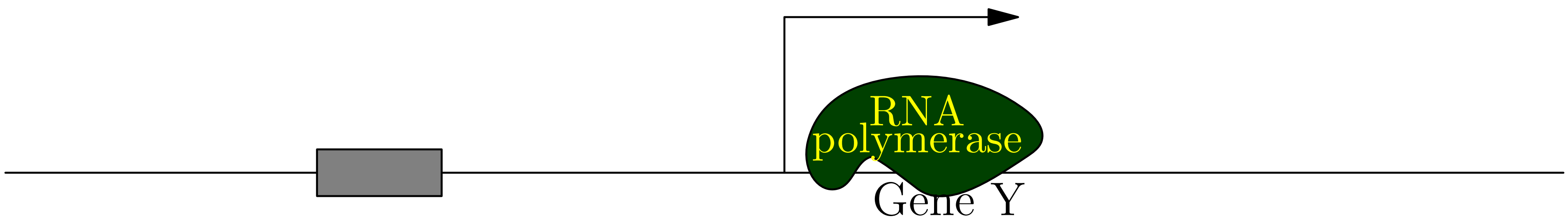

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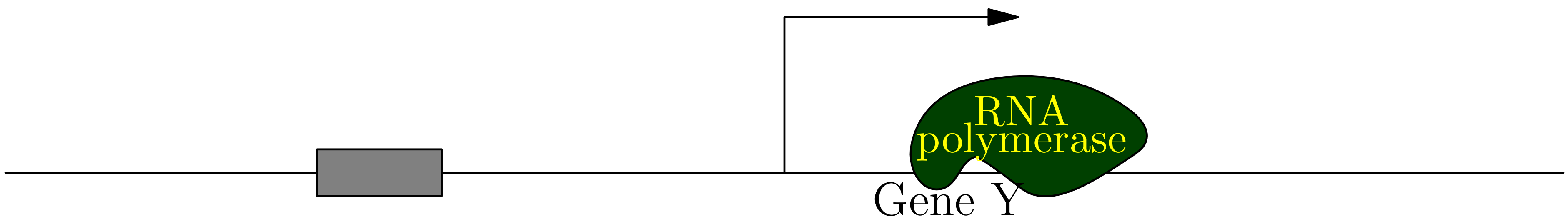

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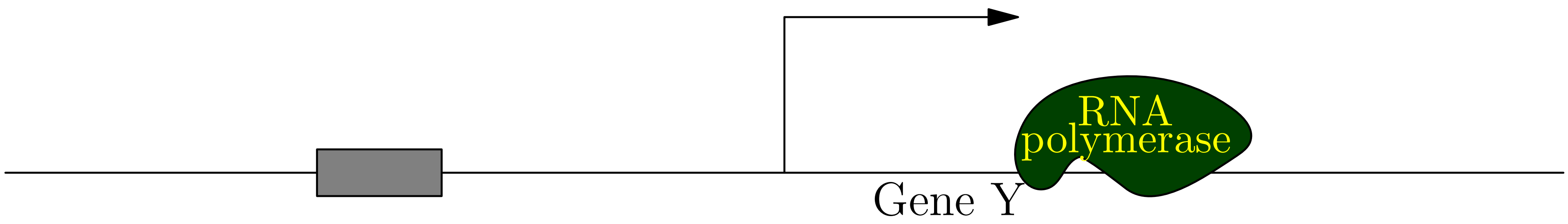

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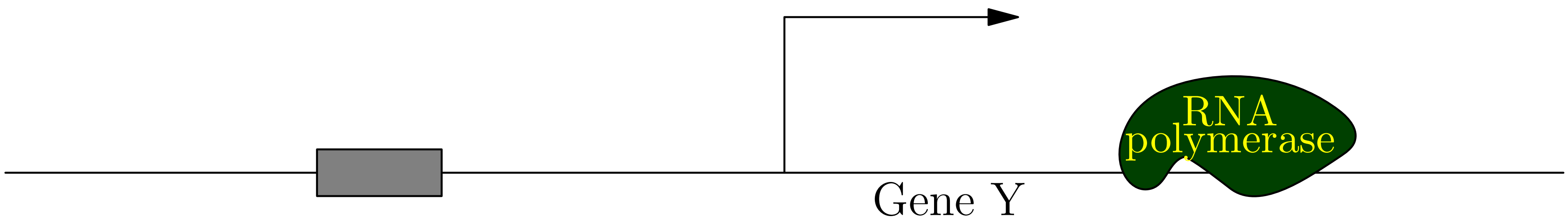

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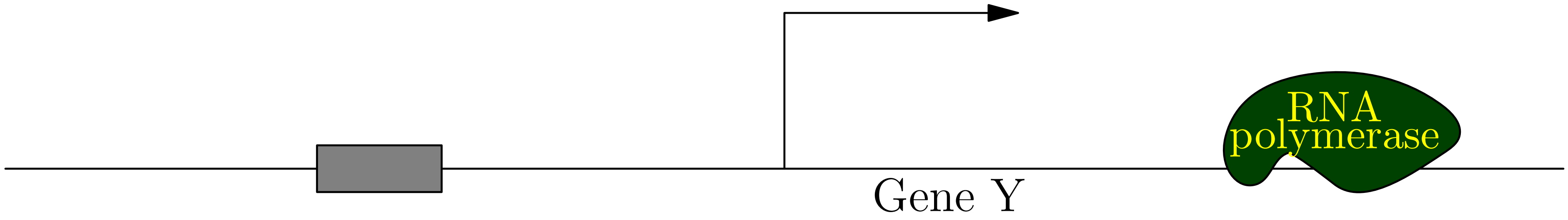

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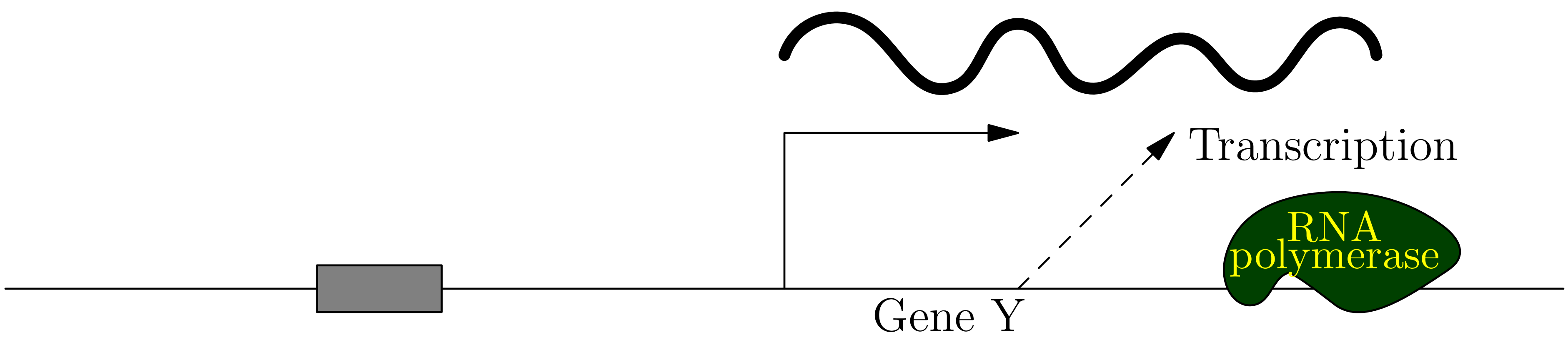

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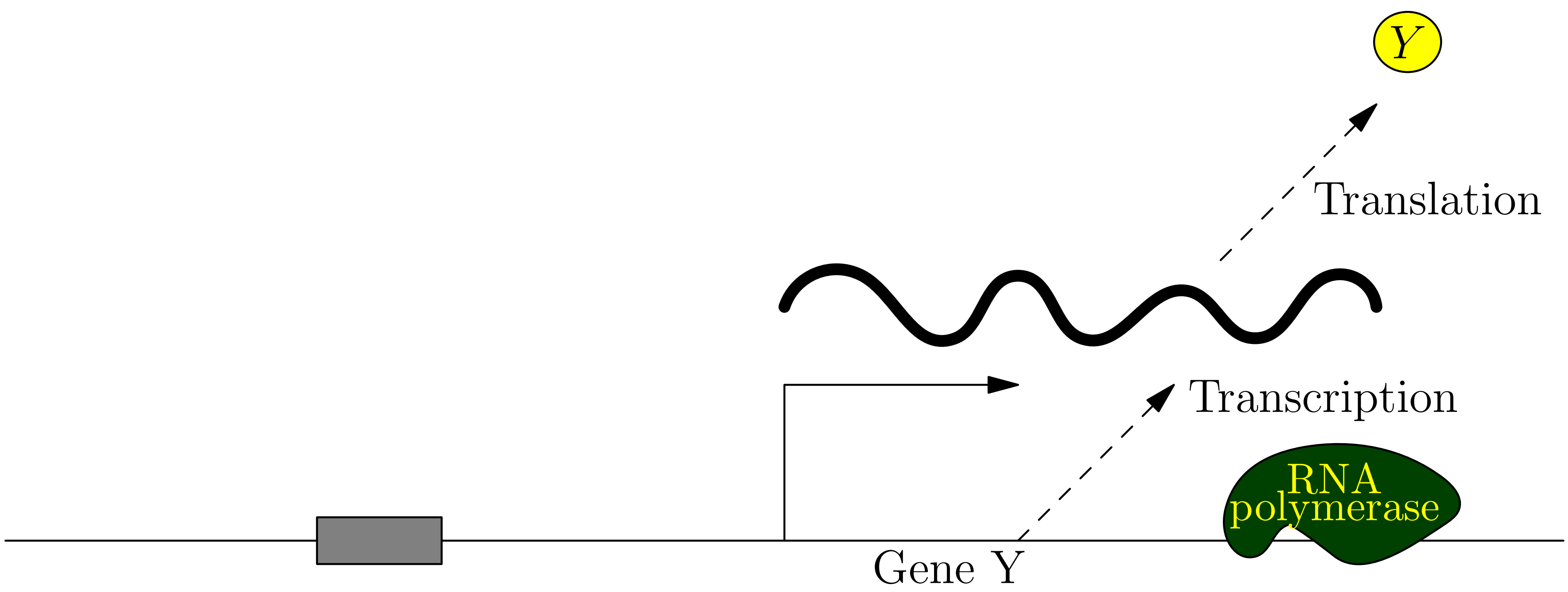

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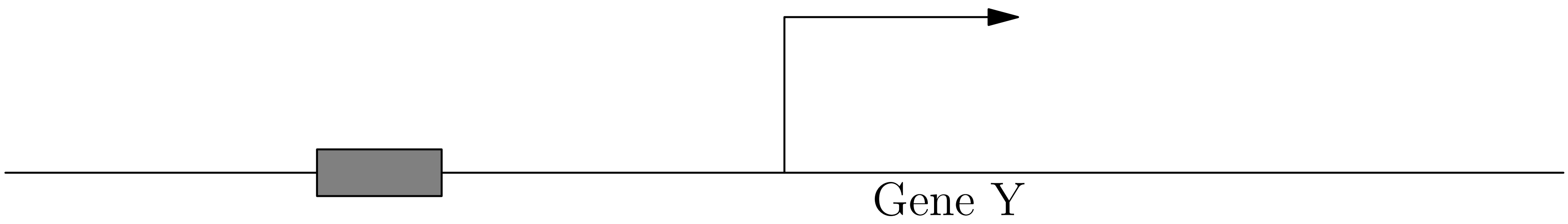

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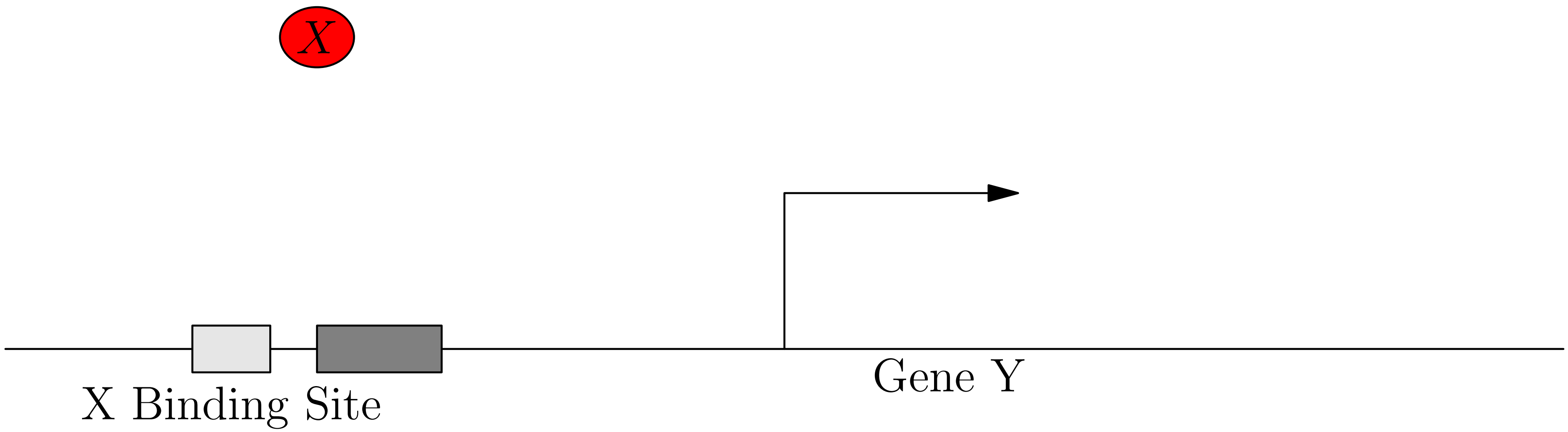

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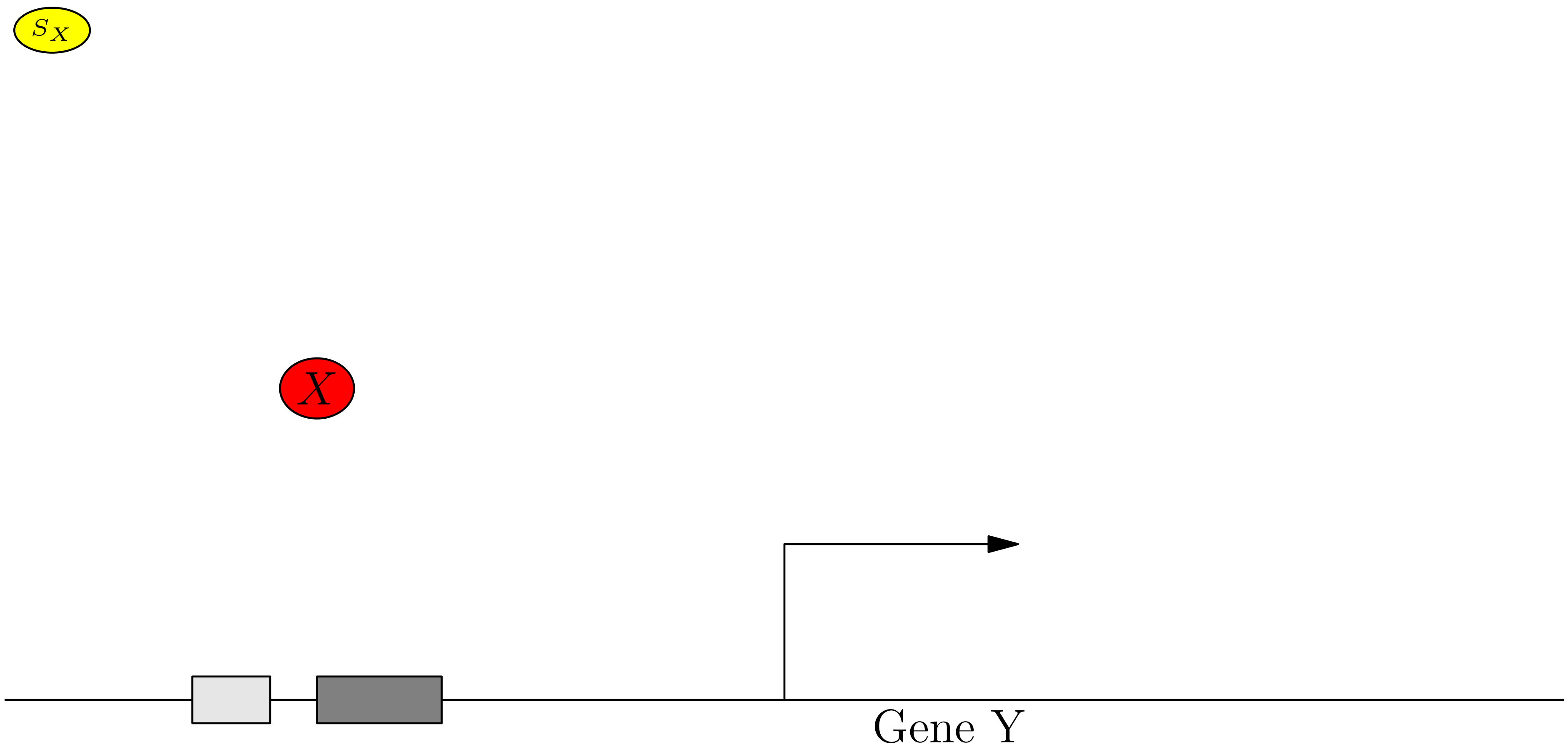

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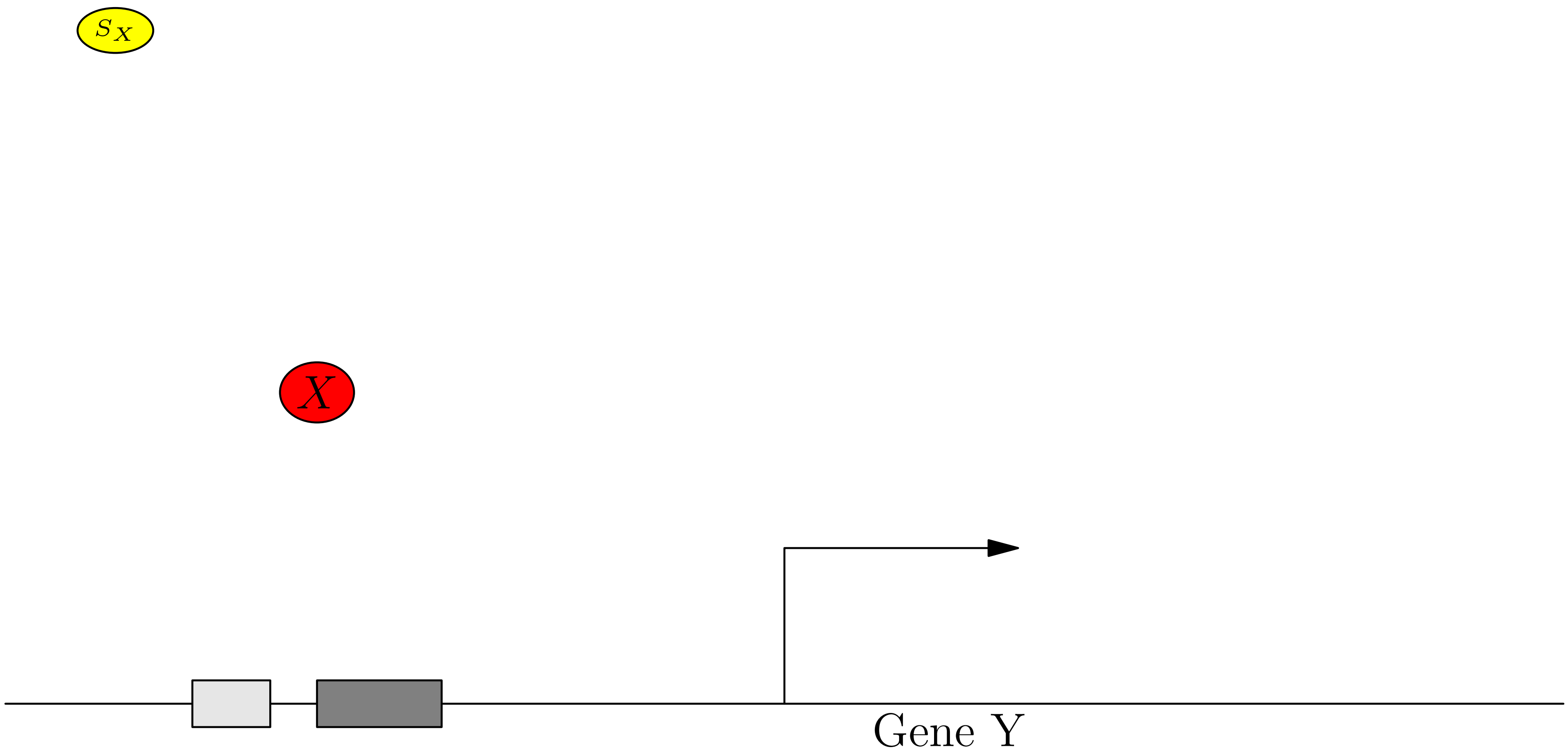

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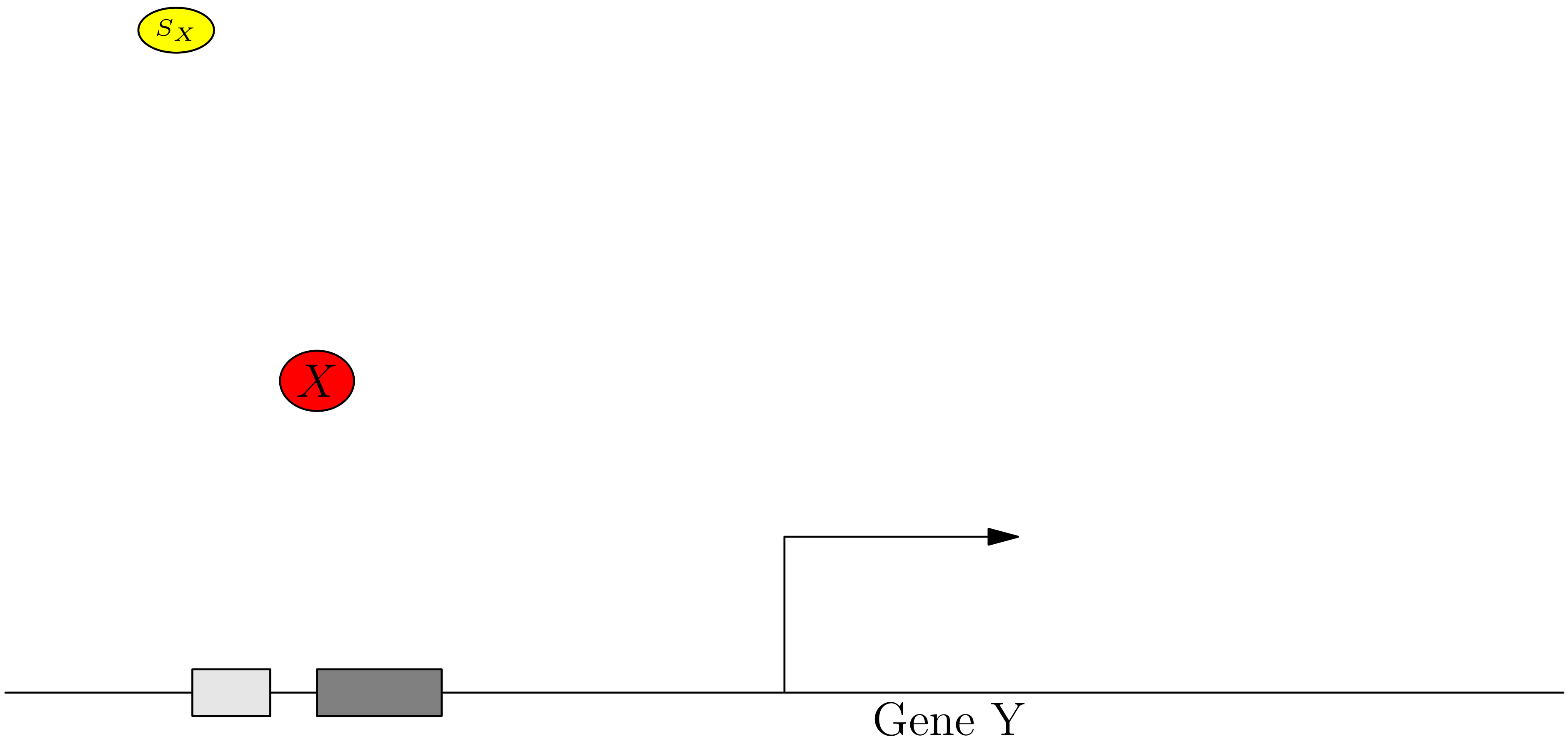

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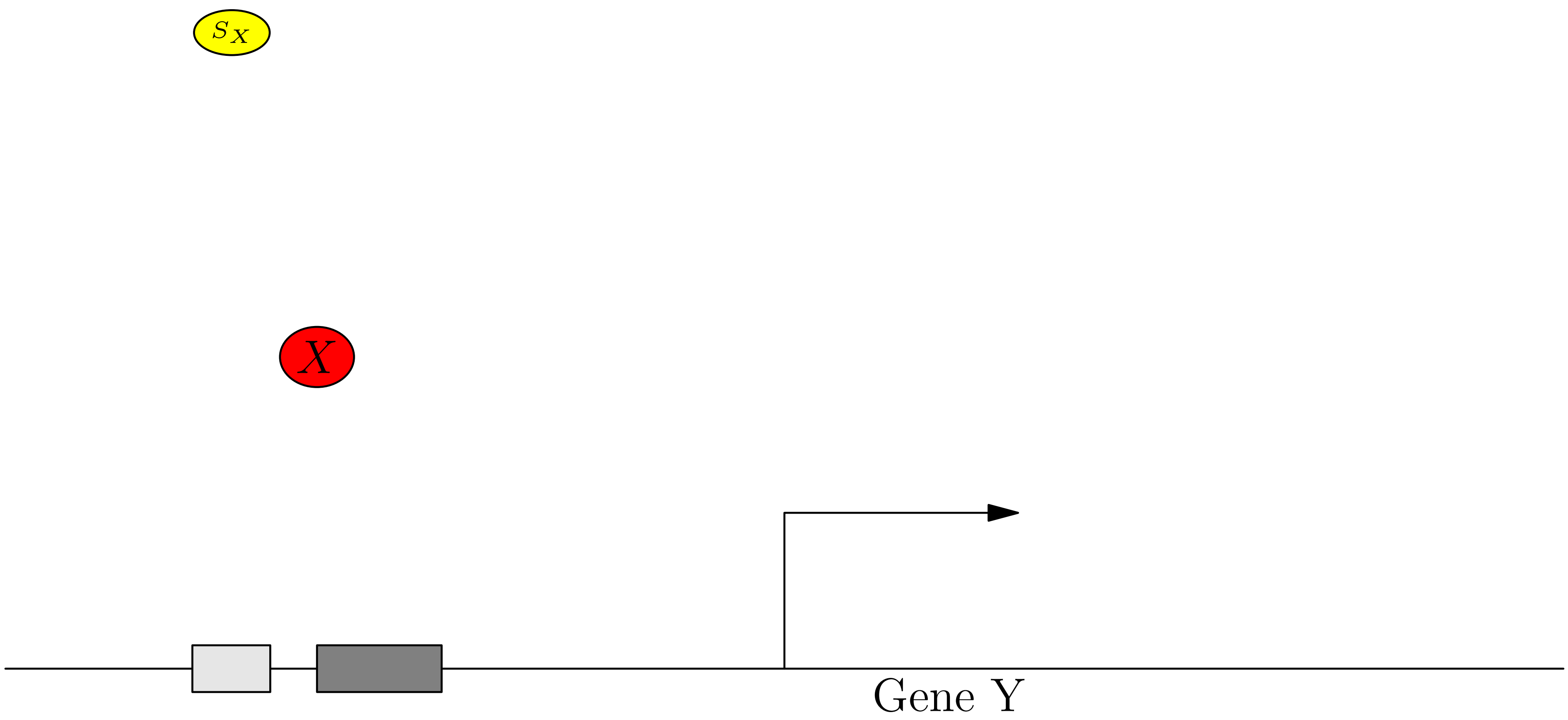

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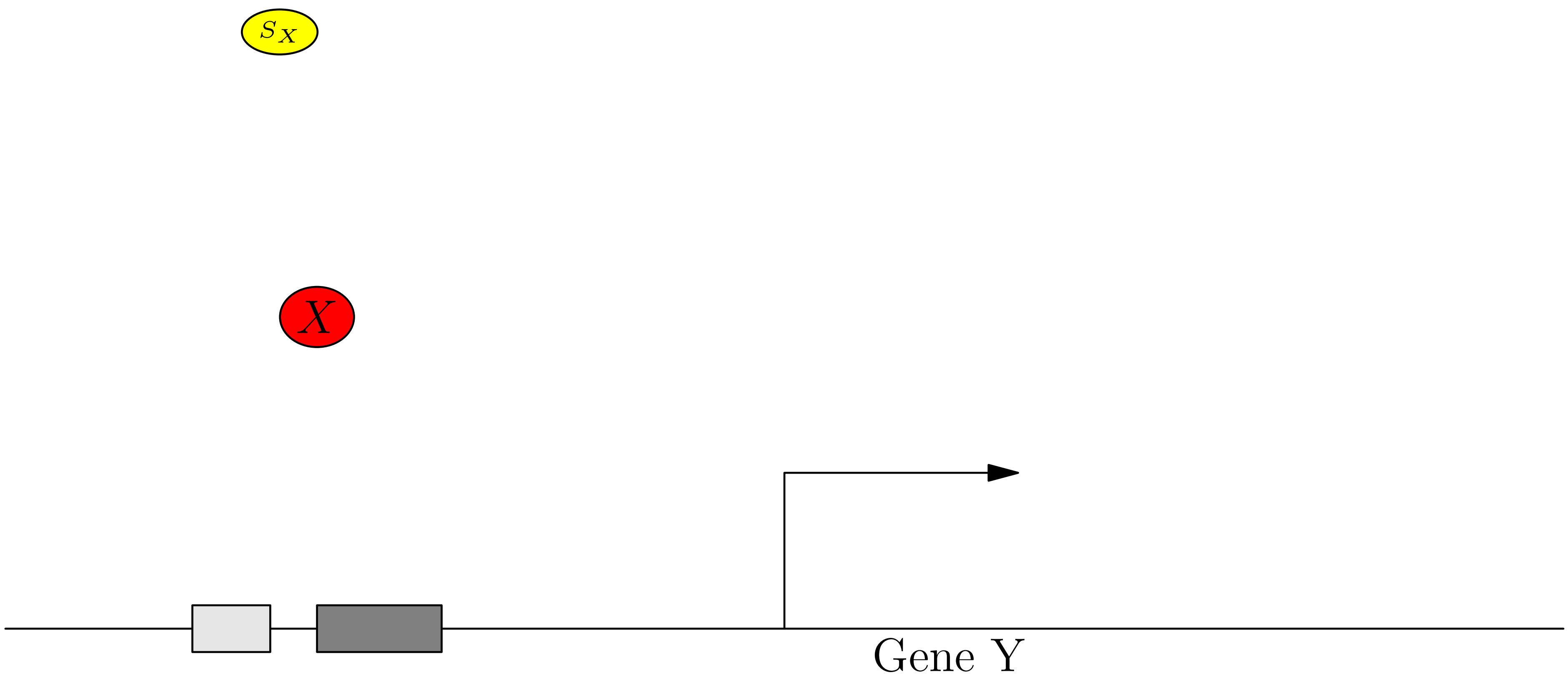

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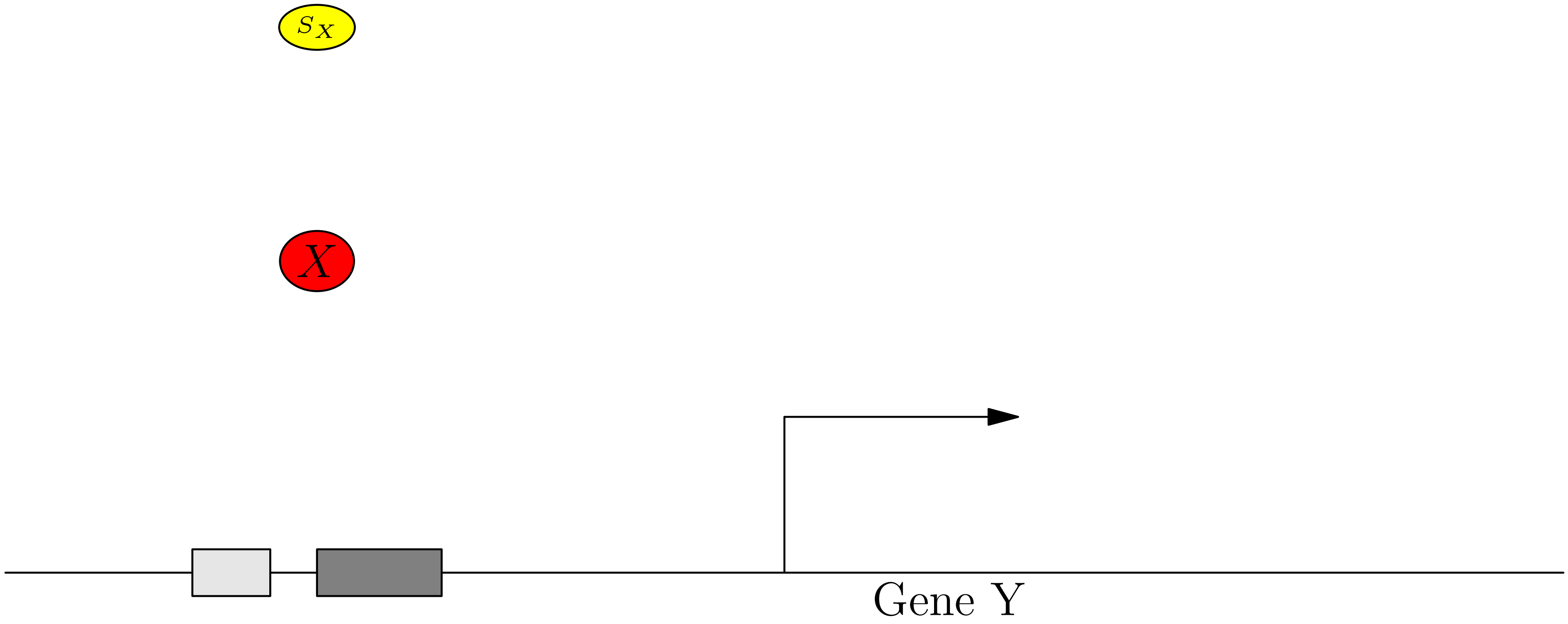

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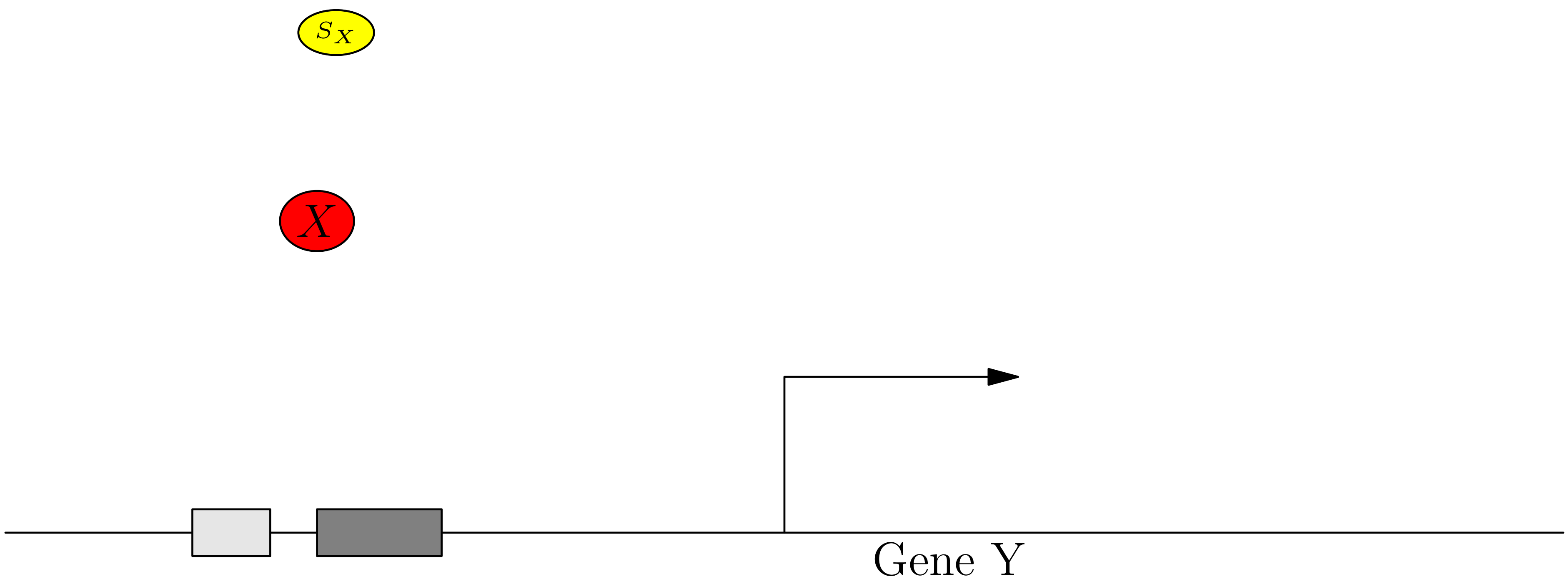

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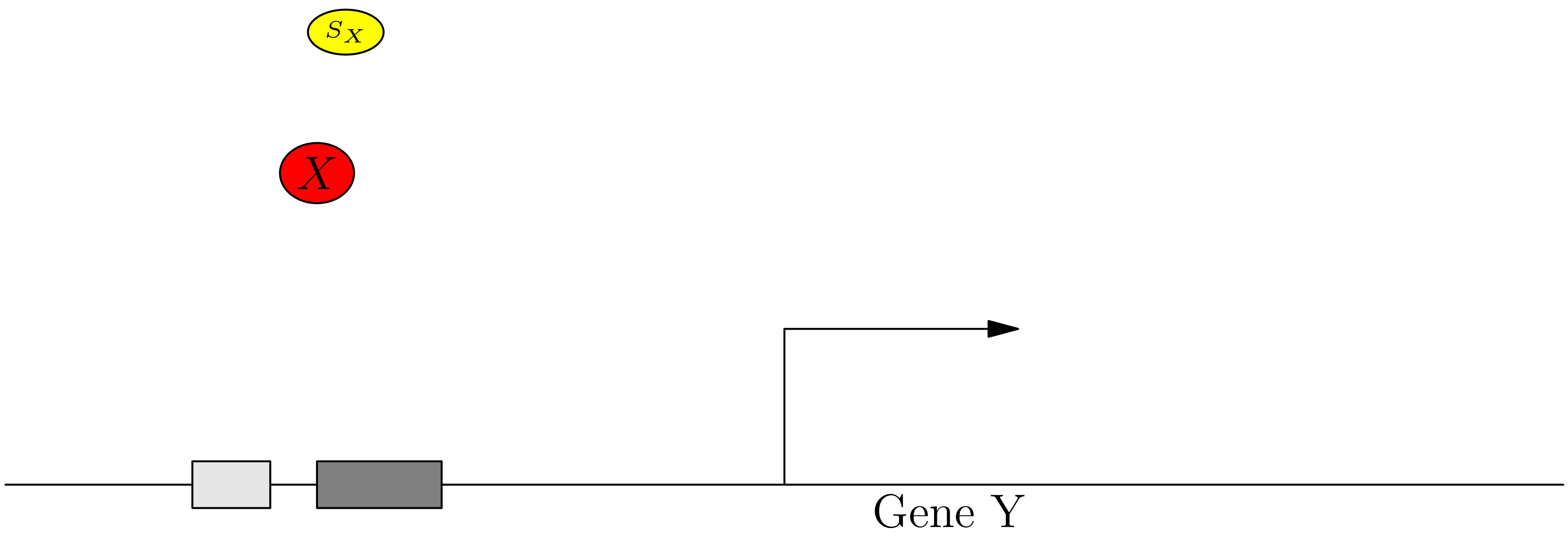

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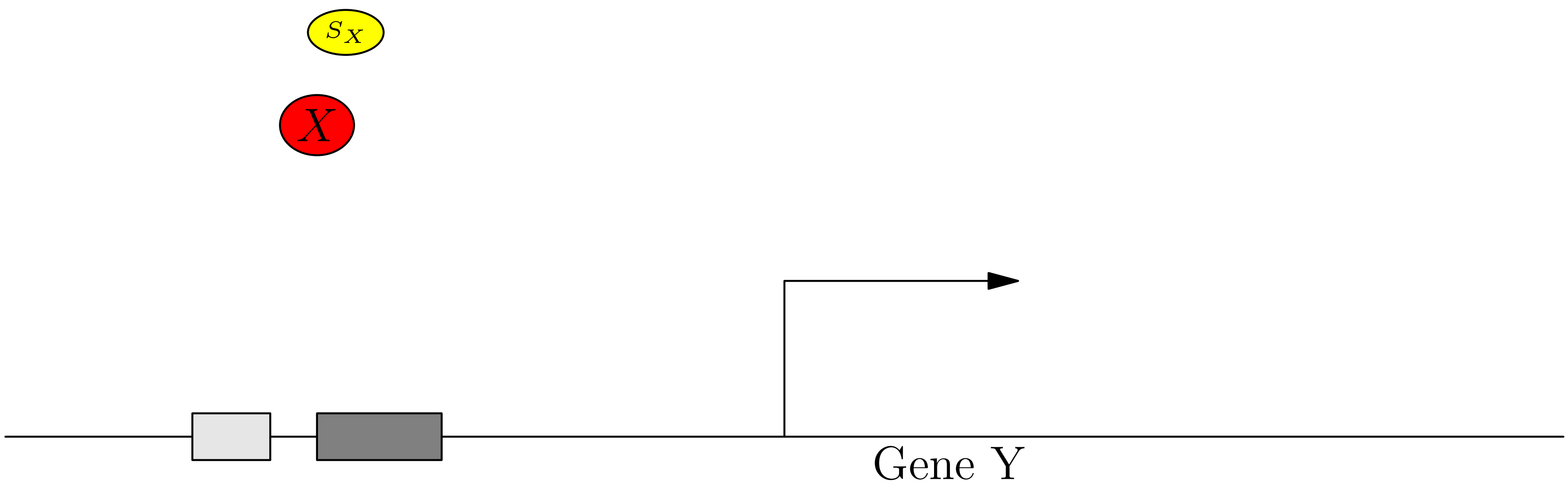

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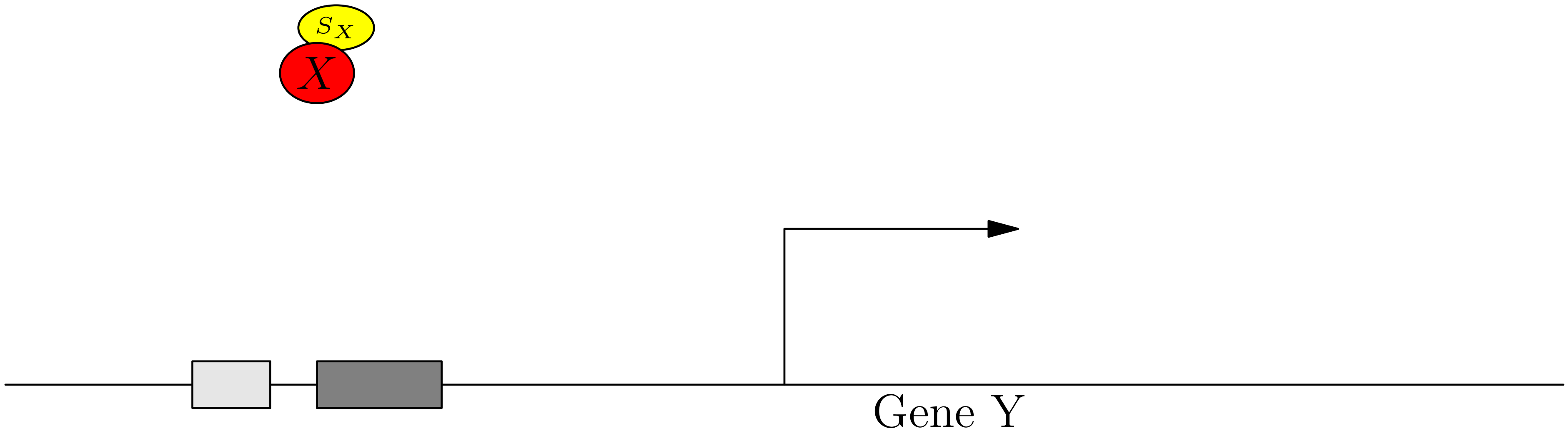

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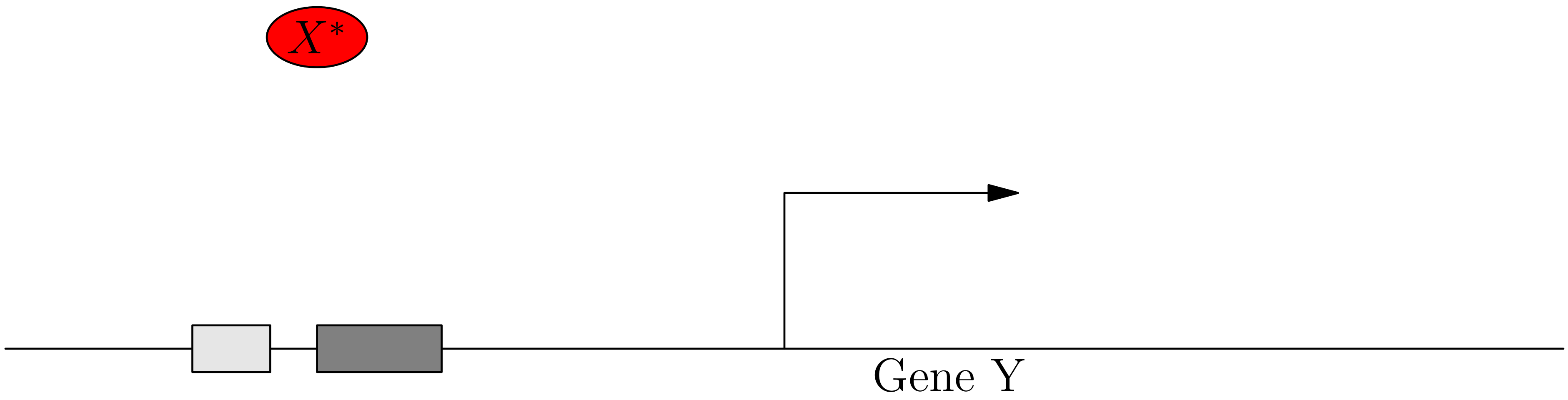

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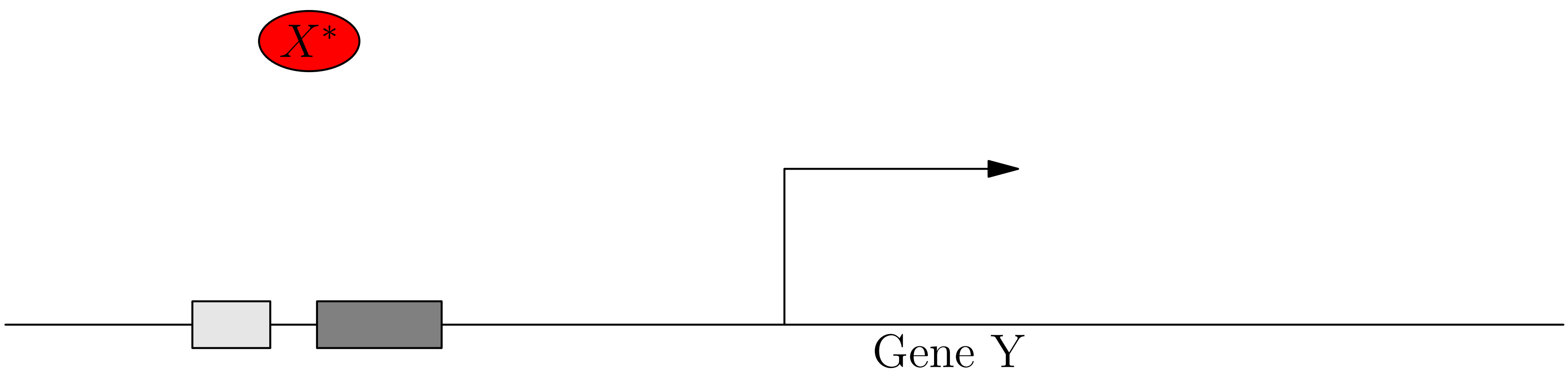

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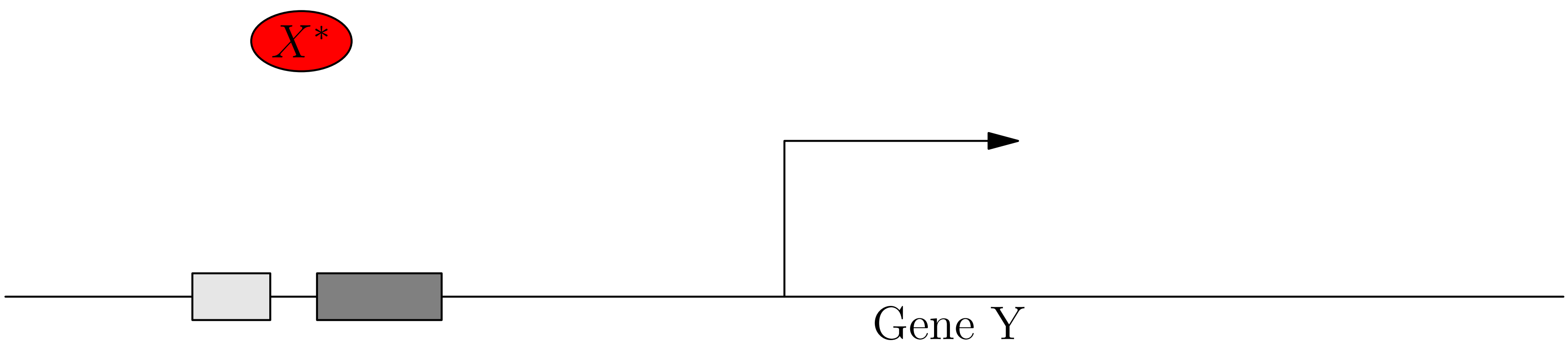

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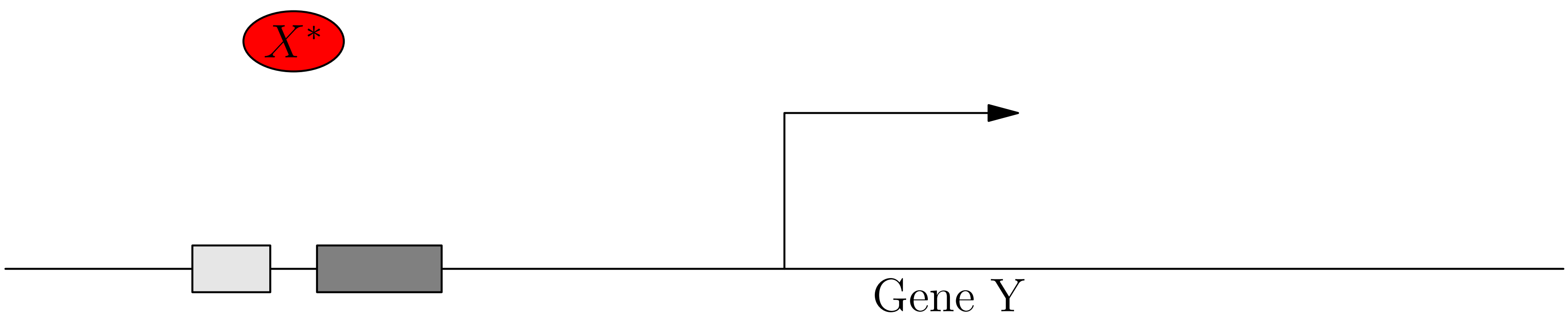

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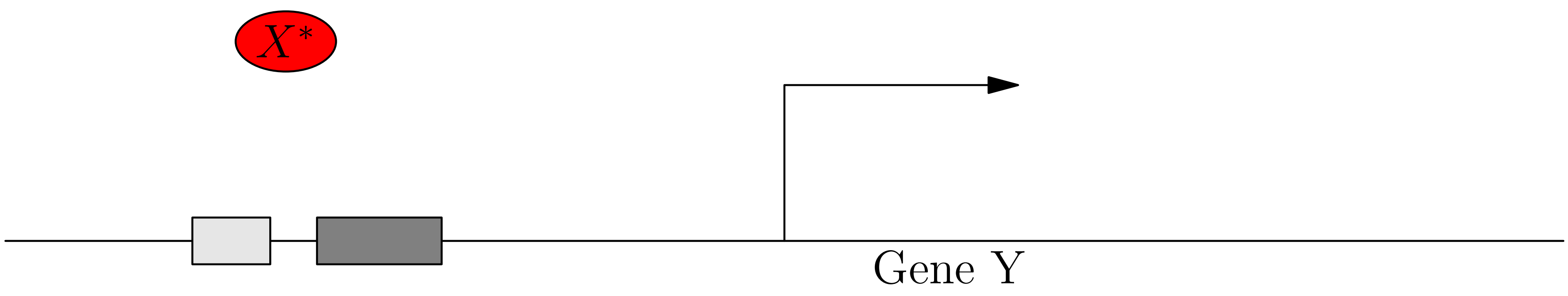

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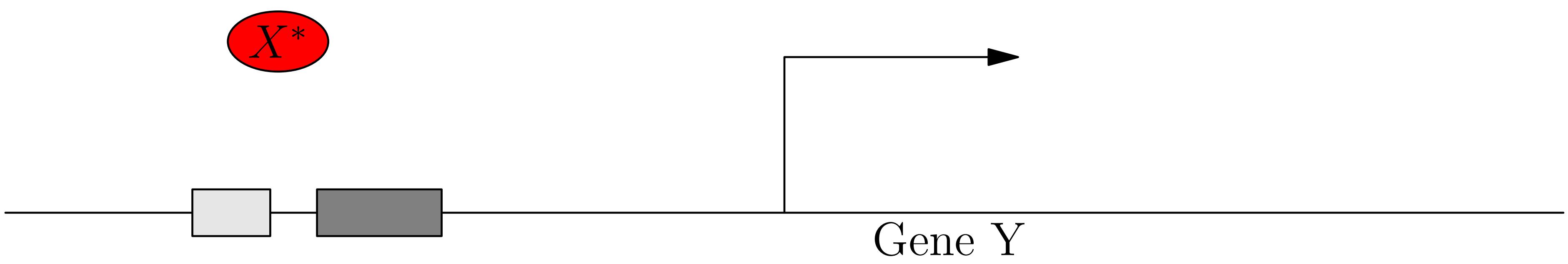

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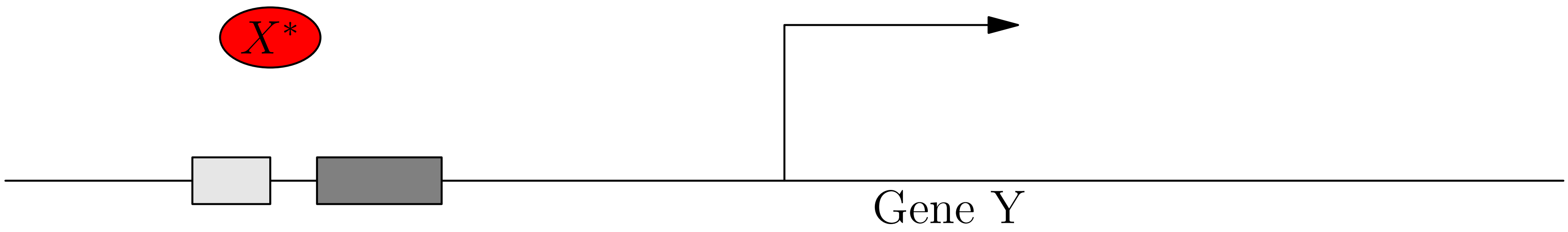

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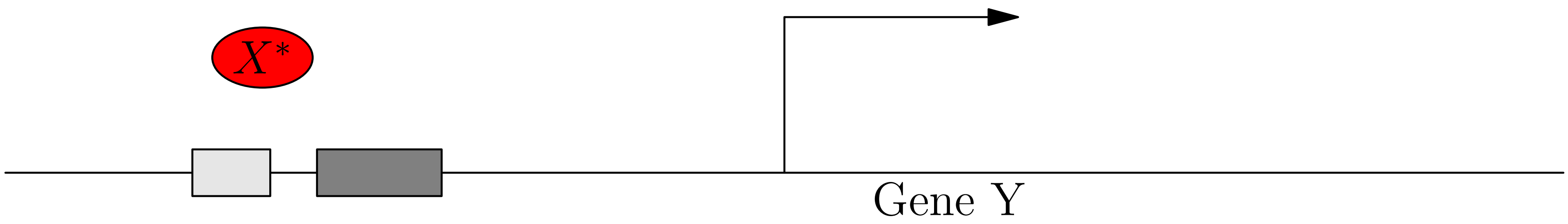

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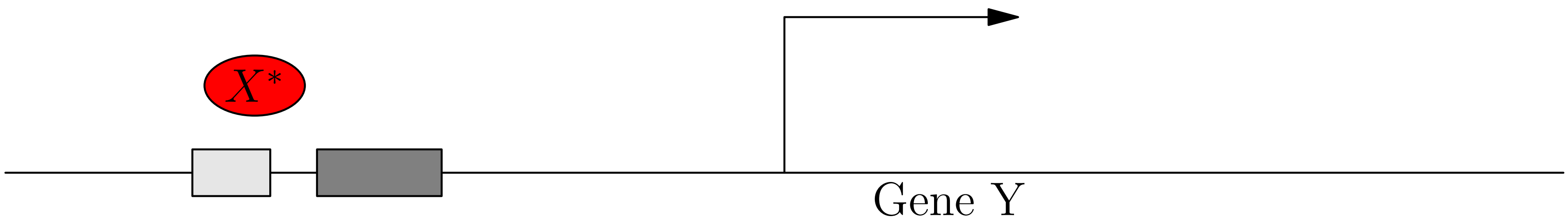

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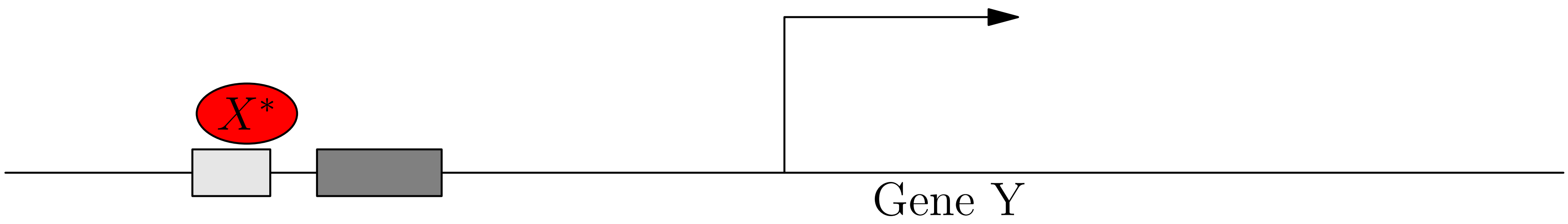

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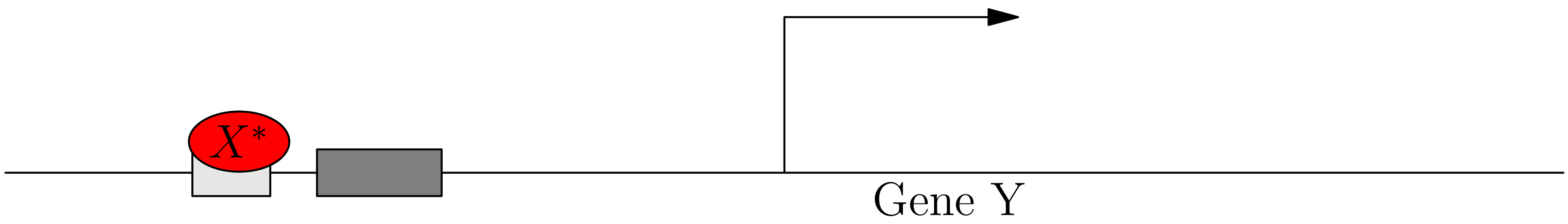

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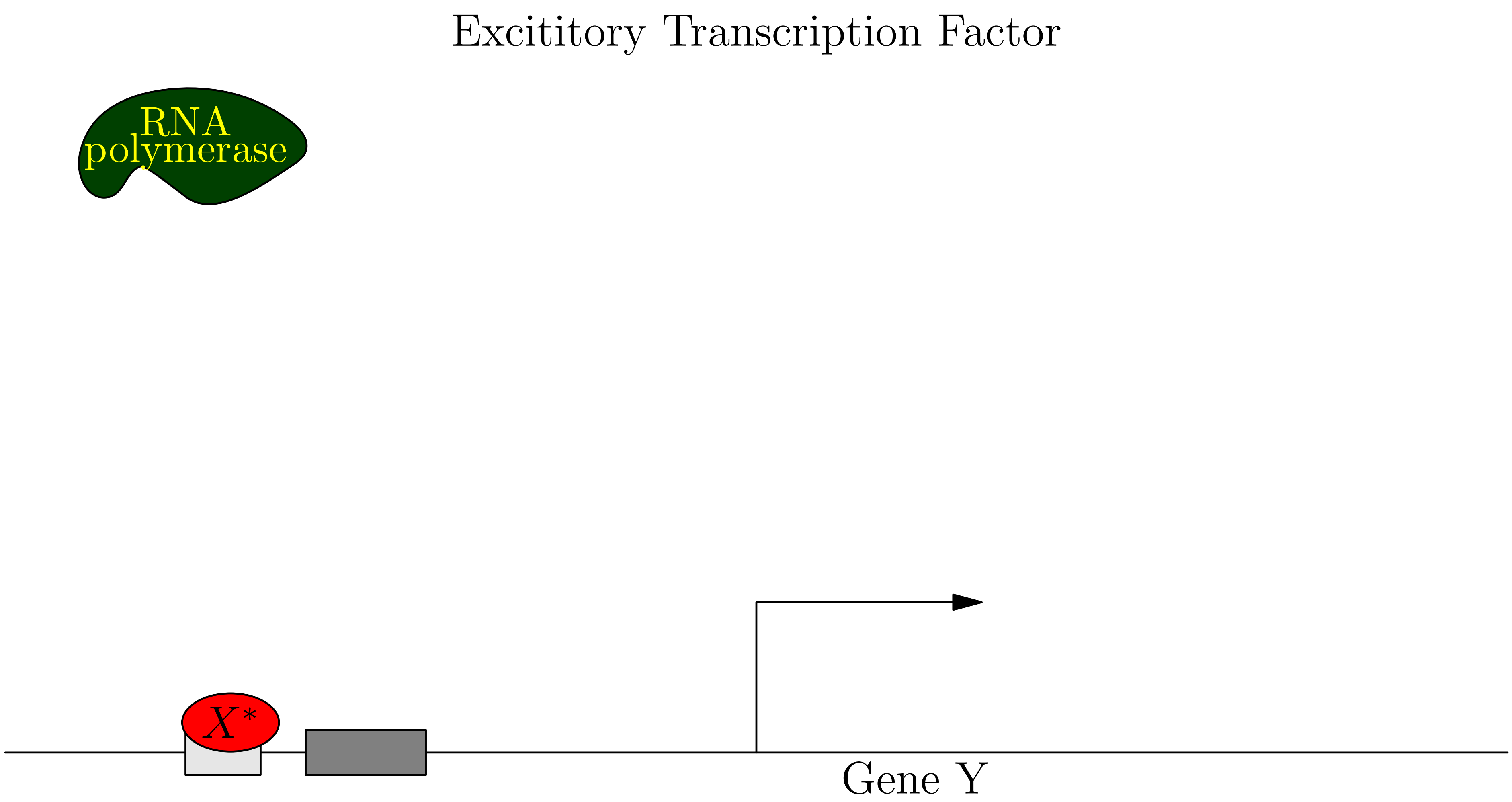
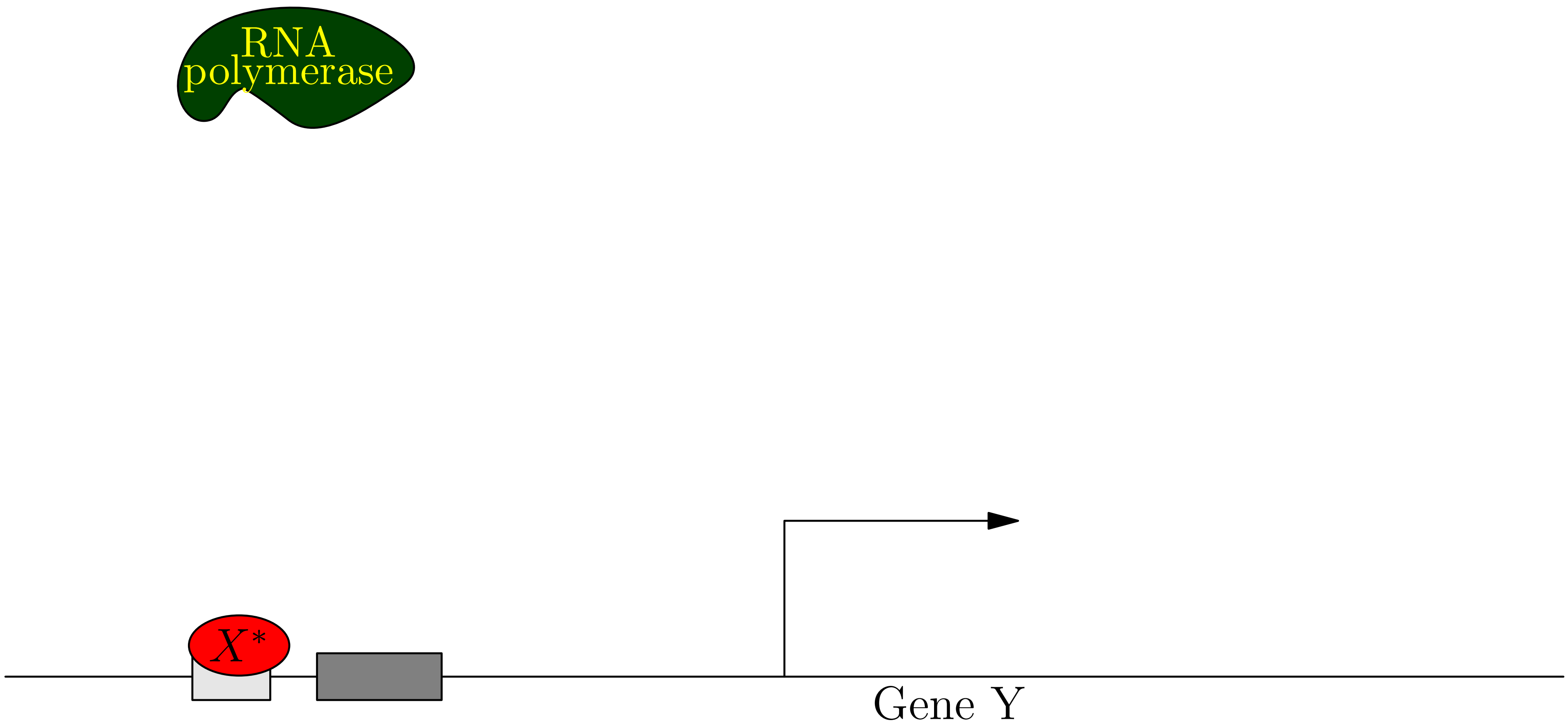


Gene Transcription Regulation

Excititory Transcription Factor



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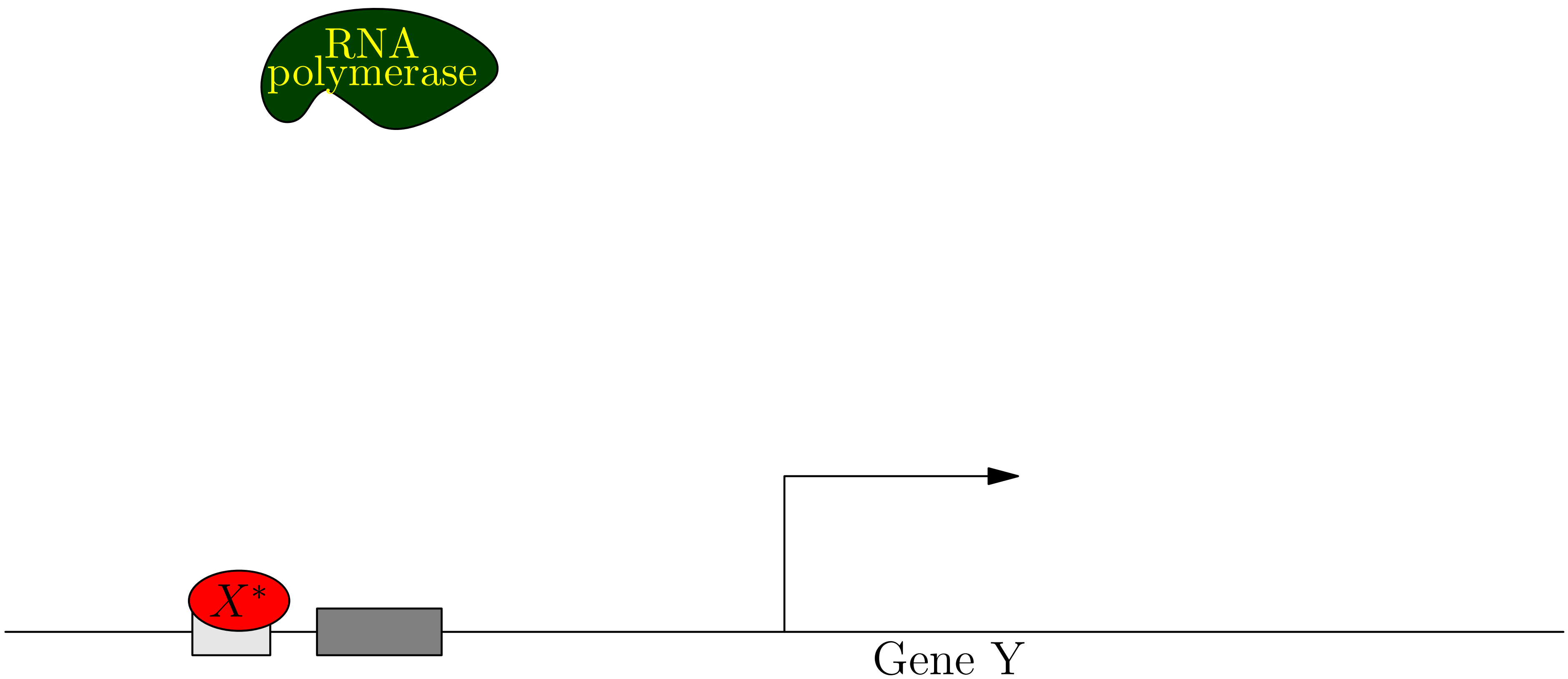

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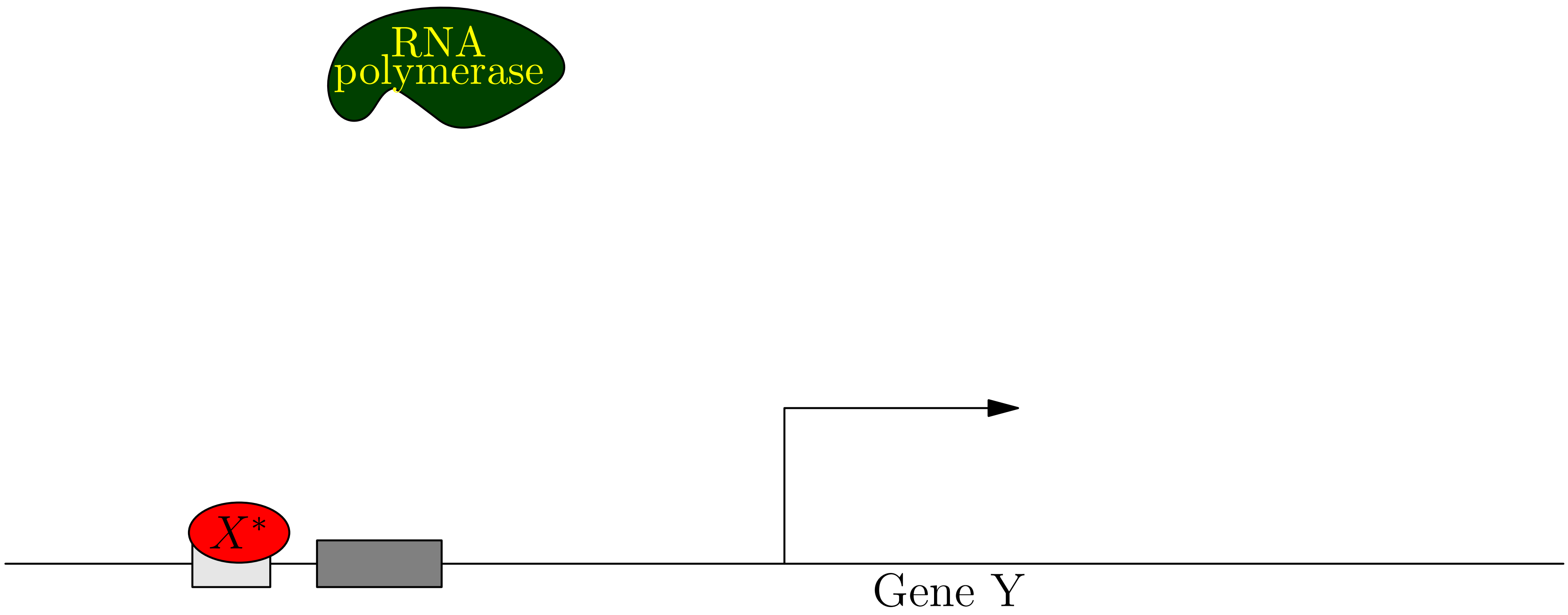

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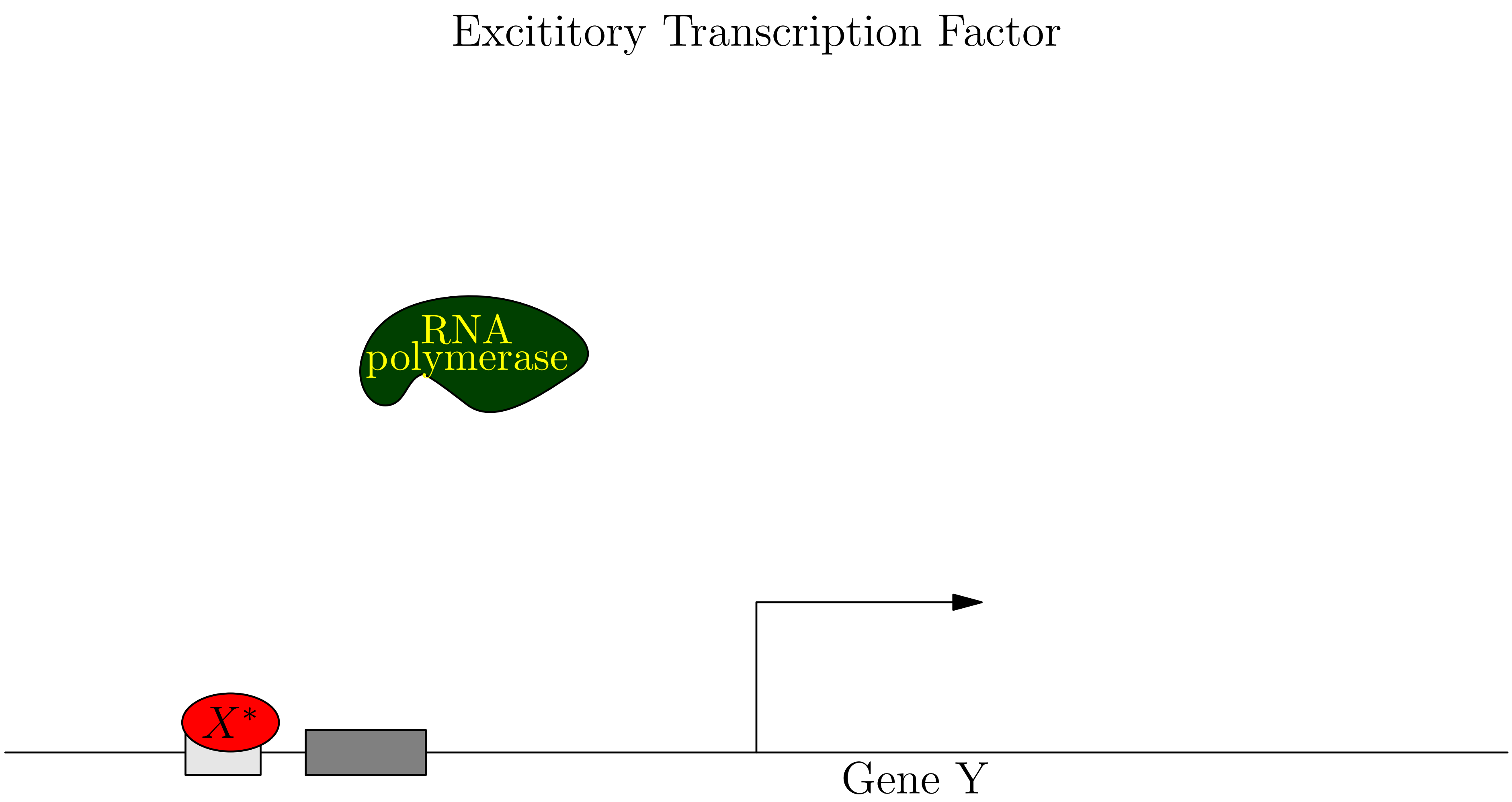

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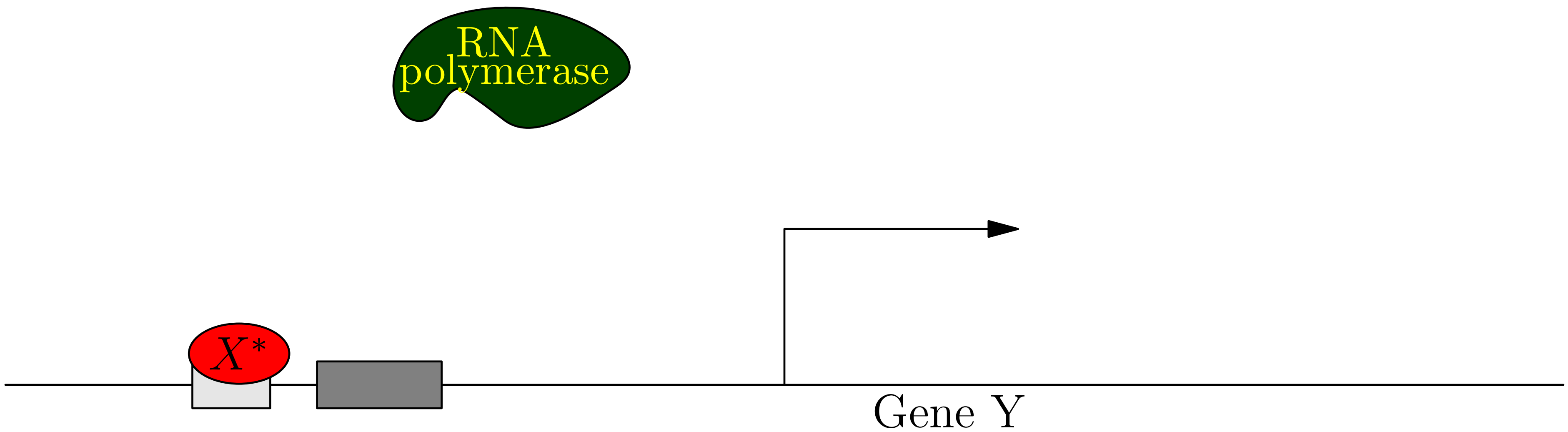

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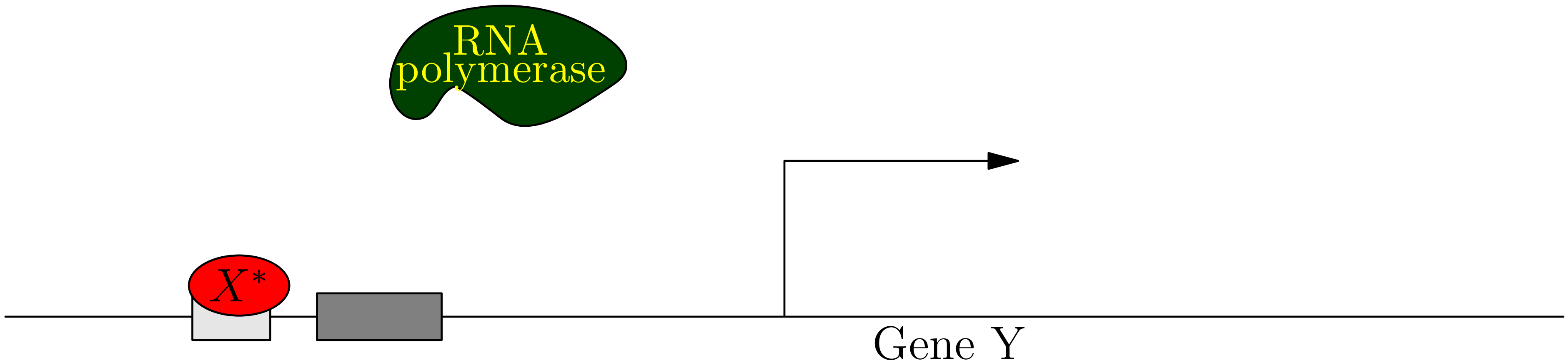

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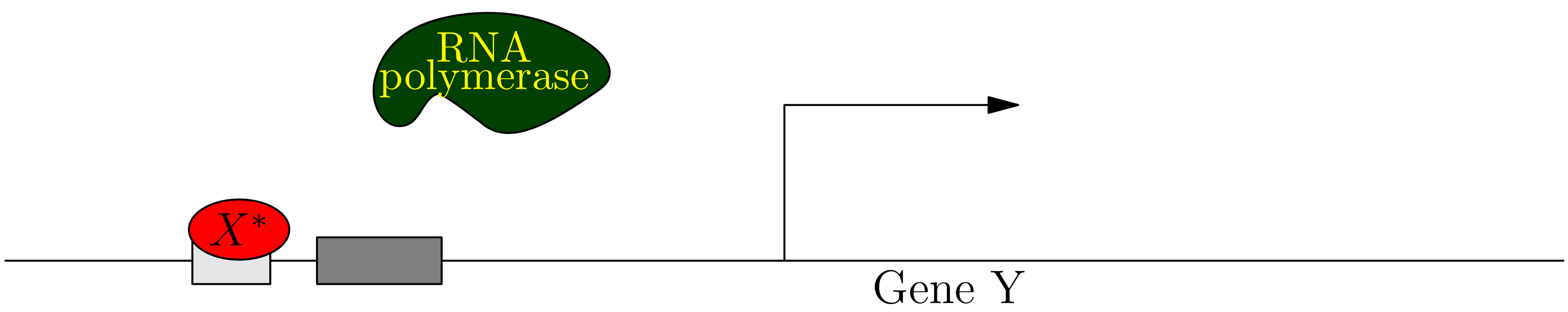

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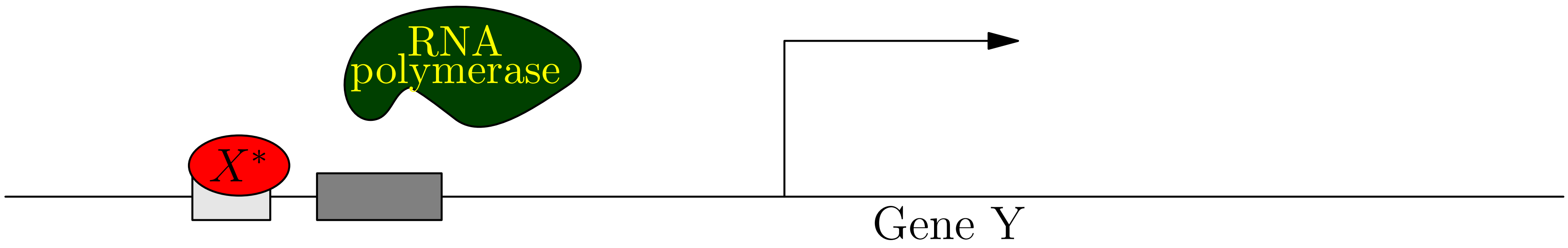

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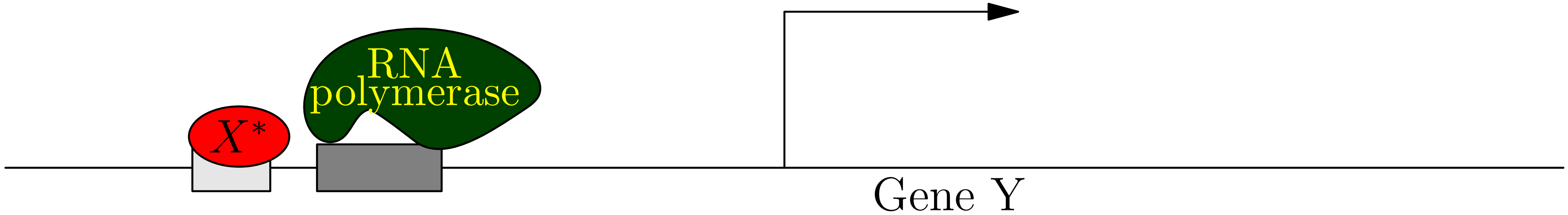

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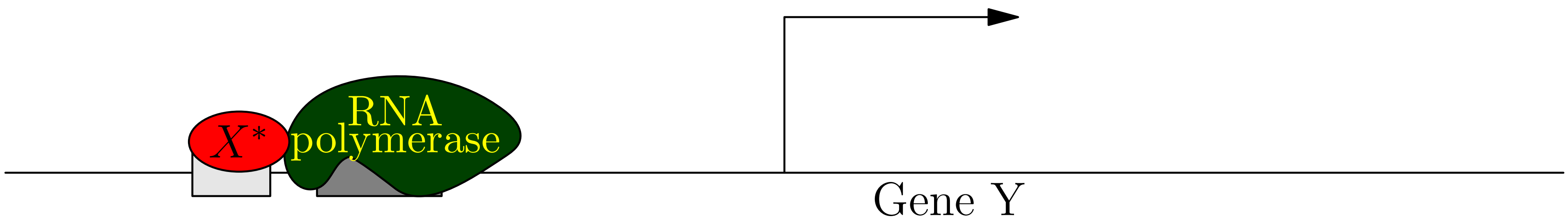

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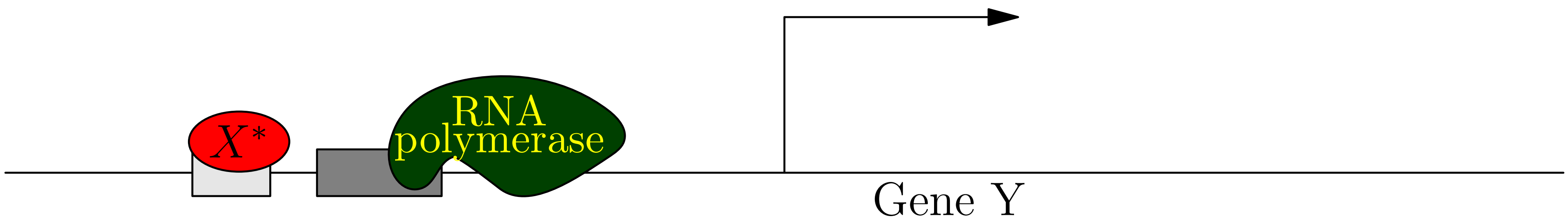

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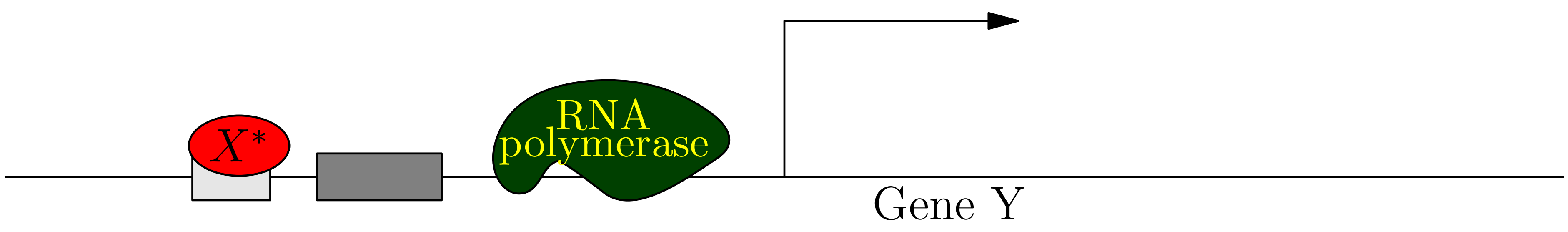

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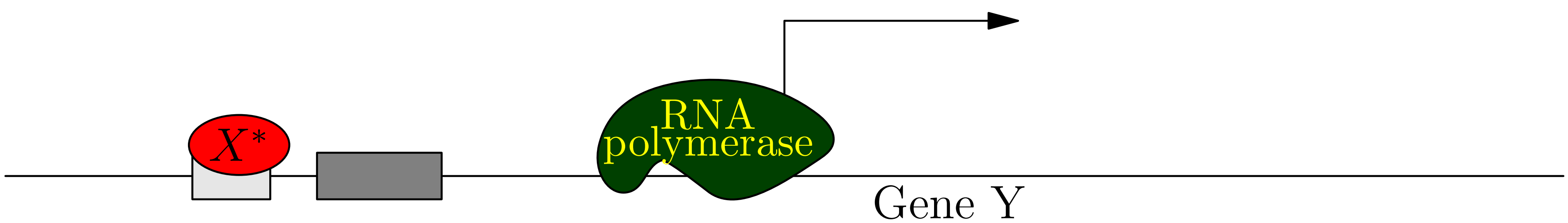

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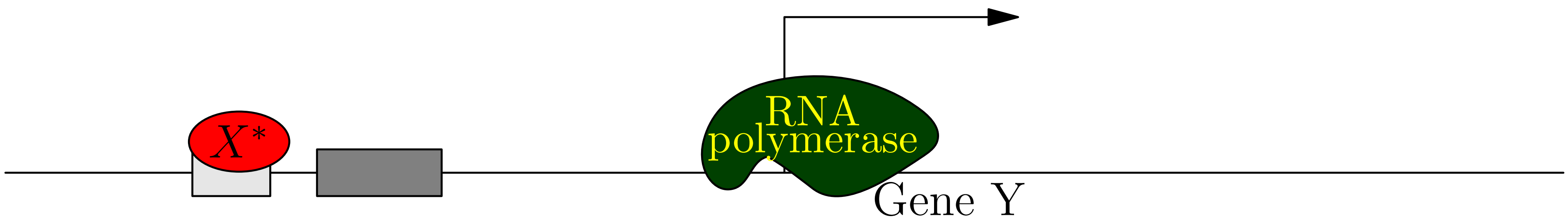

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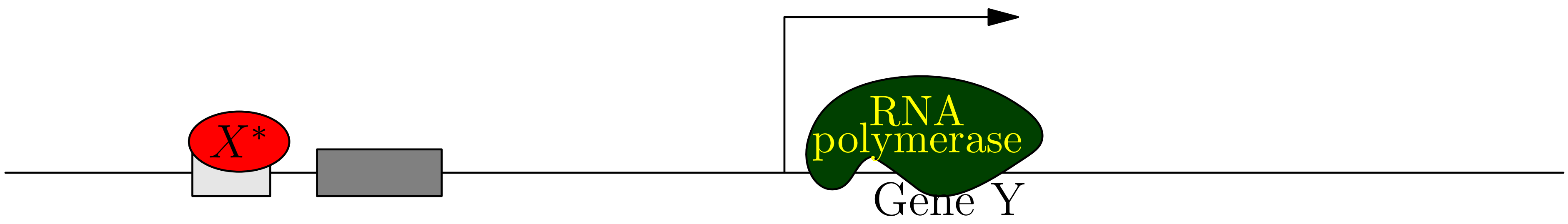

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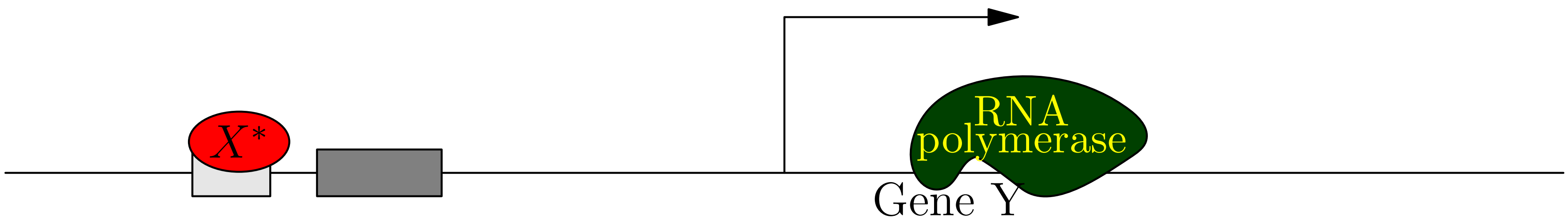

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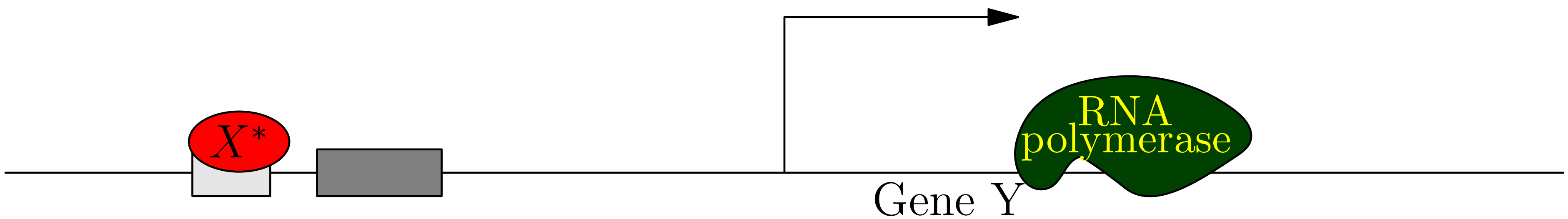

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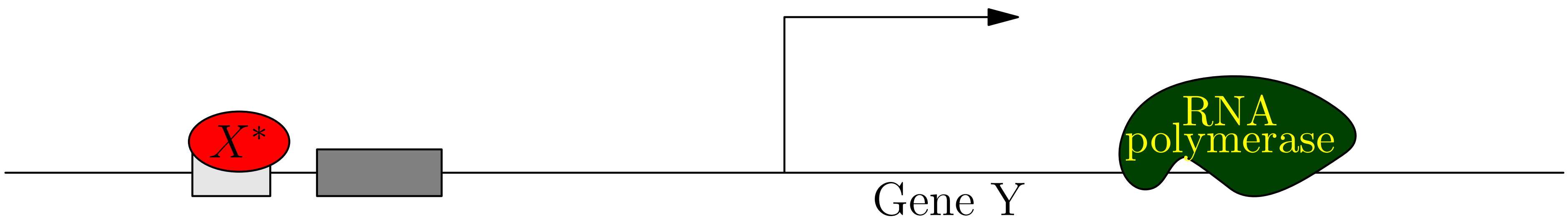

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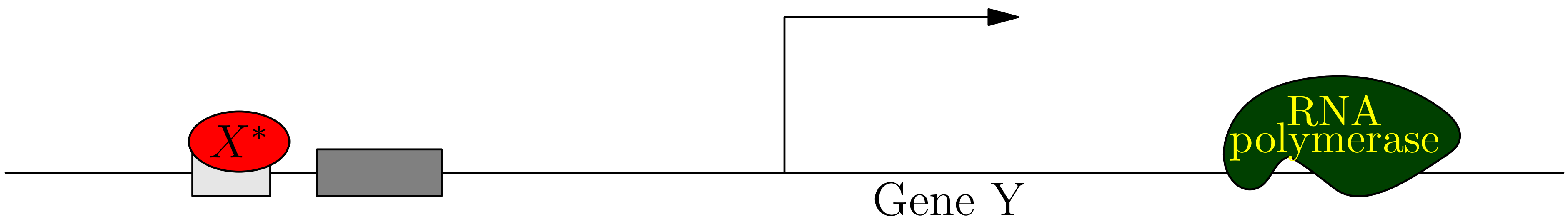

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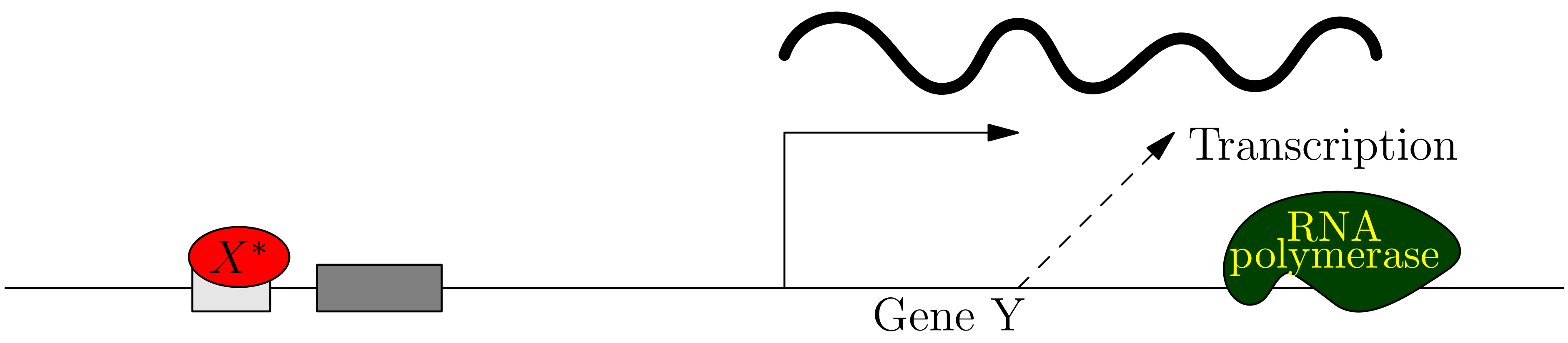

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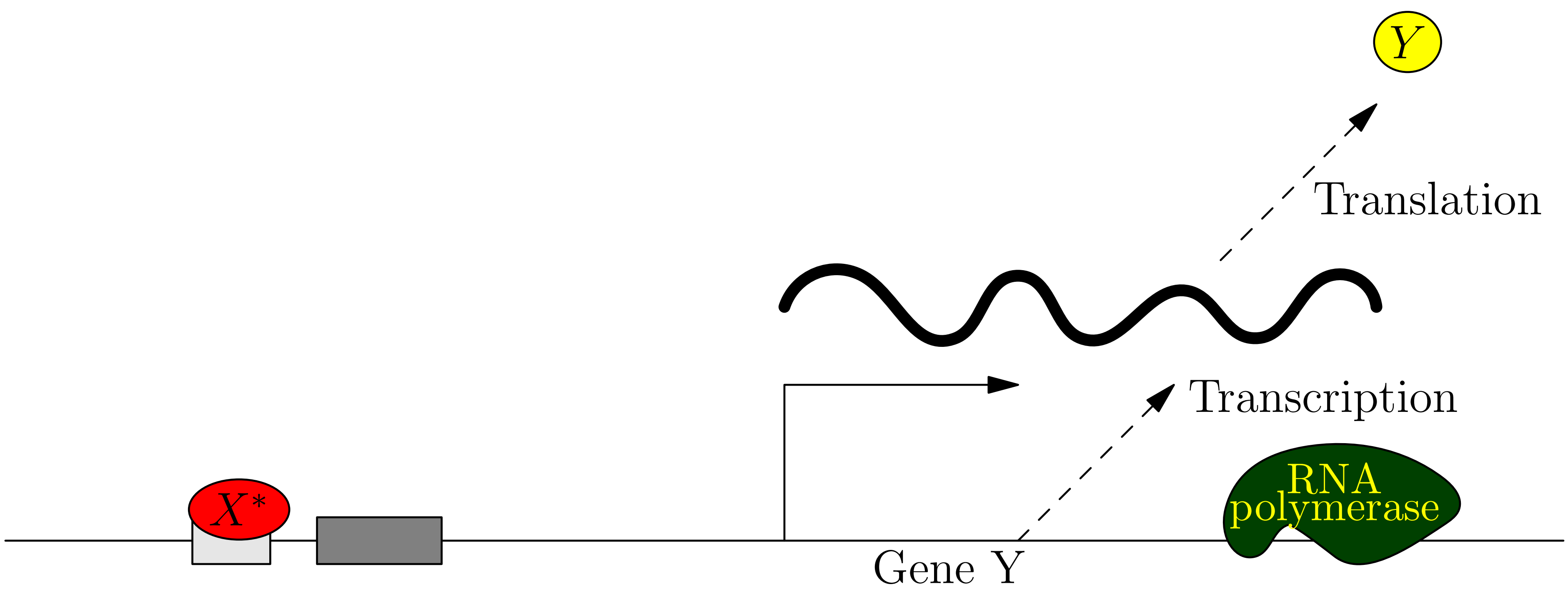

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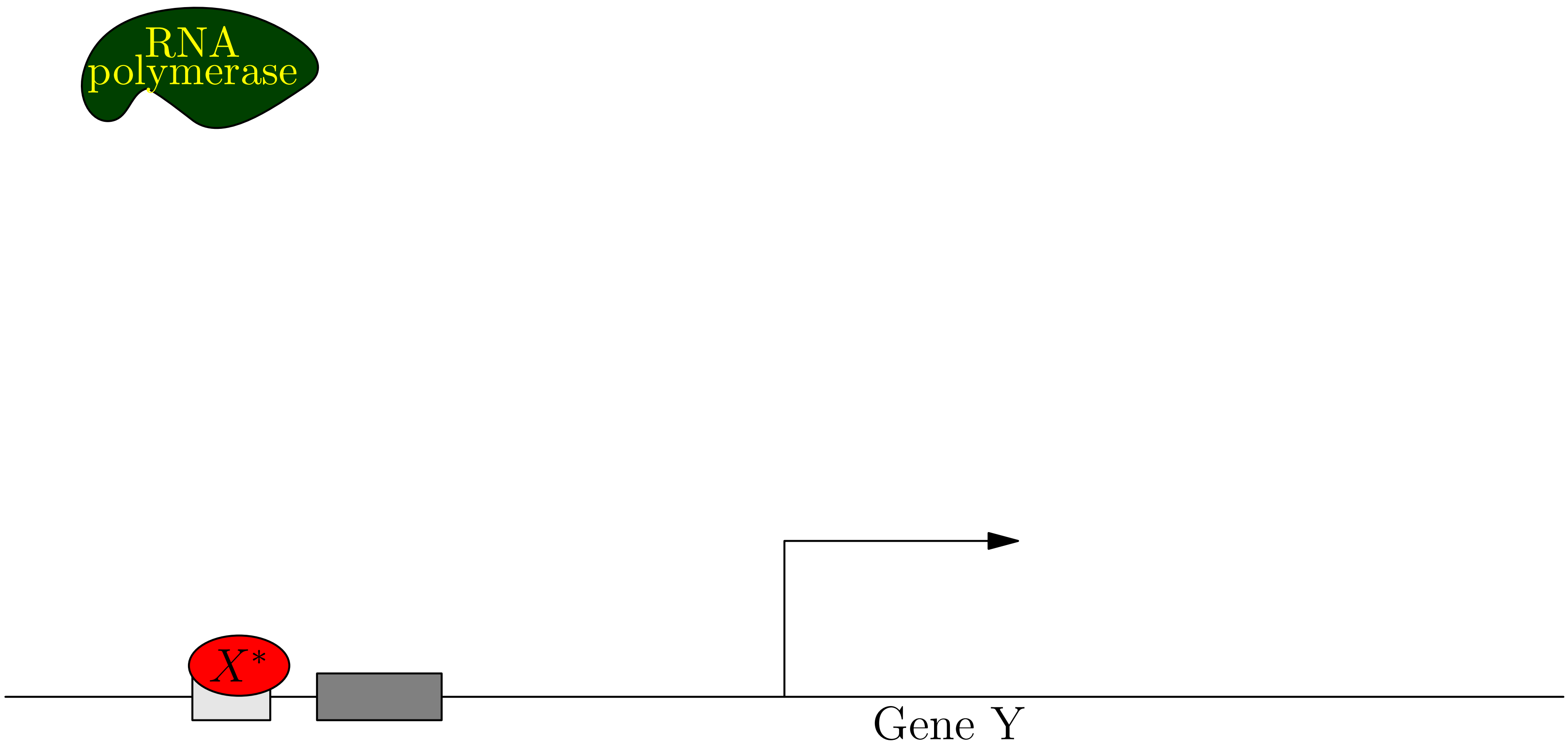

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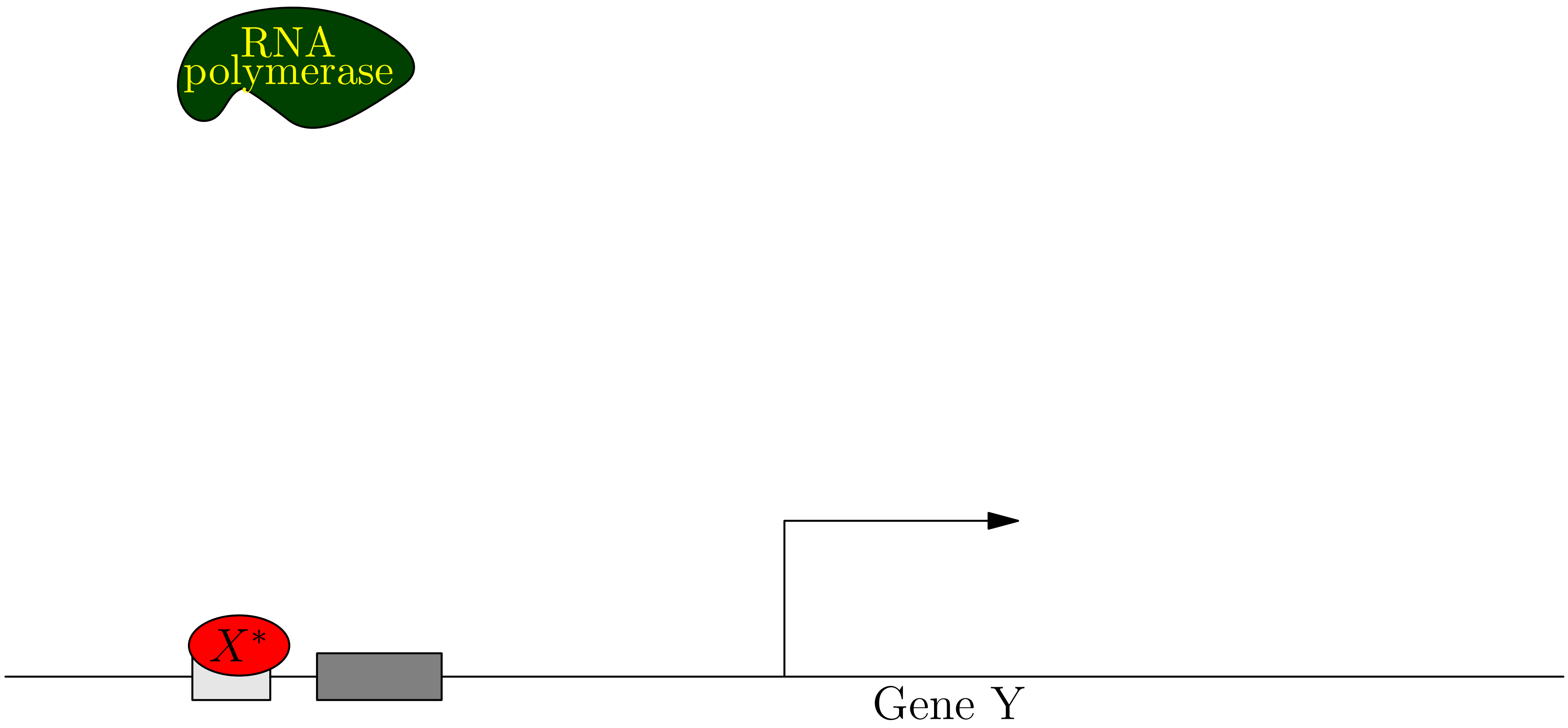

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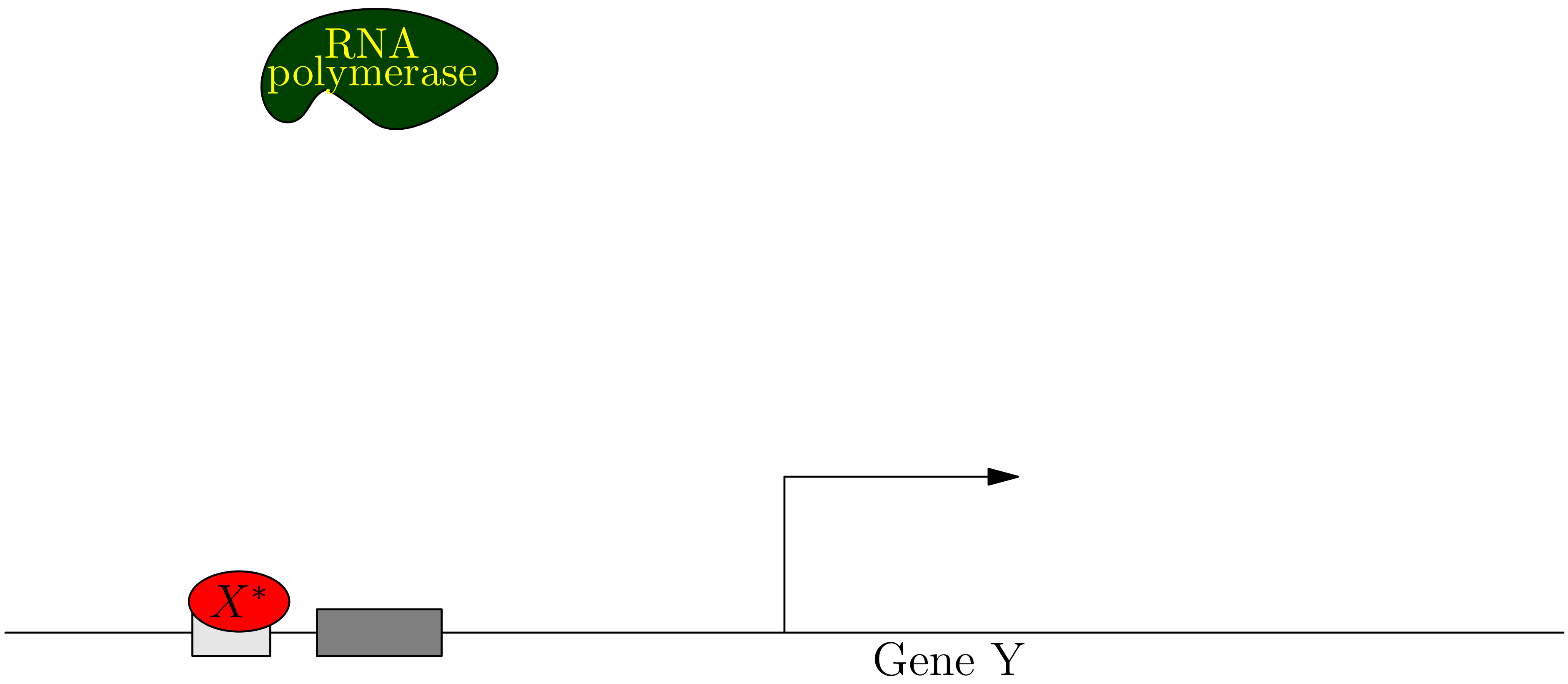

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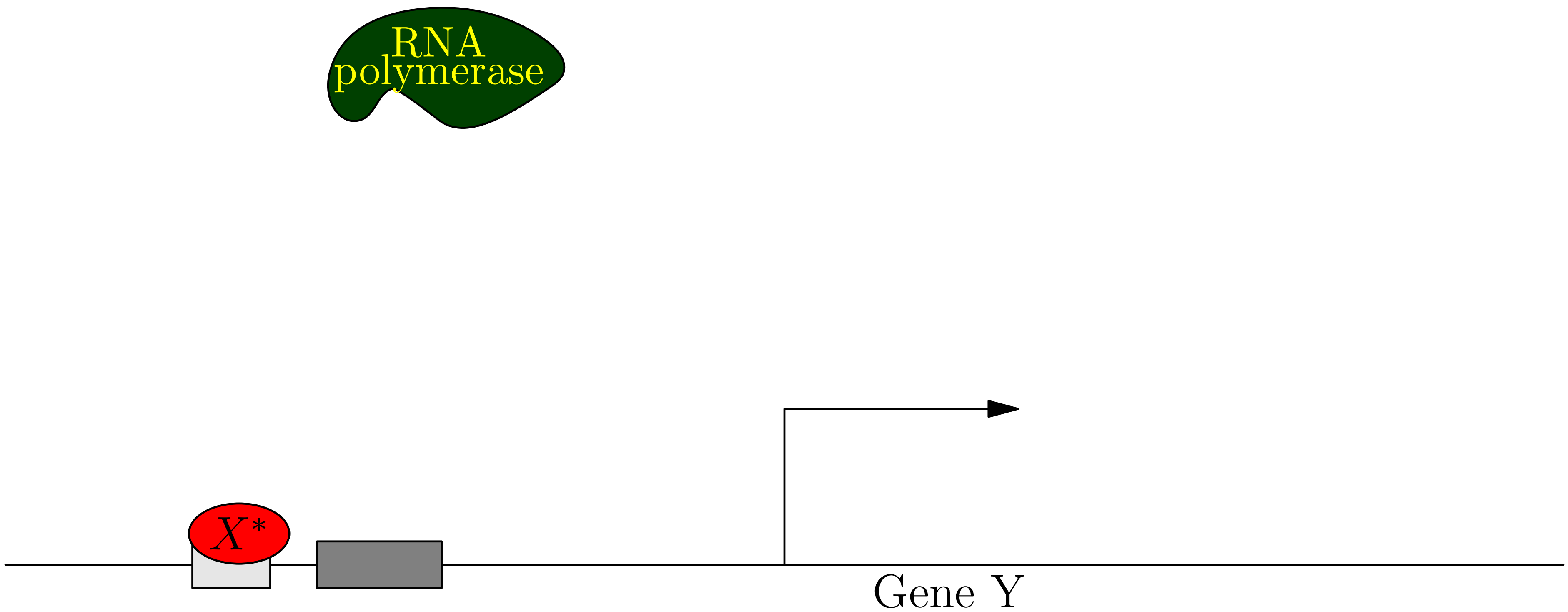

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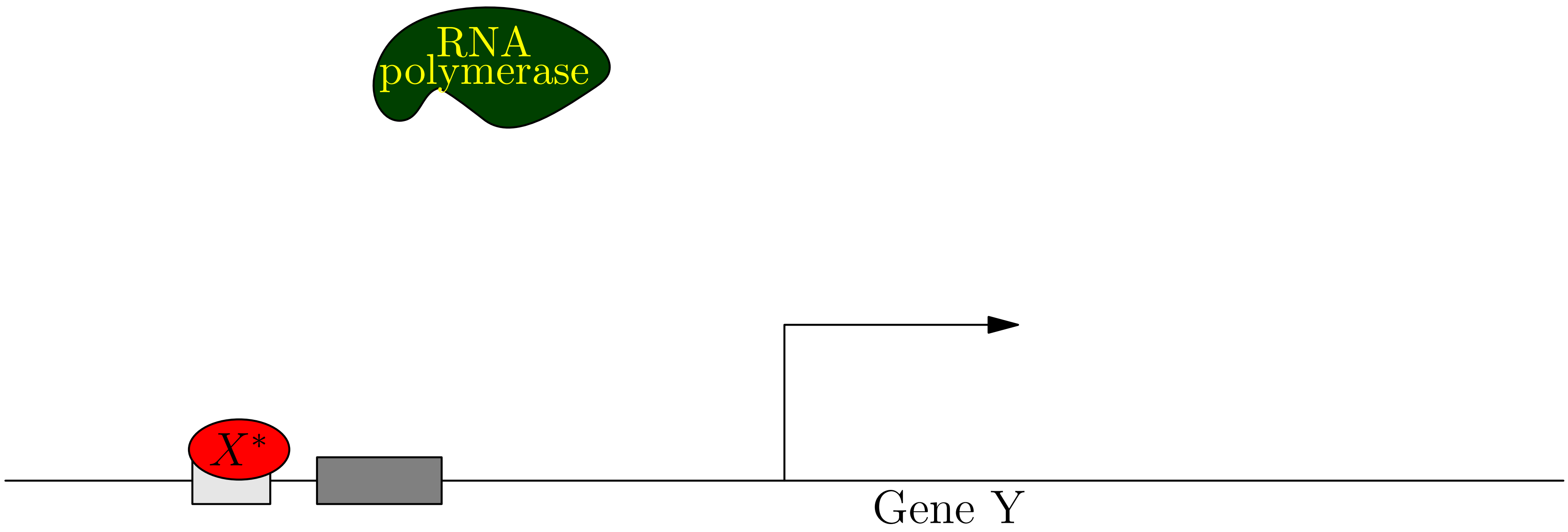

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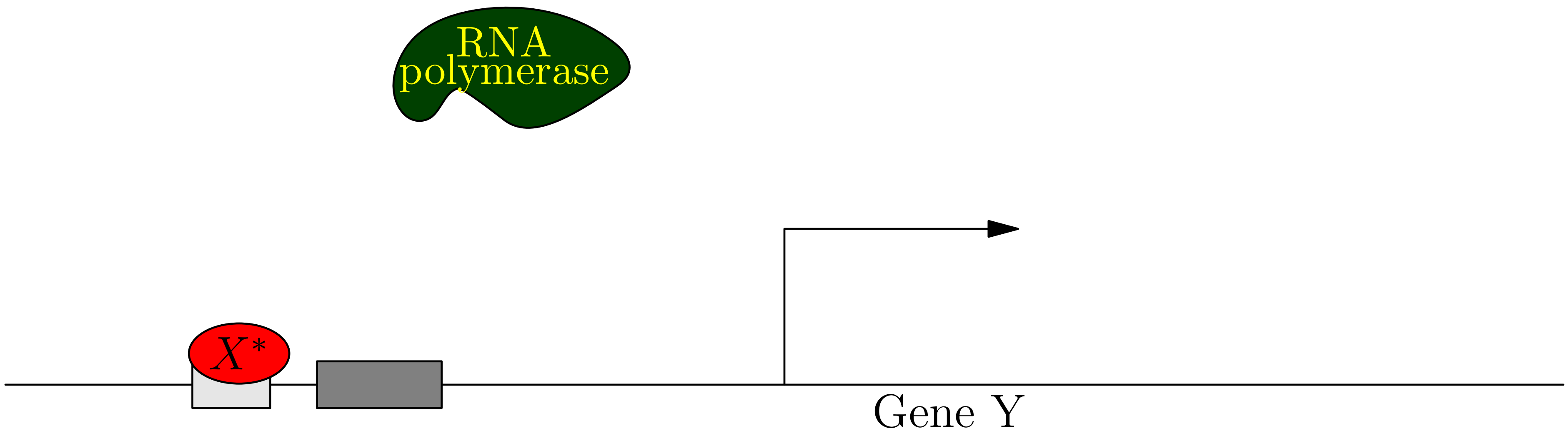

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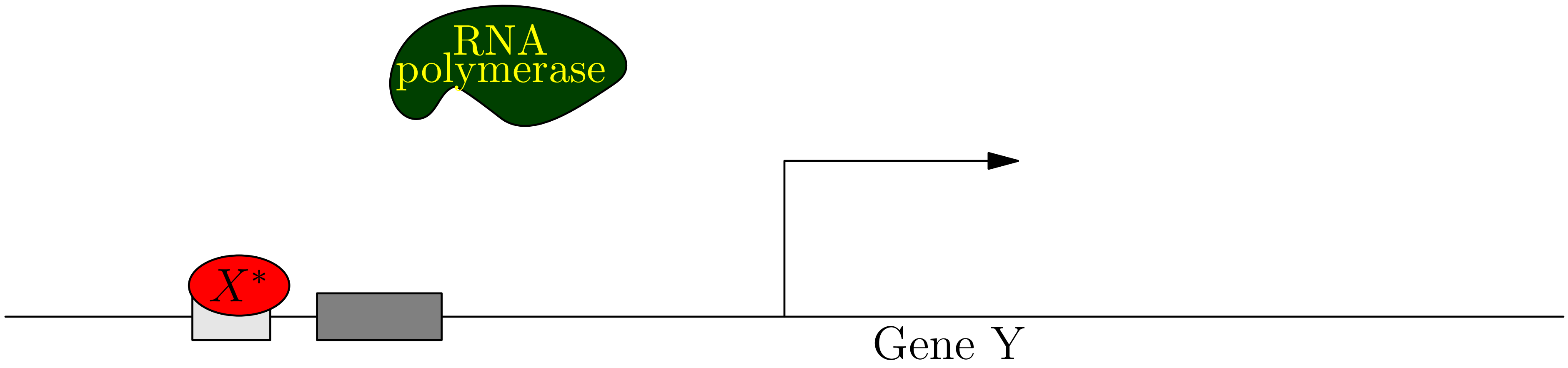

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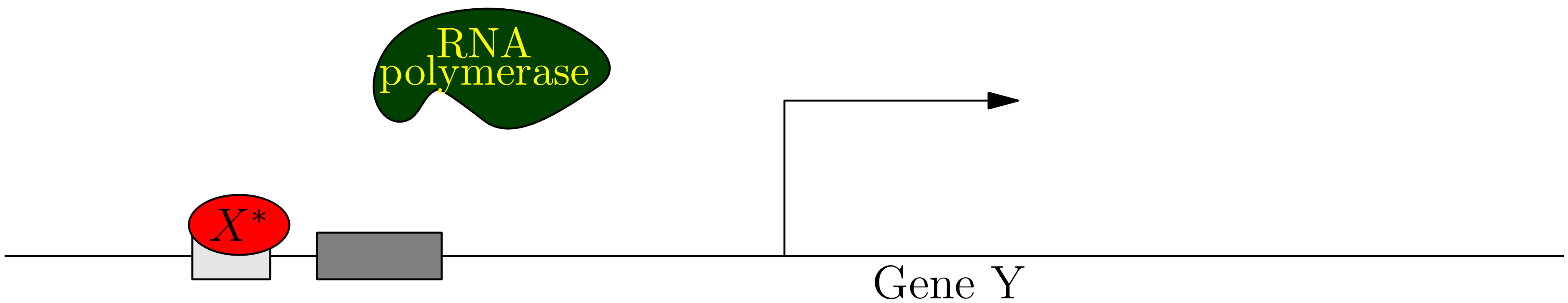

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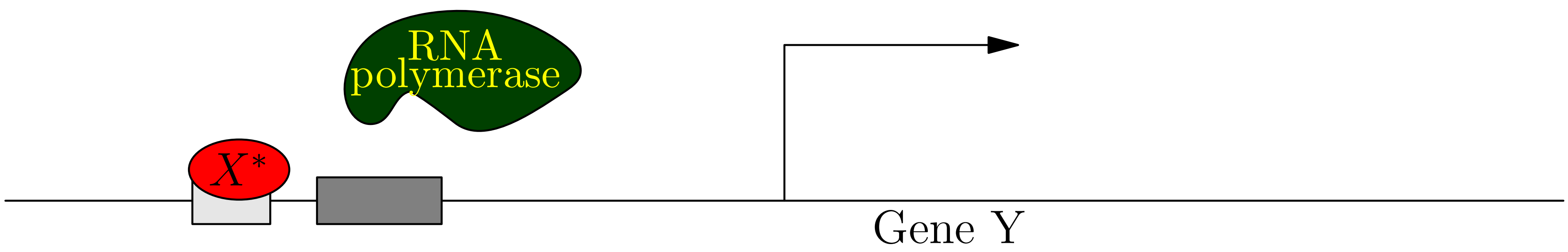

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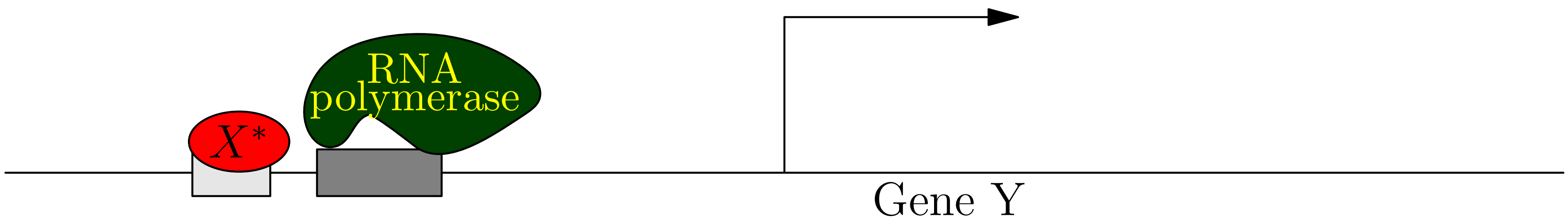

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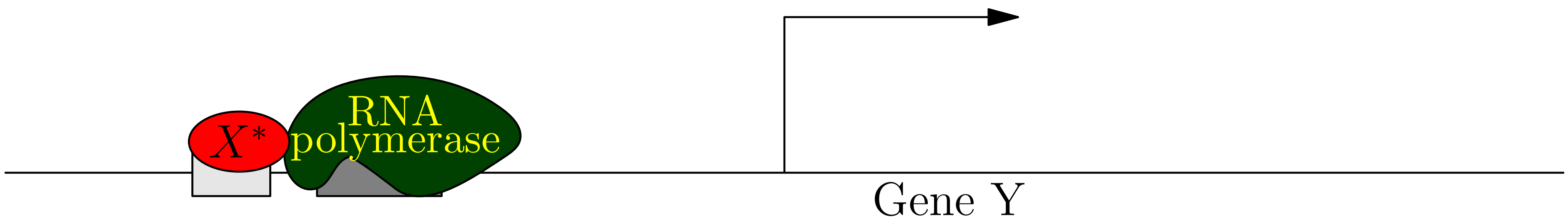

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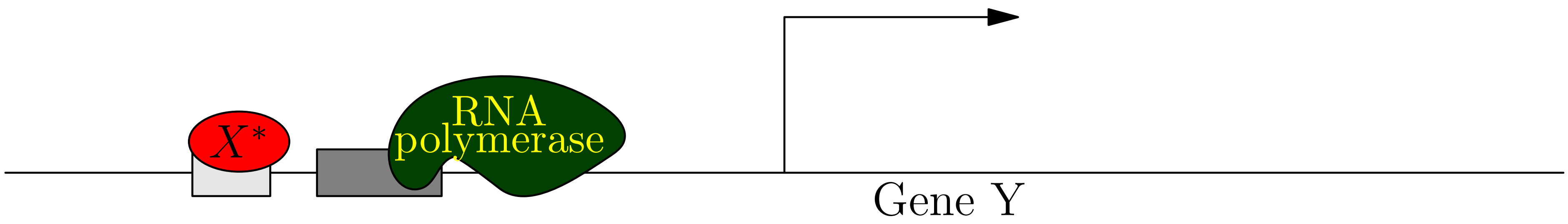

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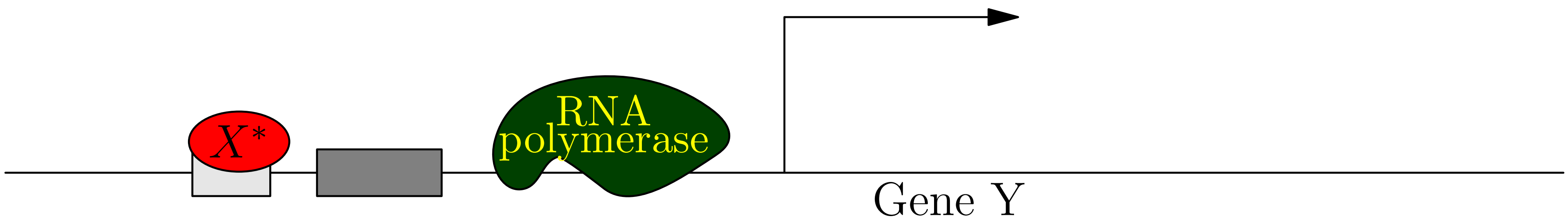

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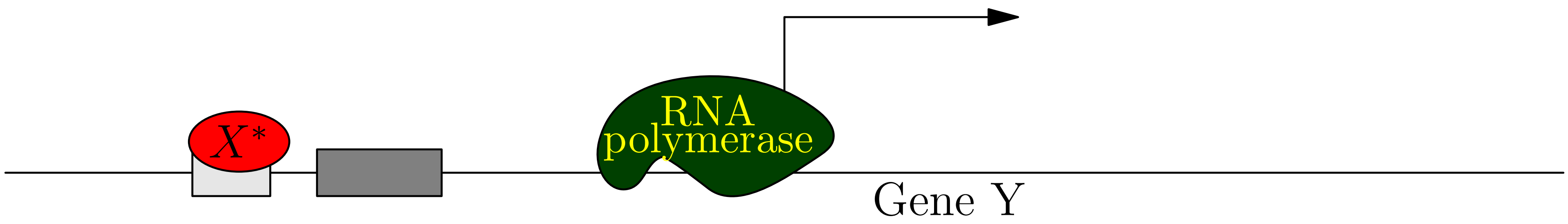

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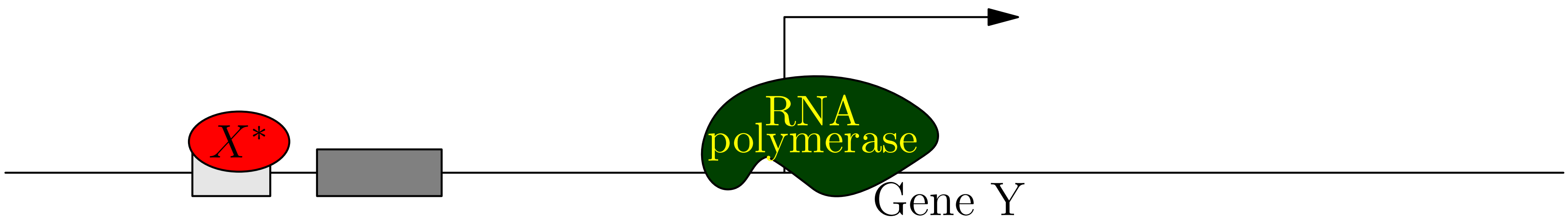

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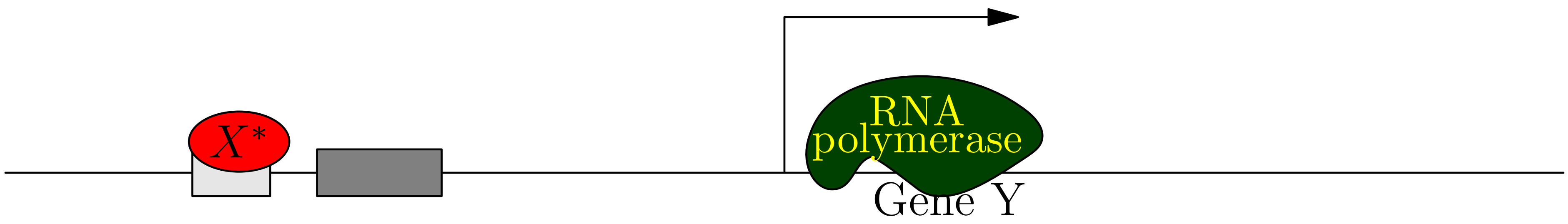

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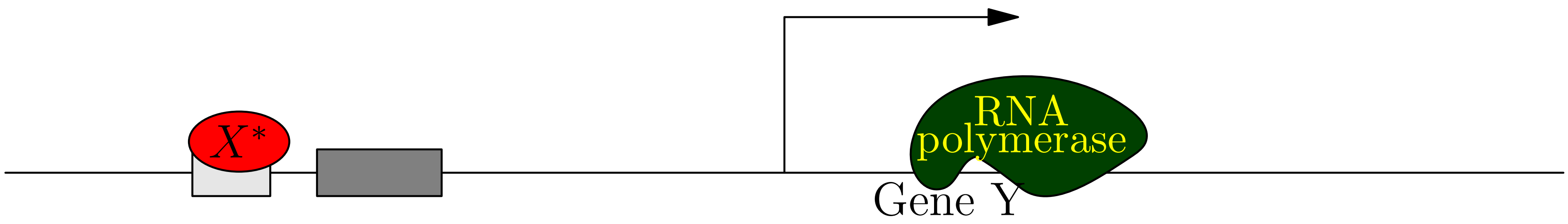

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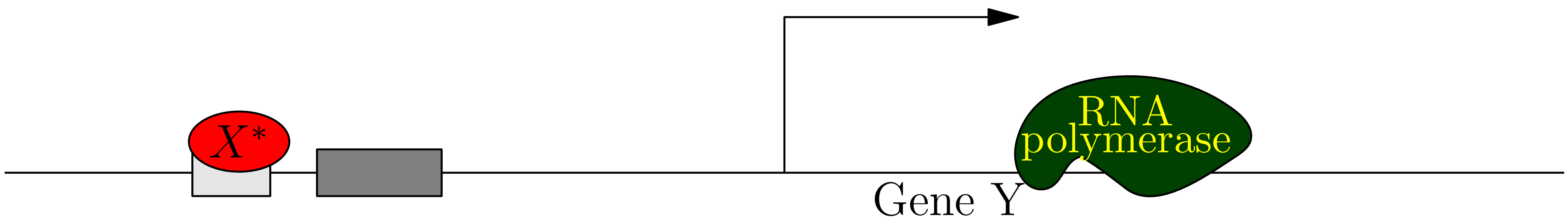

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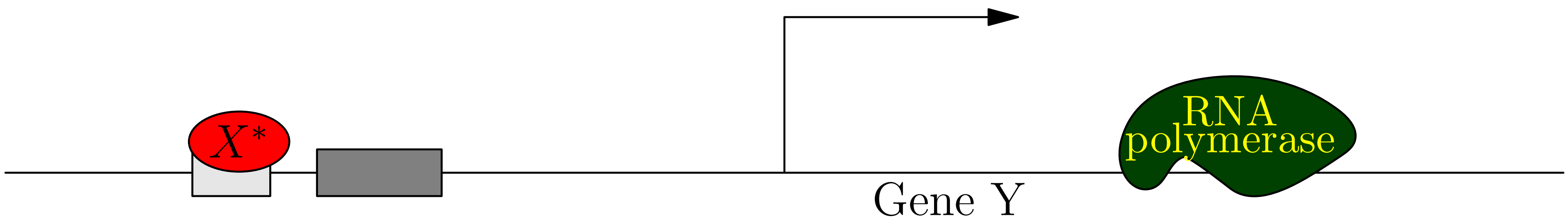

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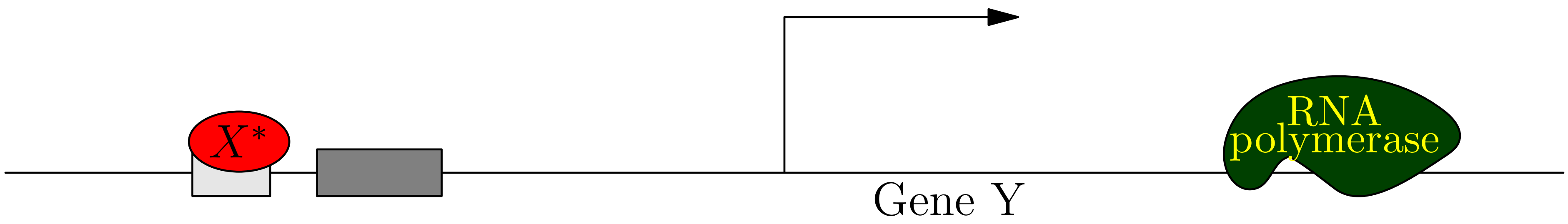

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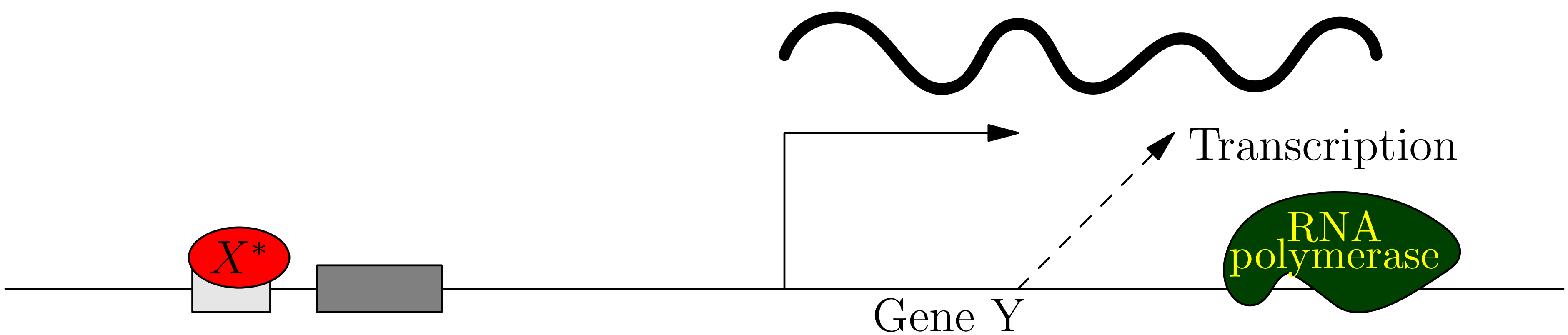

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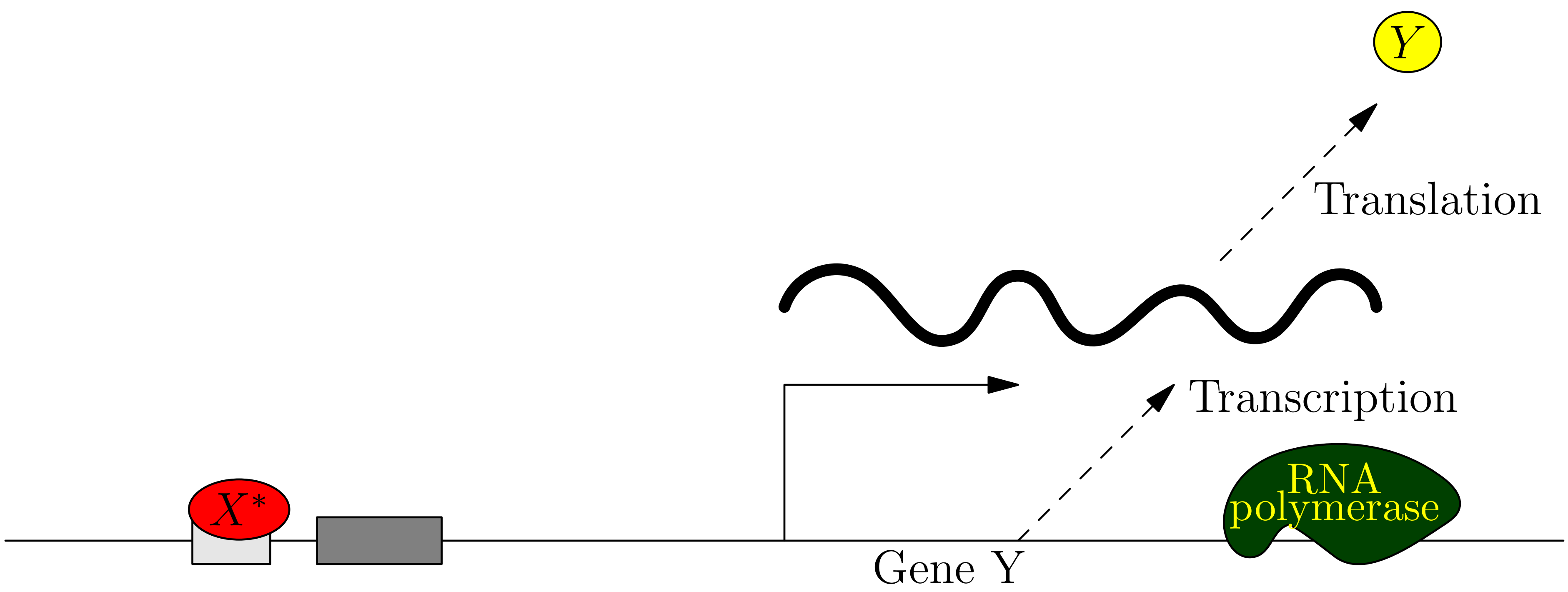

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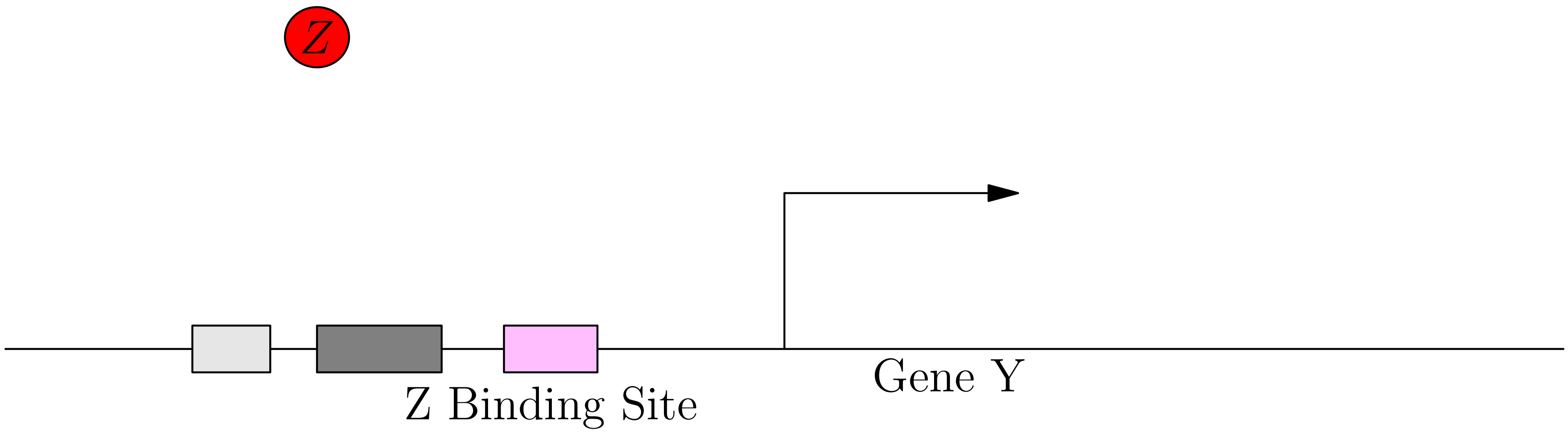

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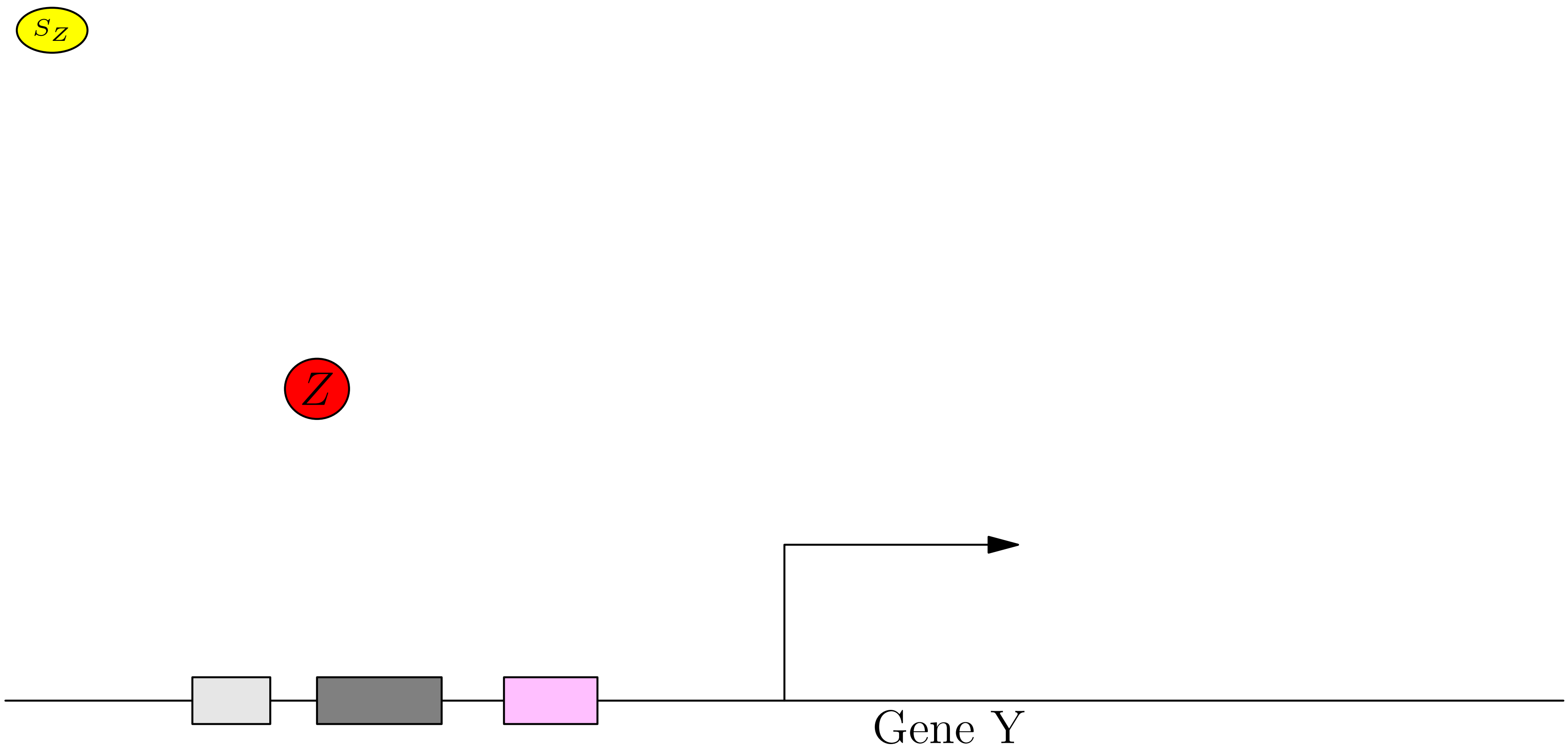

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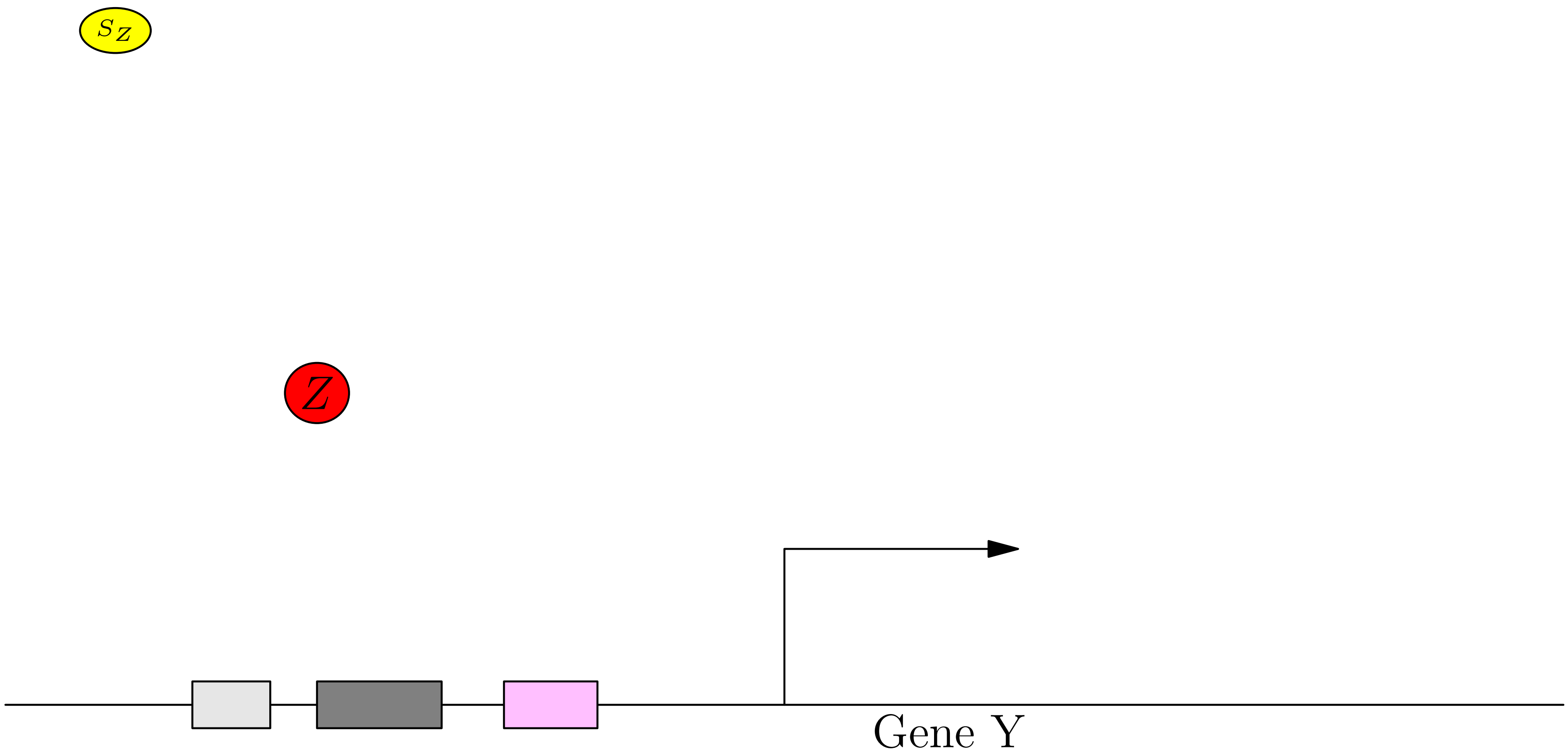

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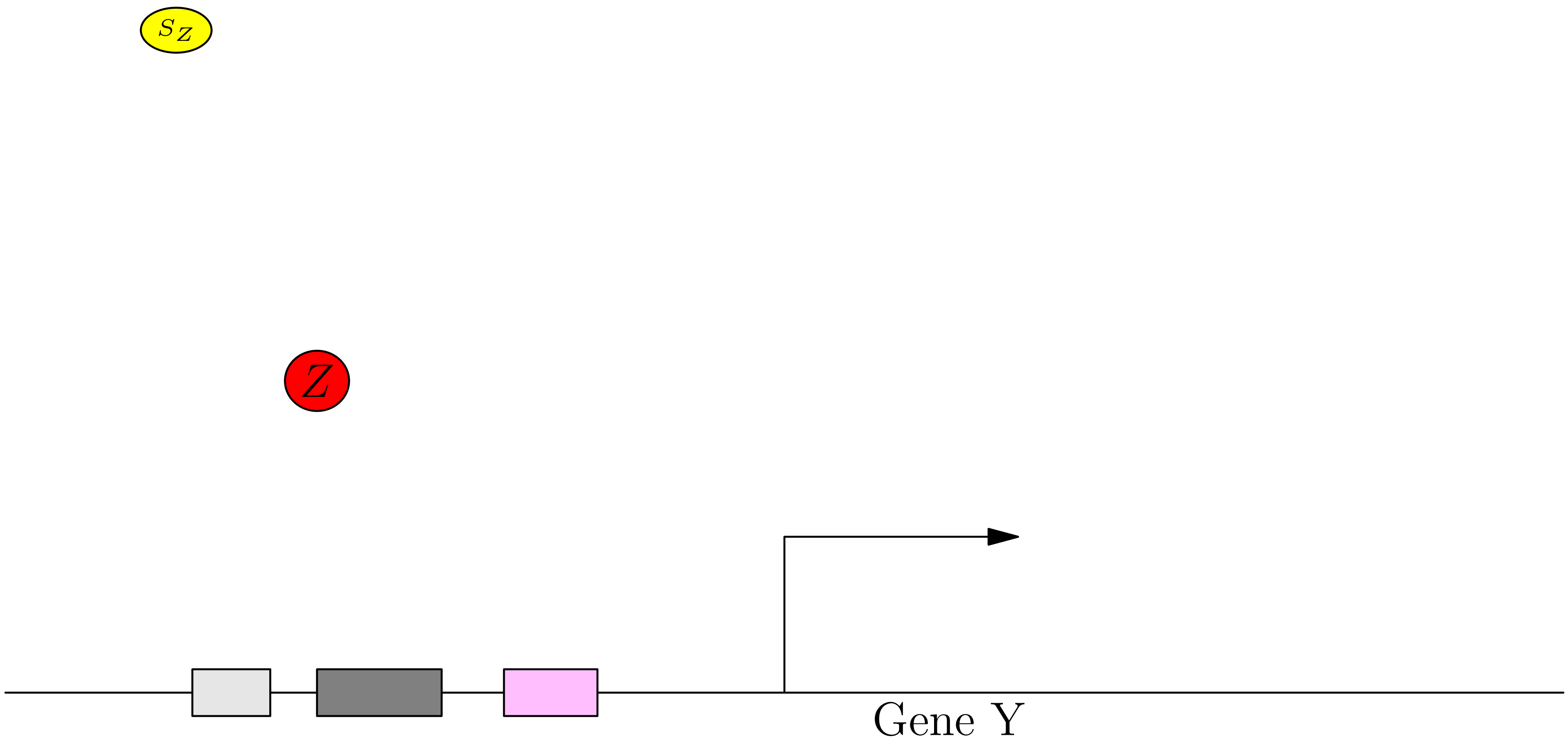

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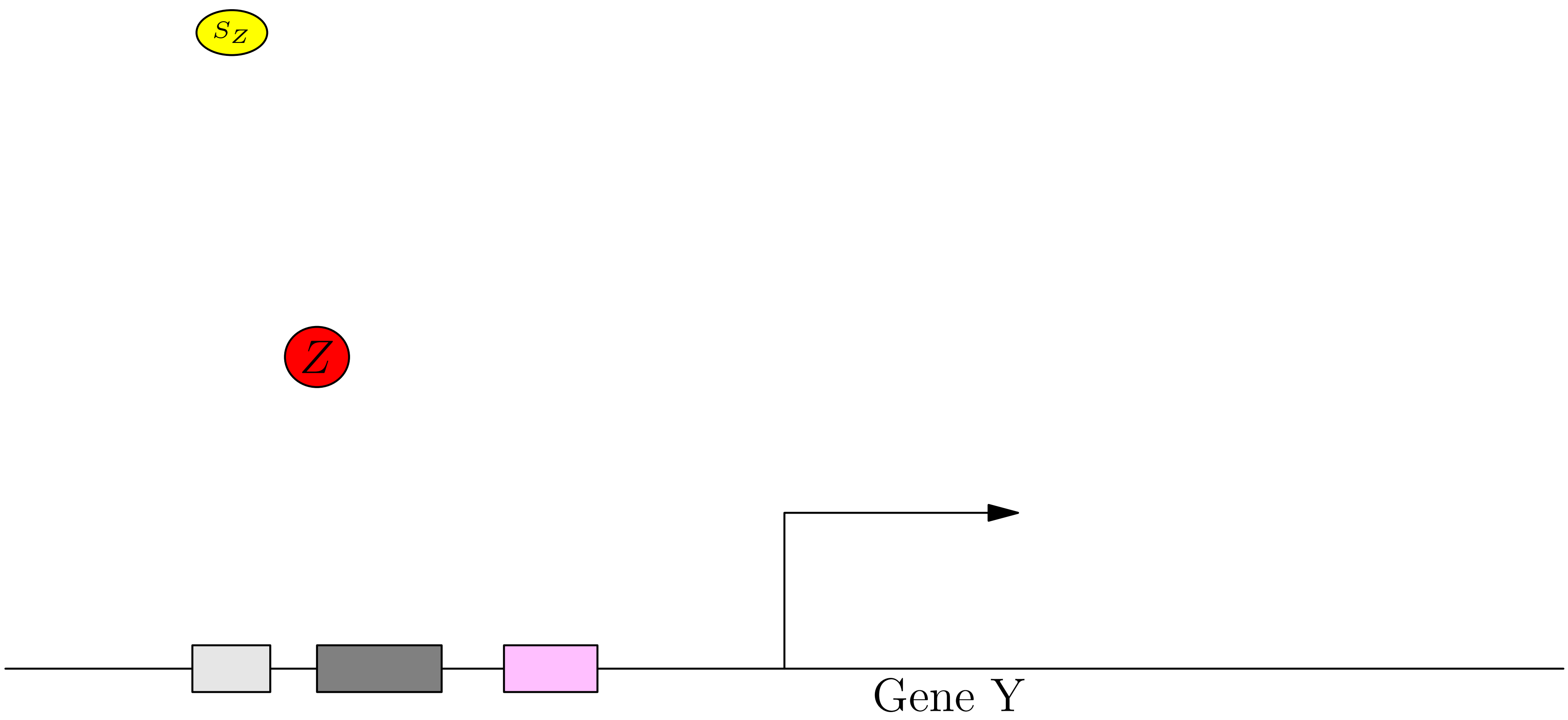

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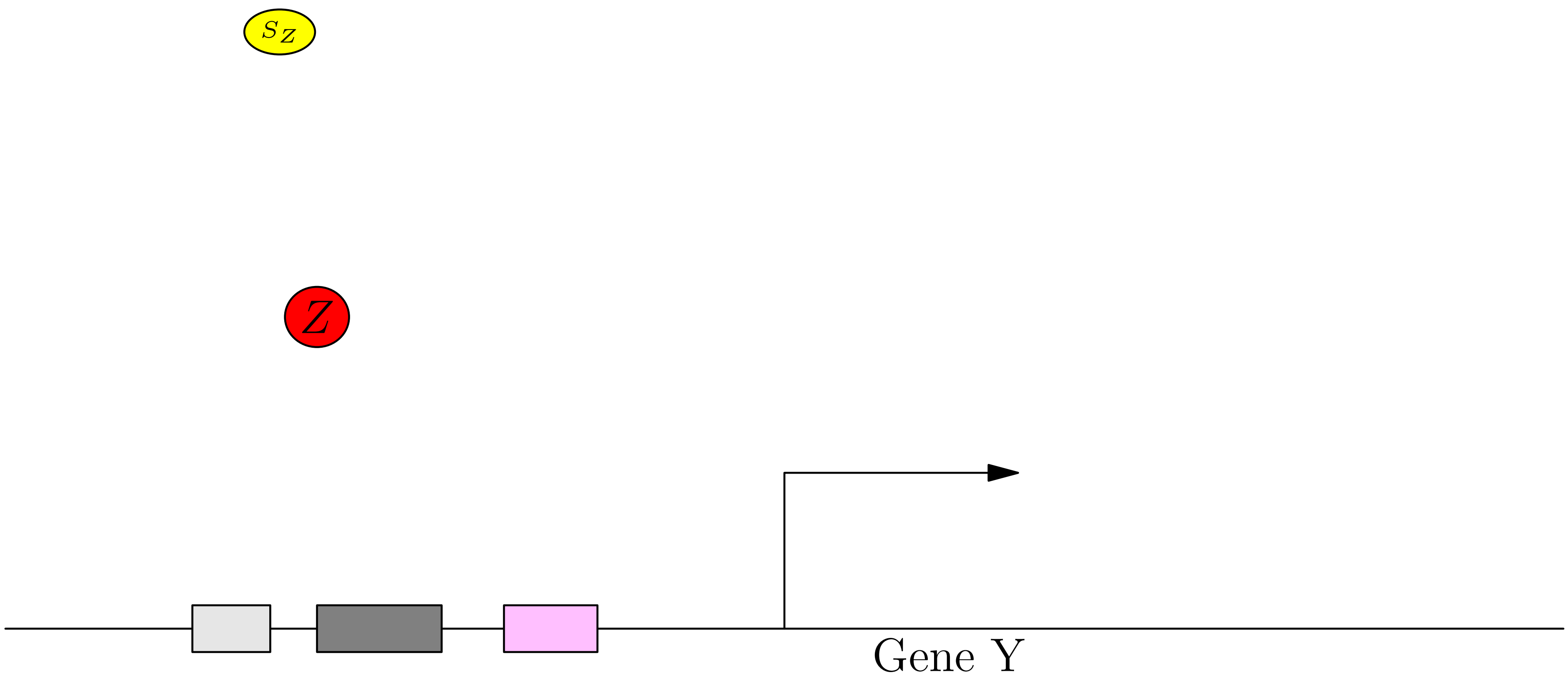

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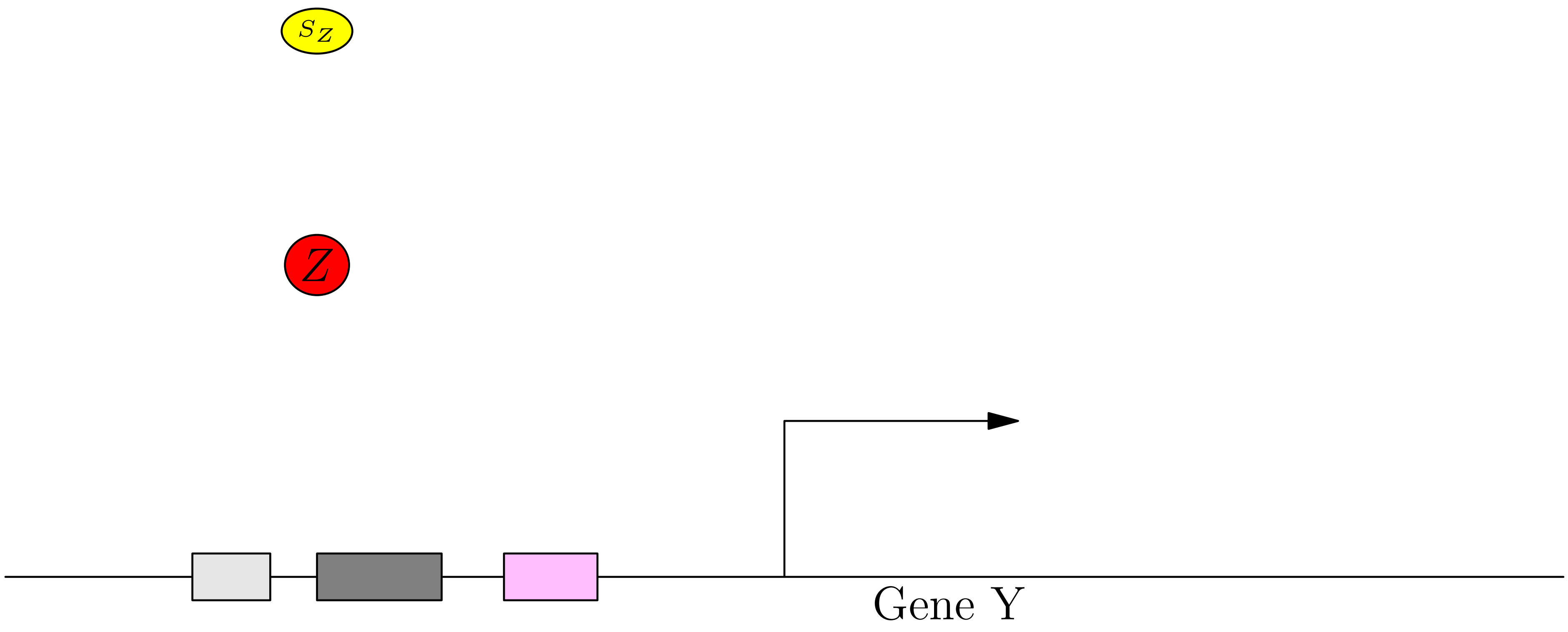

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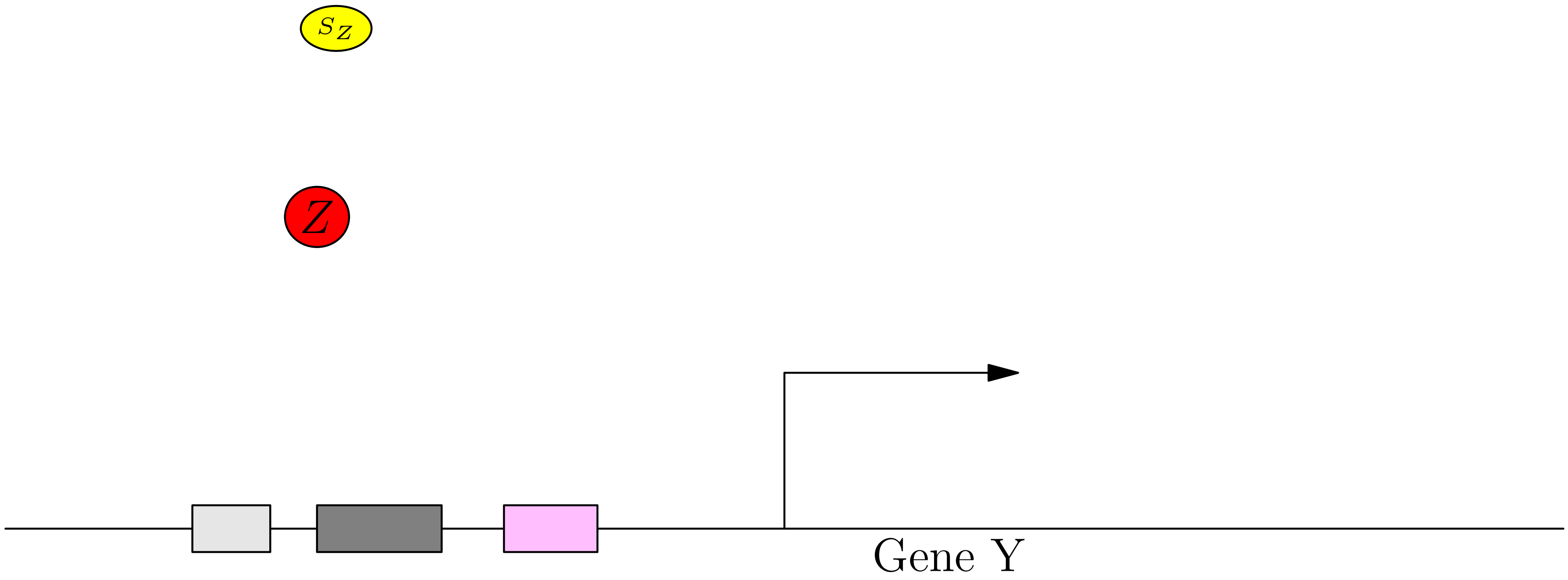

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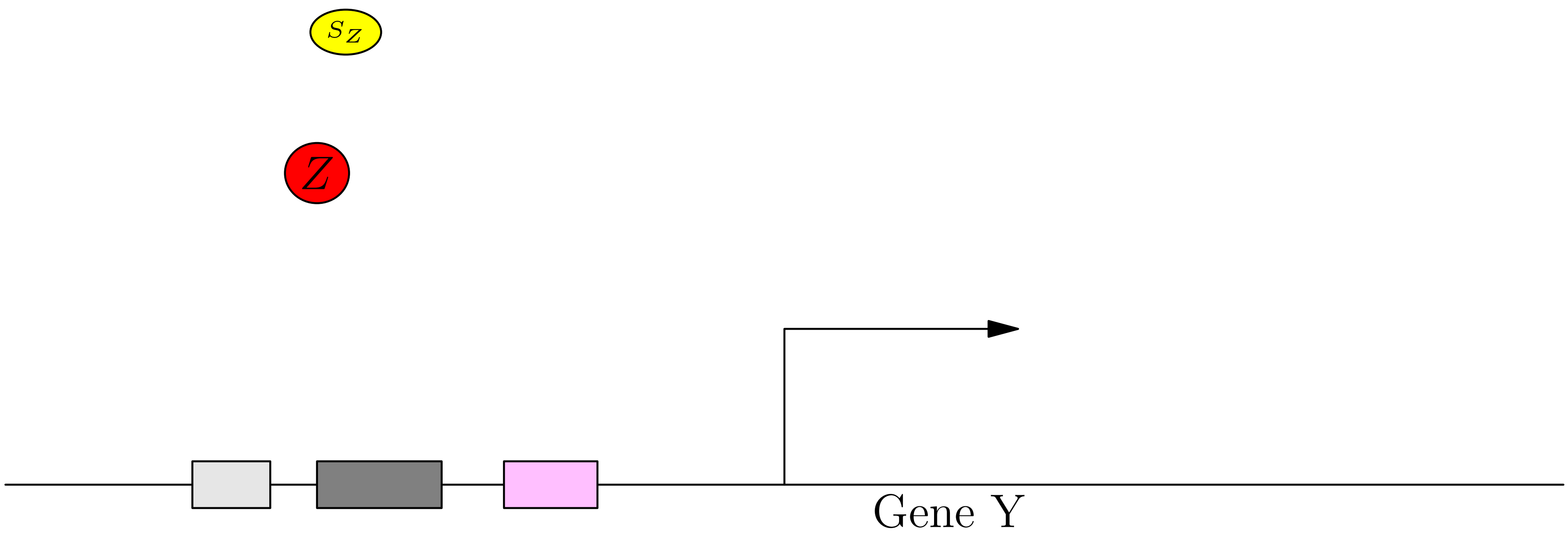

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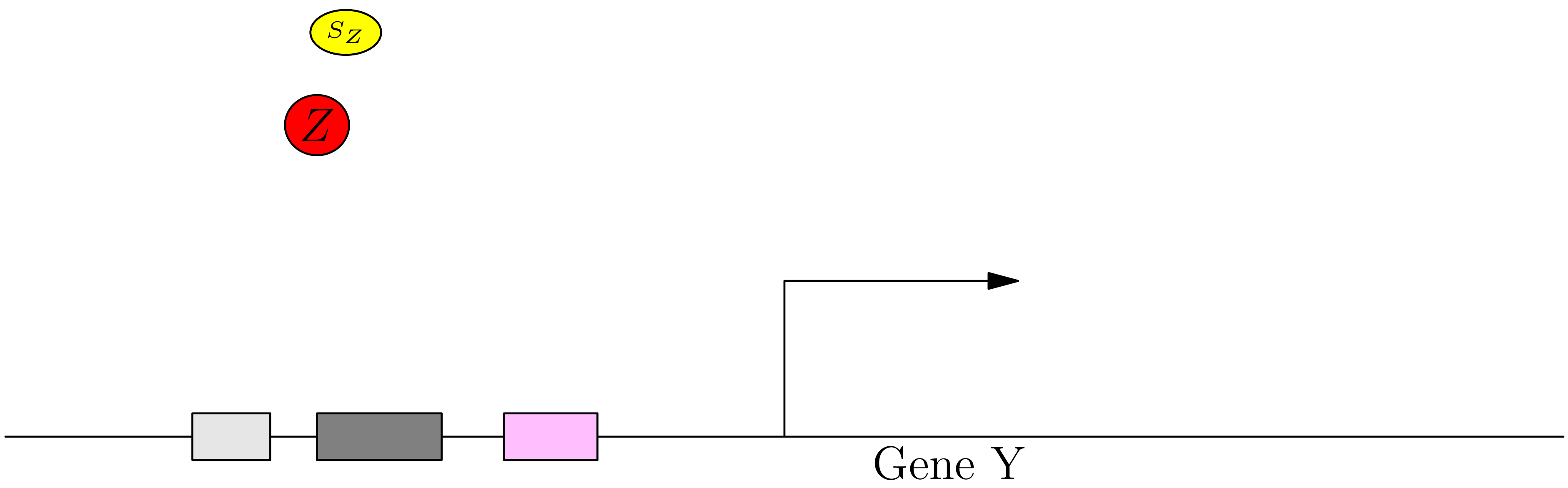

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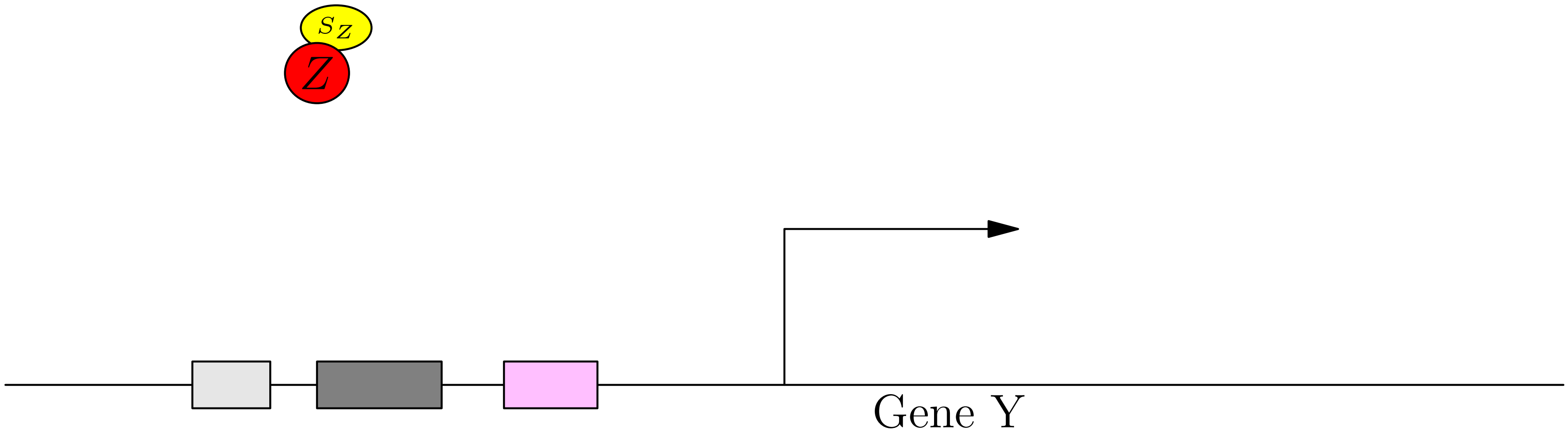

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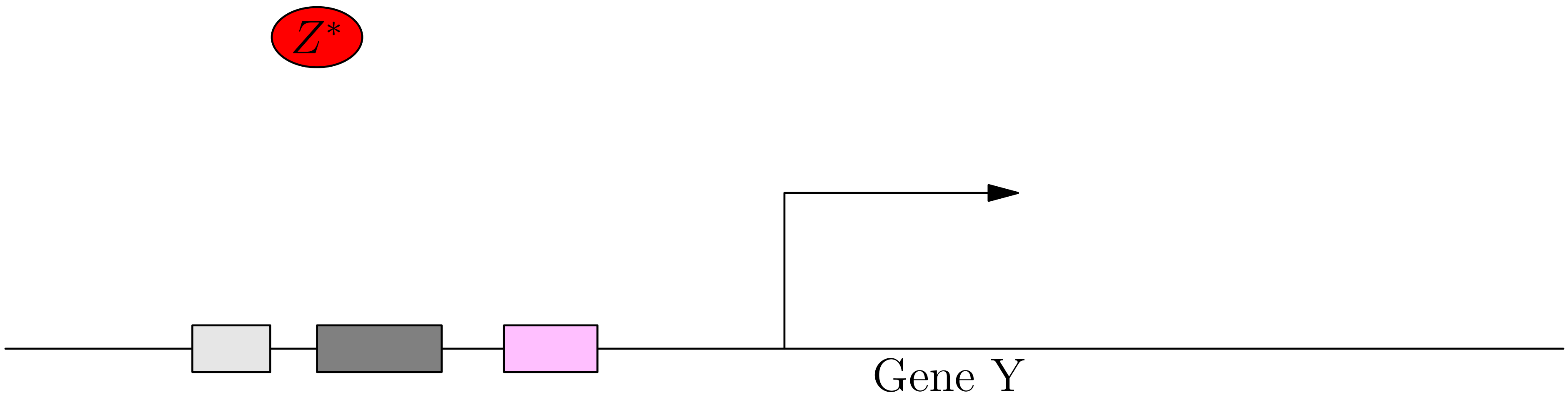

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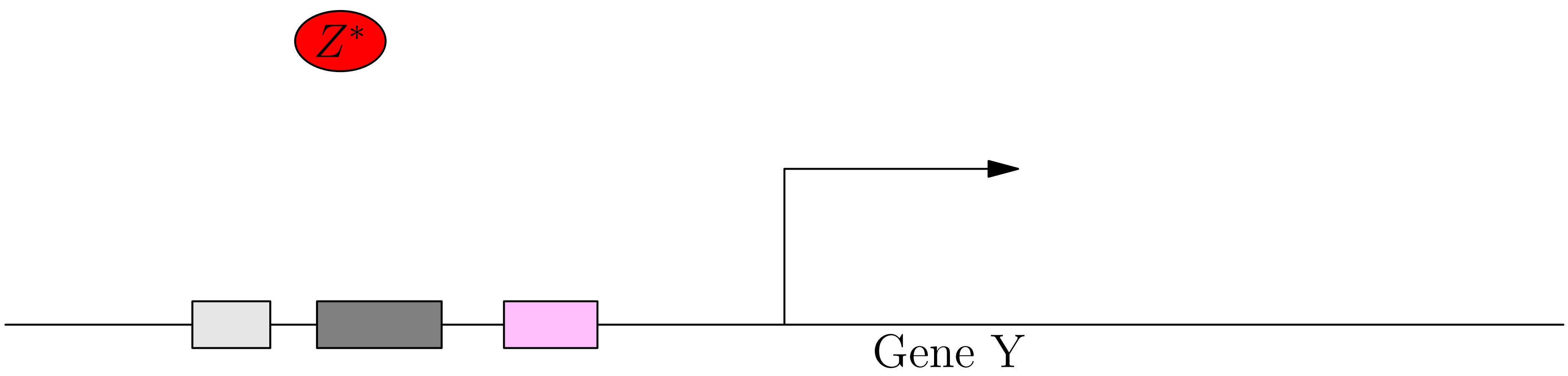

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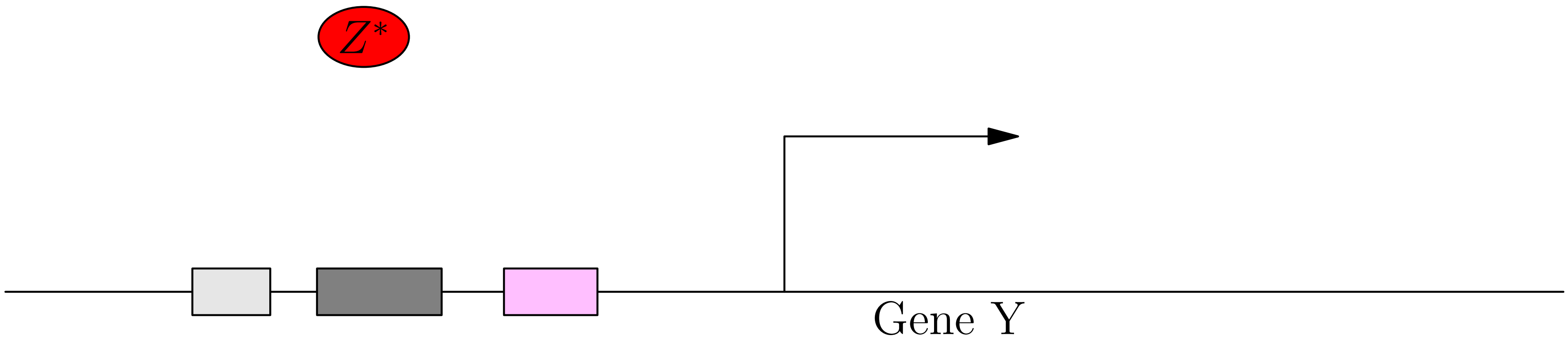

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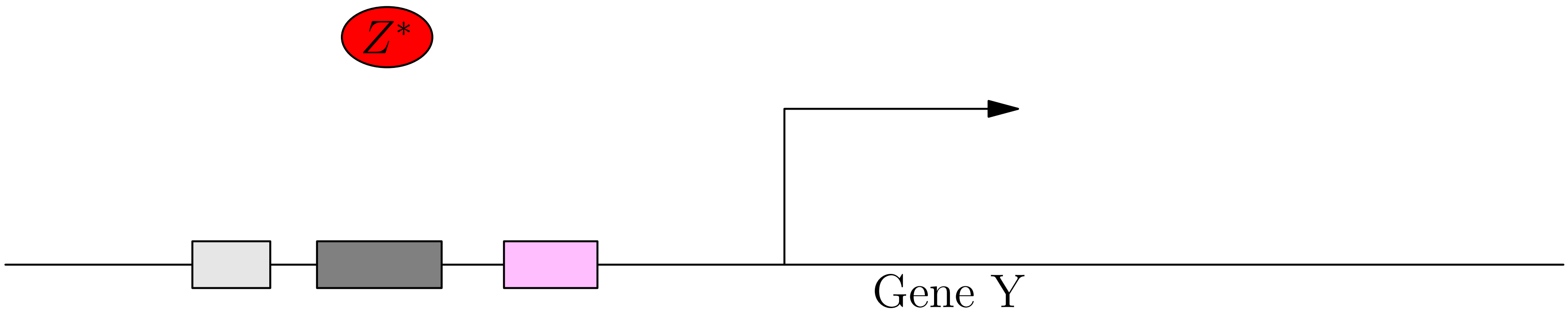

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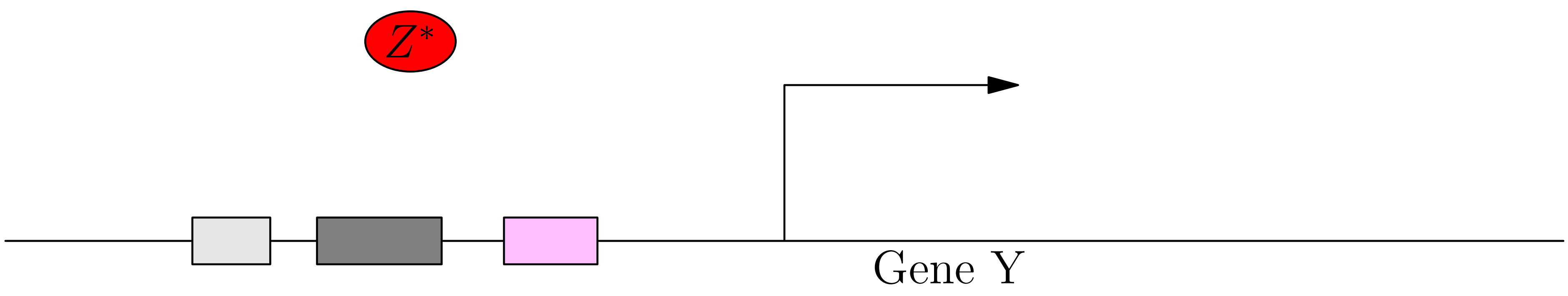

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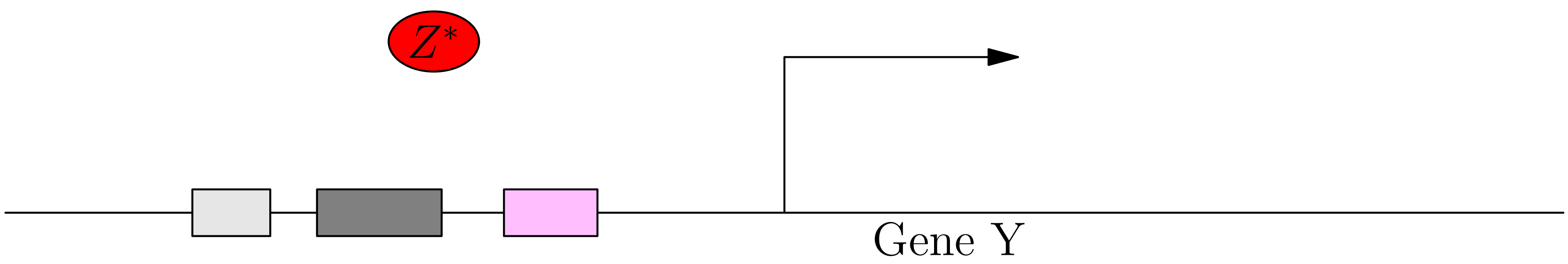

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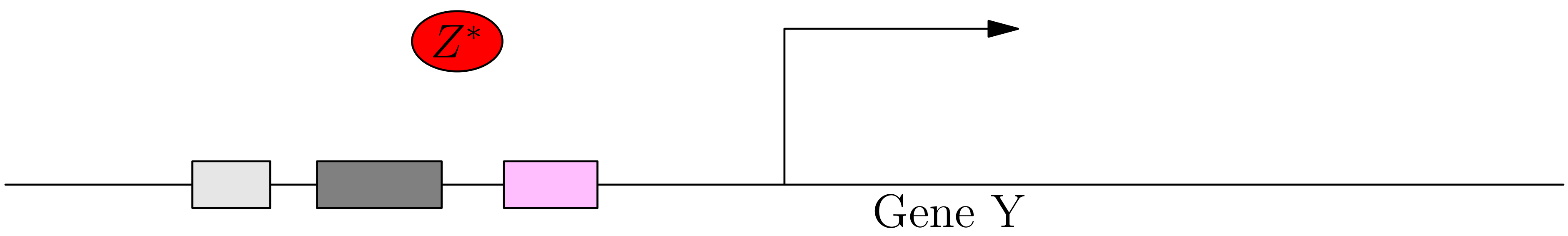

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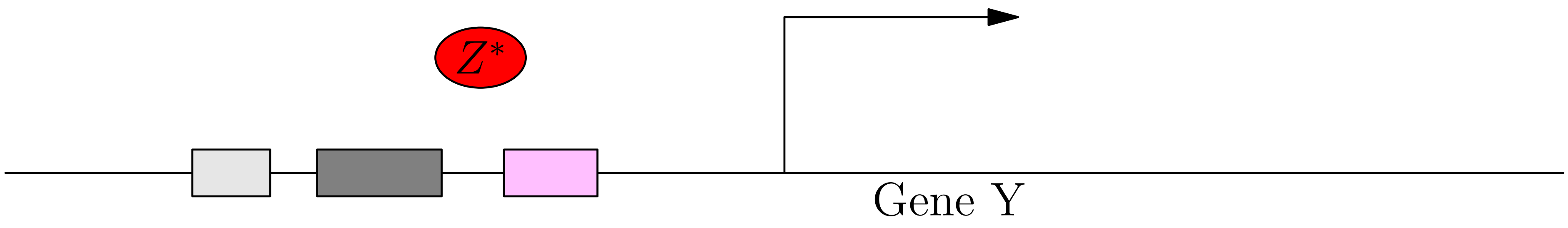

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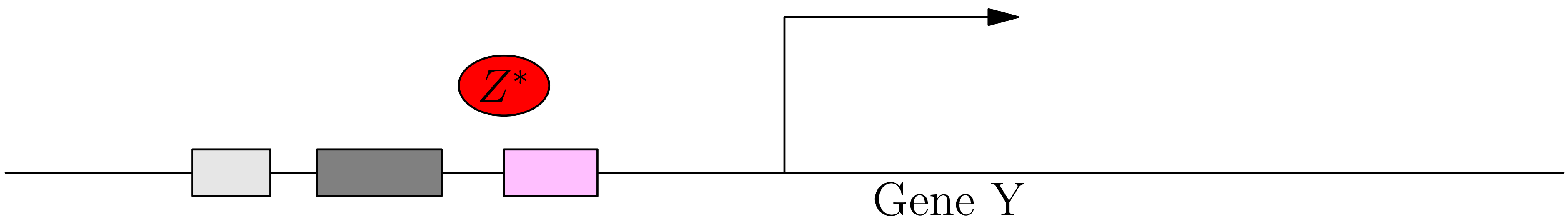

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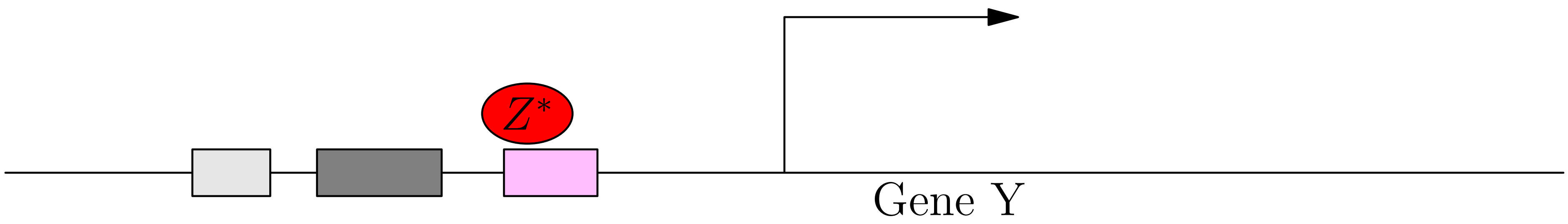

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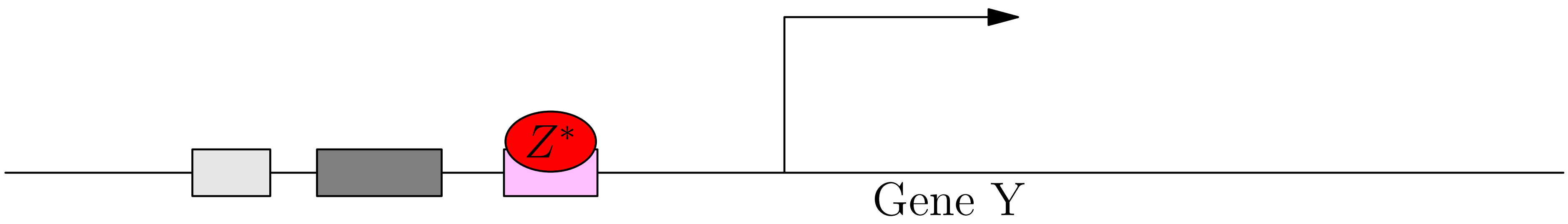

Gene Transcription Regulation

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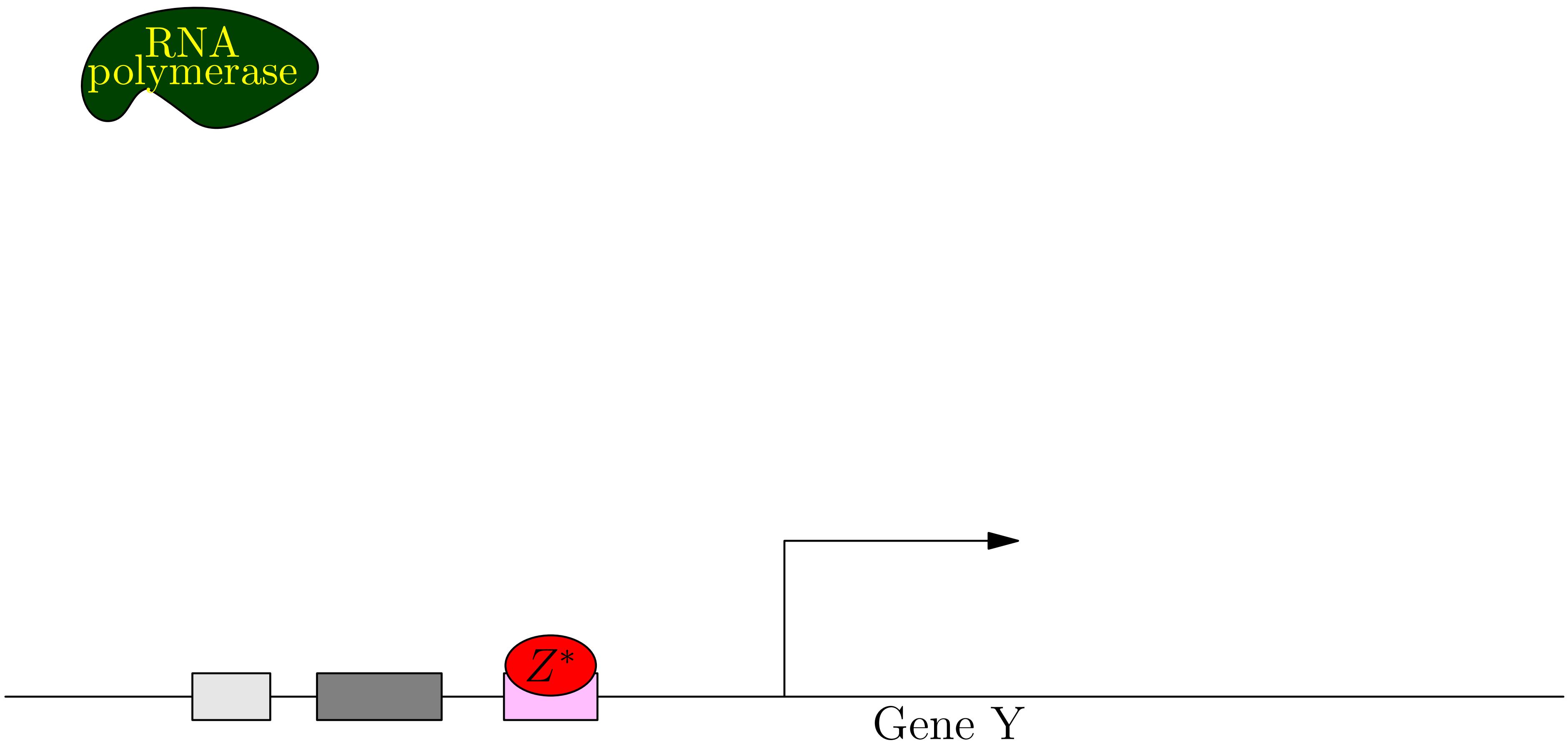

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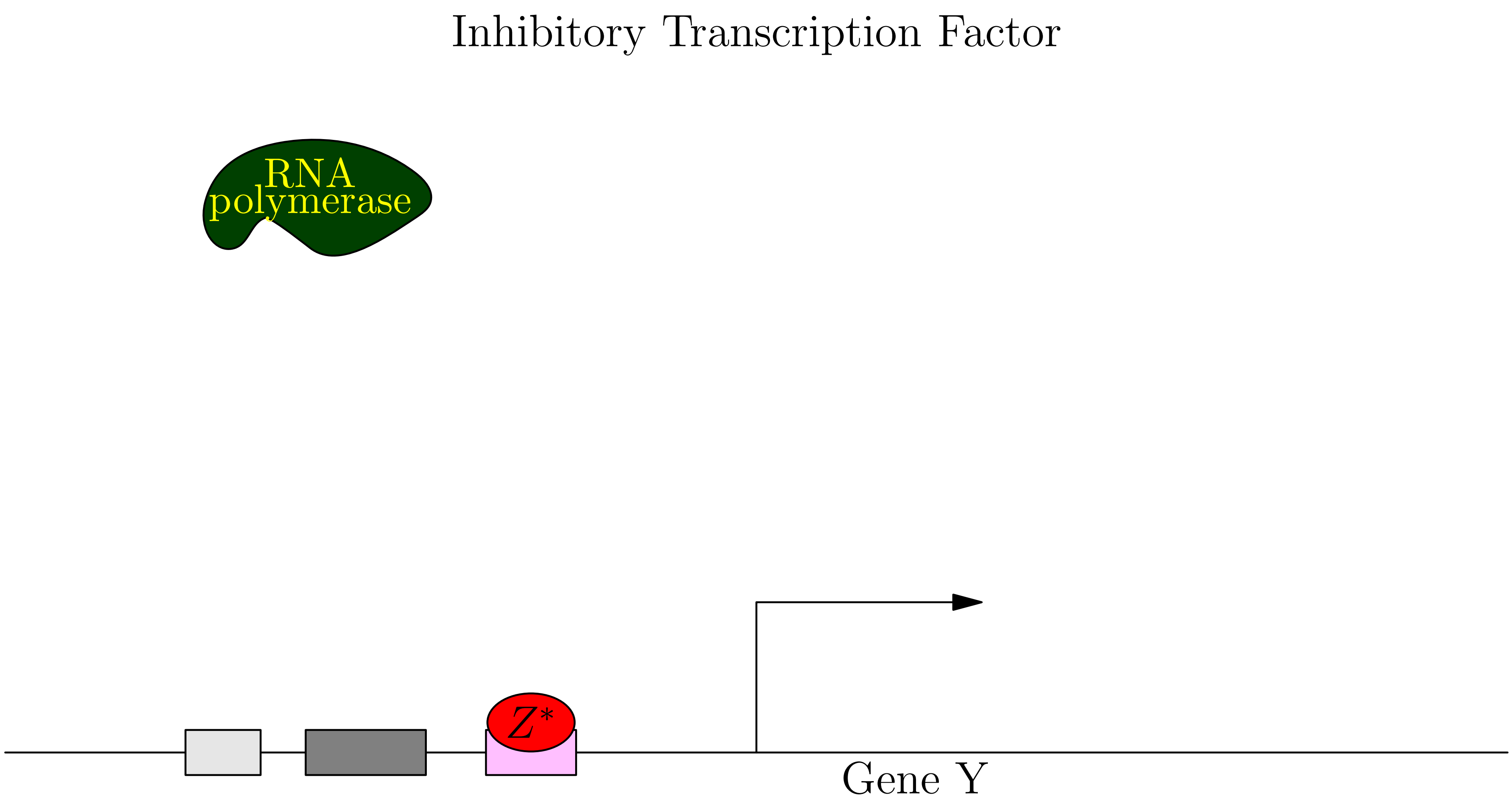

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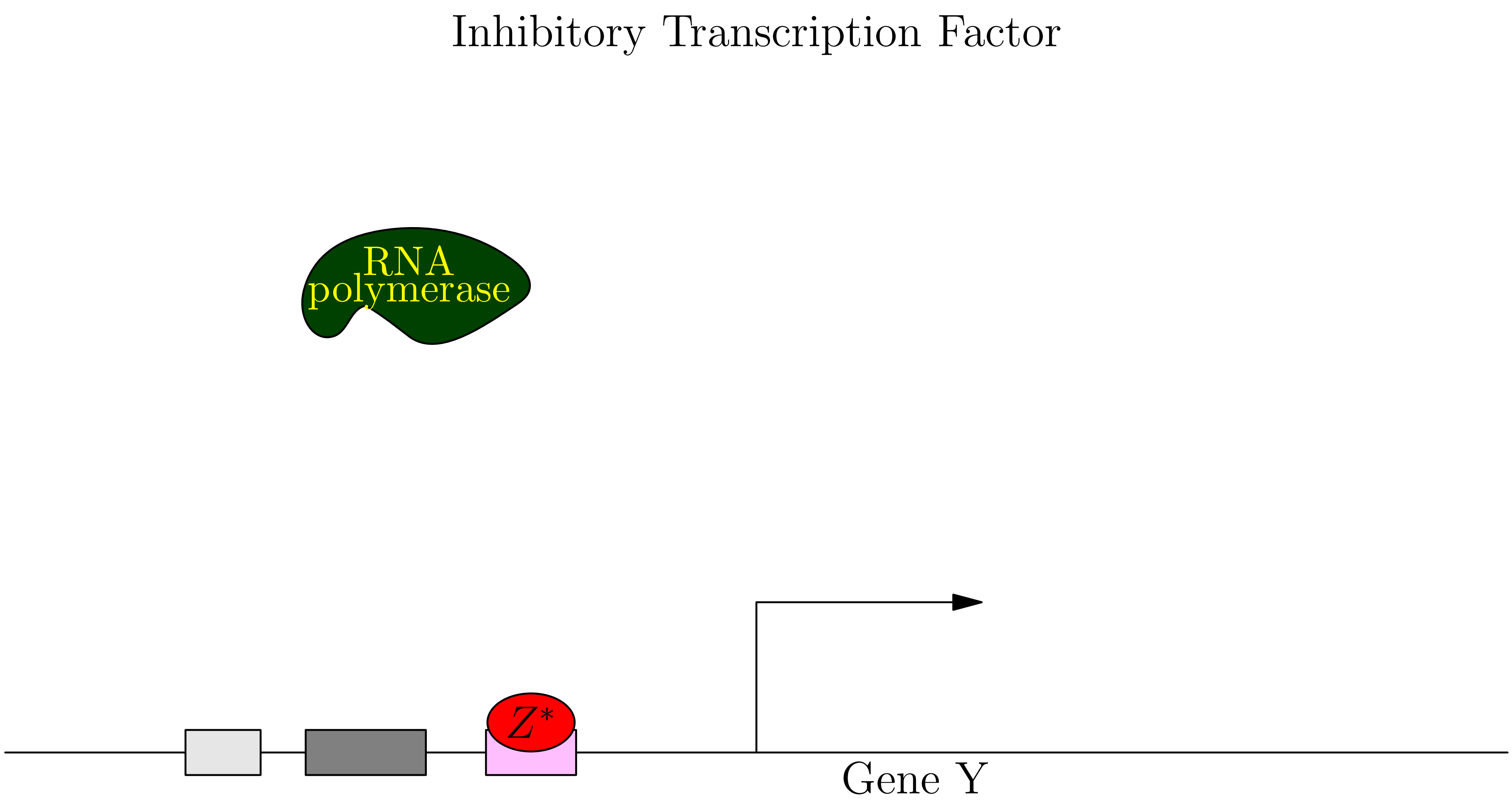

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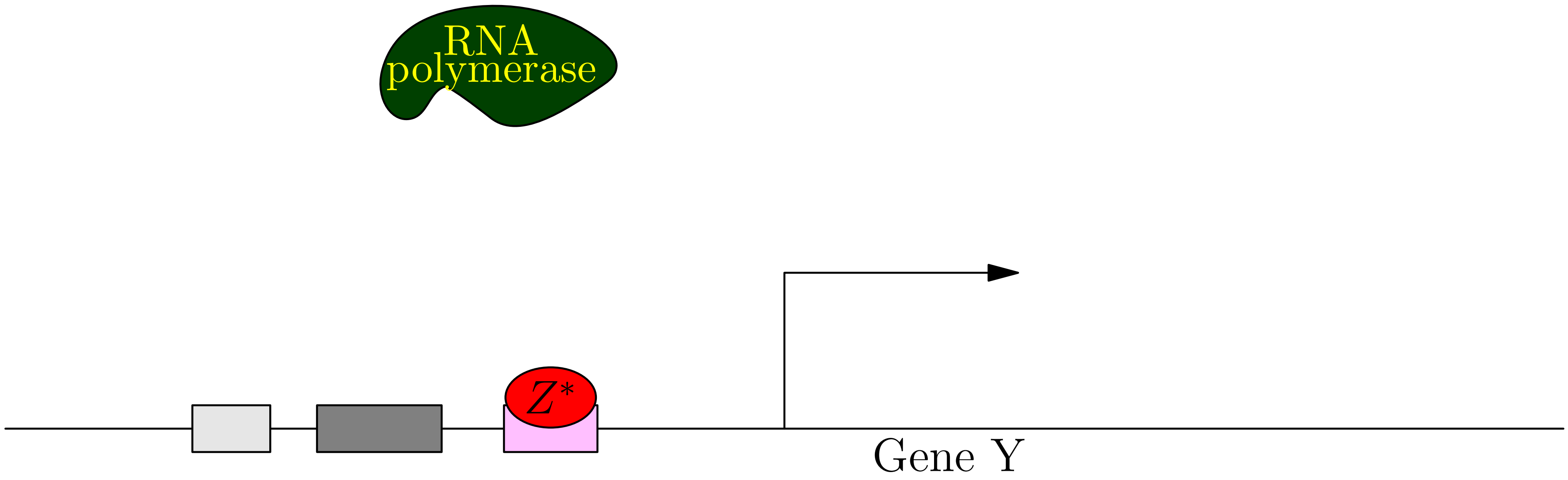

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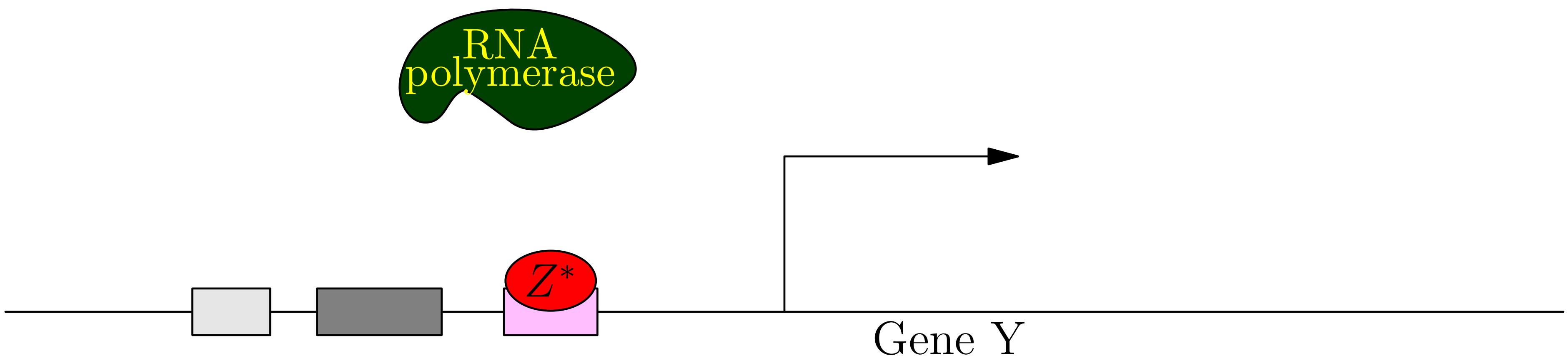

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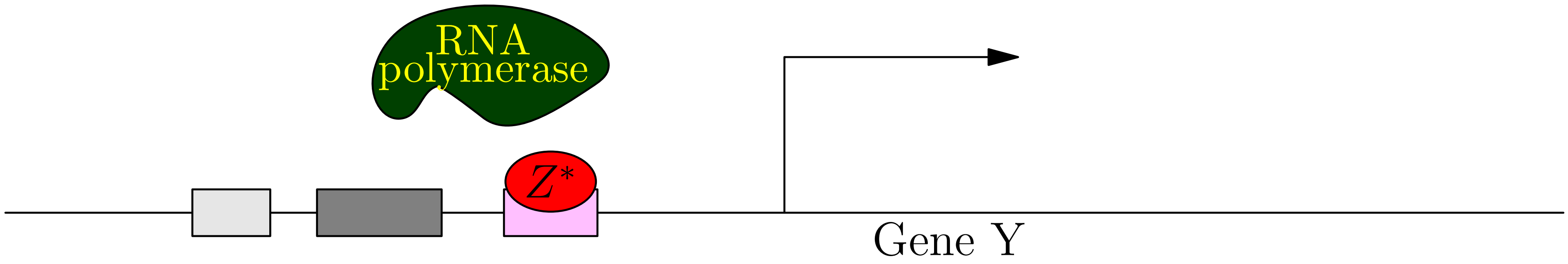

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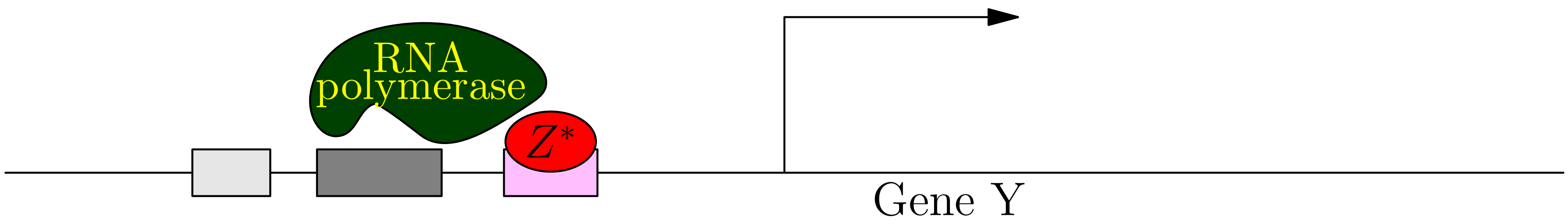

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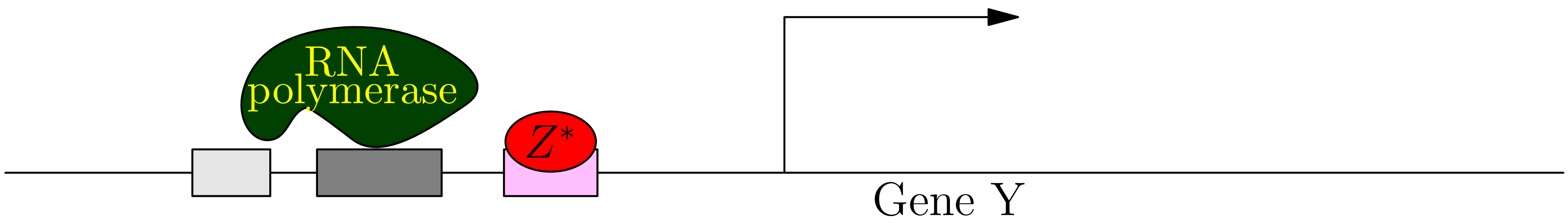

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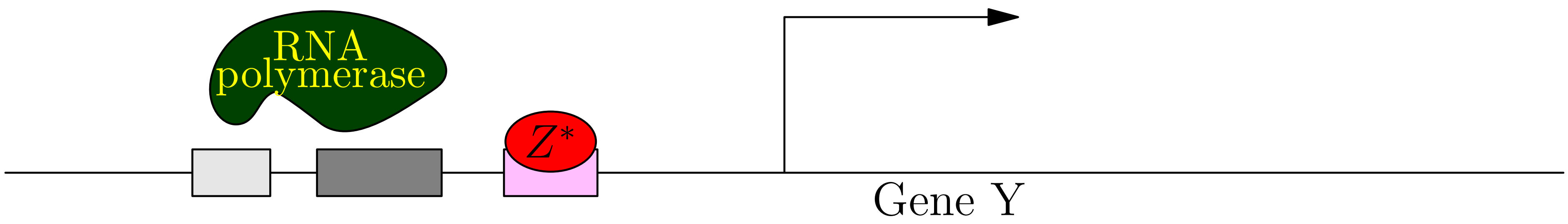

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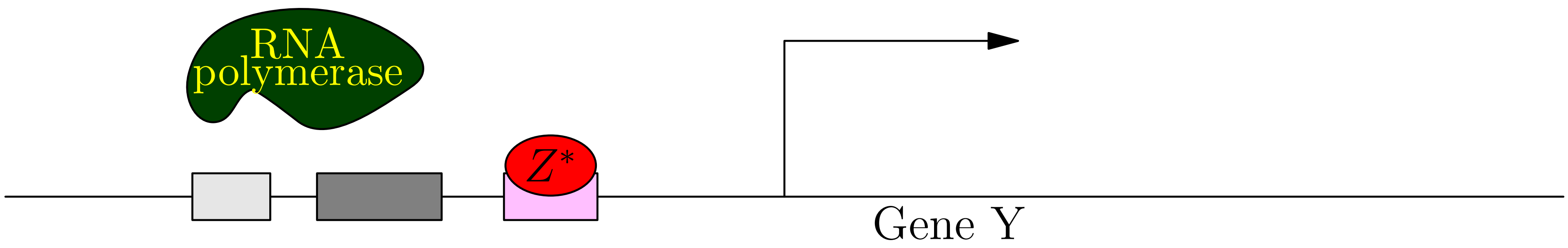

Time Scales

- Reactions happen on different time scales
  - ⋆ Binding of small (signalling) molecules: ~ 1 msec
  - ⋆ Binding of active transcription factor: ~ 1 sec
  - ⋆ Transciption $+$ translation of gene: ~ 5 min
  - ⋆ 50% change in concentration of protein: ~ 1 hour
- We can assume all events at earlier time scales happen instantly

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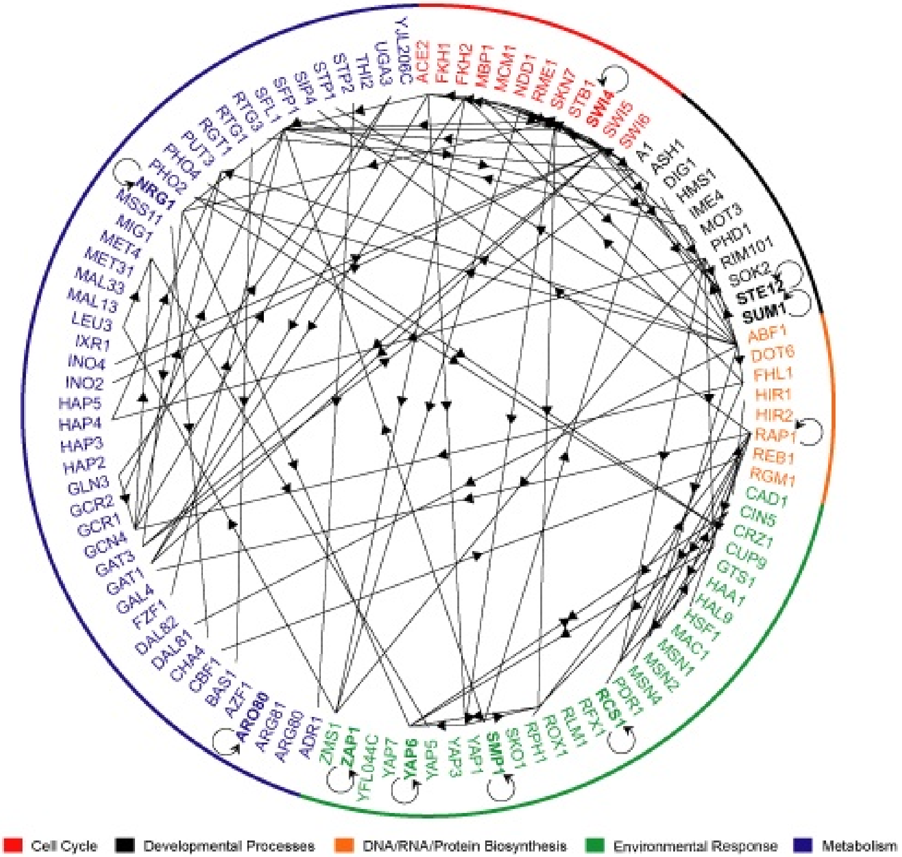
Outline

1. Gene Regulation
2. **Chemical Equations**
3. Motifs
4. Quiz

All Factors



Chemical Equations

- We can write down the mechanism that underlie gene regulation in terms of chemical equations

$$X \xrightarrow{k_1} X + RNA_Y$$

$$RNA_Y \xrightarrow{k_2} RNA_Y + Y$$

$$RNA_Y \xrightarrow{k_3} \varnothing$$

- These are massively simplified (but we can ignore reactions that happen quickly)

- The rate of reaction depends on the number/concentration of the molecules on the left-hand side and the reaction rate k_i

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Changes on Average

- Chemical equations happen at random, so the rate equations describe the average changes

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- Let $[X]$ denote the (mean) concentration of X then

$$\frac{\mathrm{d}[RNA_Y]}{\mathrm{d}t} = k_1\,[X] - k_3\,[RNA_Y]$$

$$\frac{\mathrm{d}[Y]}{\mathrm{d}t} = k_2\,[RNA_Y]$$

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Reaction Rates and Concentrations

- When we consider reaction involving multiple molecular species, e.g. two proteins forming a complex

$$X + Y \xrightarrow{k_4} C$$

- The reaction rates depend on concentrations of X and Y (how likely is it for two molecules to collide)
- To find the concentration we have to divide by the volume of the cell, V
- To obtain concentrations in moles (we also have to divide by Avogadro's number A)

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Changes on Average

- Thus

$$X + Y \xrightarrow{k_4} C$$

- Implies on average

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = \frac{\mathrm{d}[Y]}{\mathrm{d}t} = -k_4\,[X]\,[Y]$$
$$\frac{\mathrm{d}[C]}{\mathrm{d}t} = k_4\,[X]\,[Y]$$

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- Thus

$$X + Y \xrightarrow{k_4} C$$

- Implies on average

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Changes on Average

- Thus

$$X + Y \xrightarrow{k_4} C$$

- Implies on average

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = \frac{\mathrm{d}[Y]}{\mathrm{d}t} = -k_4\,[X]\,[Y] \qquad \frac{\mathrm{d}X}{\mathrm{d}t} = \frac{\mathrm{d}Y}{\mathrm{d}t} = -\frac{k_4}{\Omega}X\,Y$$

$$\frac{\mathrm{d}[C]}{\mathrm{d}t} = k_4\,[X]\,[Y] \qquad \frac{\mathrm{d}C}{\mathrm{d}t} = \frac{k_4}{\Omega}X\,Y$$

- $\Omega = VA$, $[X] = X/\Omega$ and $[Y] = Y/\Omega$

Fluctuations

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- This is often true in biology, but not always
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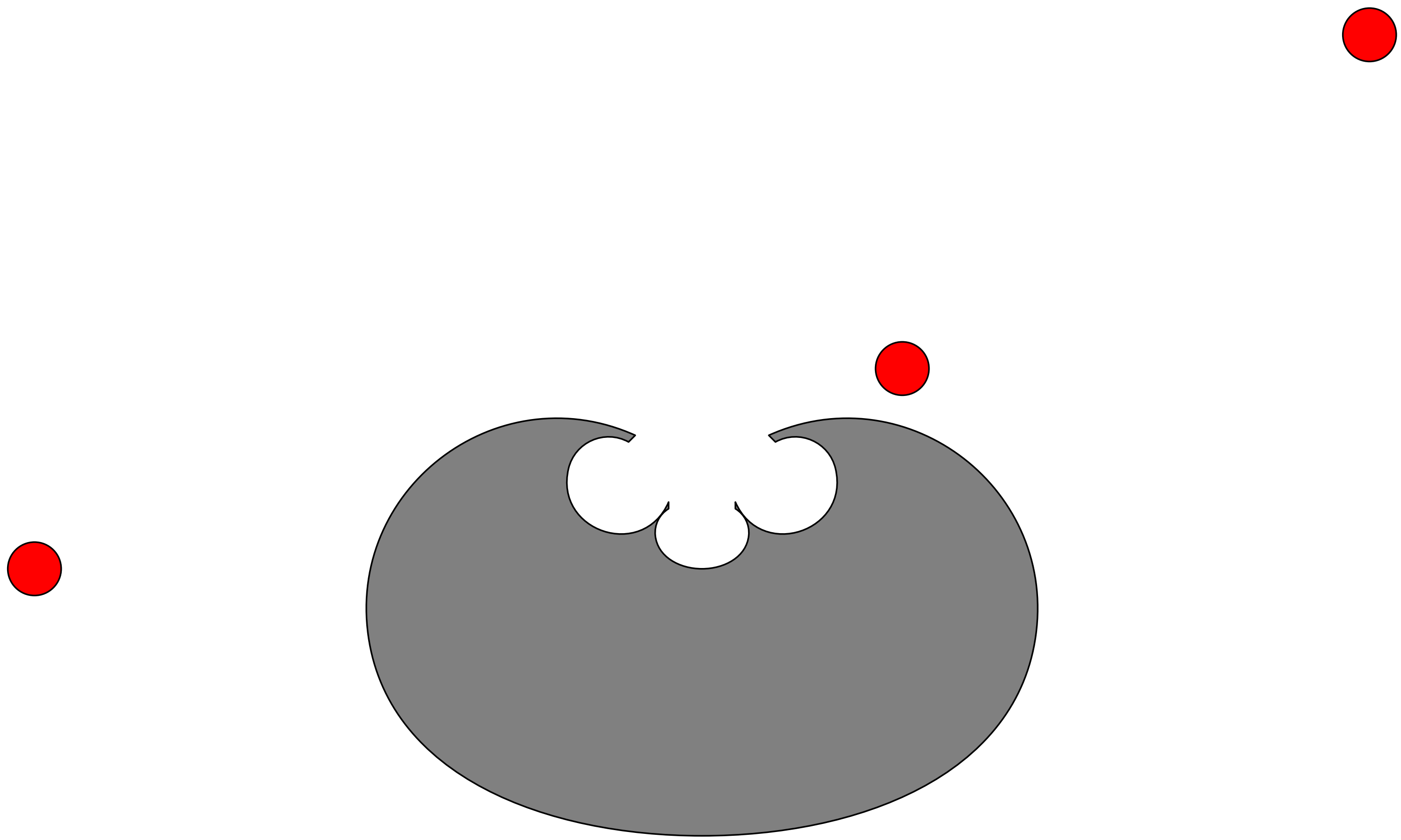
Multi-signalling



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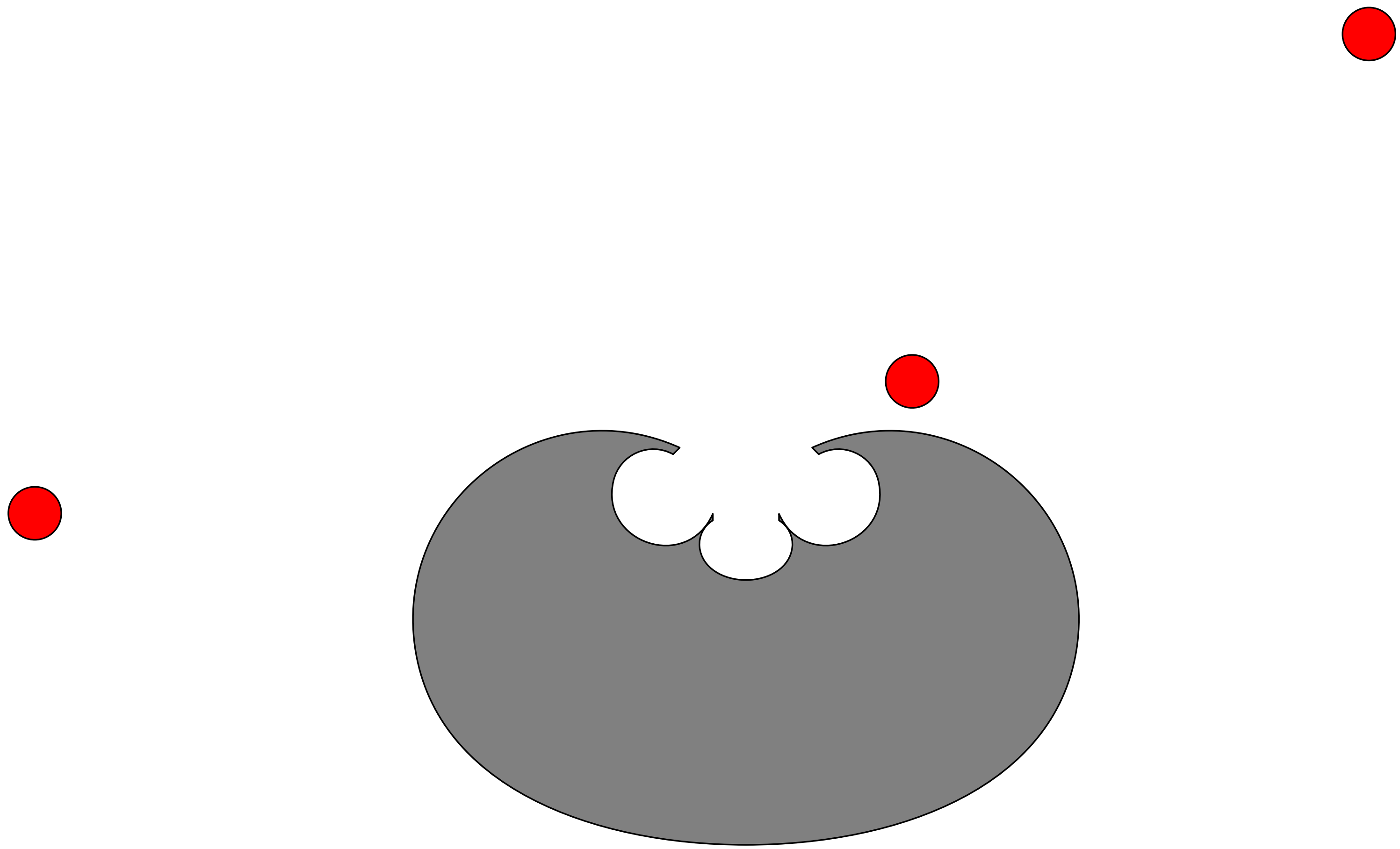

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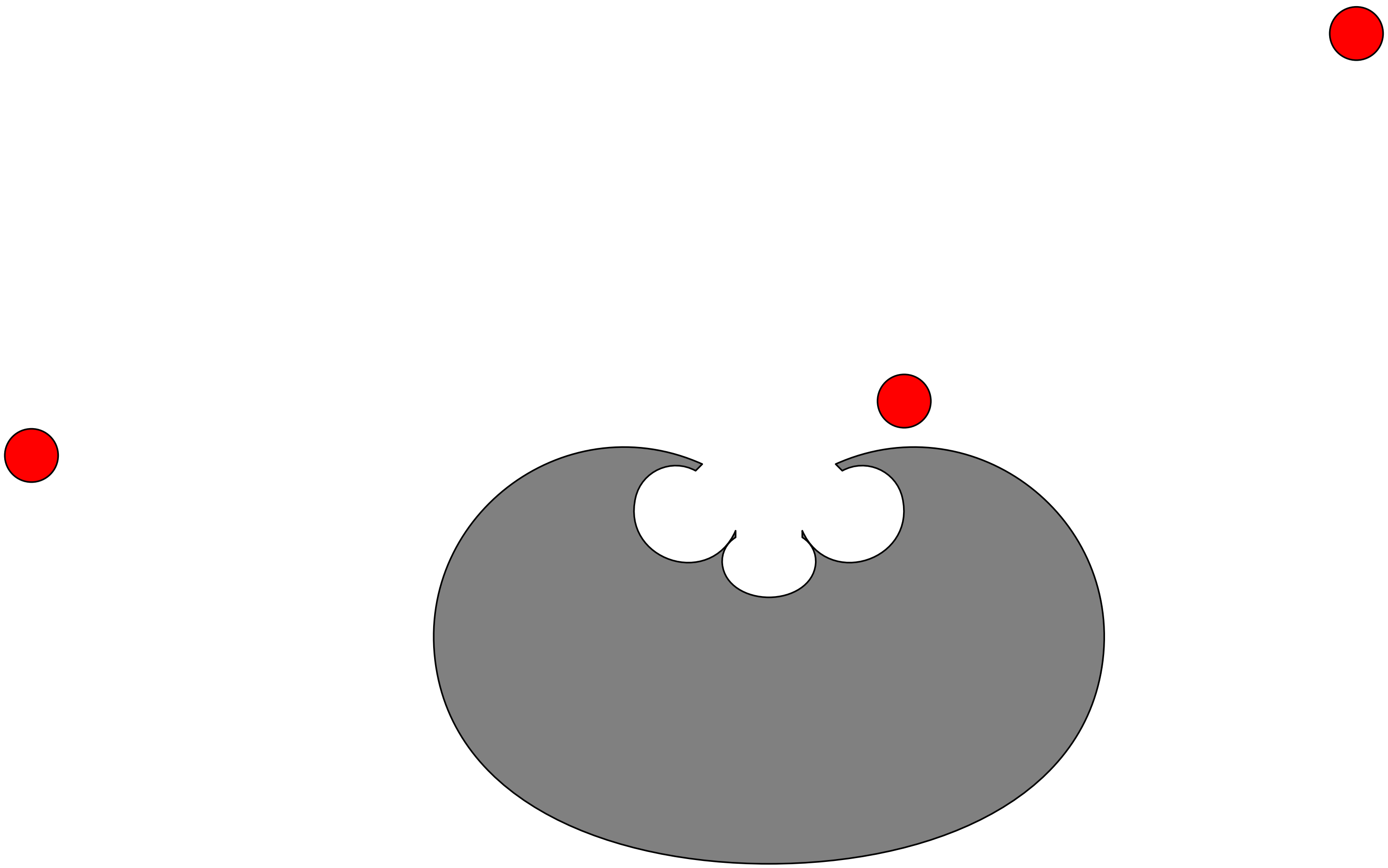

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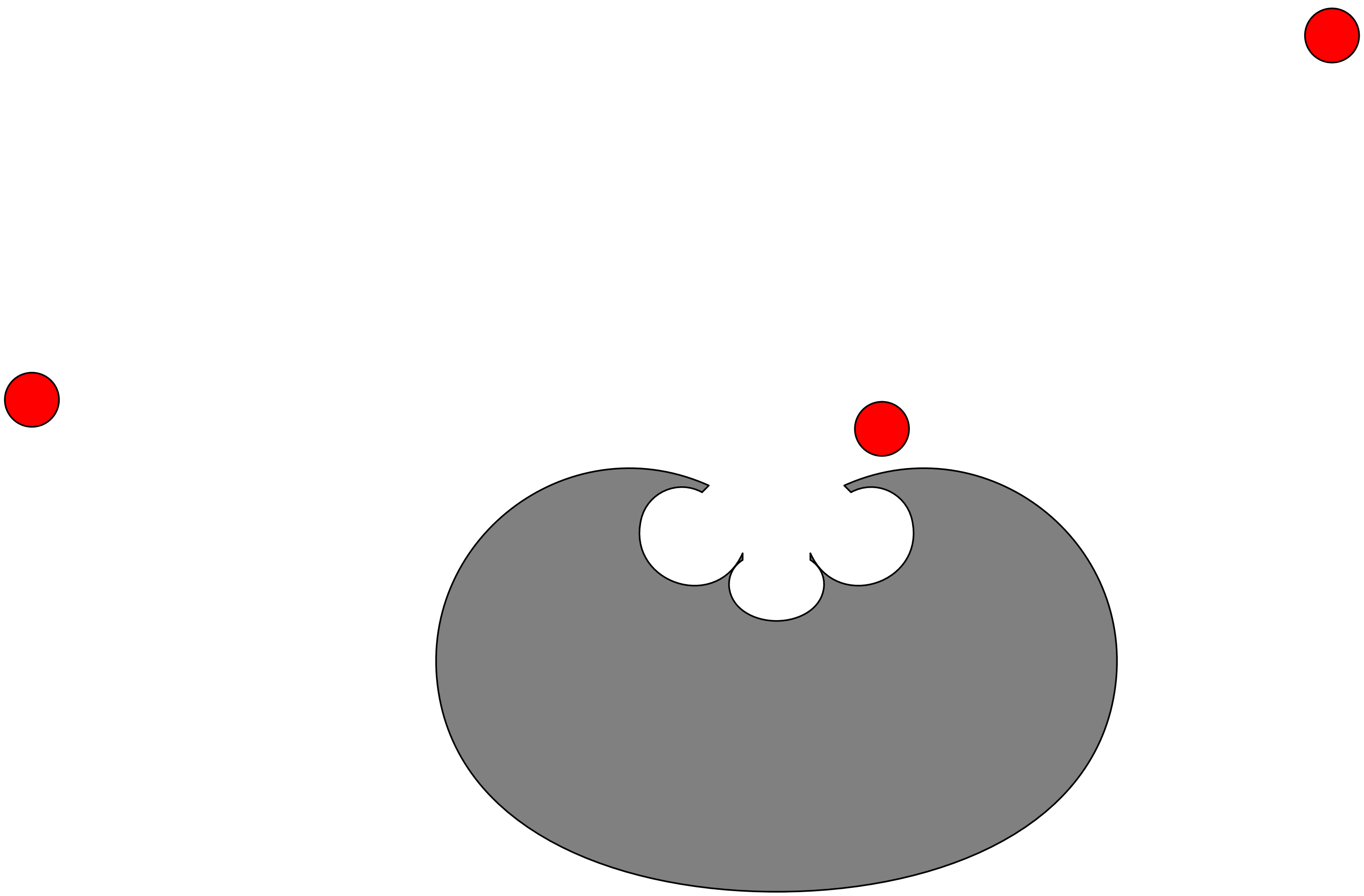

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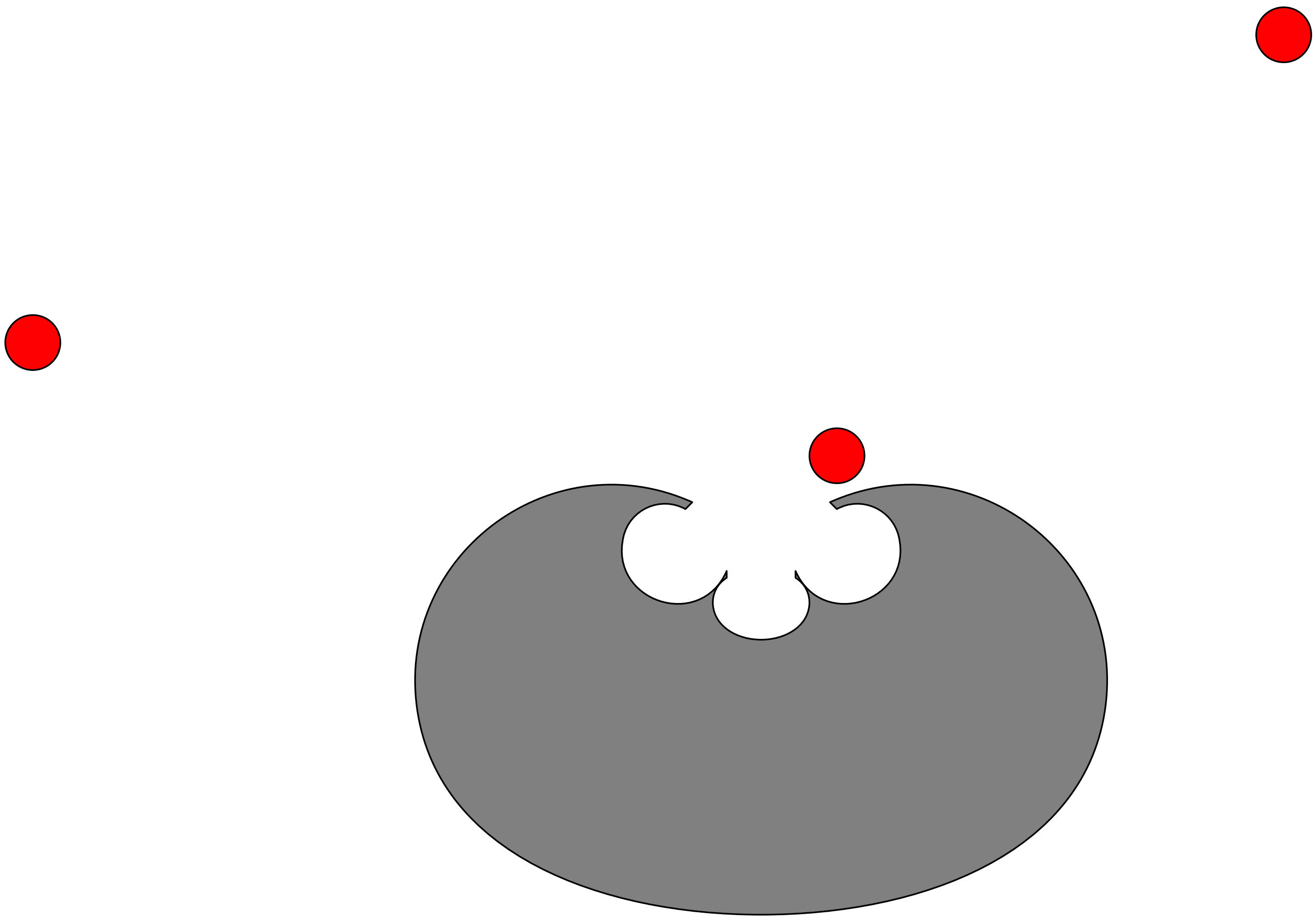

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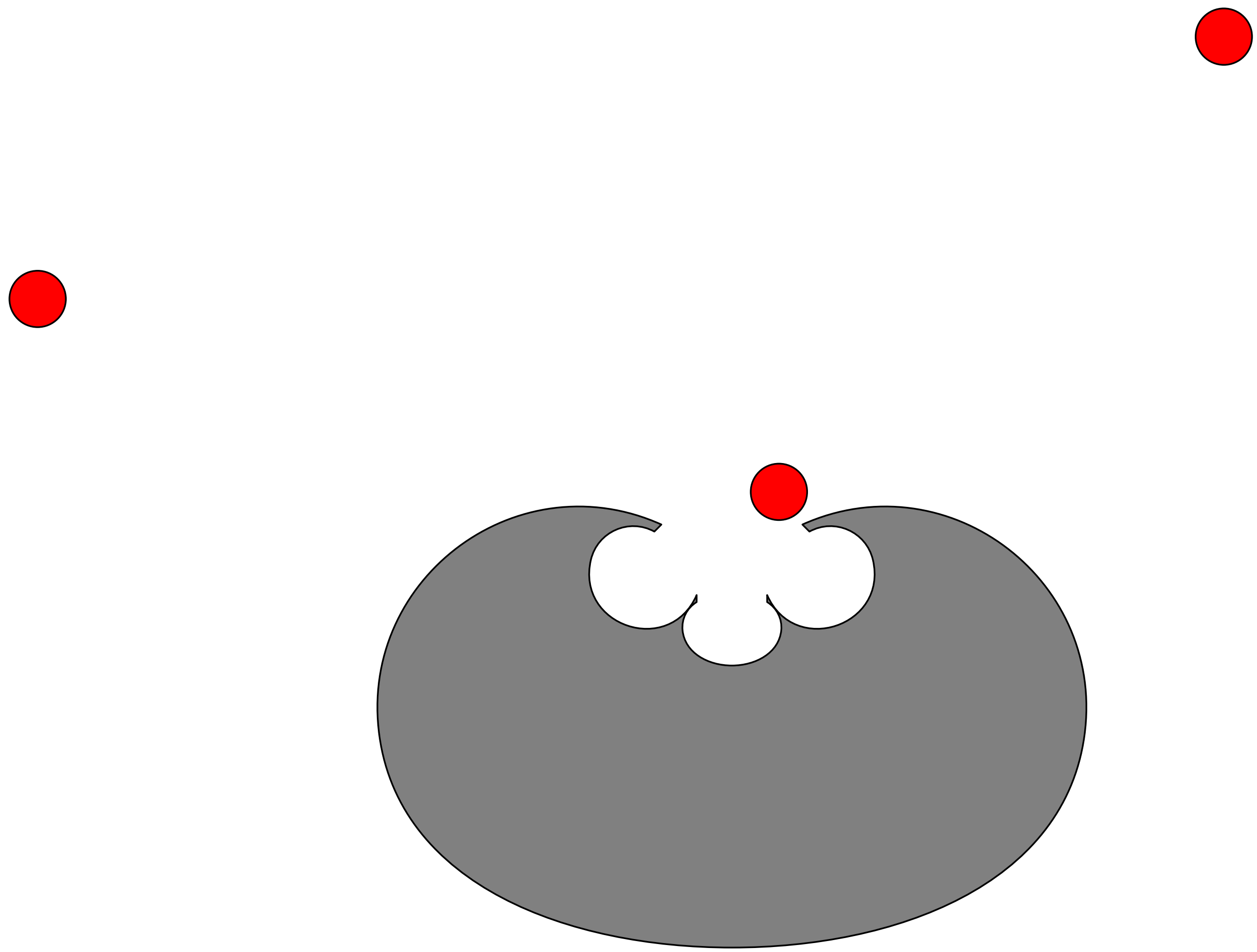

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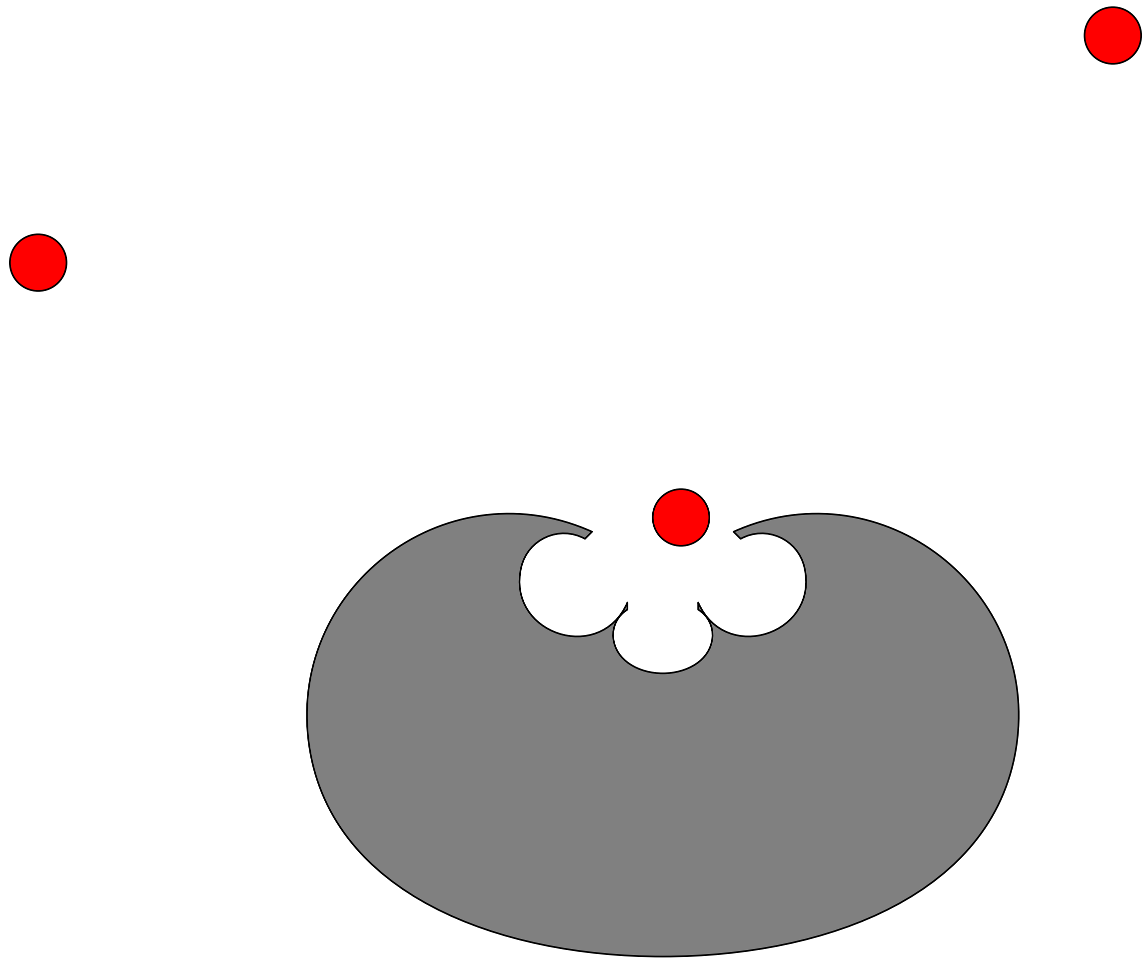

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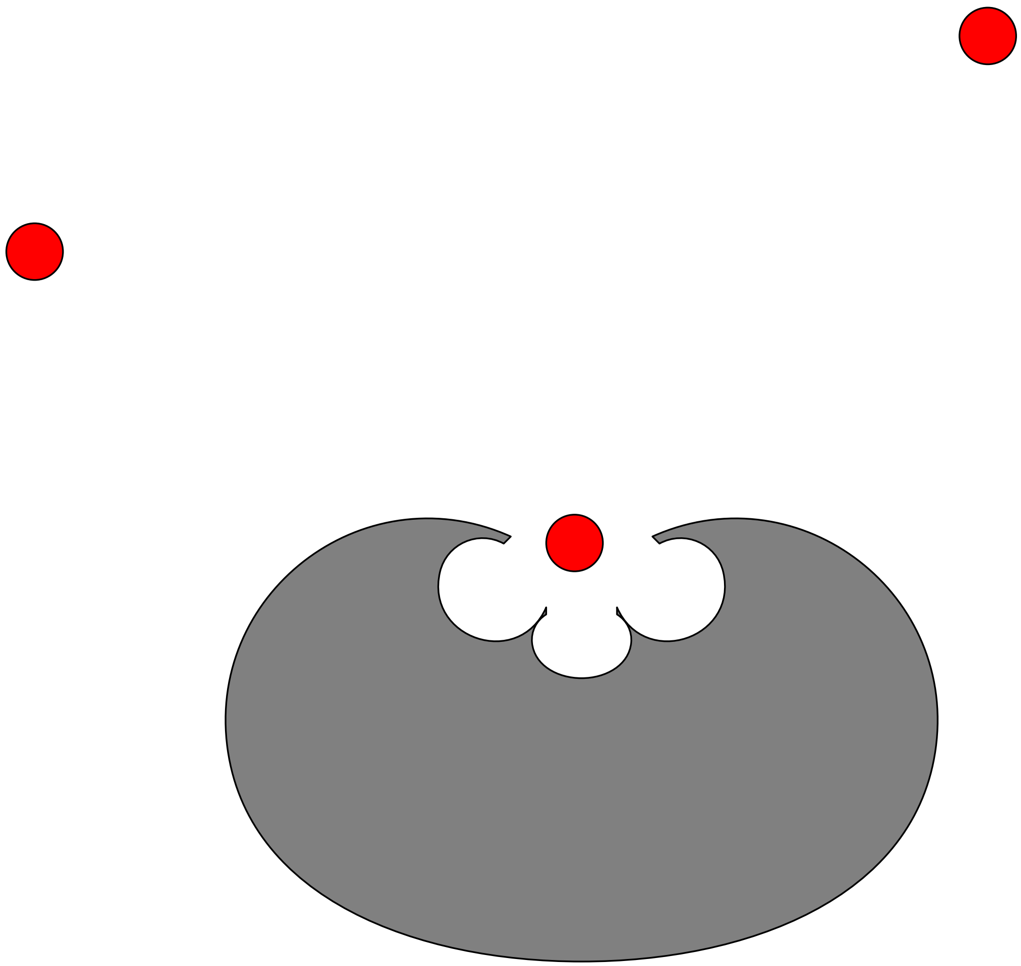

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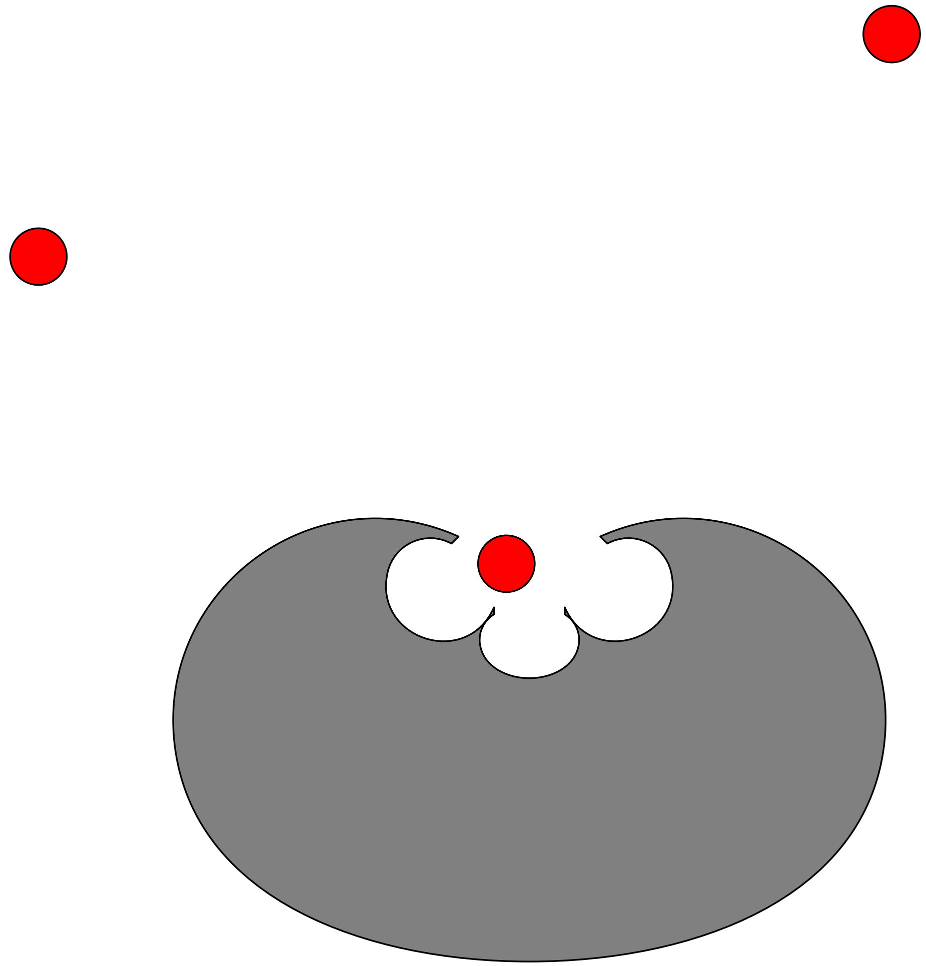

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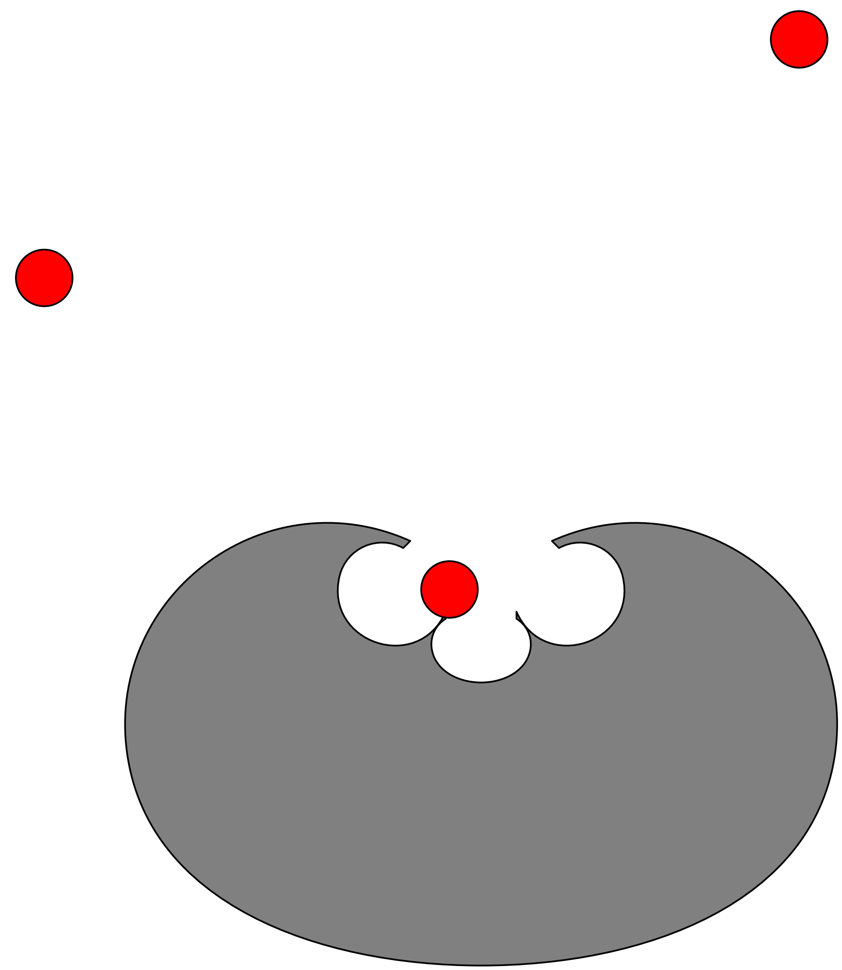

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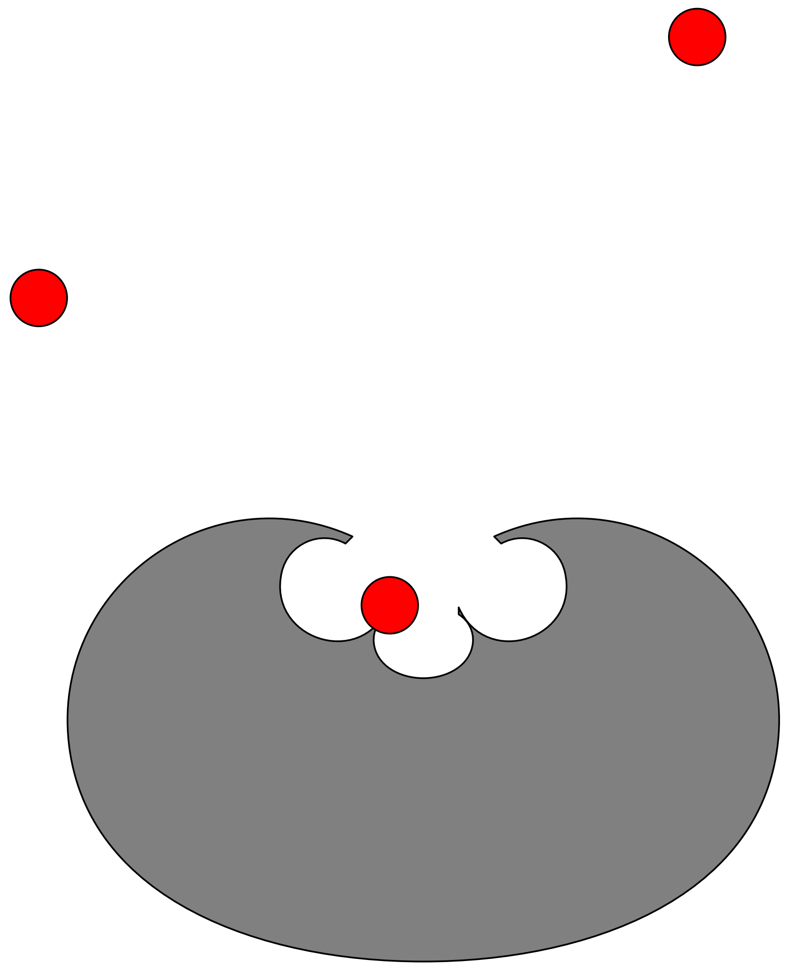

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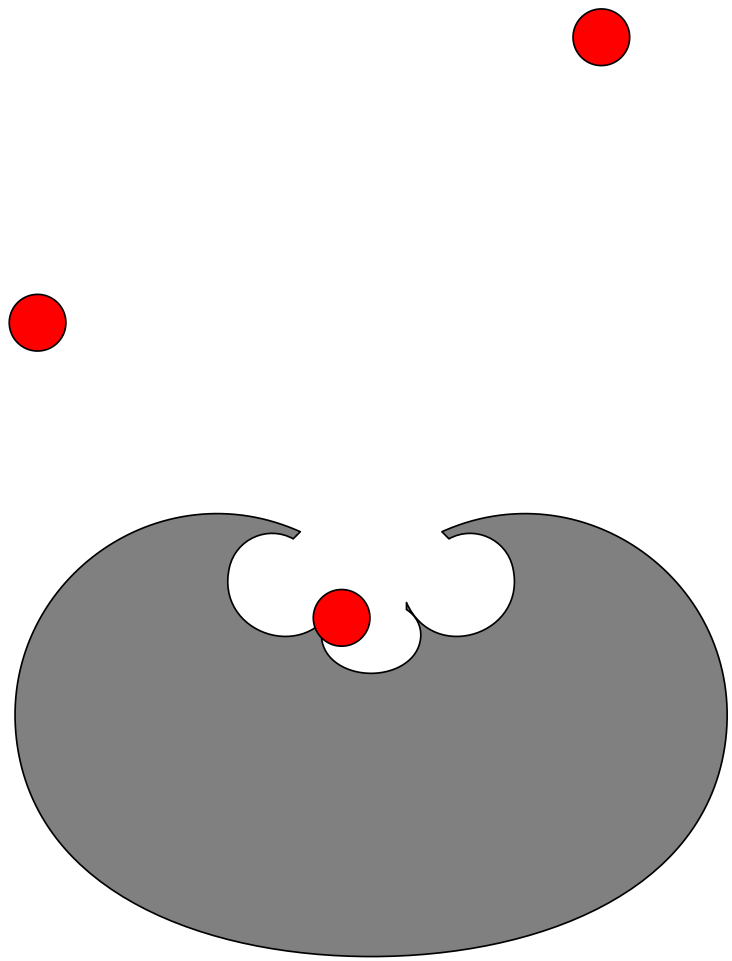

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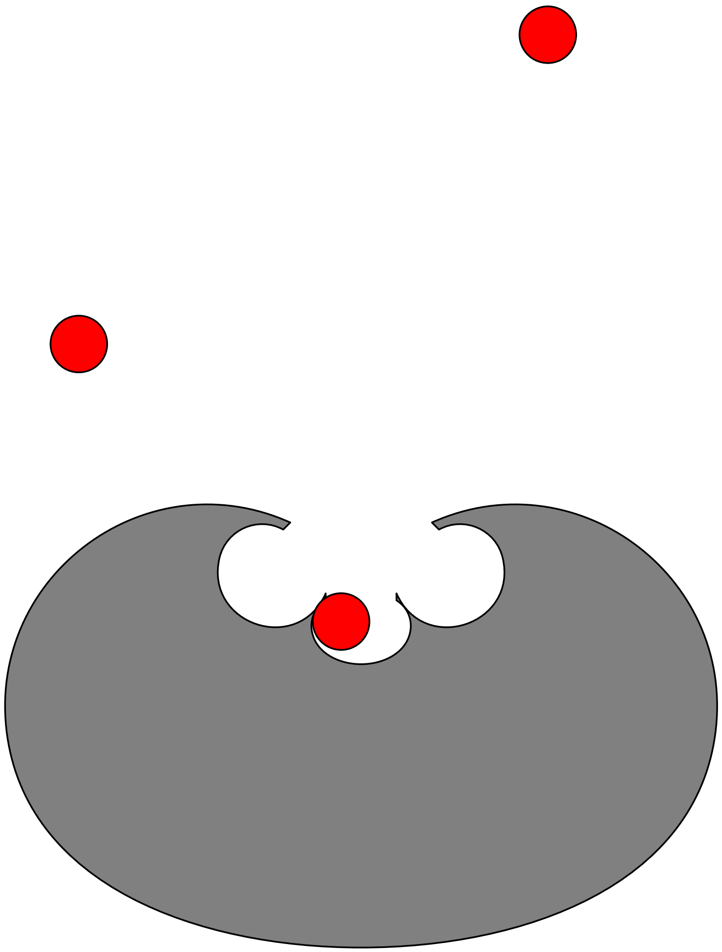

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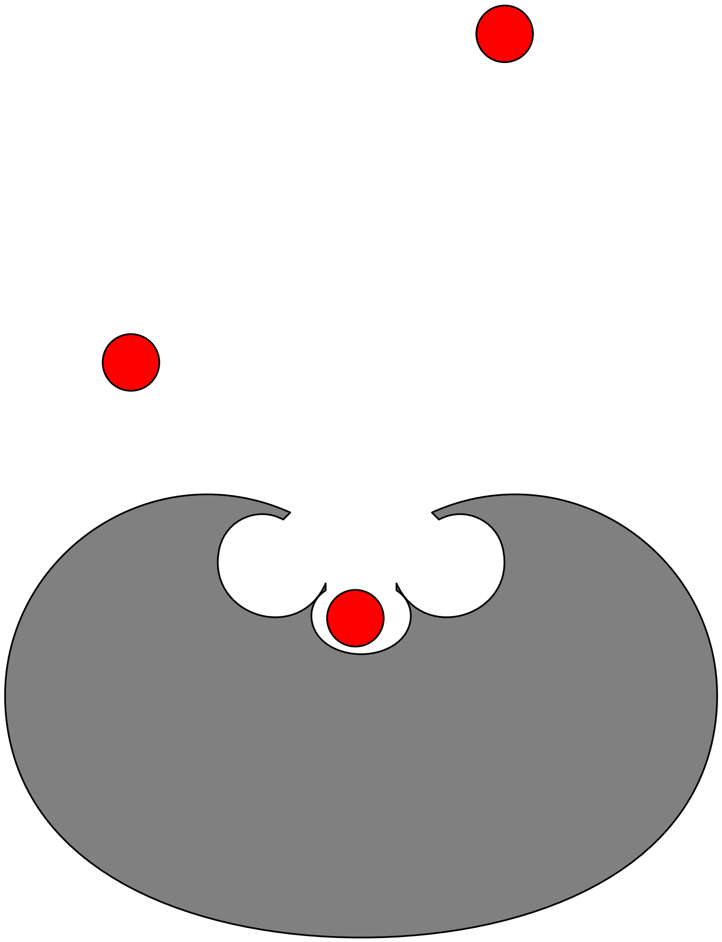

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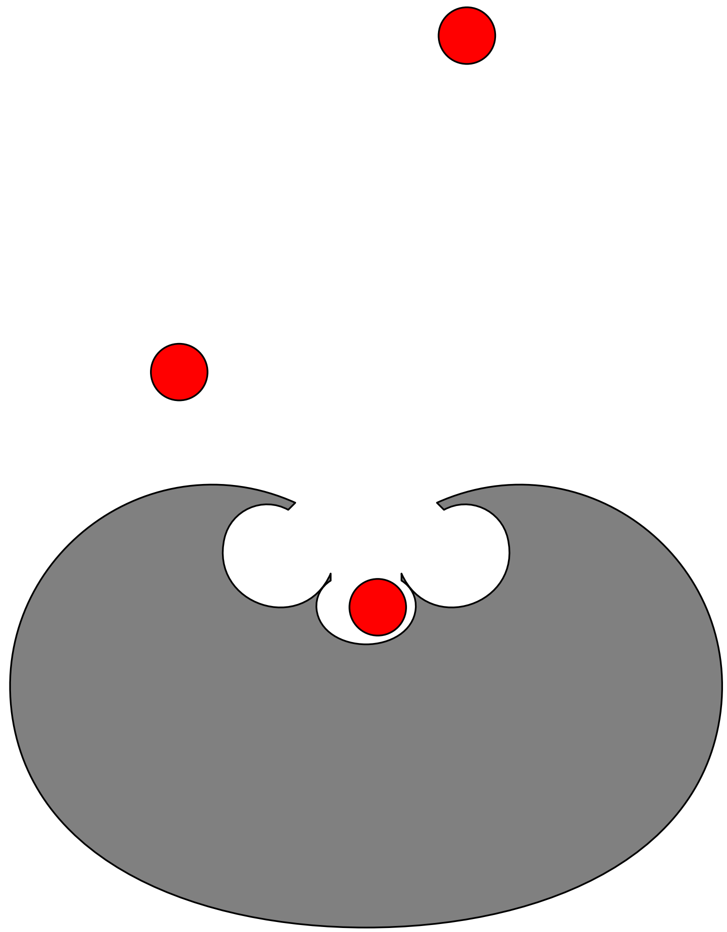

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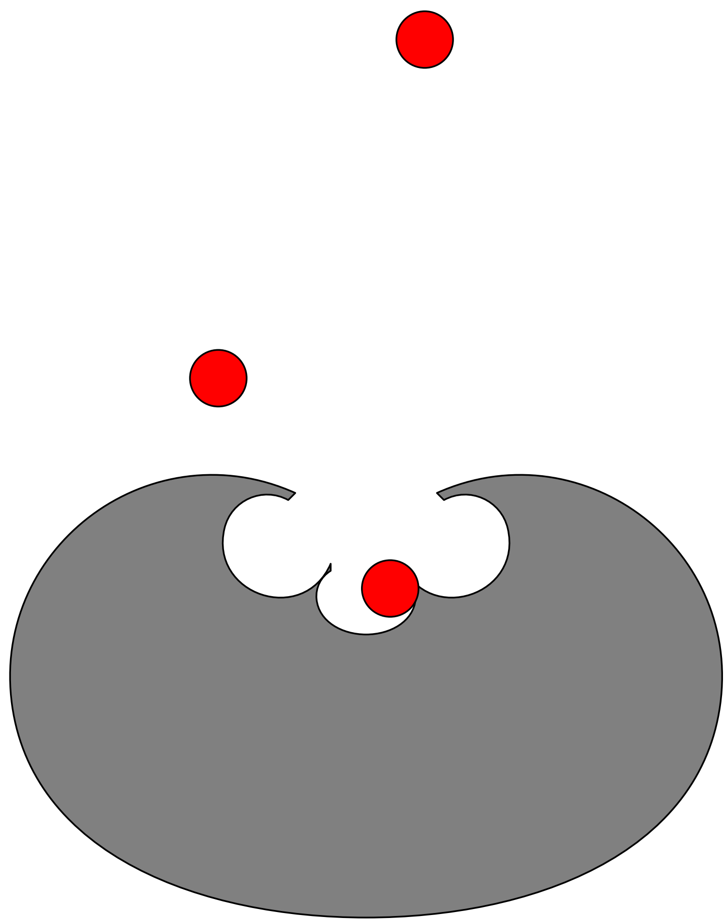

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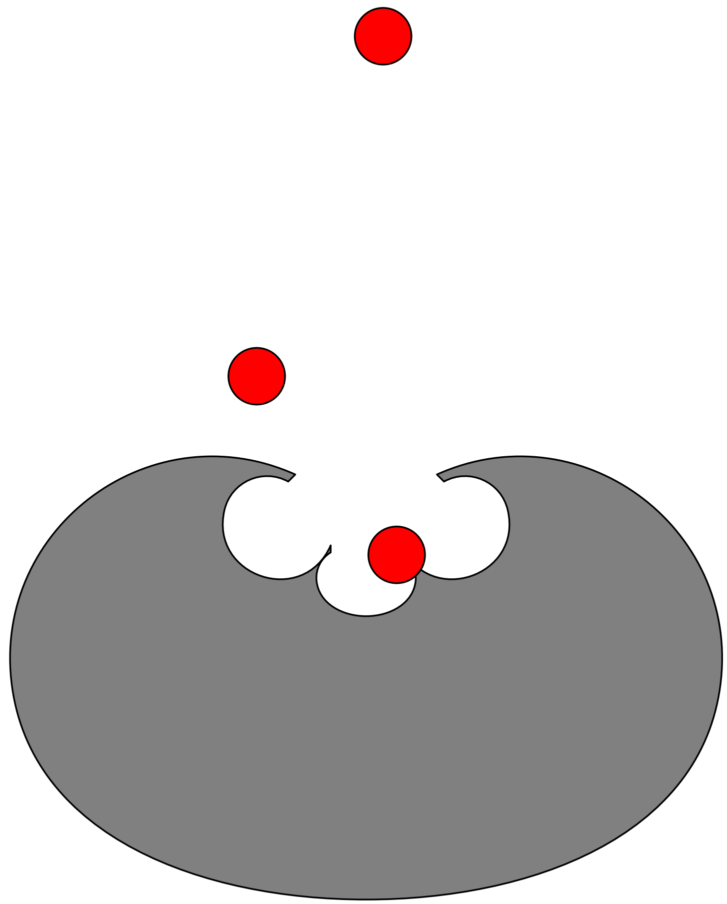

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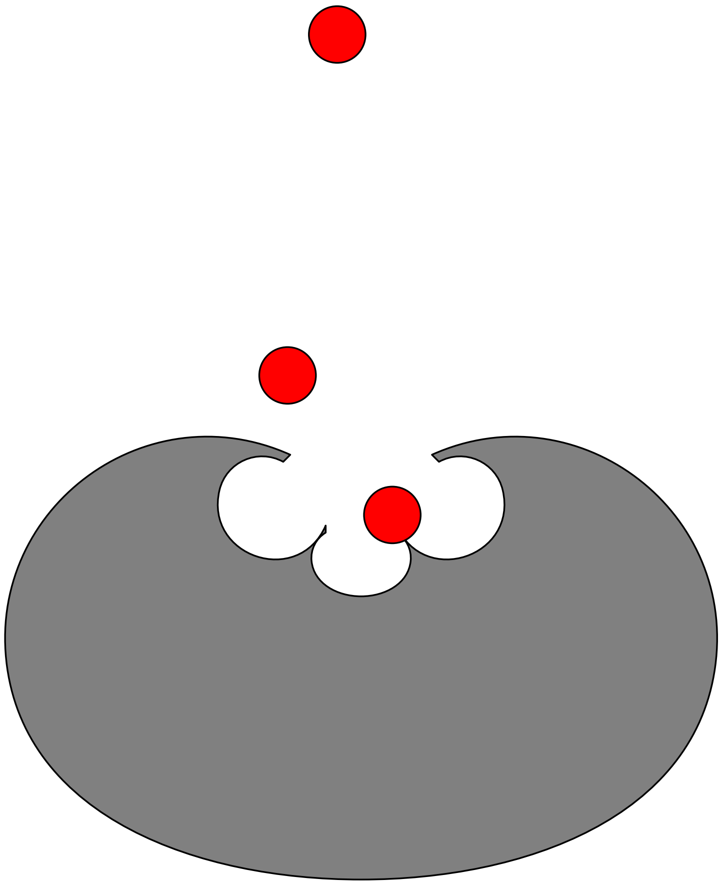

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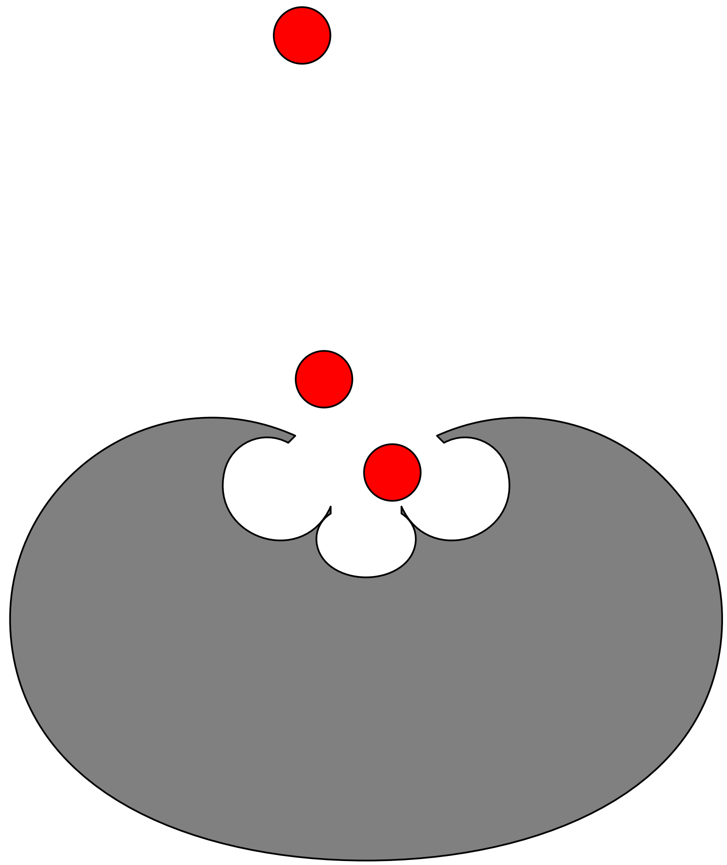

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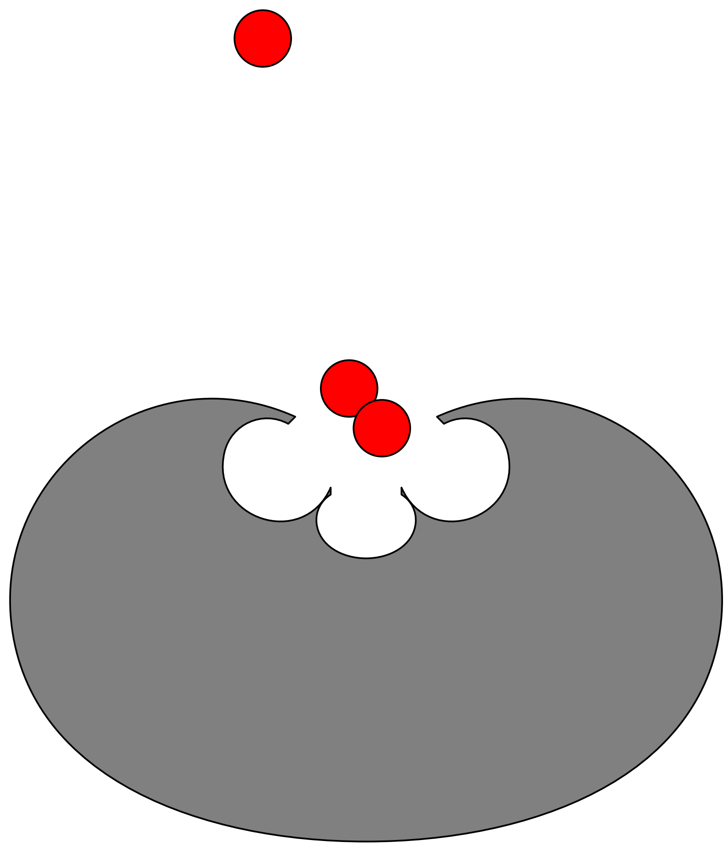

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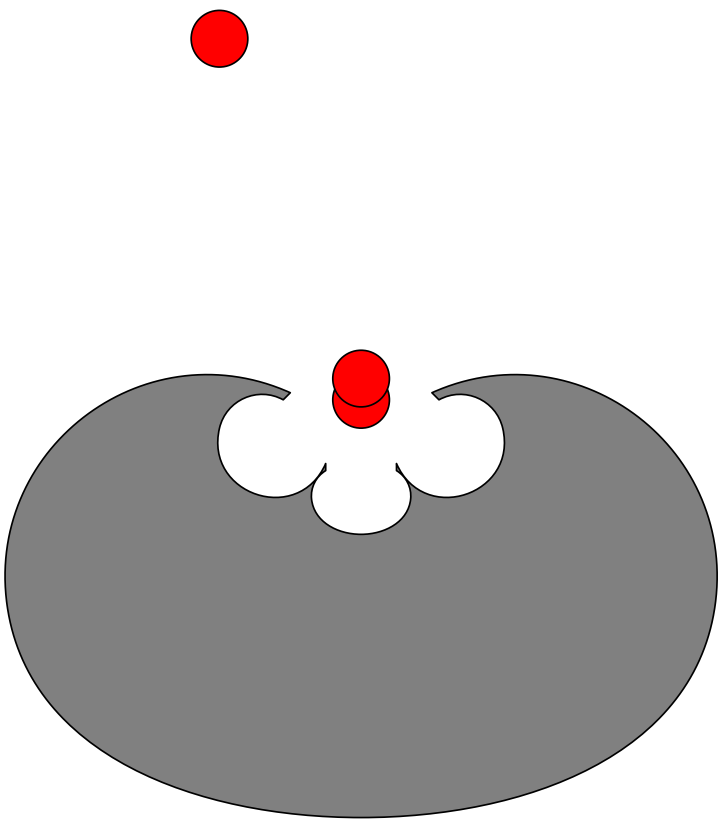

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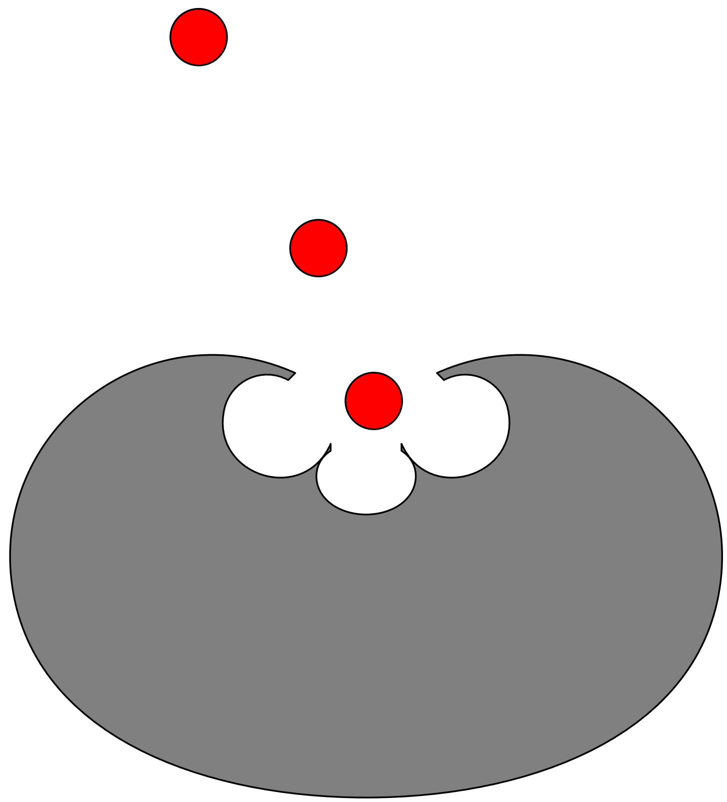

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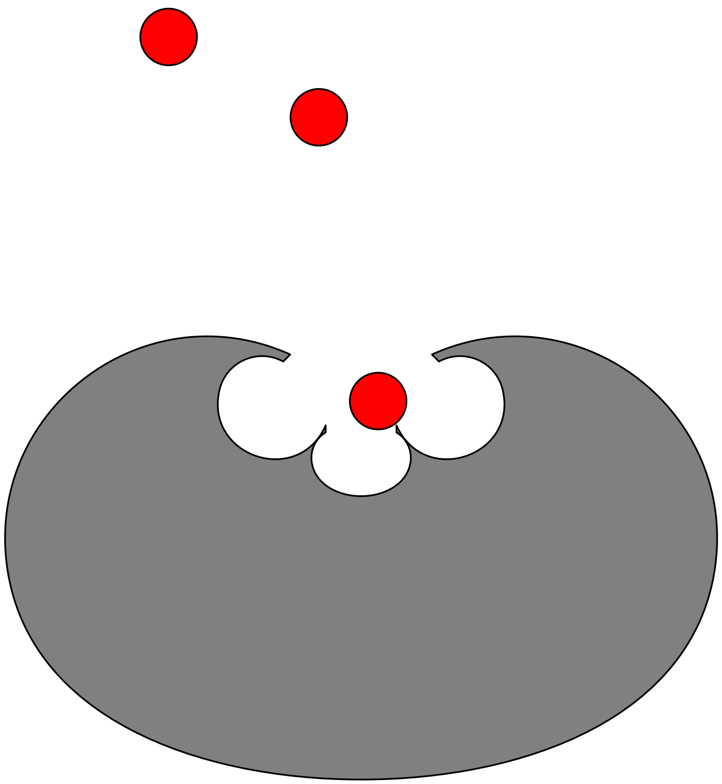

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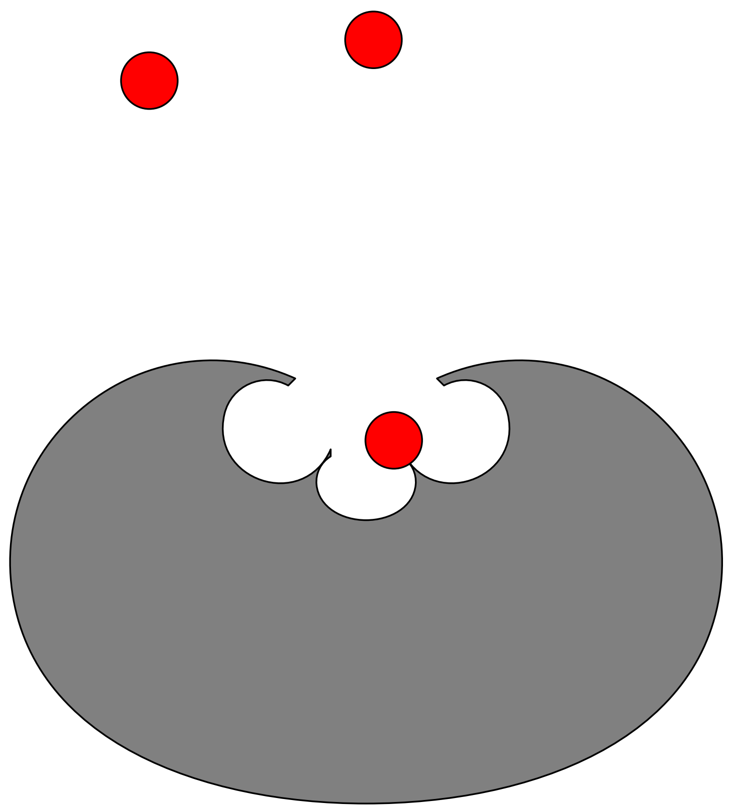

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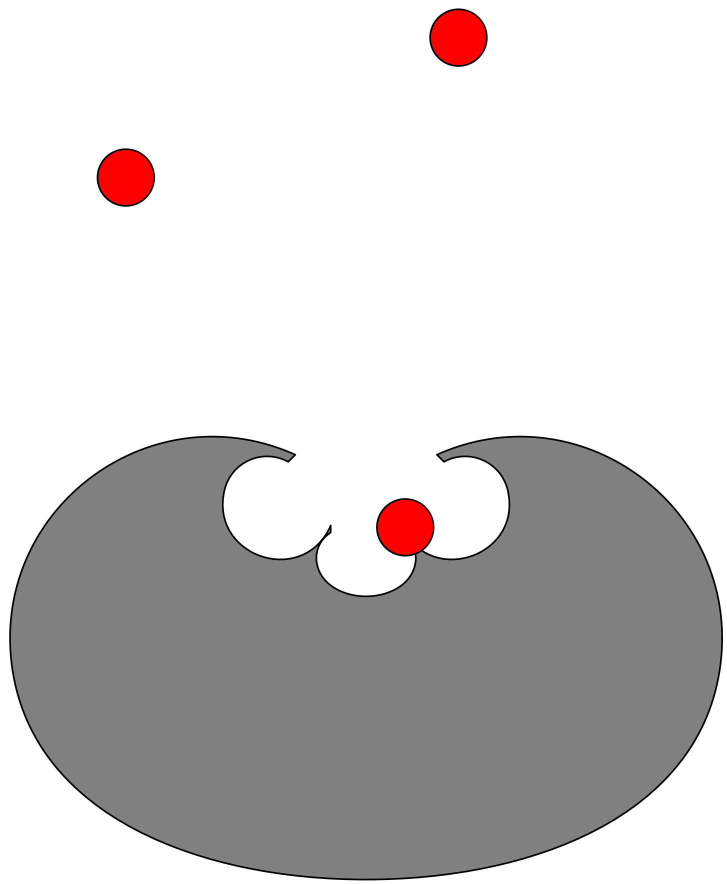

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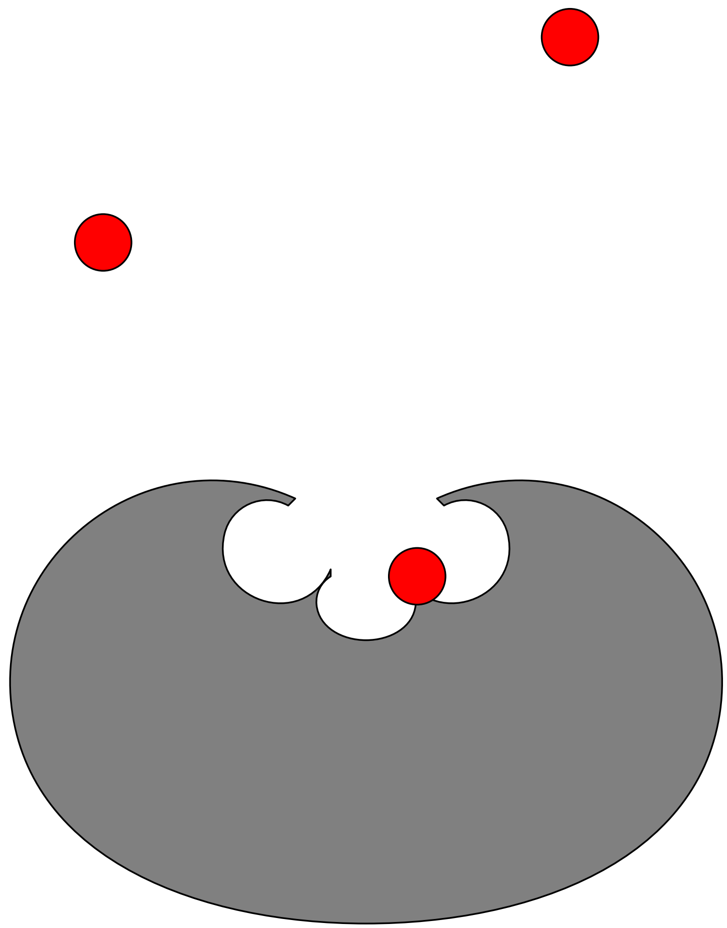

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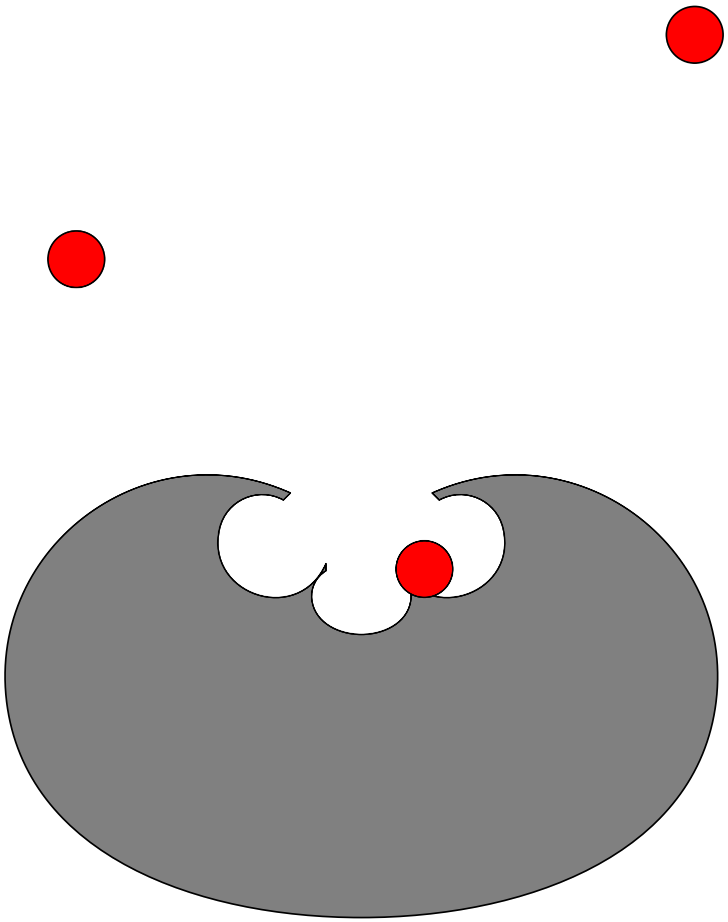

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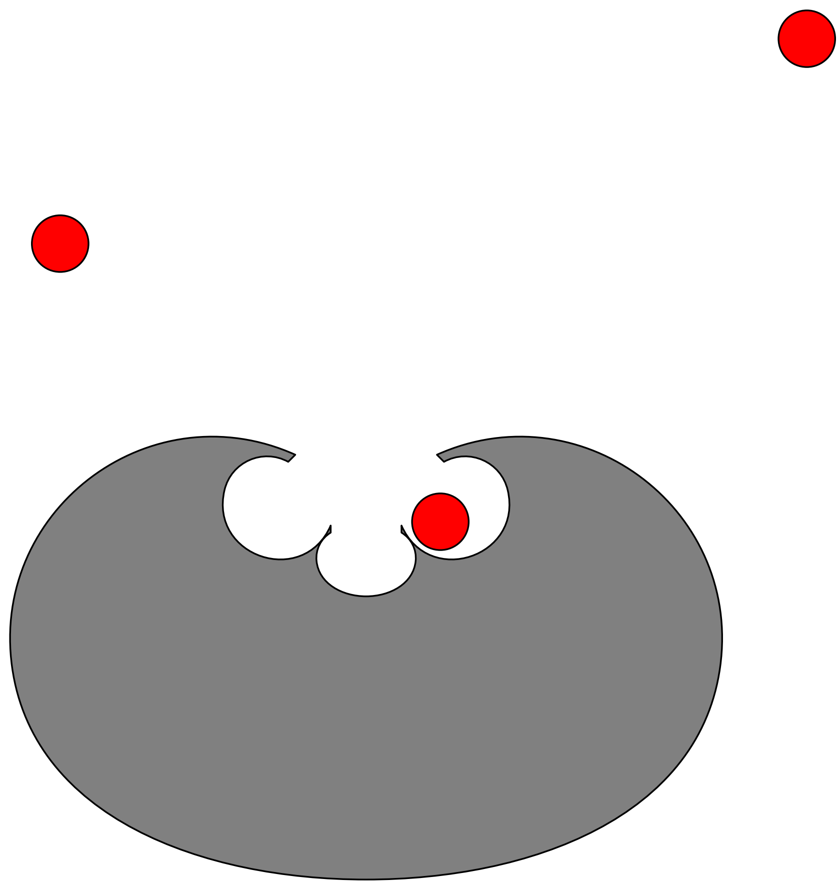

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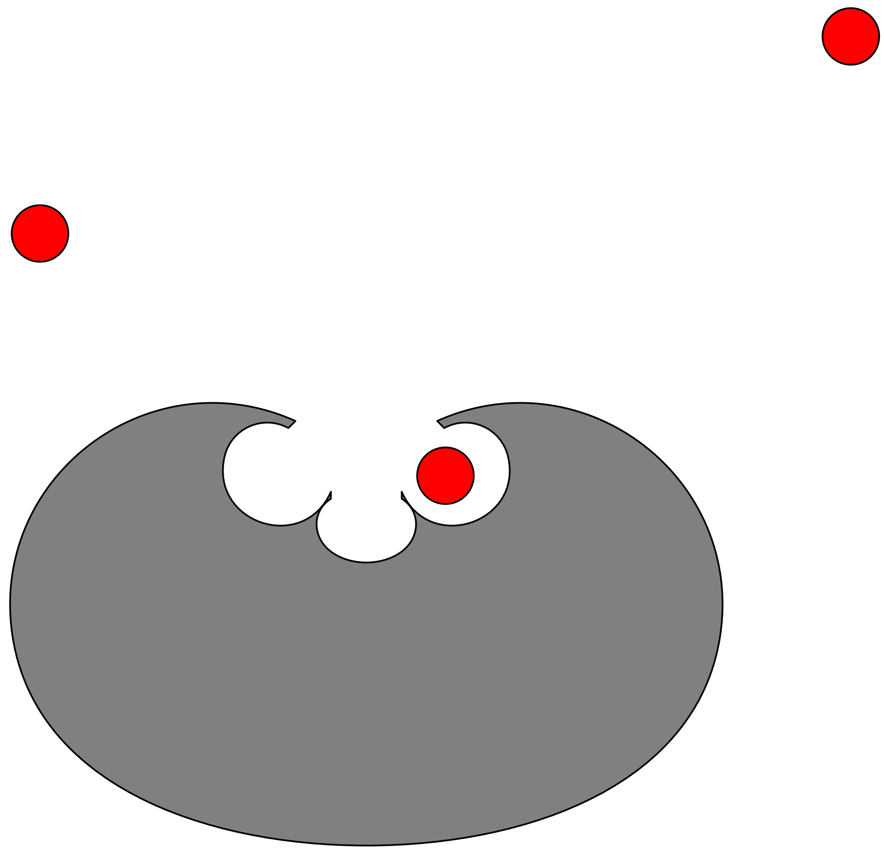

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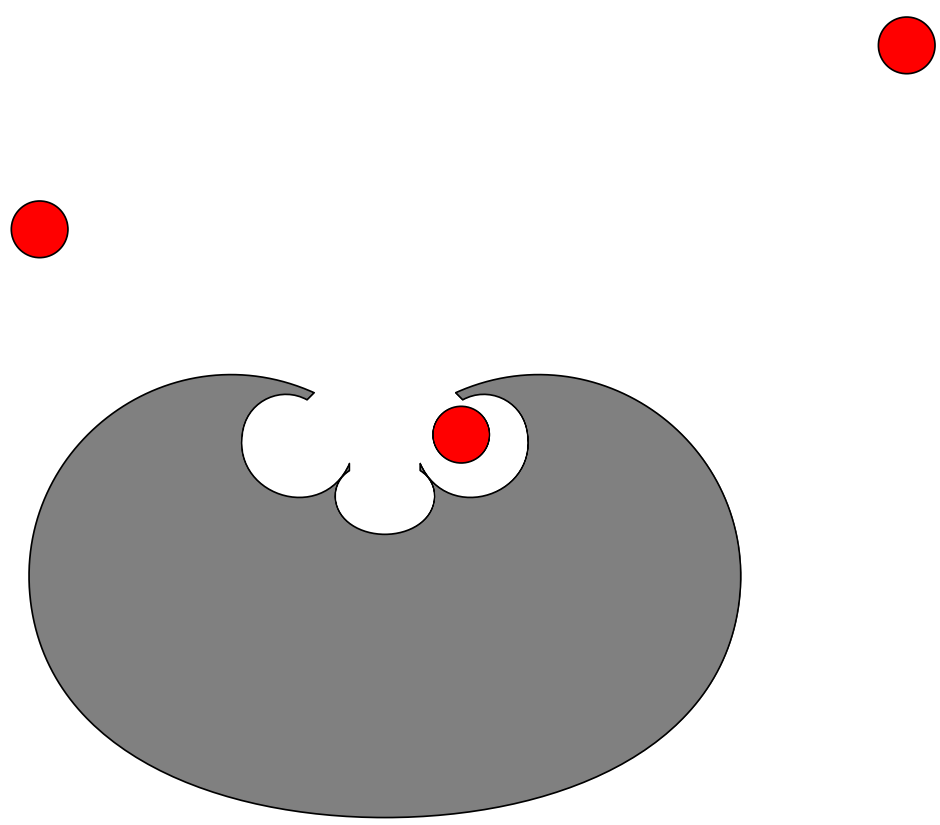

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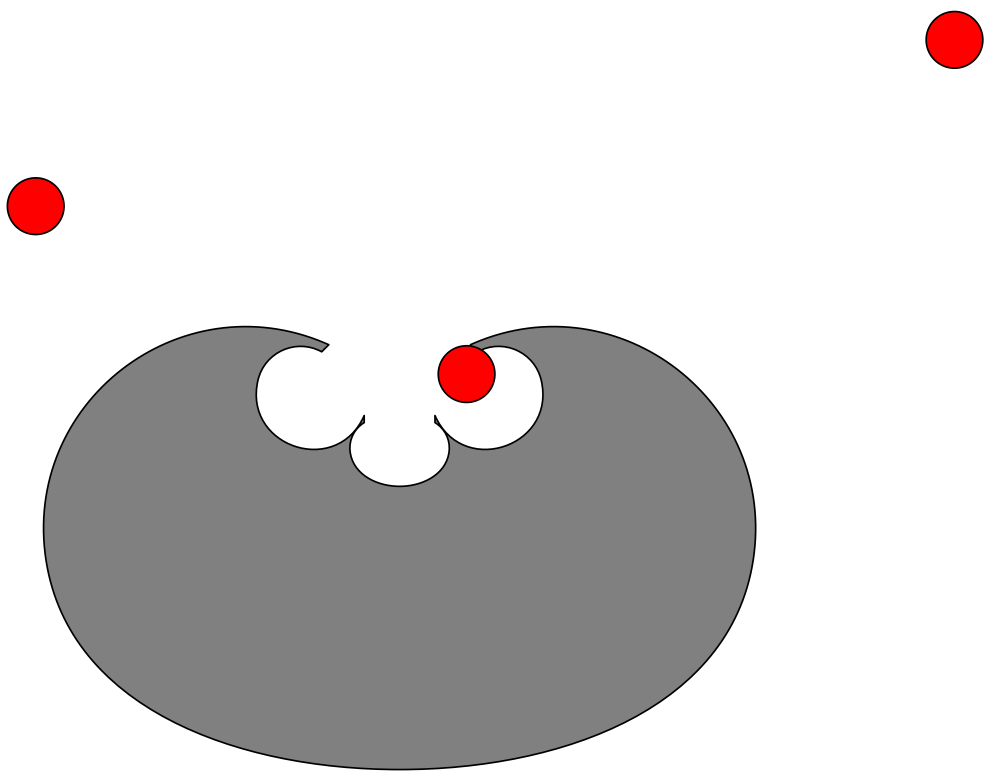

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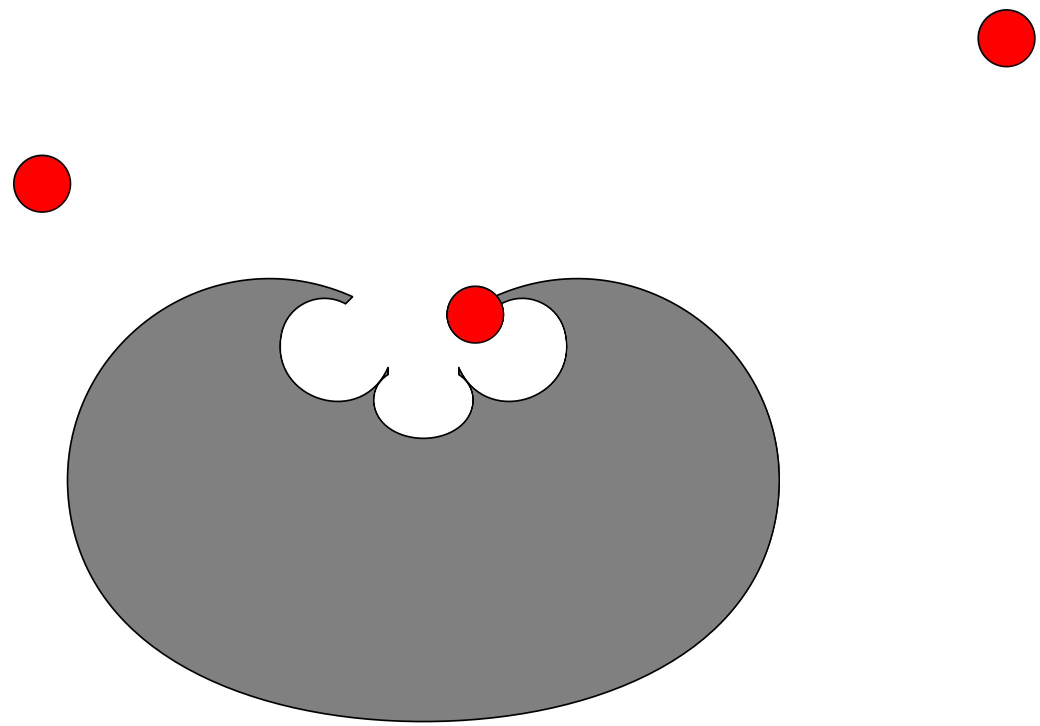

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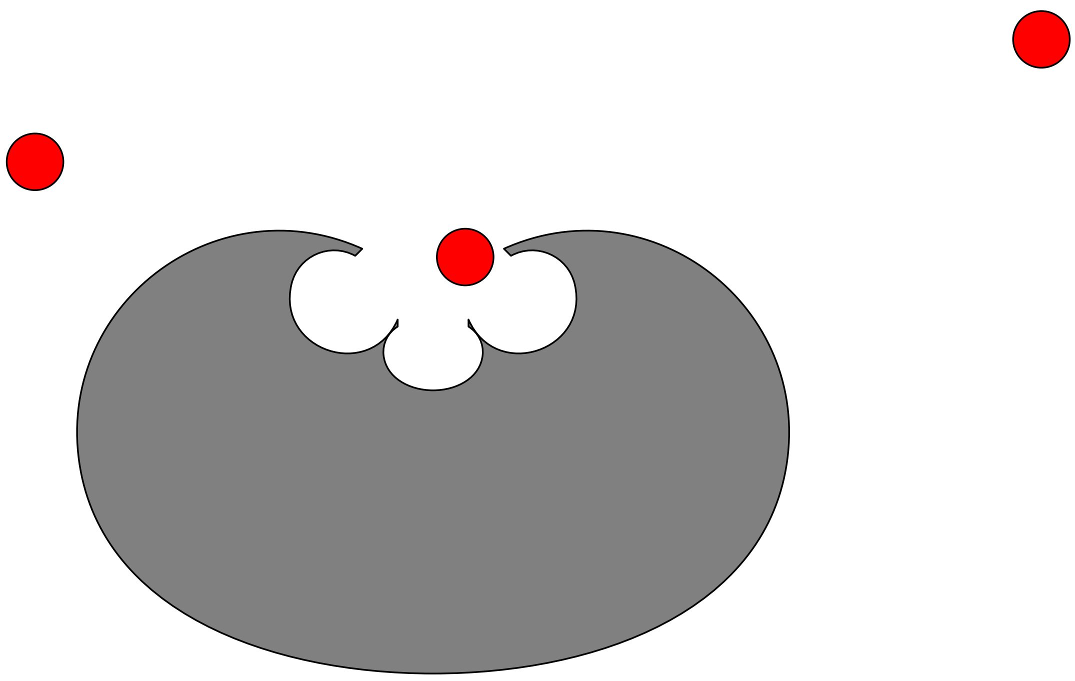

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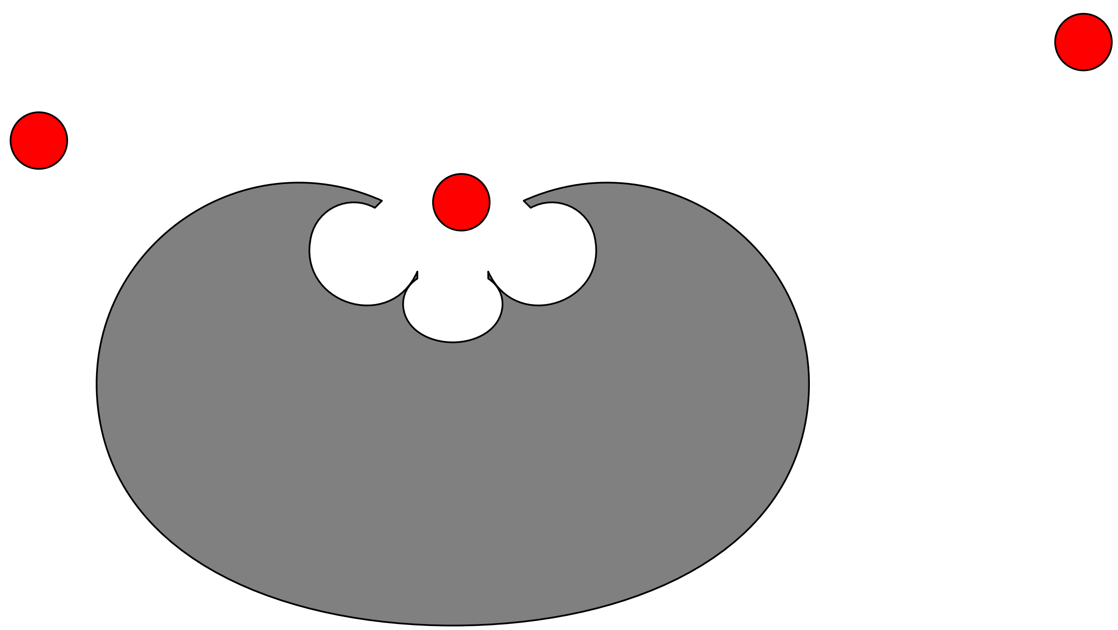

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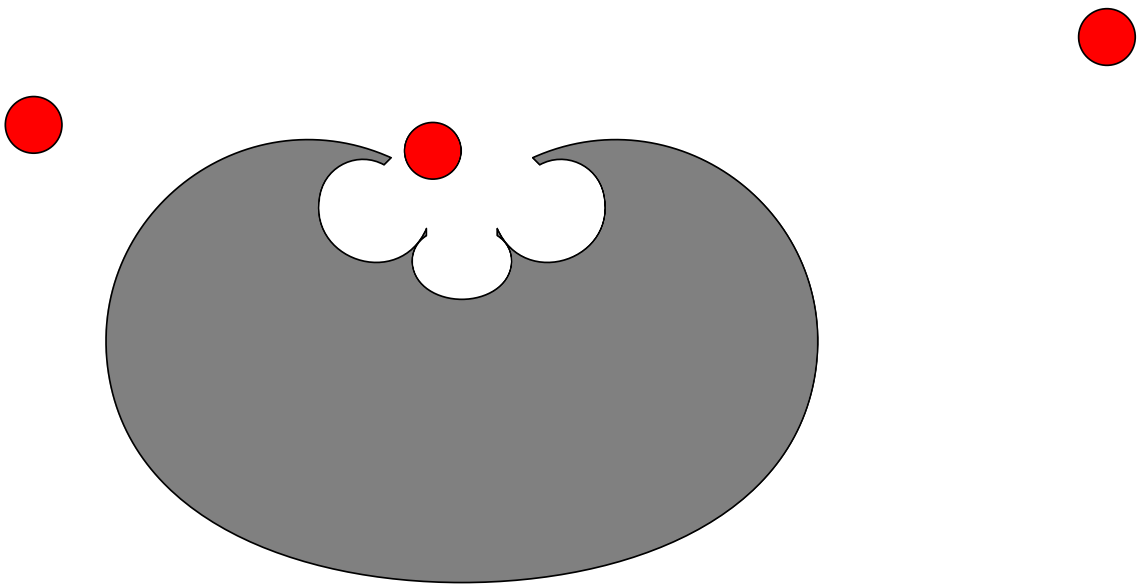

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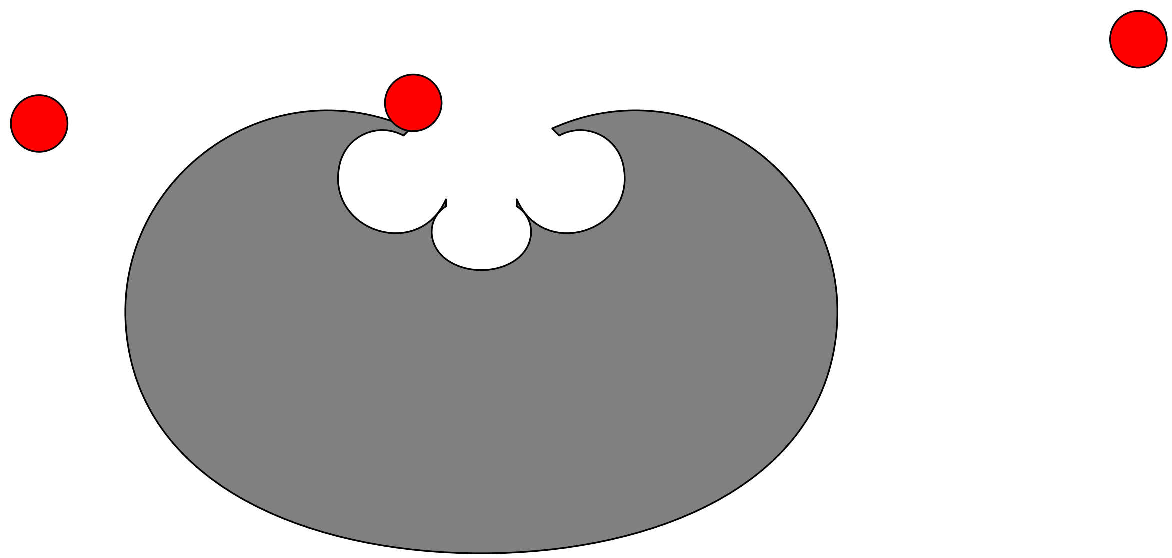

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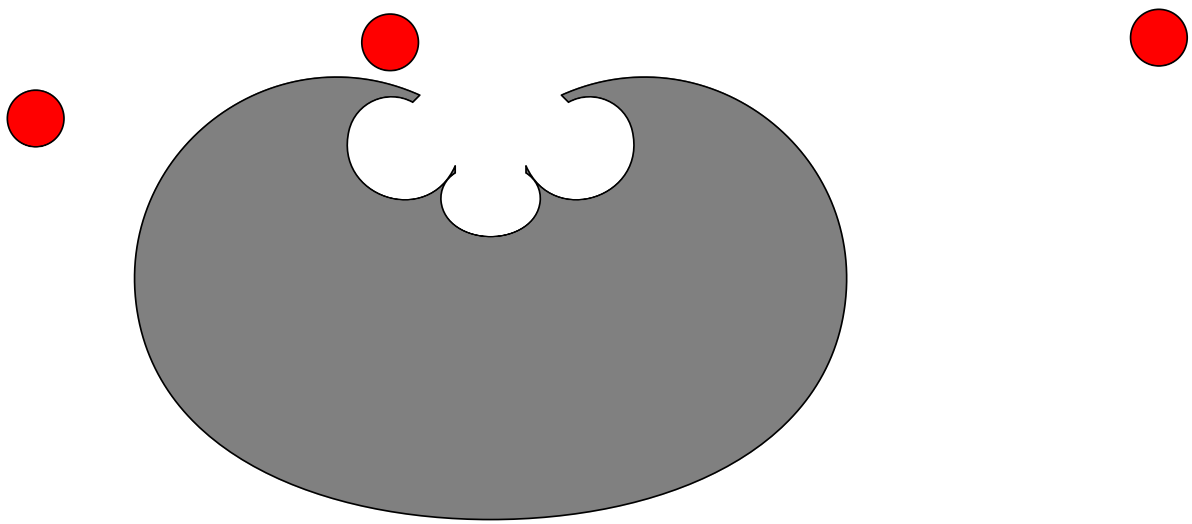

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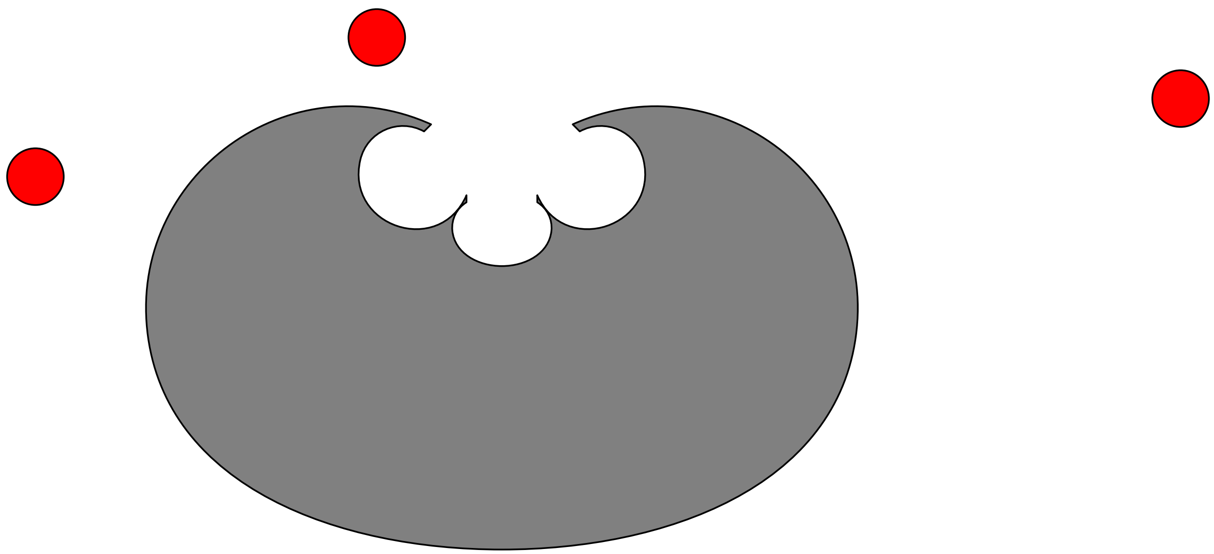

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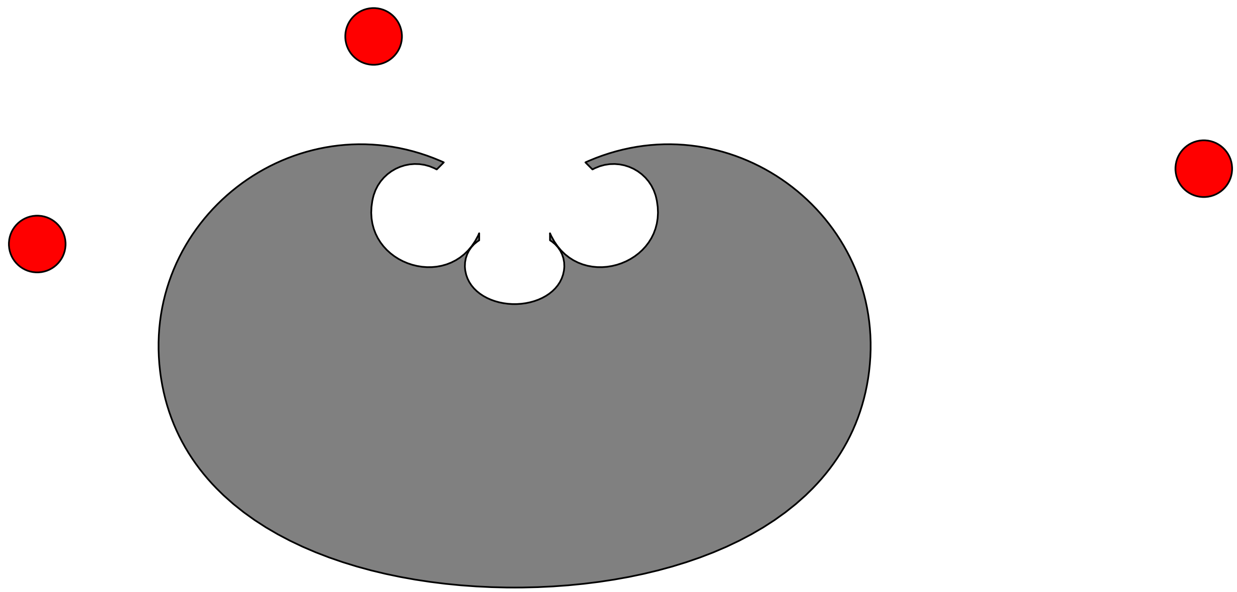

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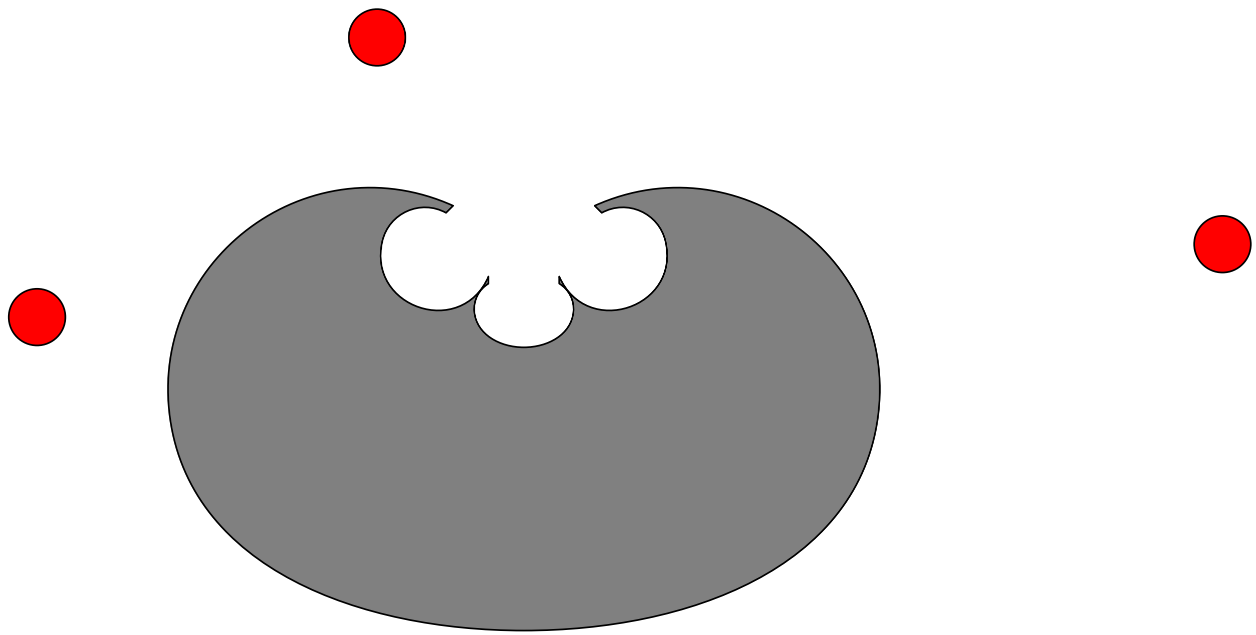

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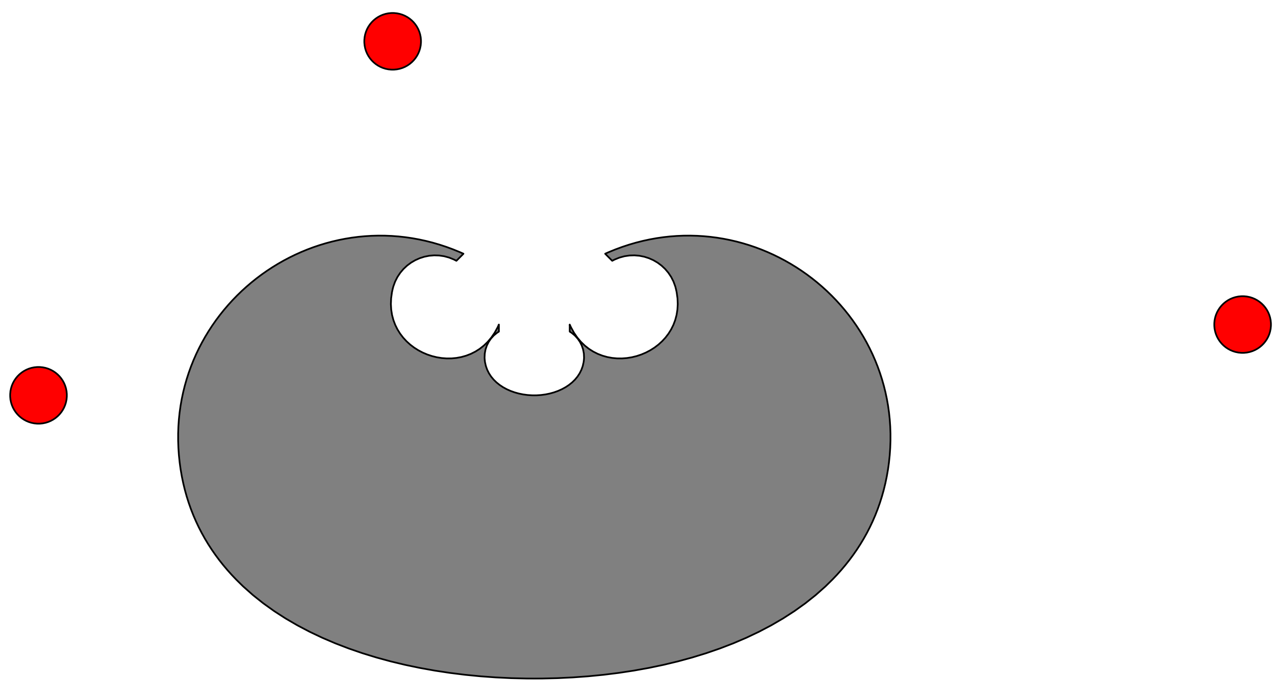

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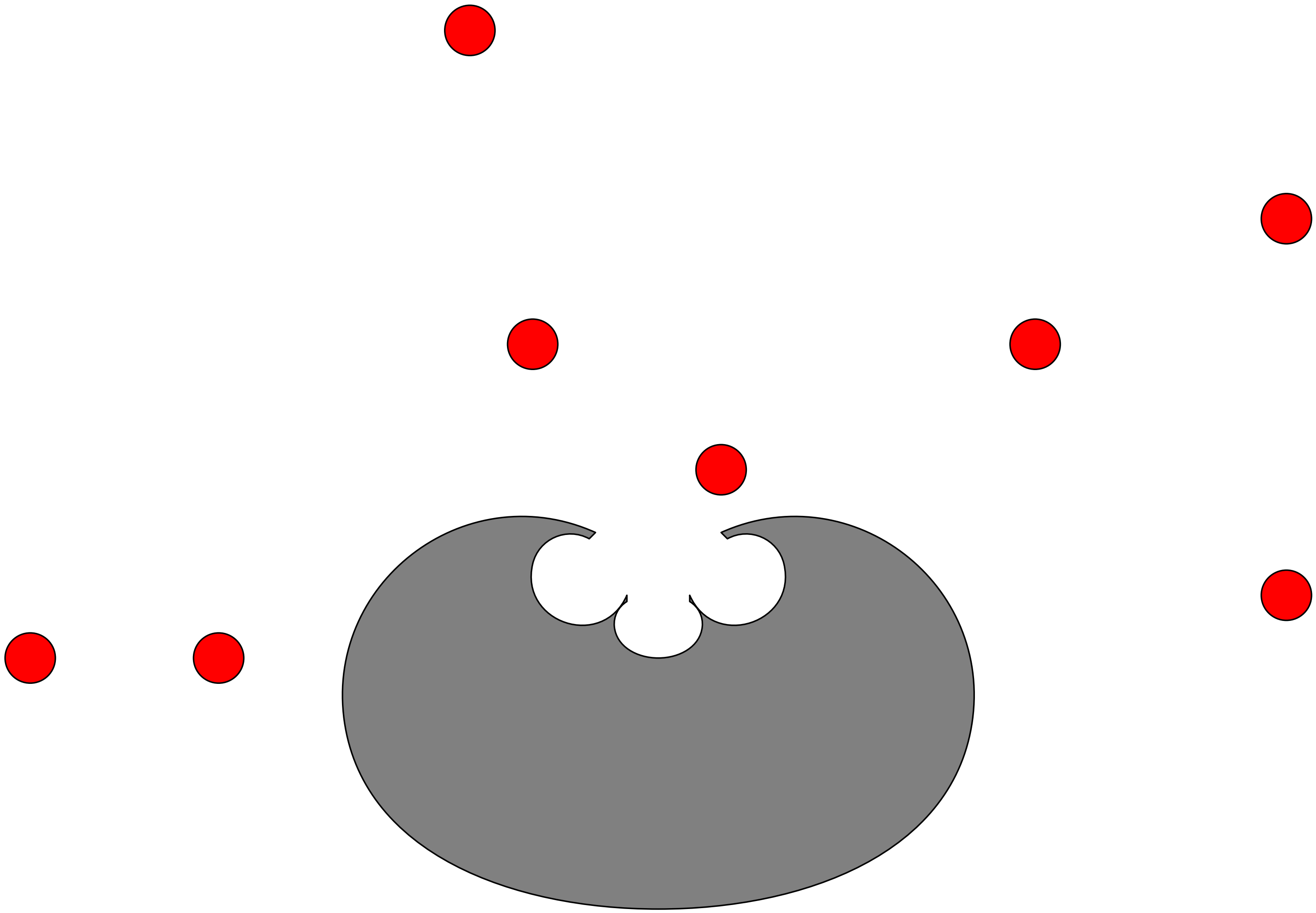

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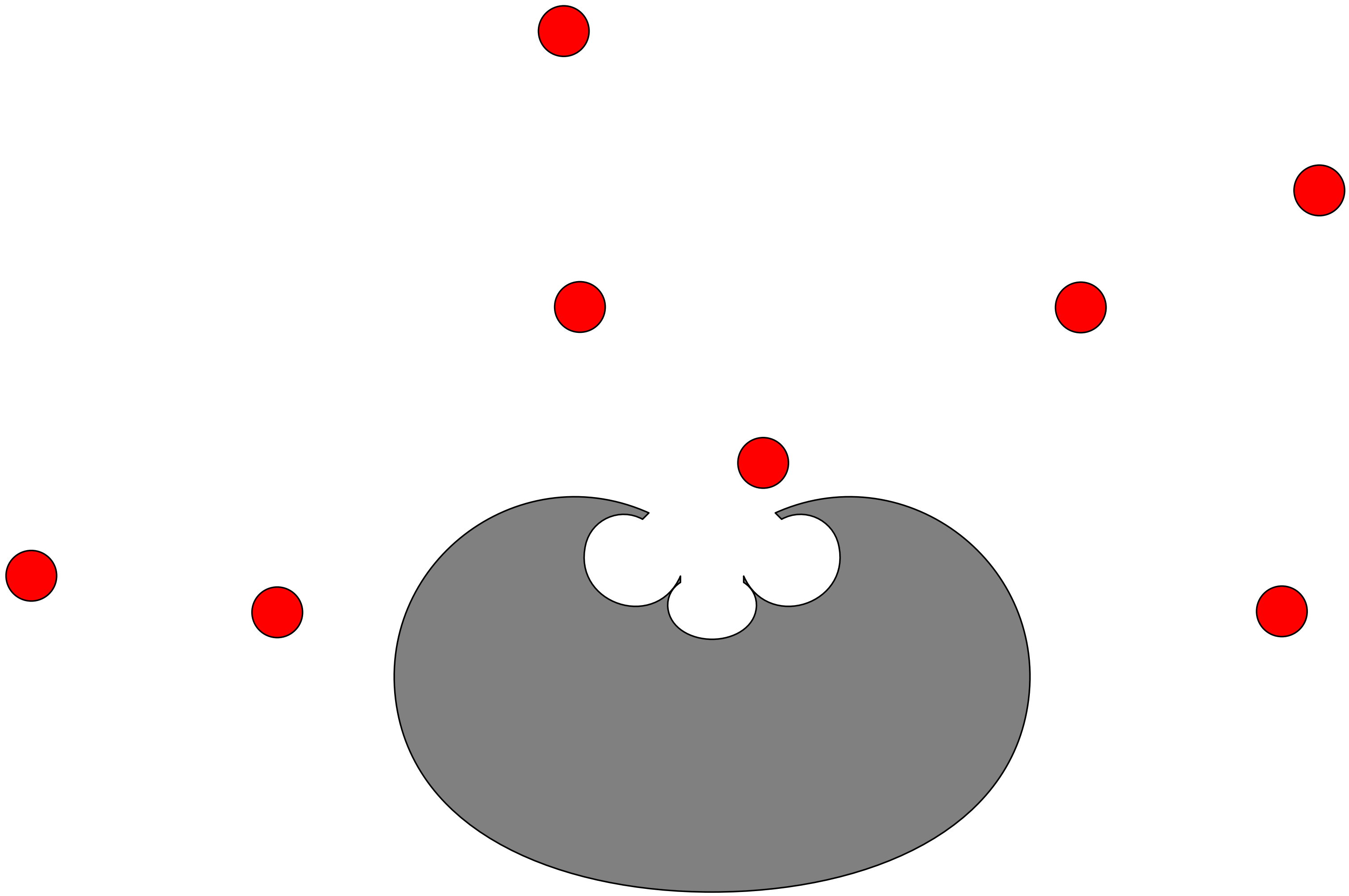

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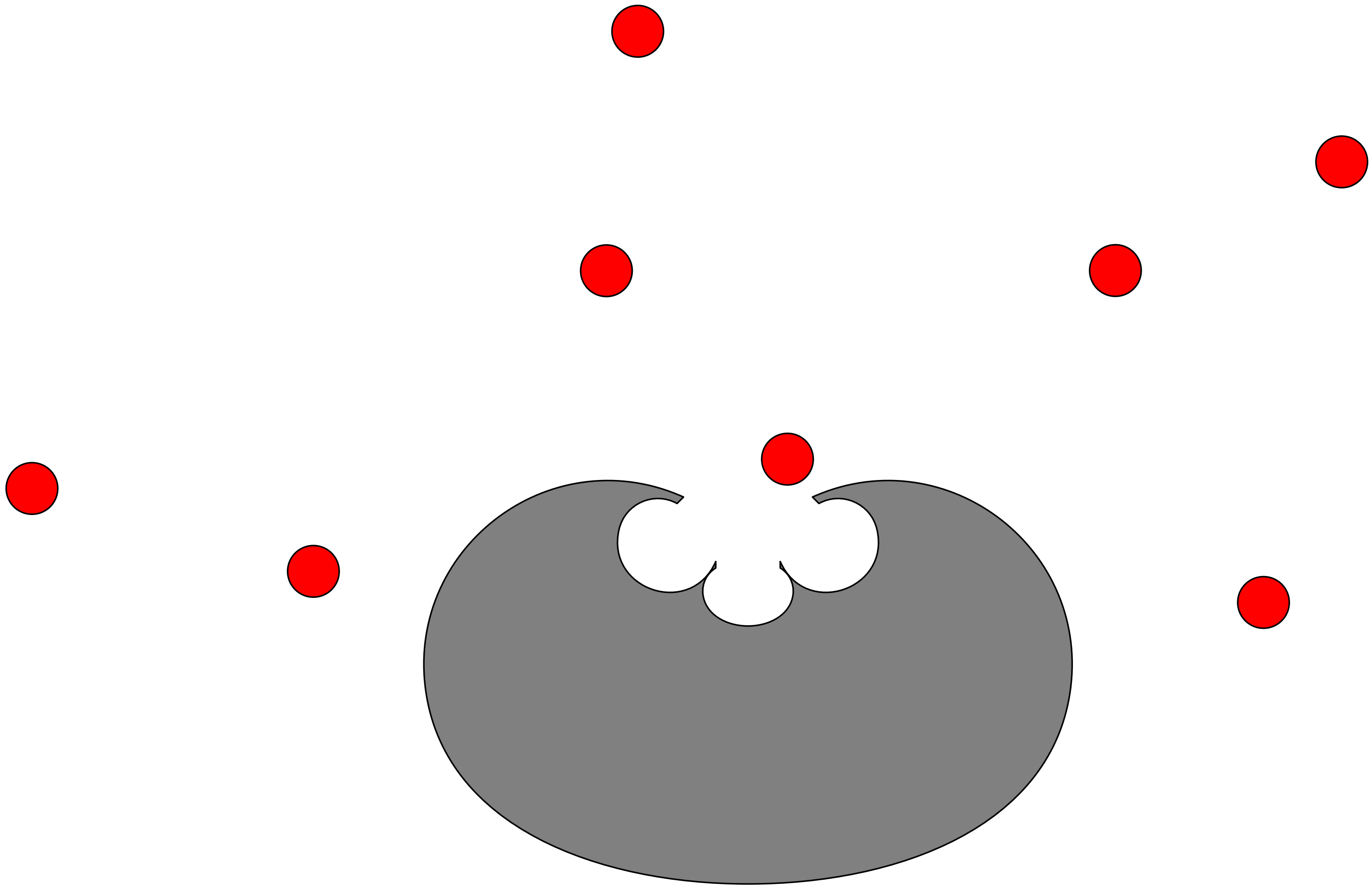

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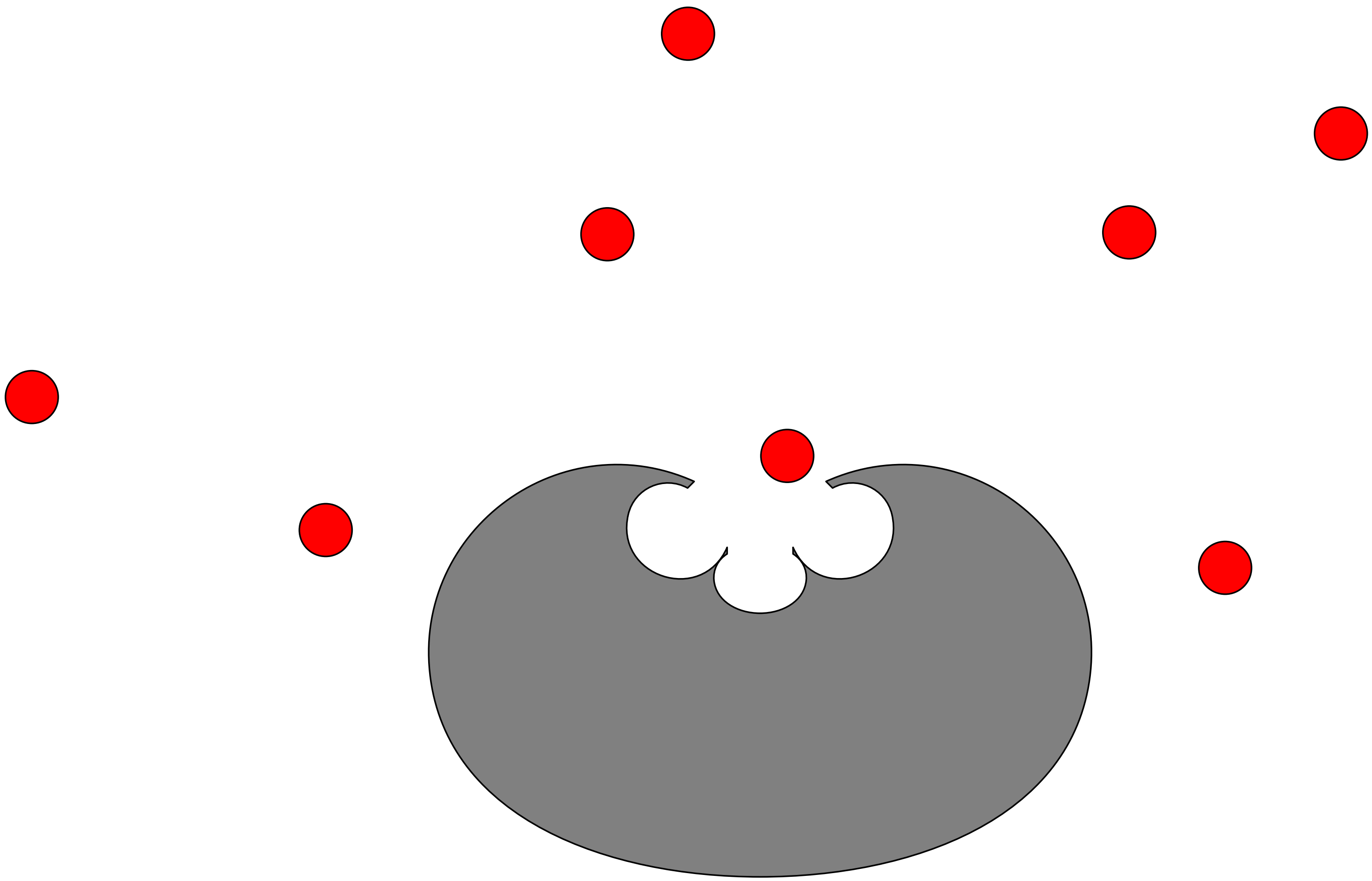

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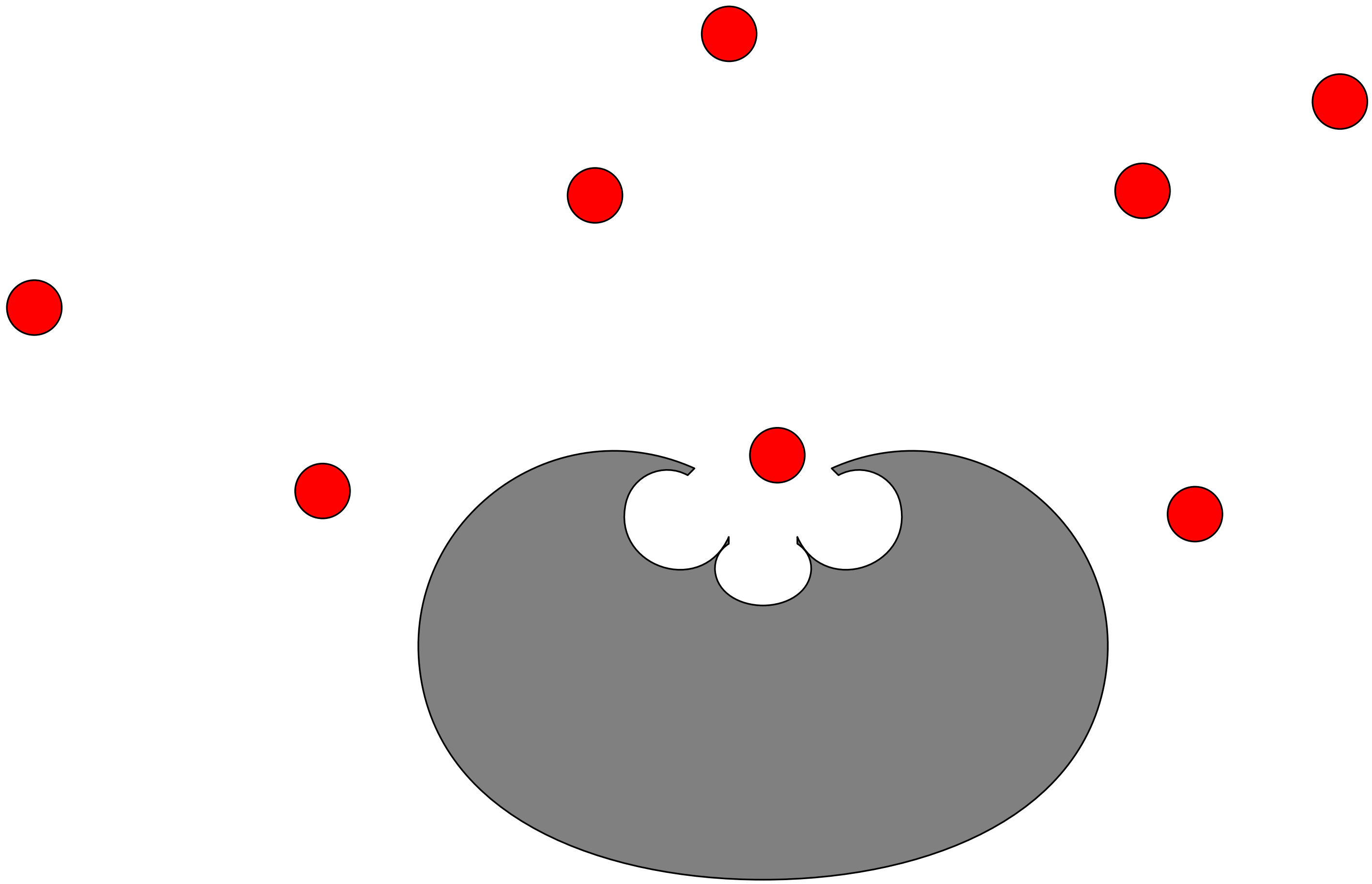

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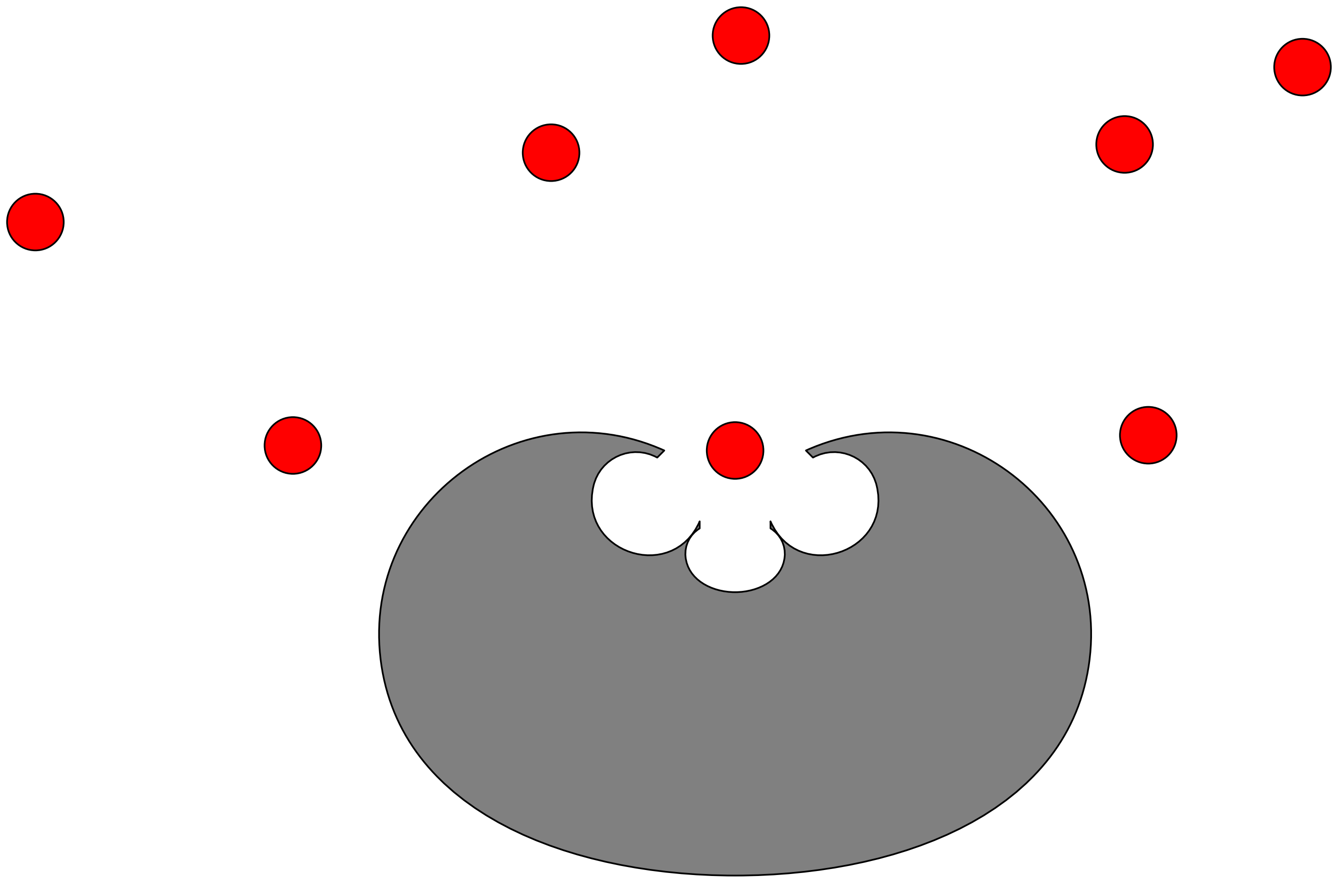

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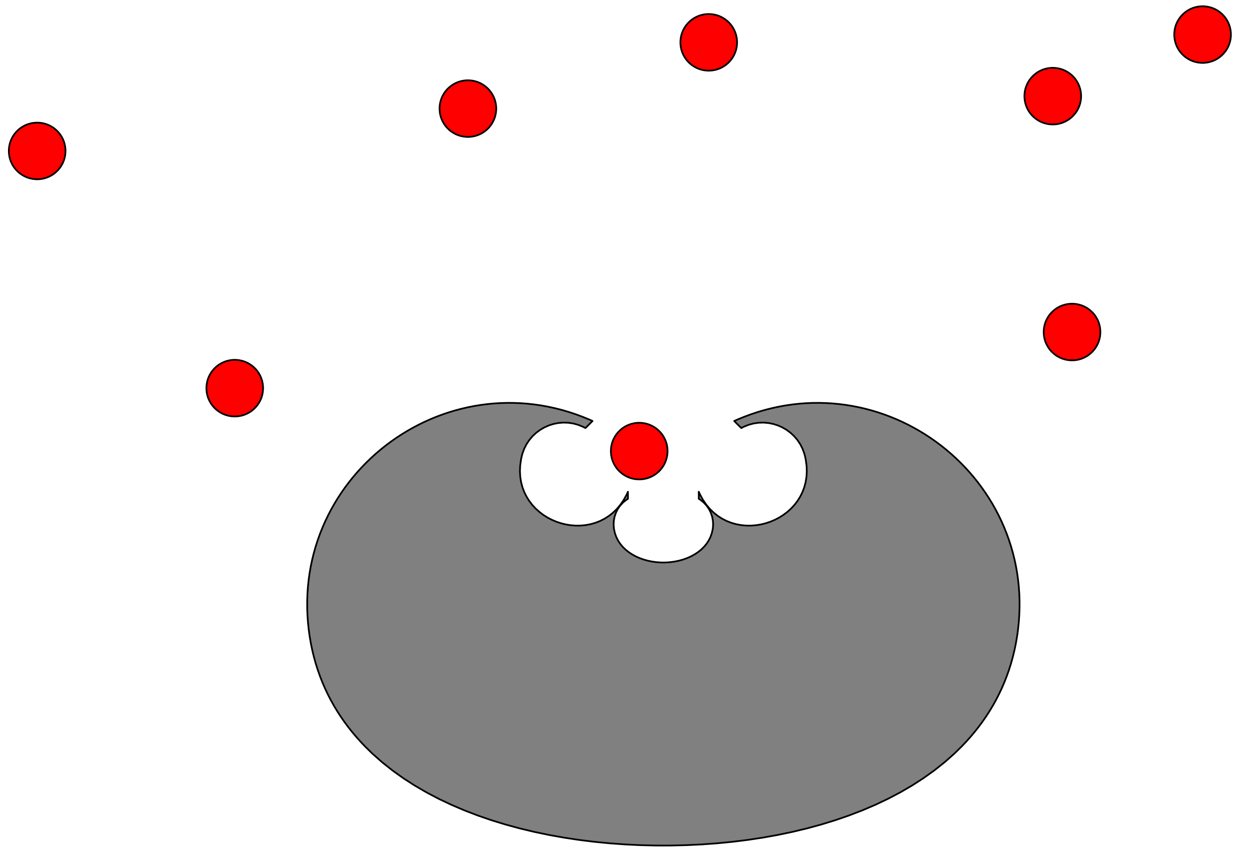

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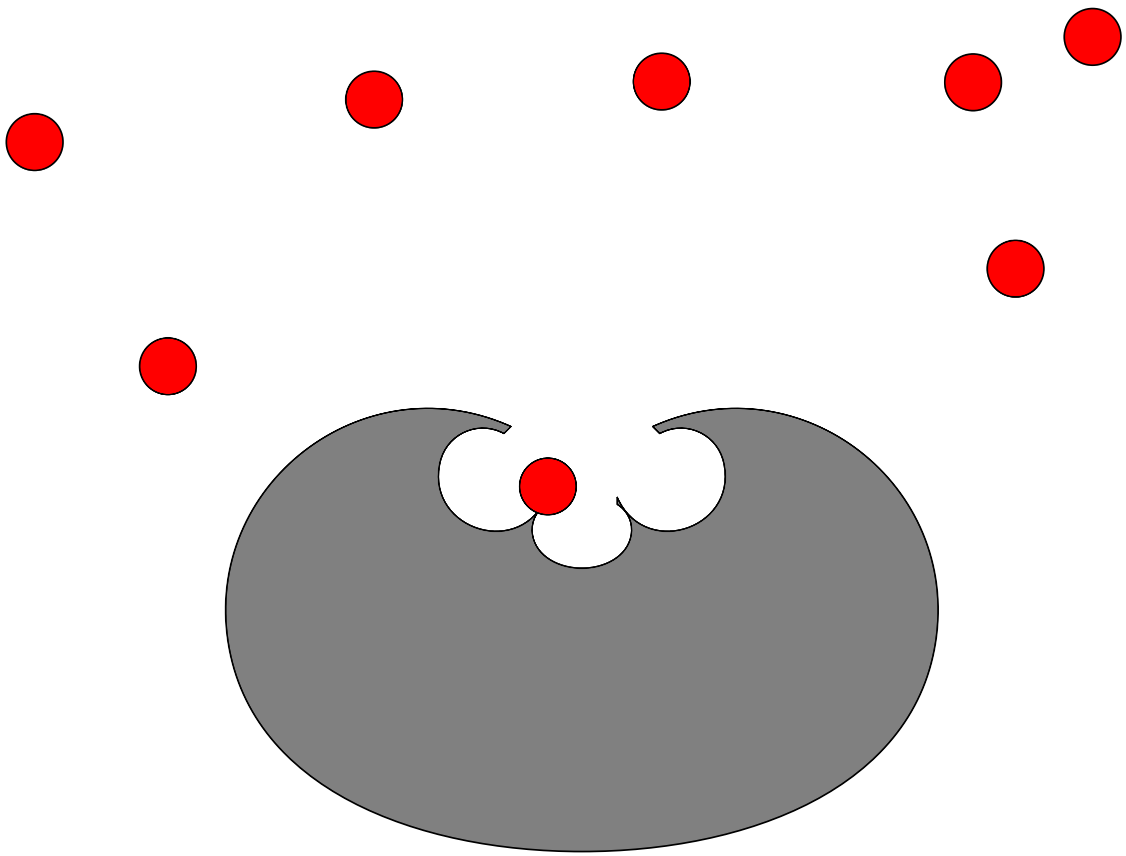

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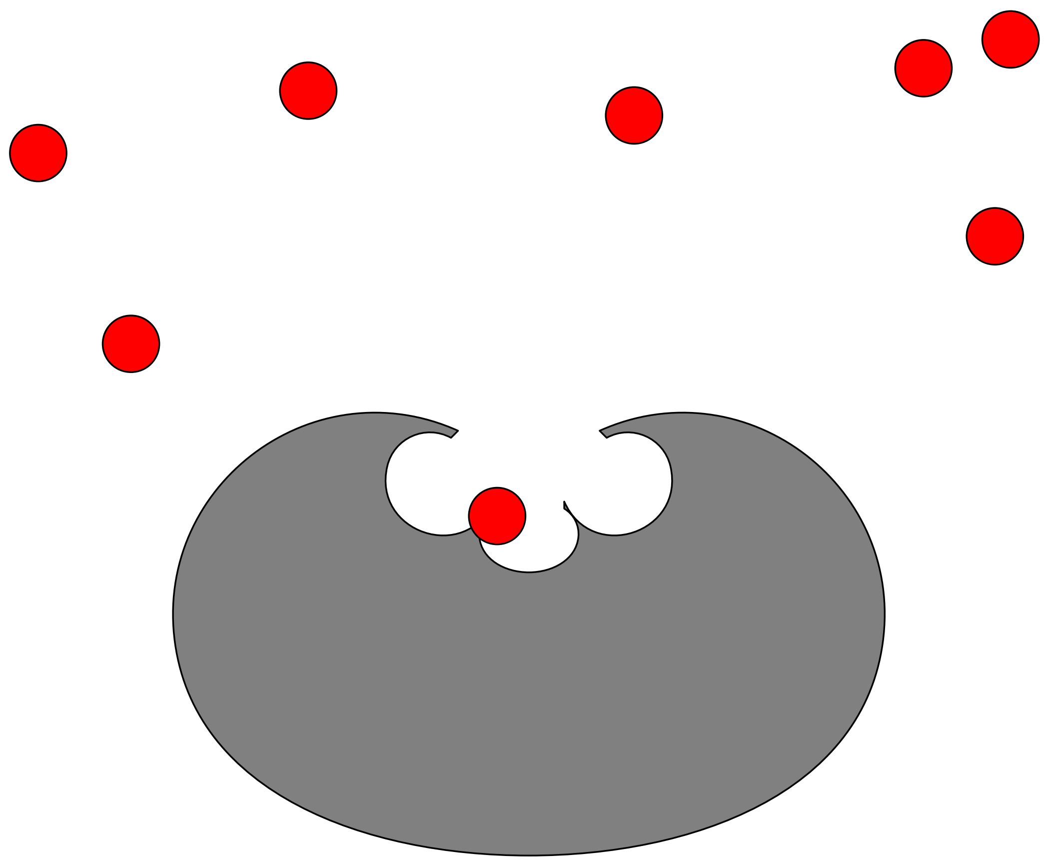

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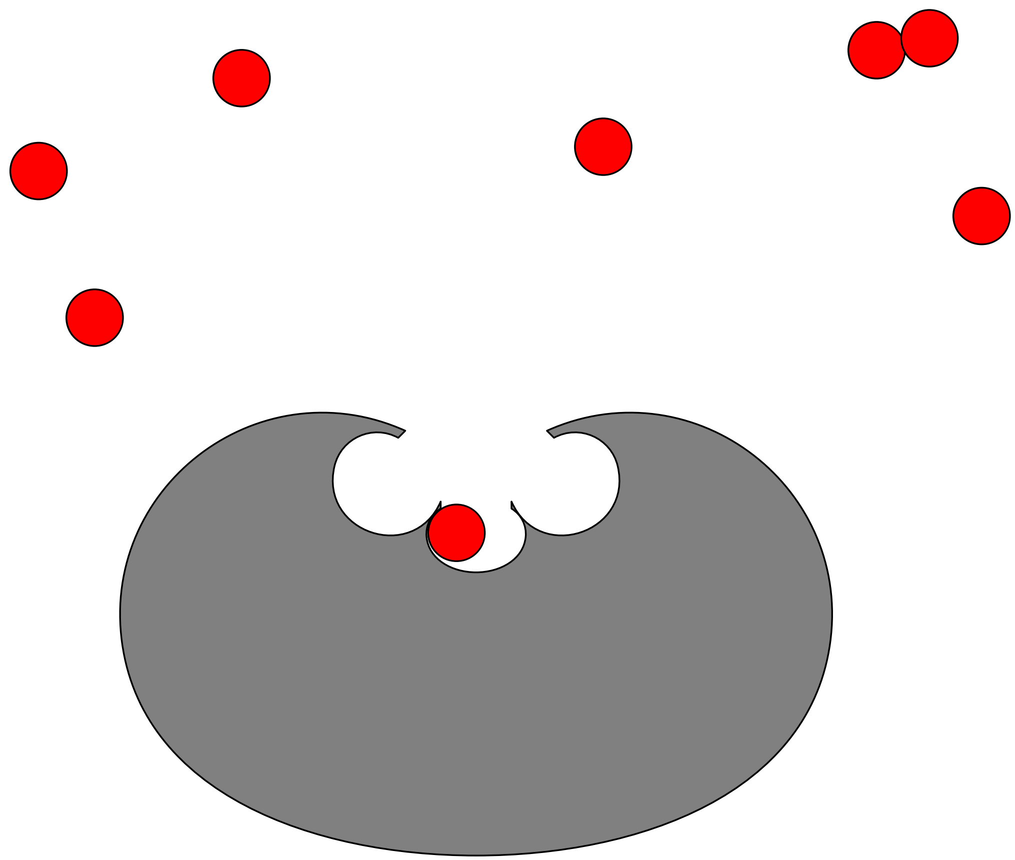

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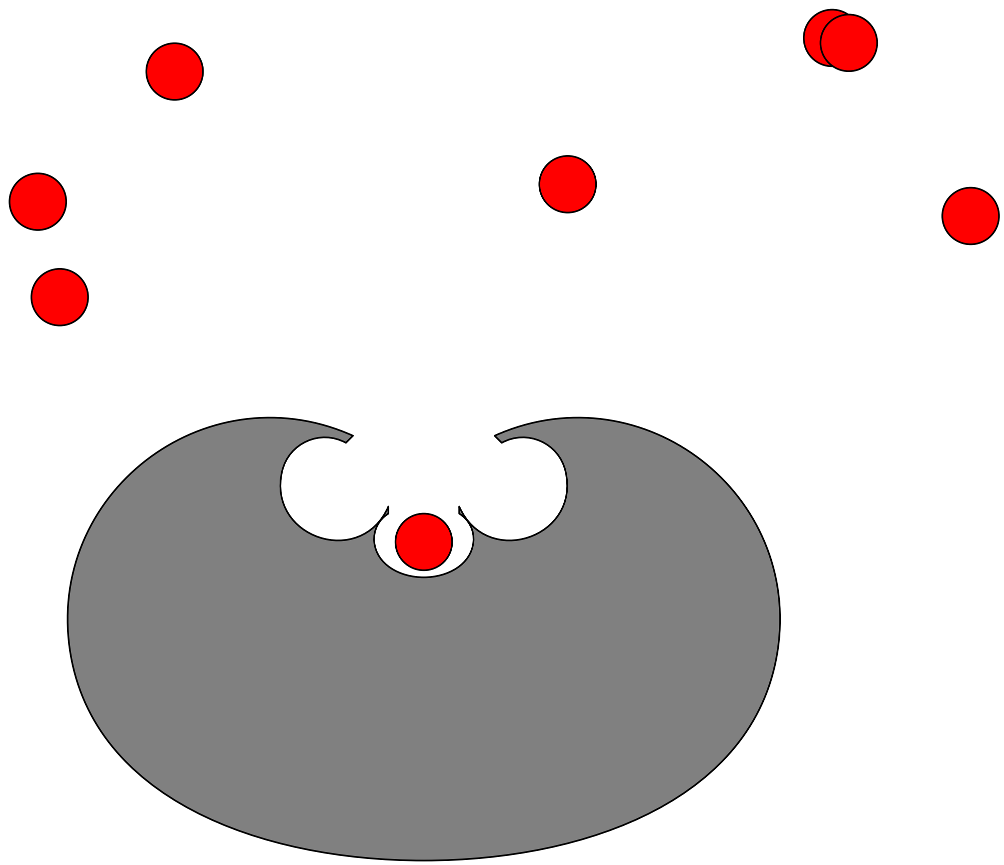

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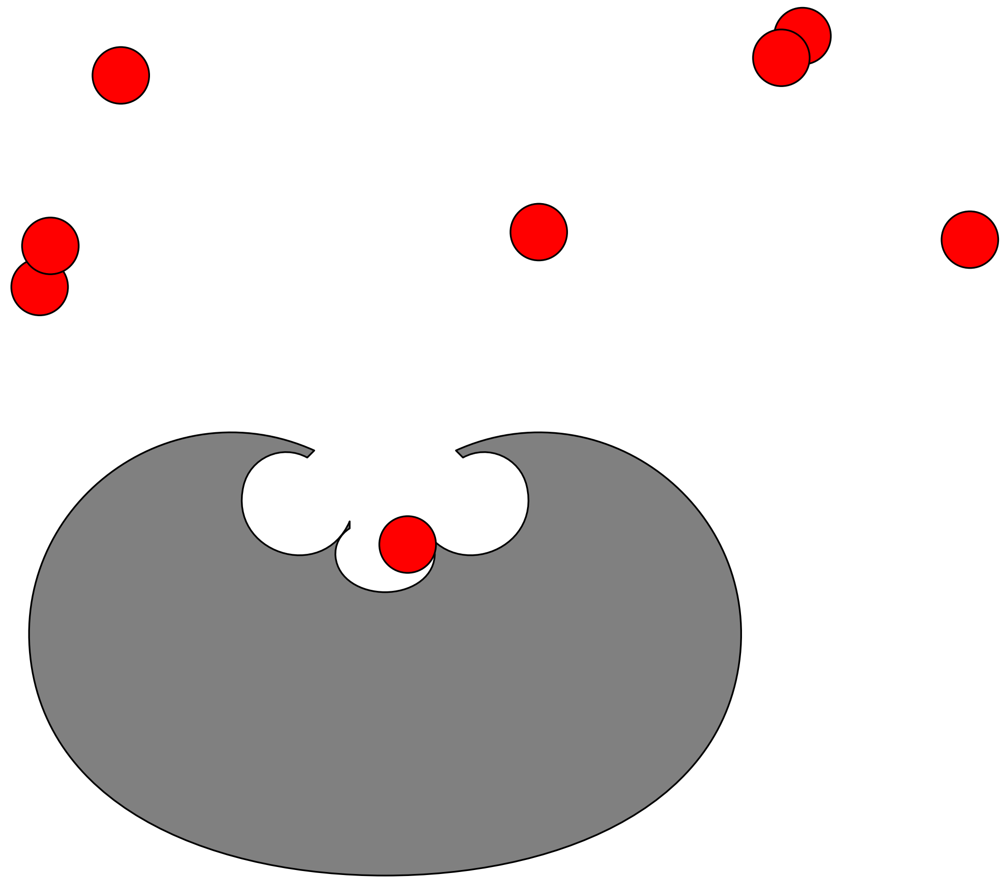

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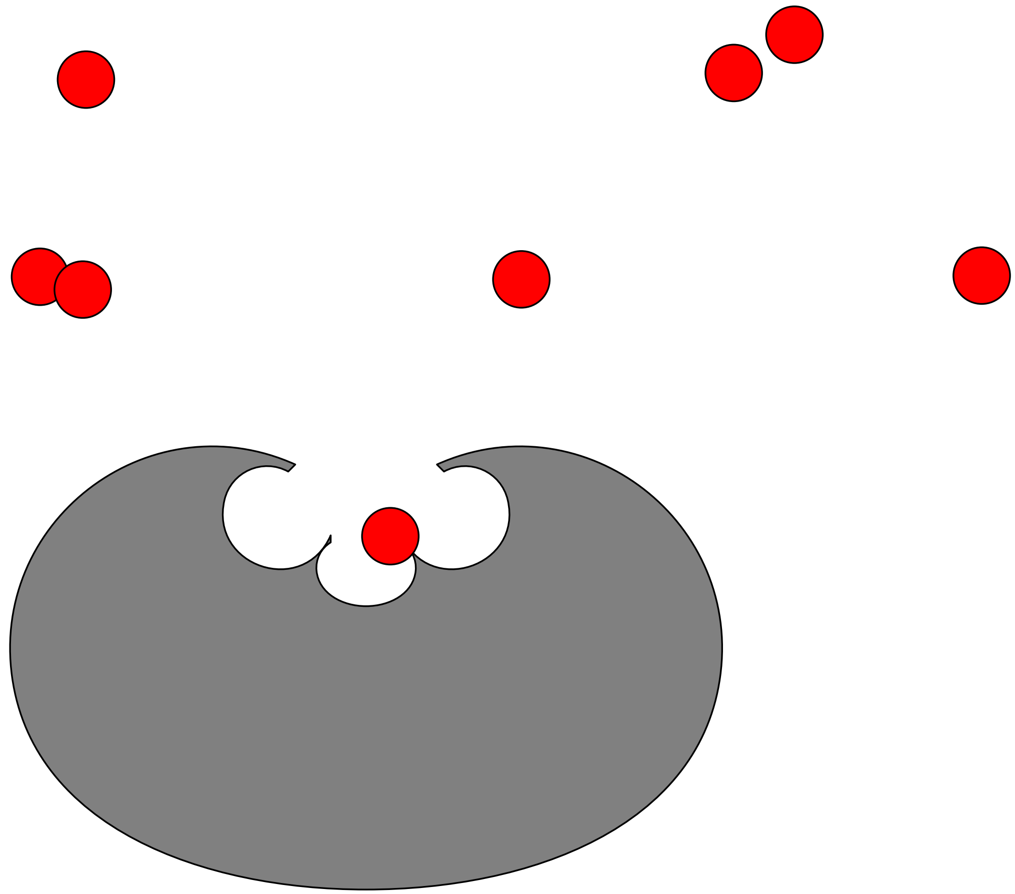

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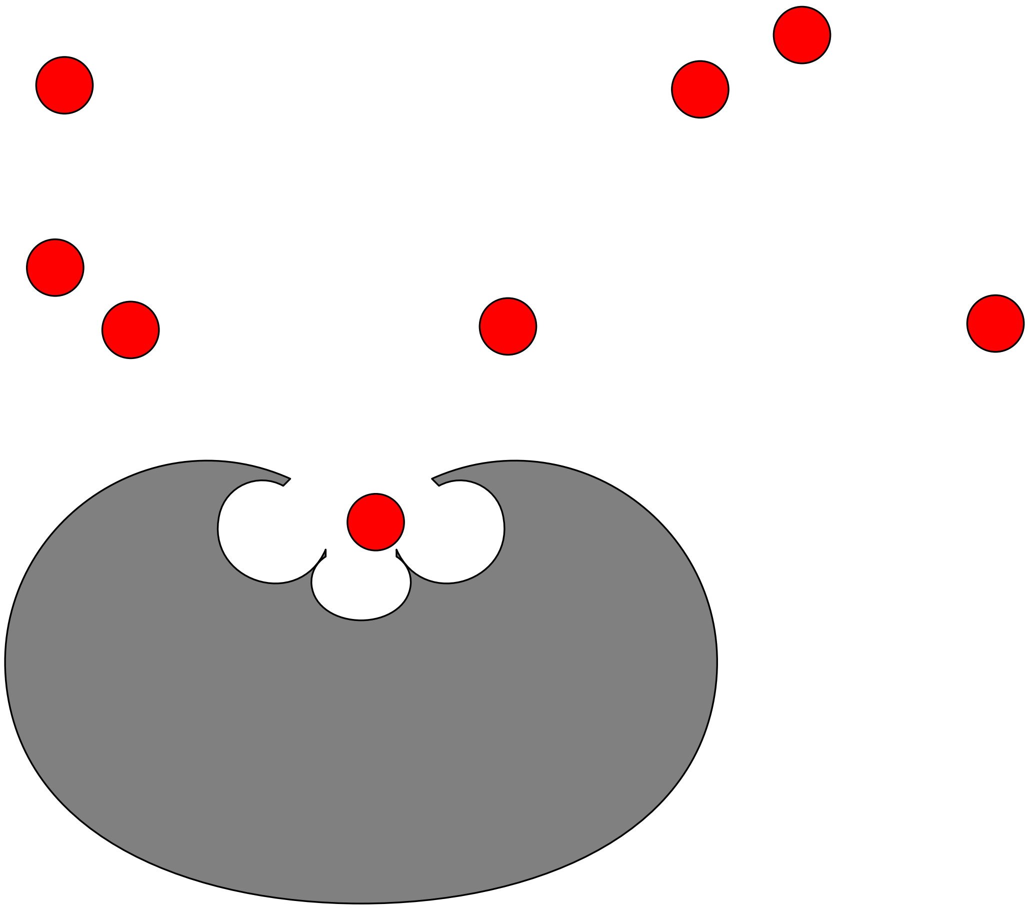

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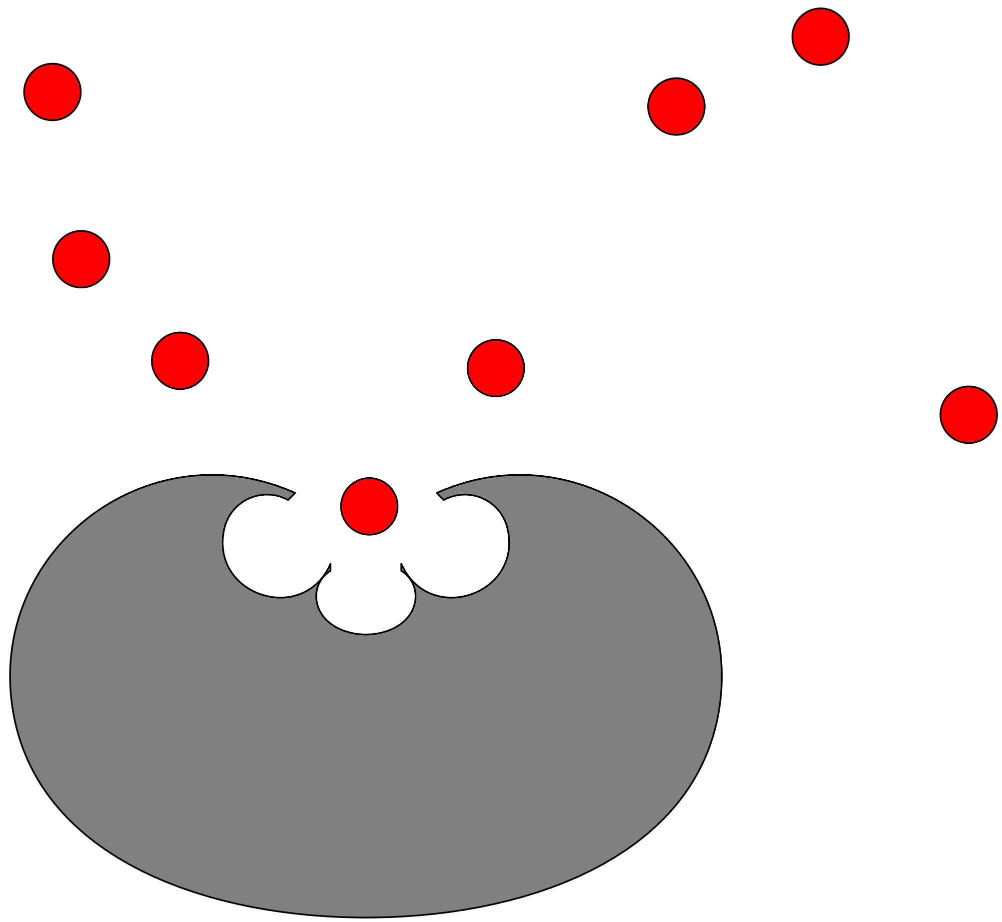

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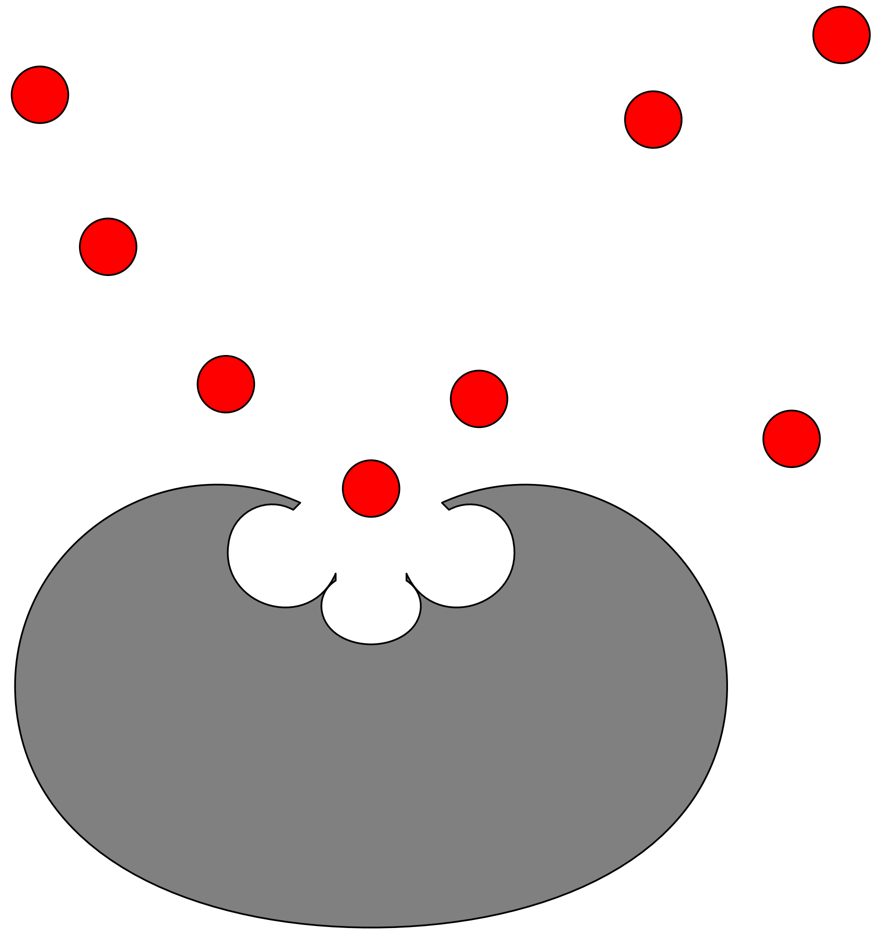

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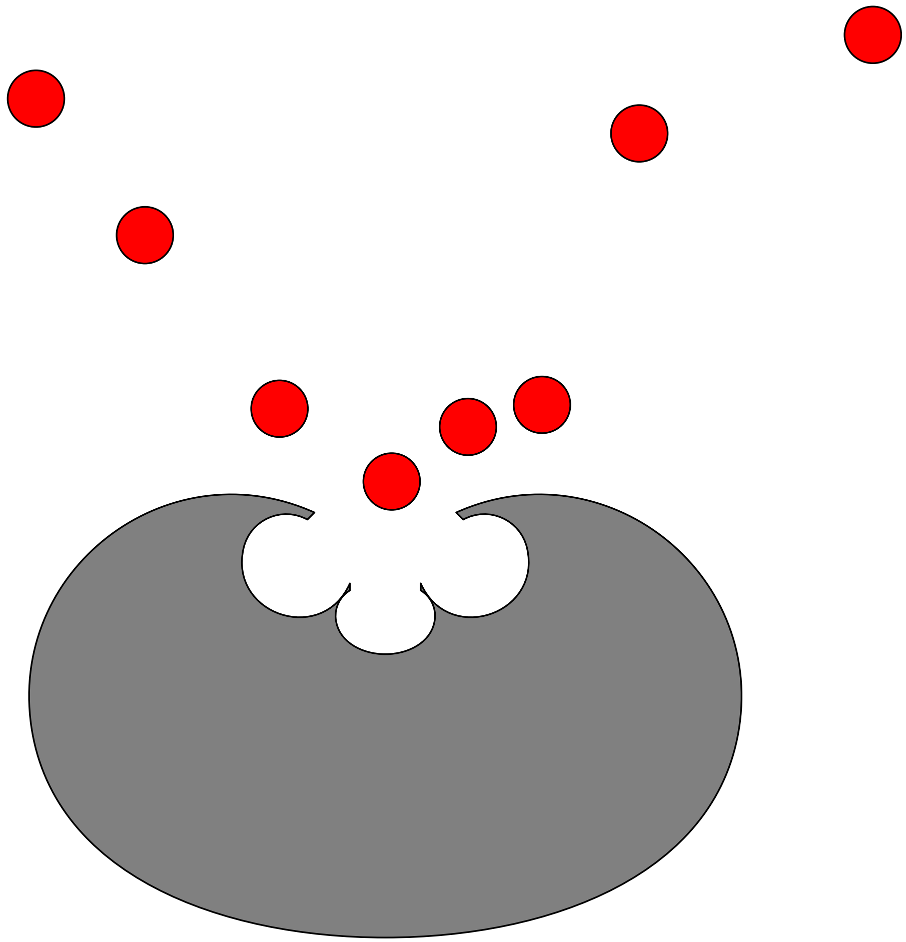

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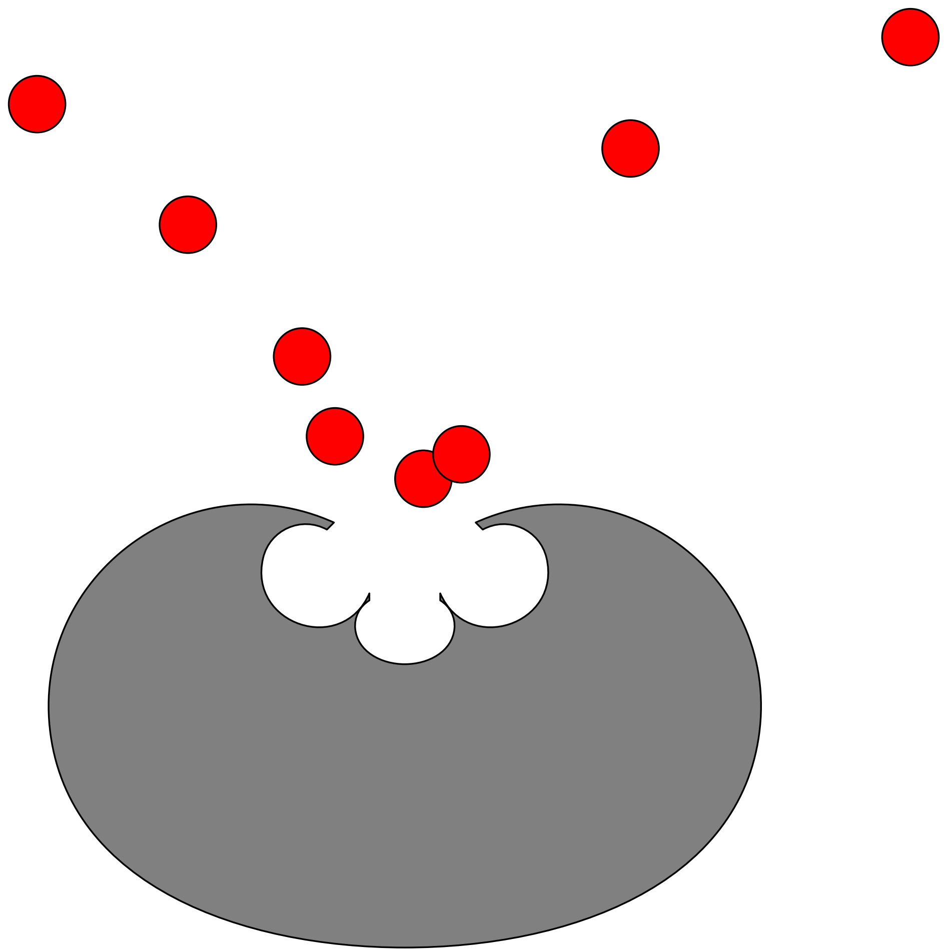

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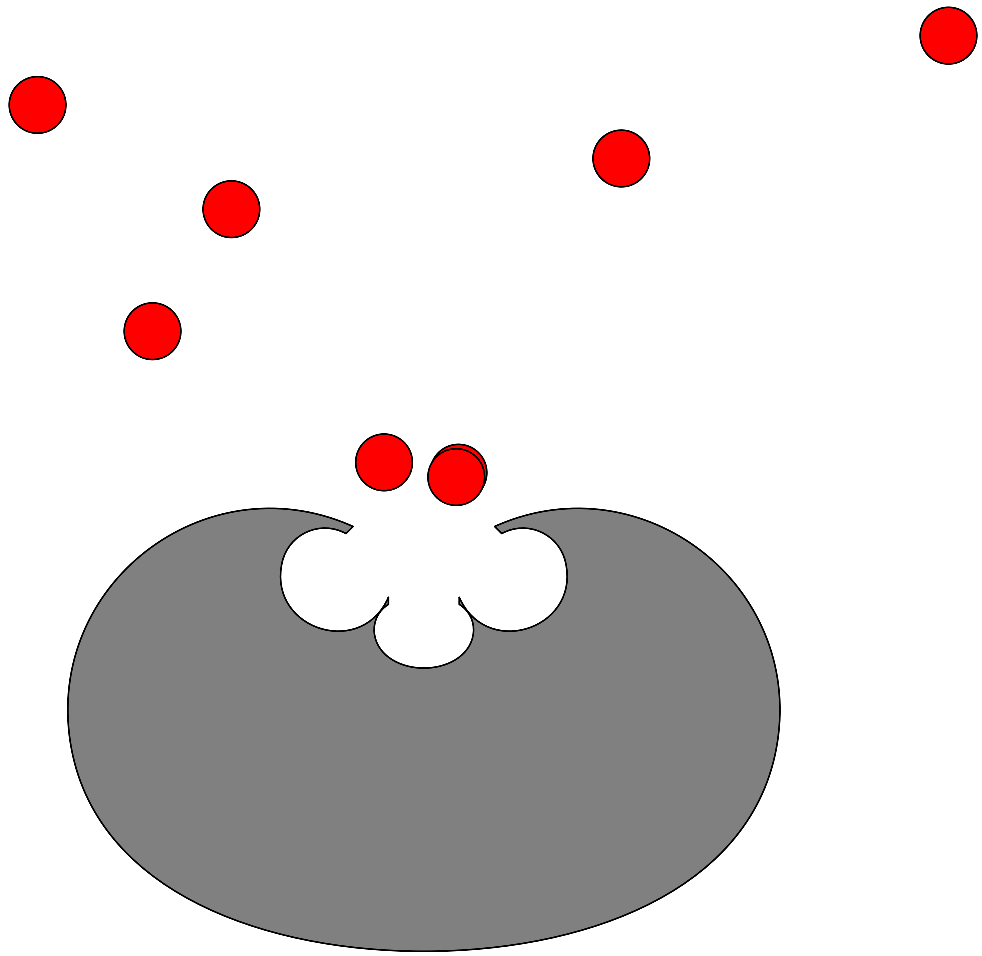

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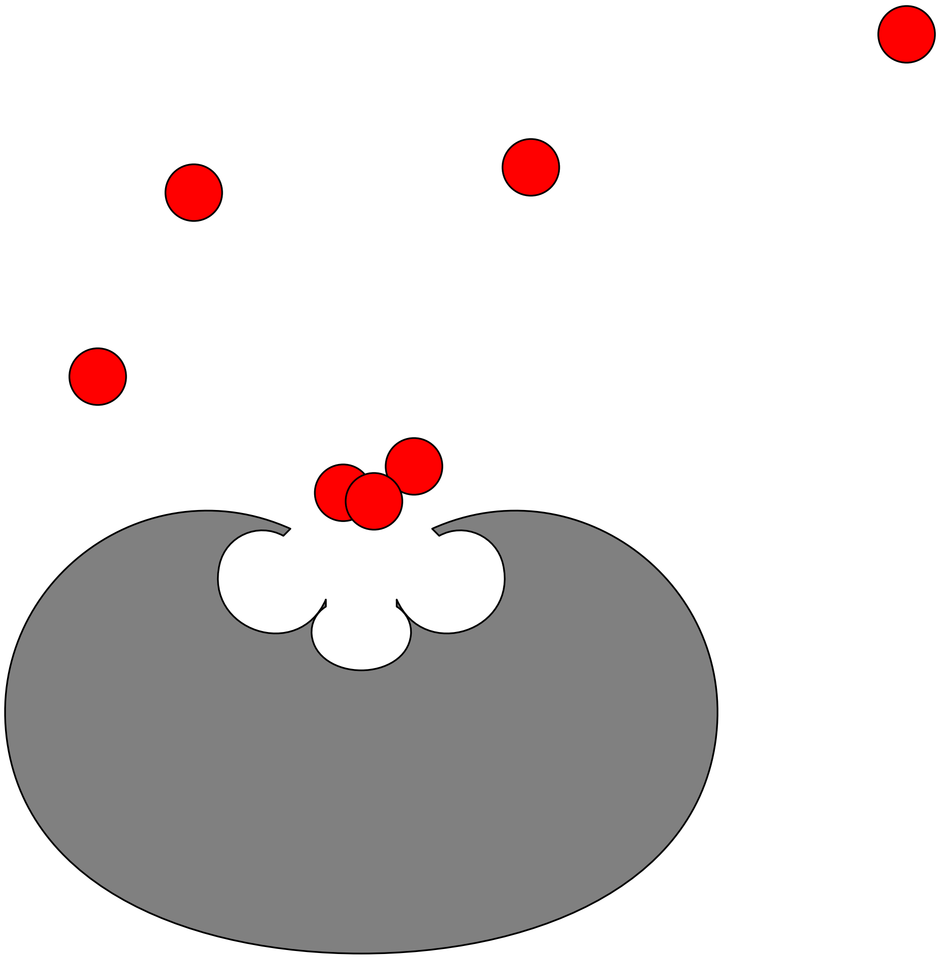

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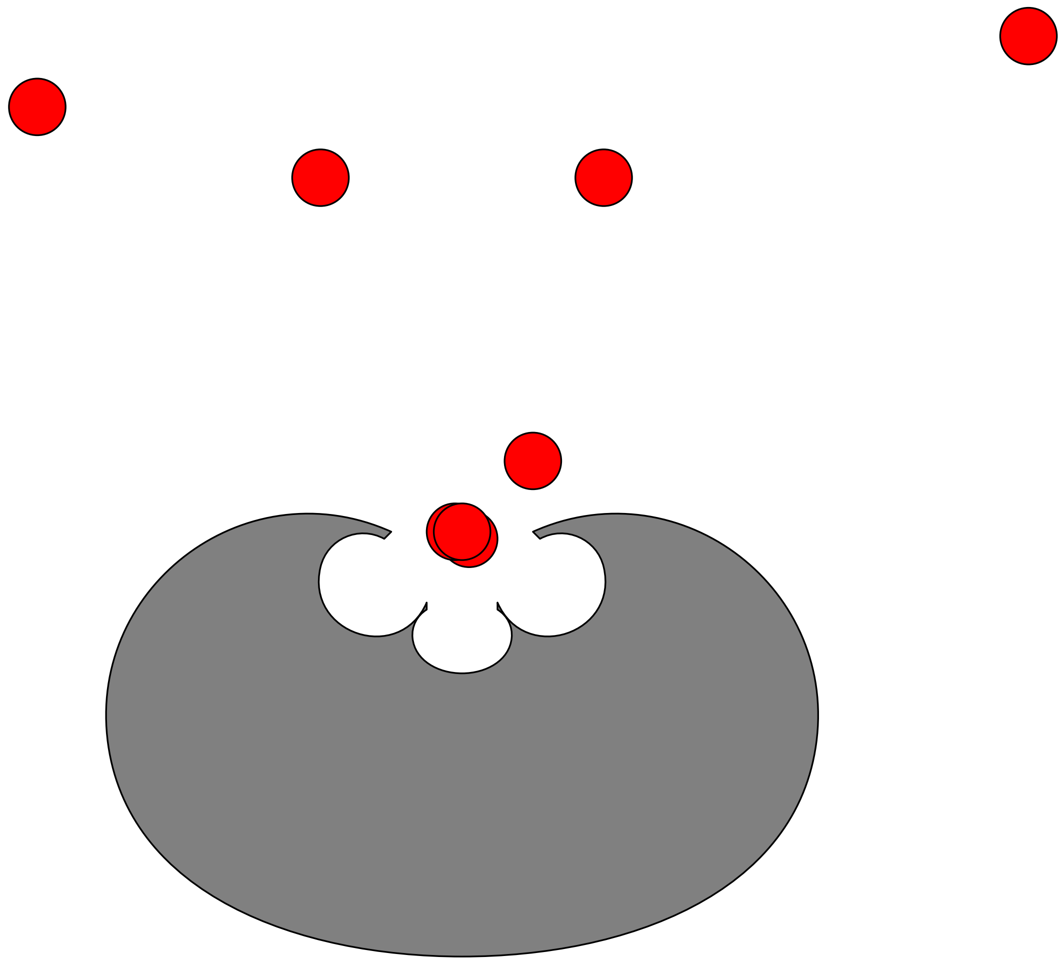

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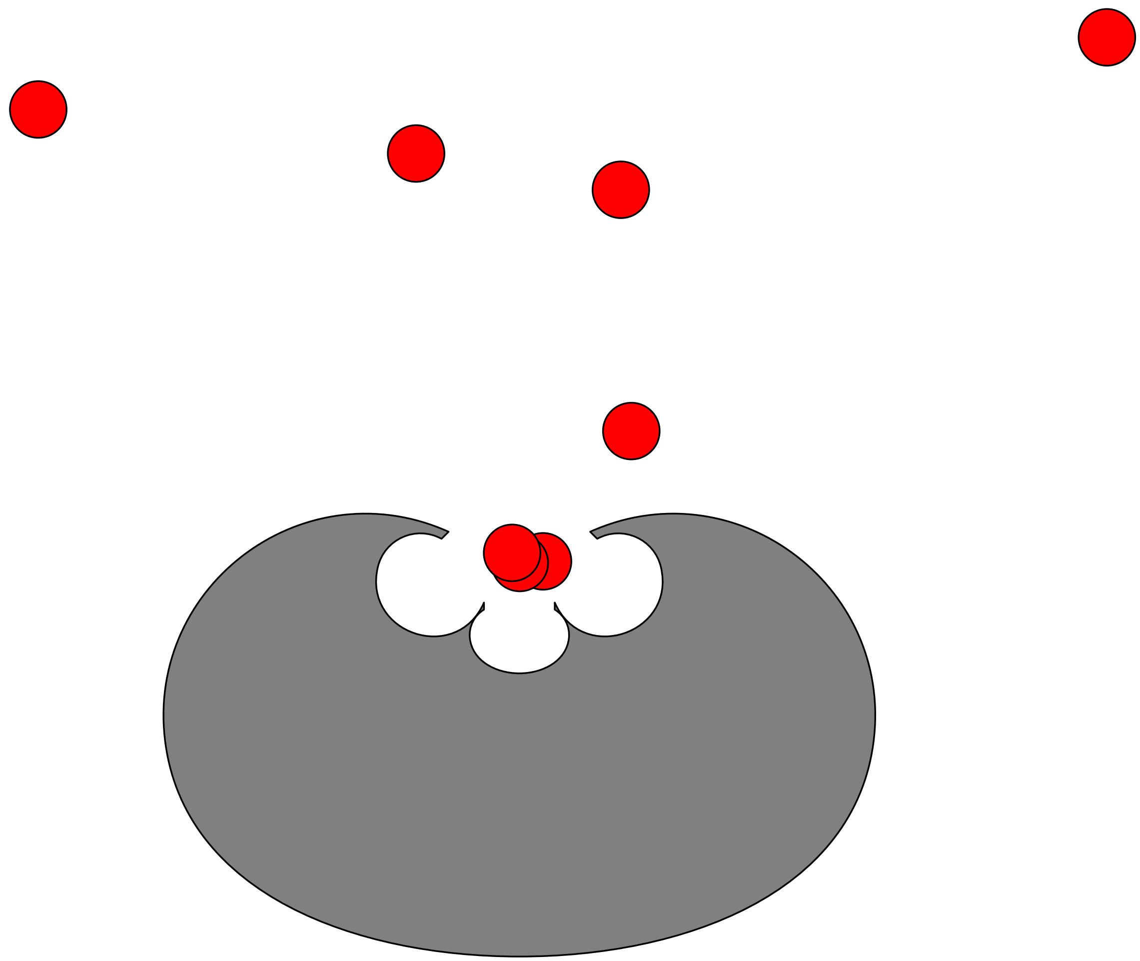

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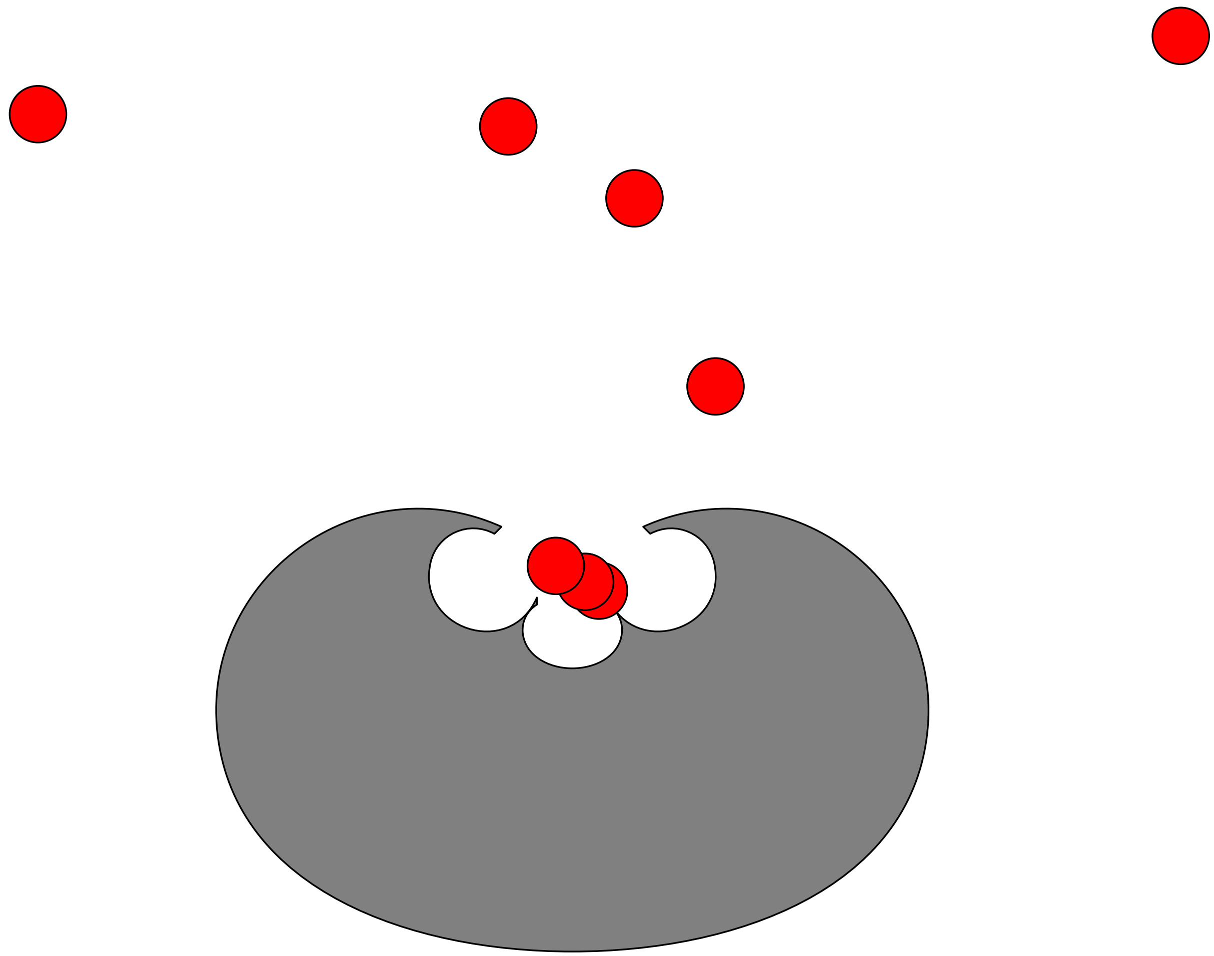

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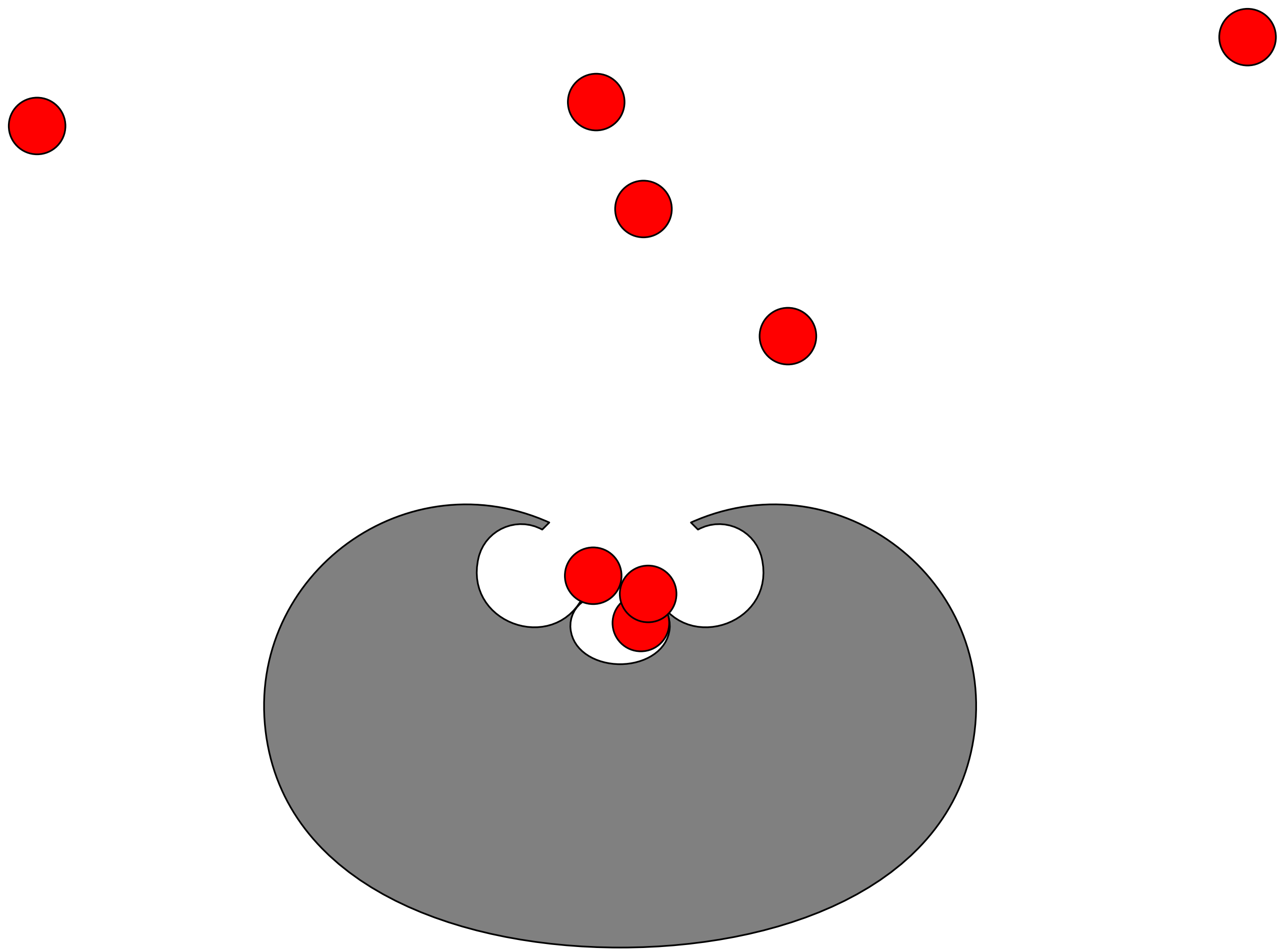

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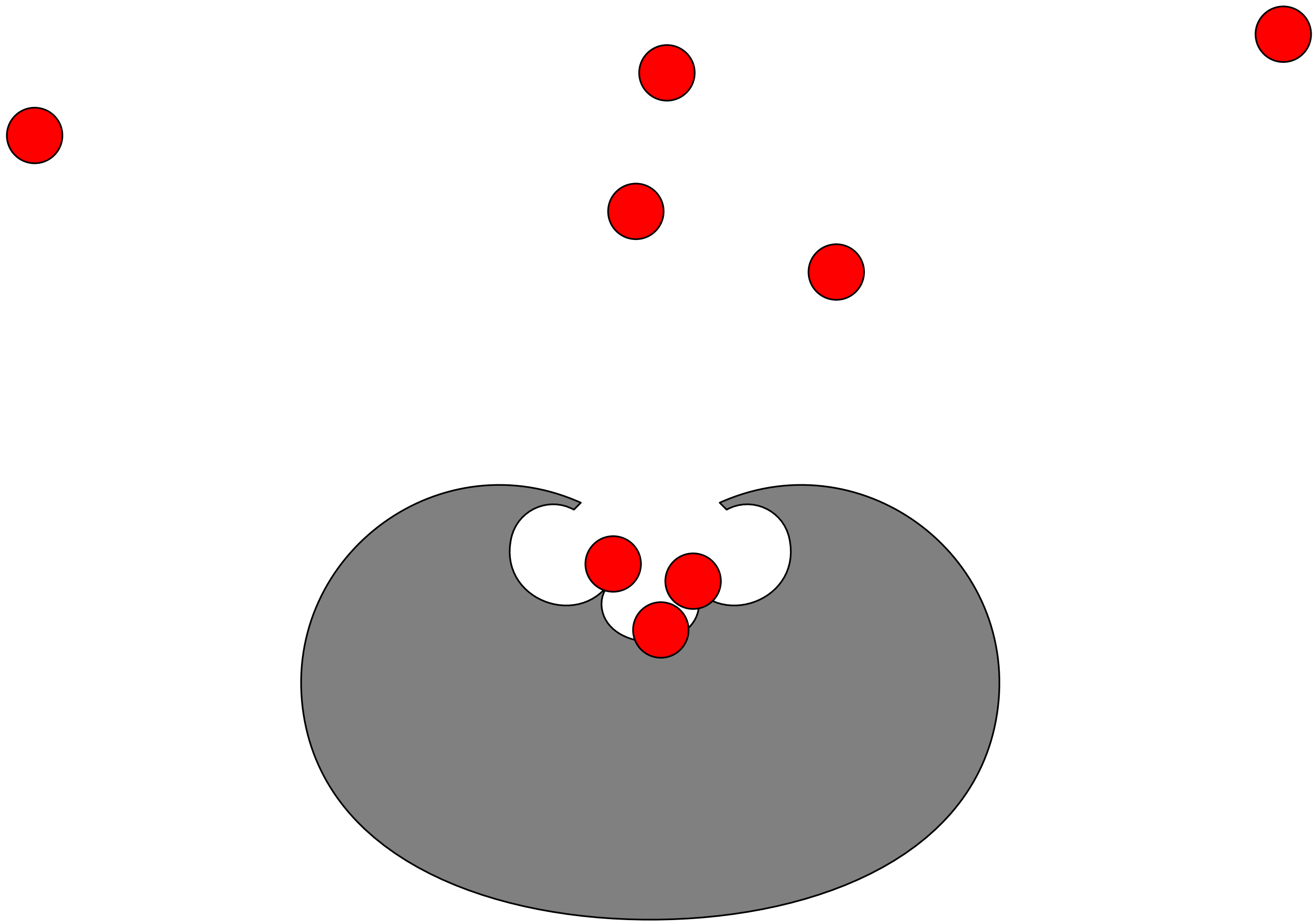

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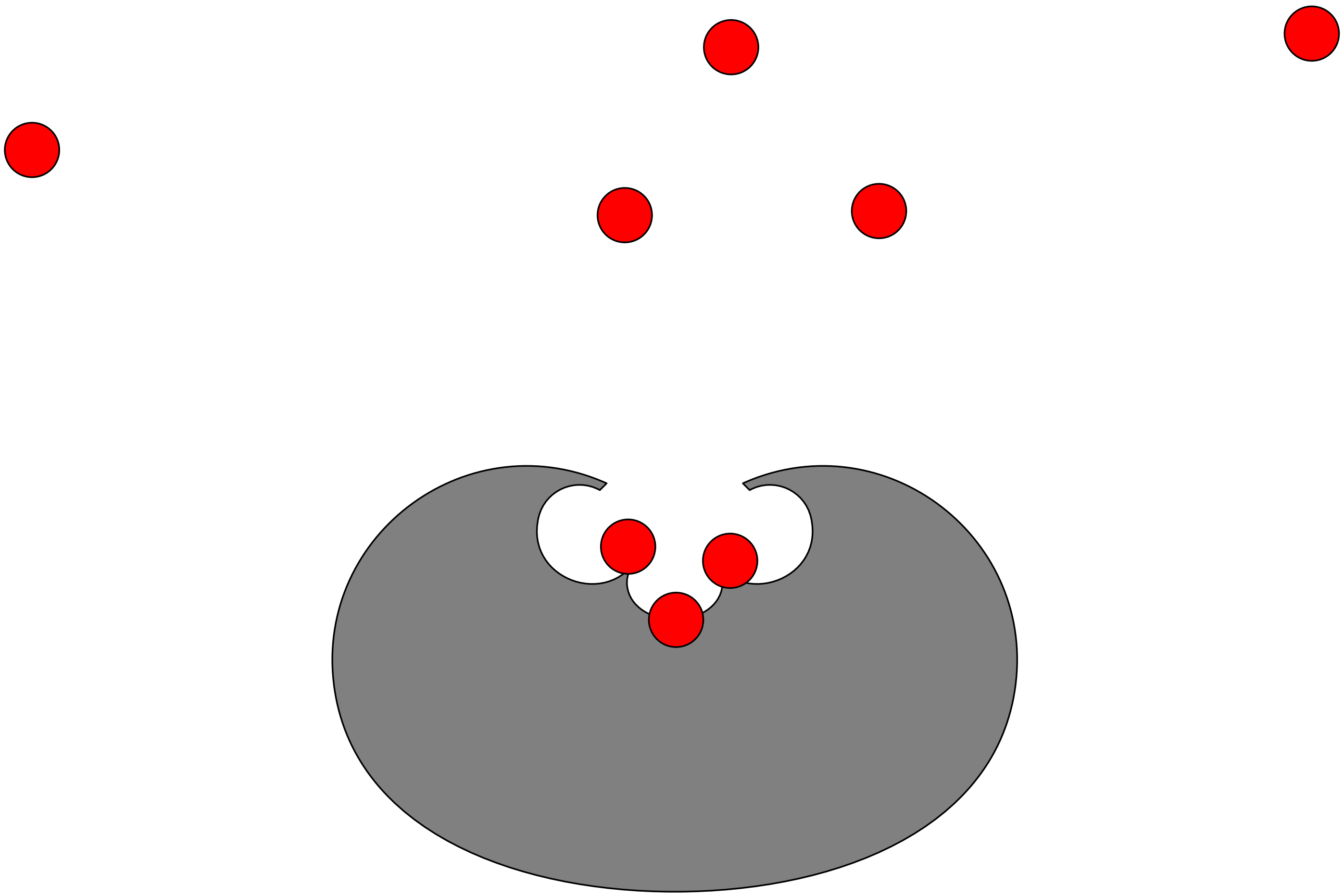

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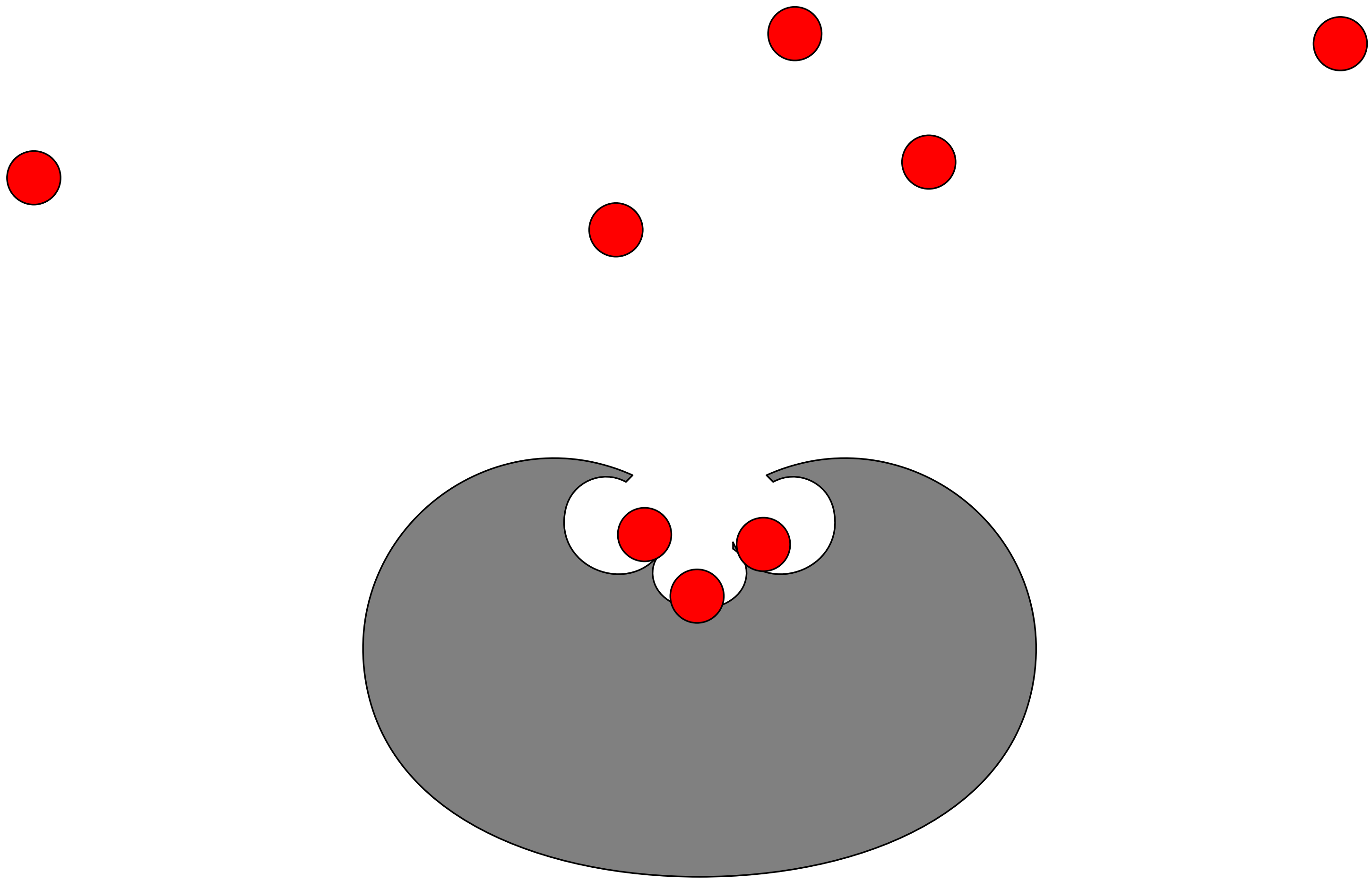

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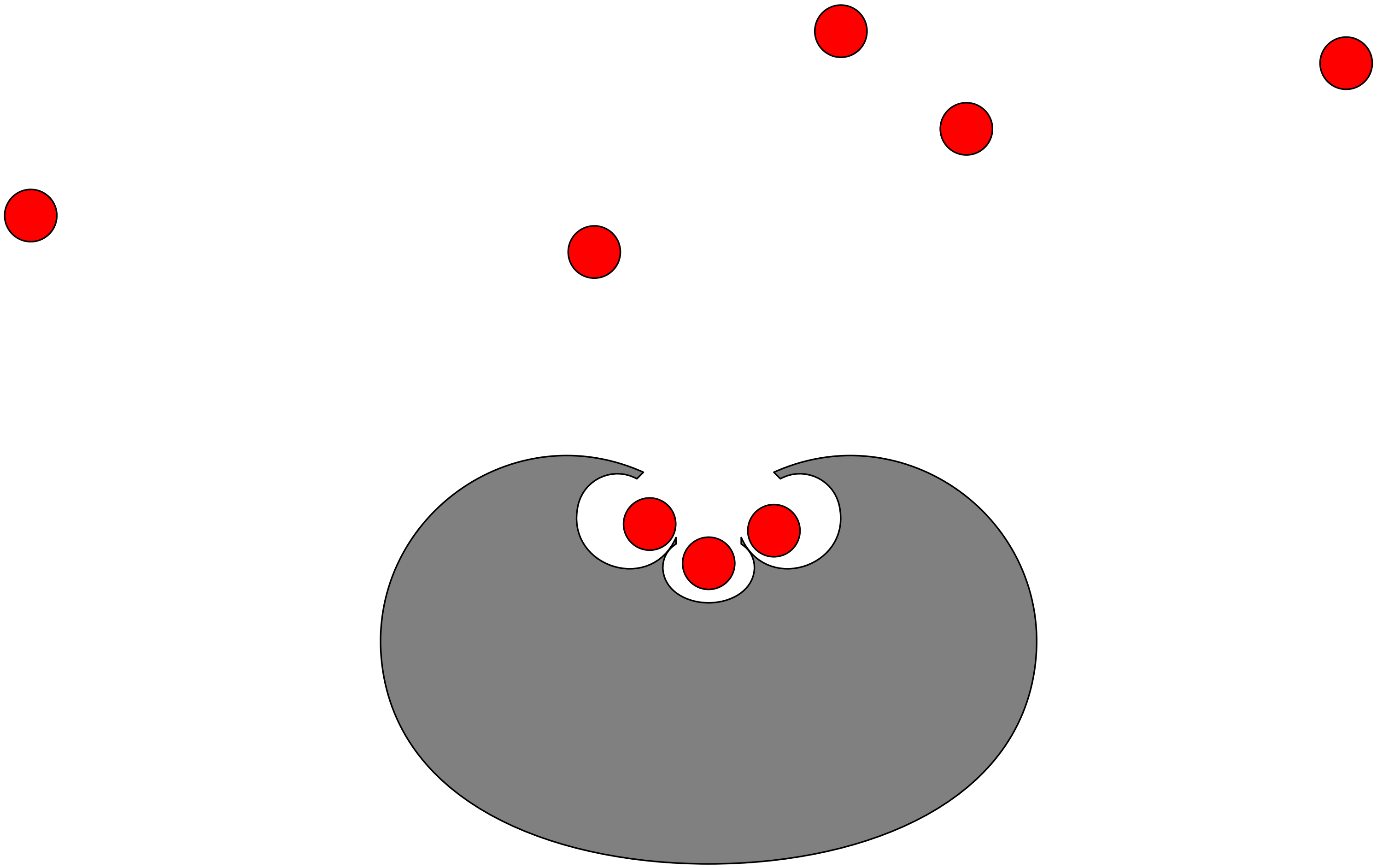

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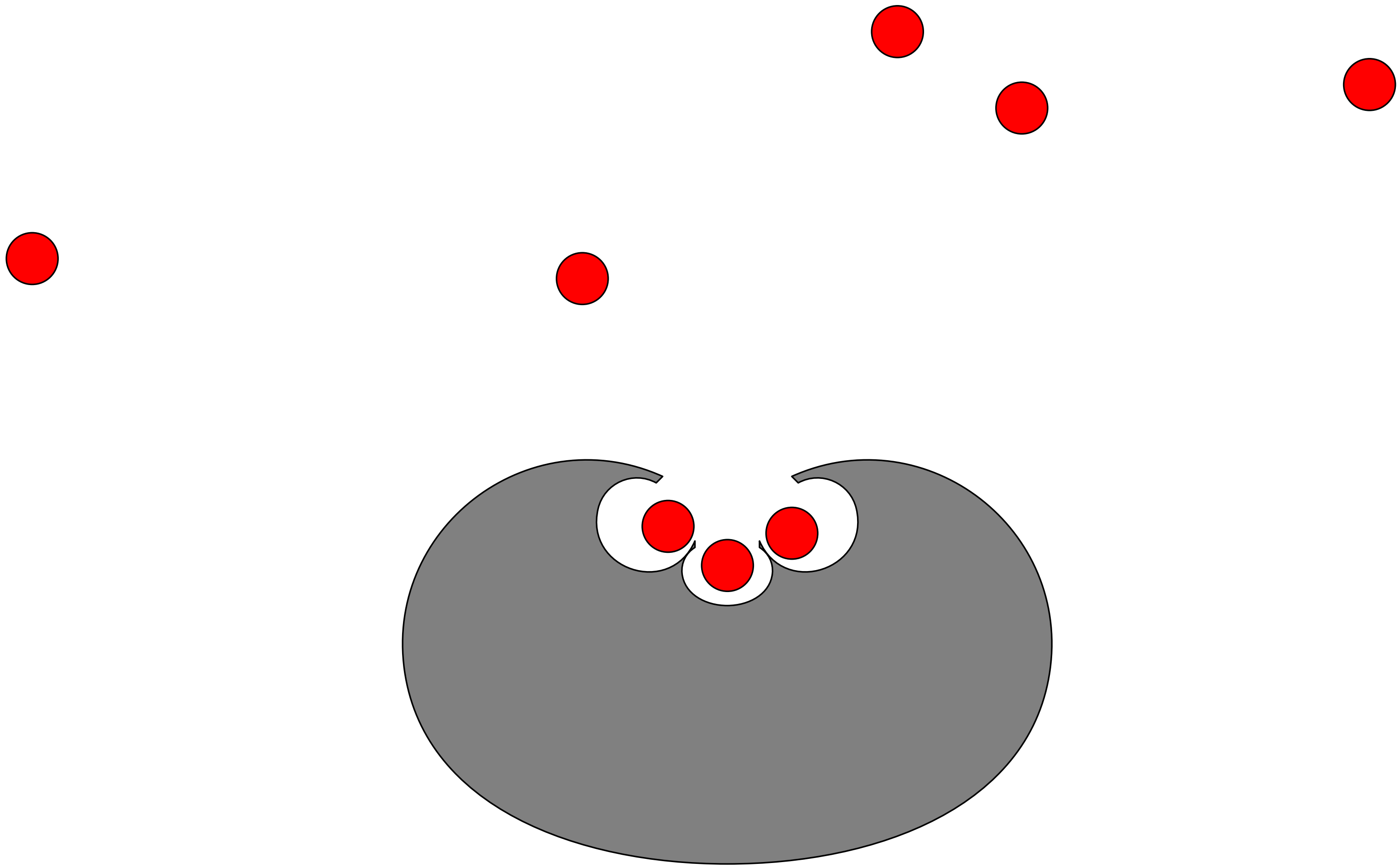

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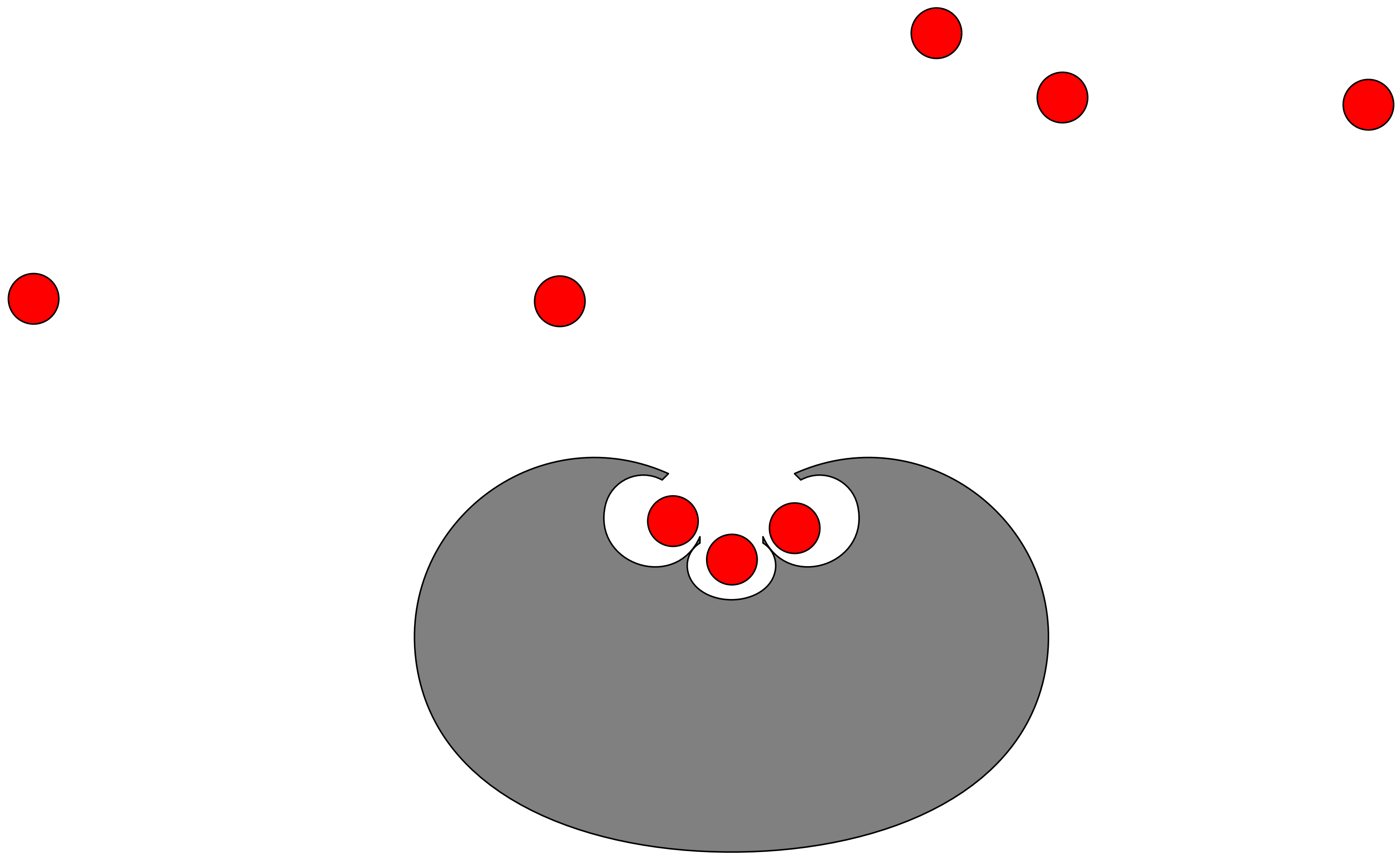

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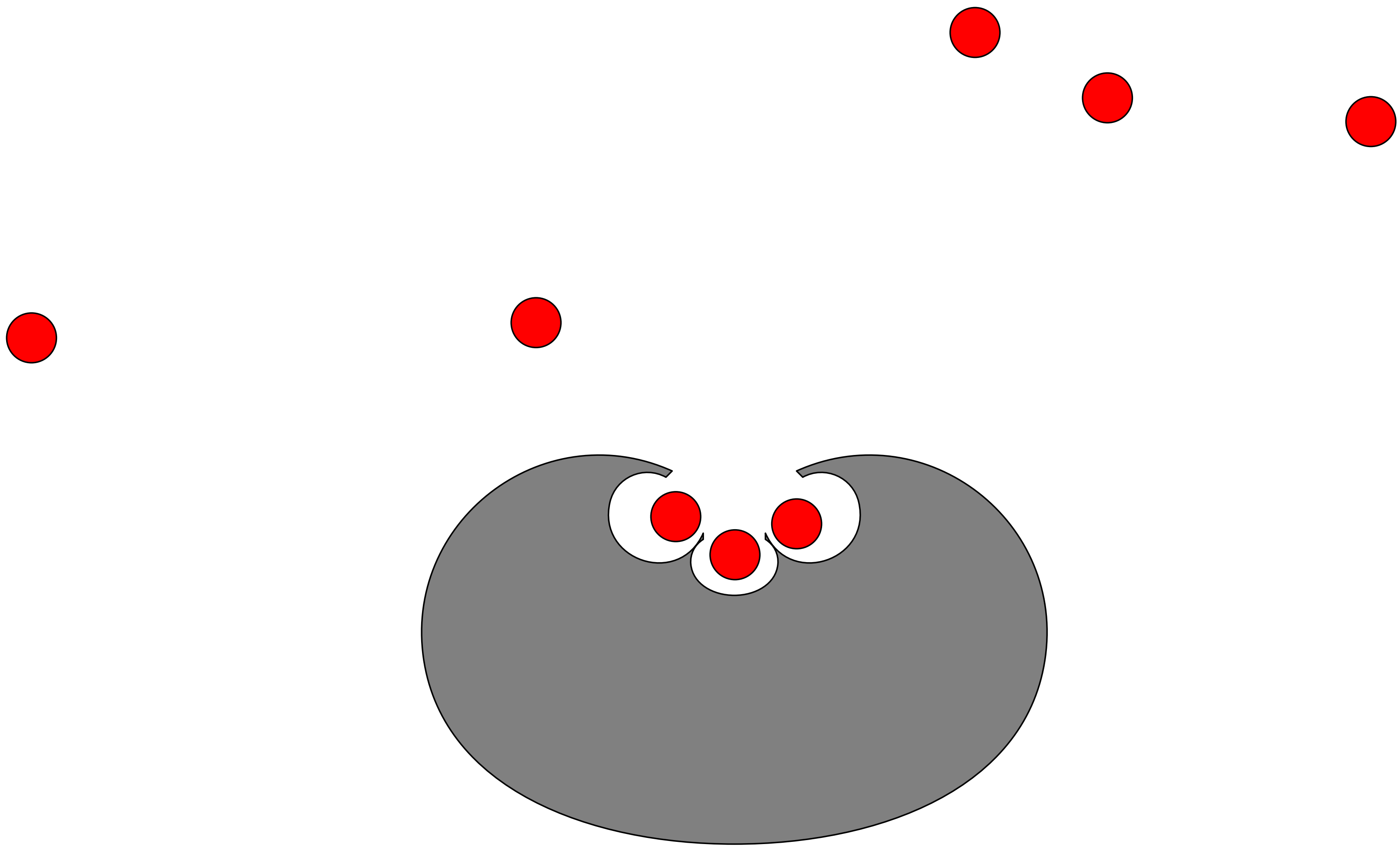

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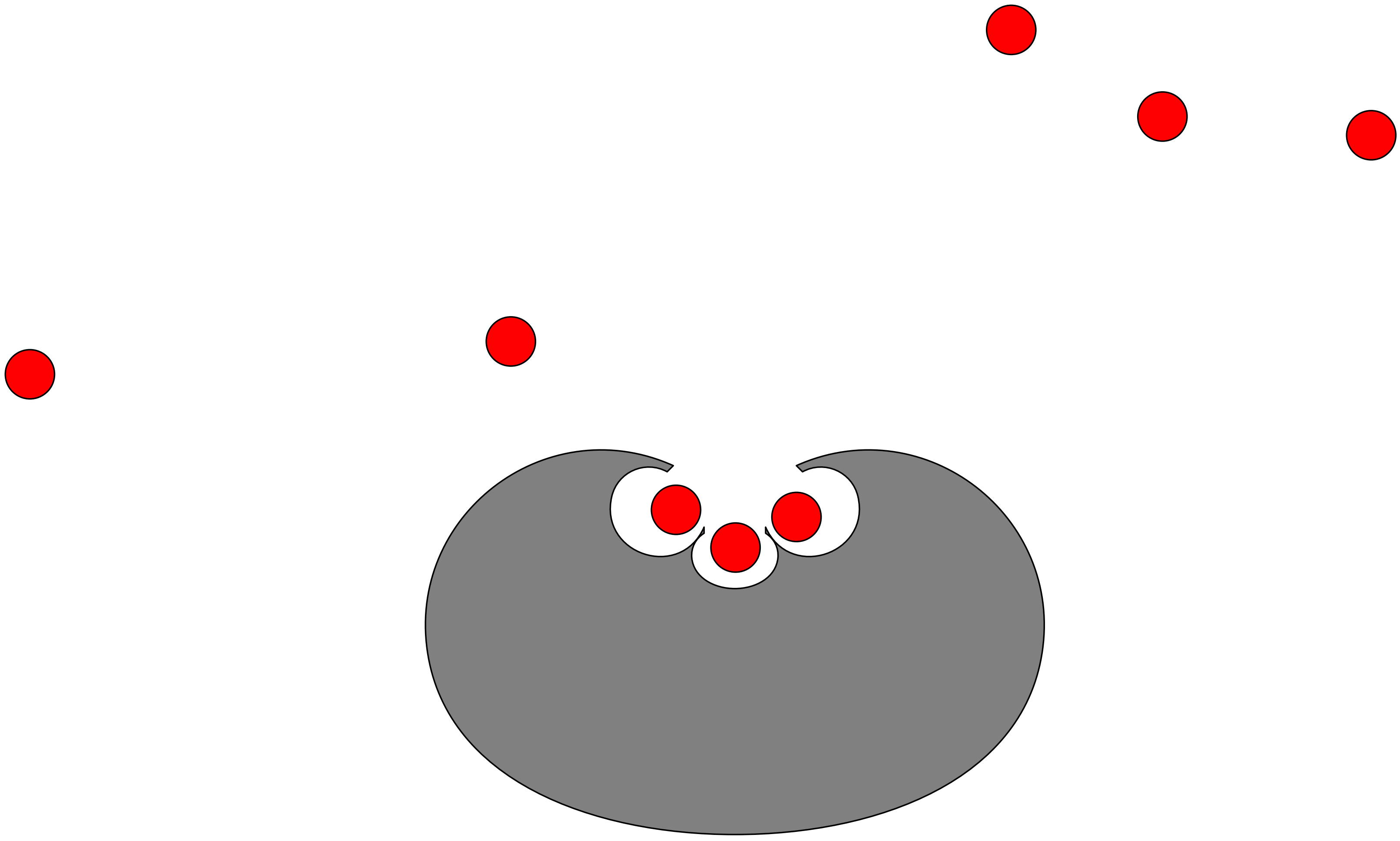

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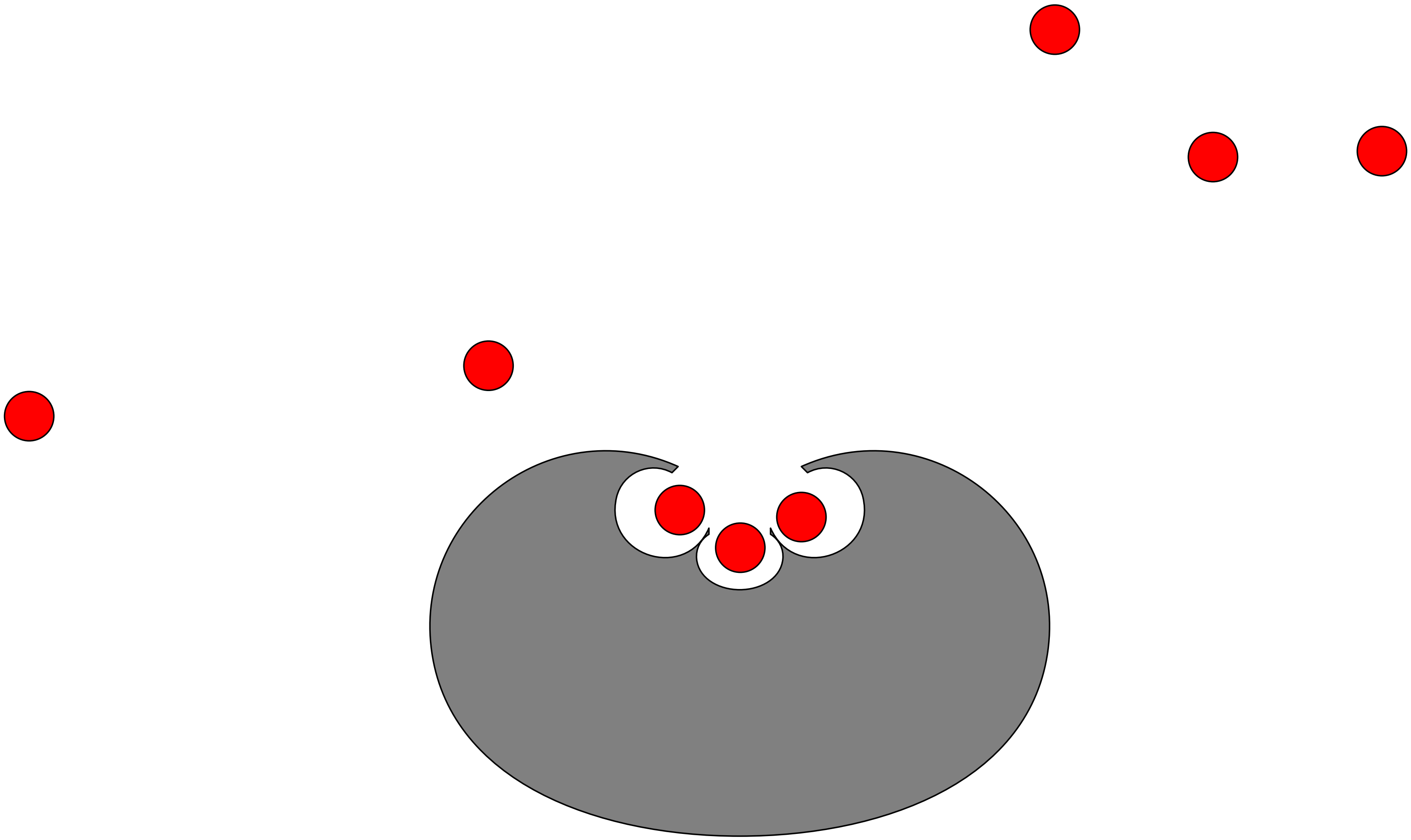

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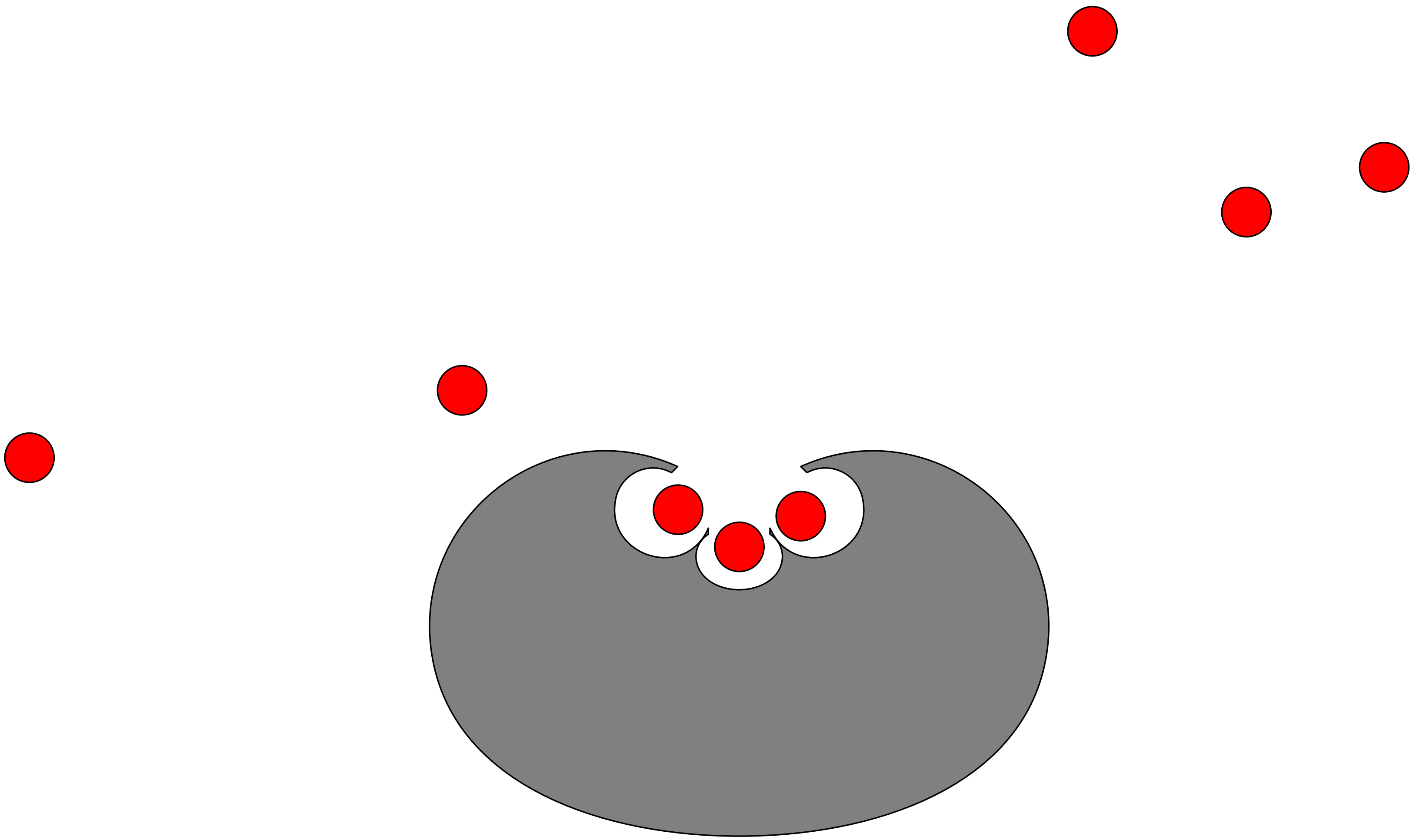

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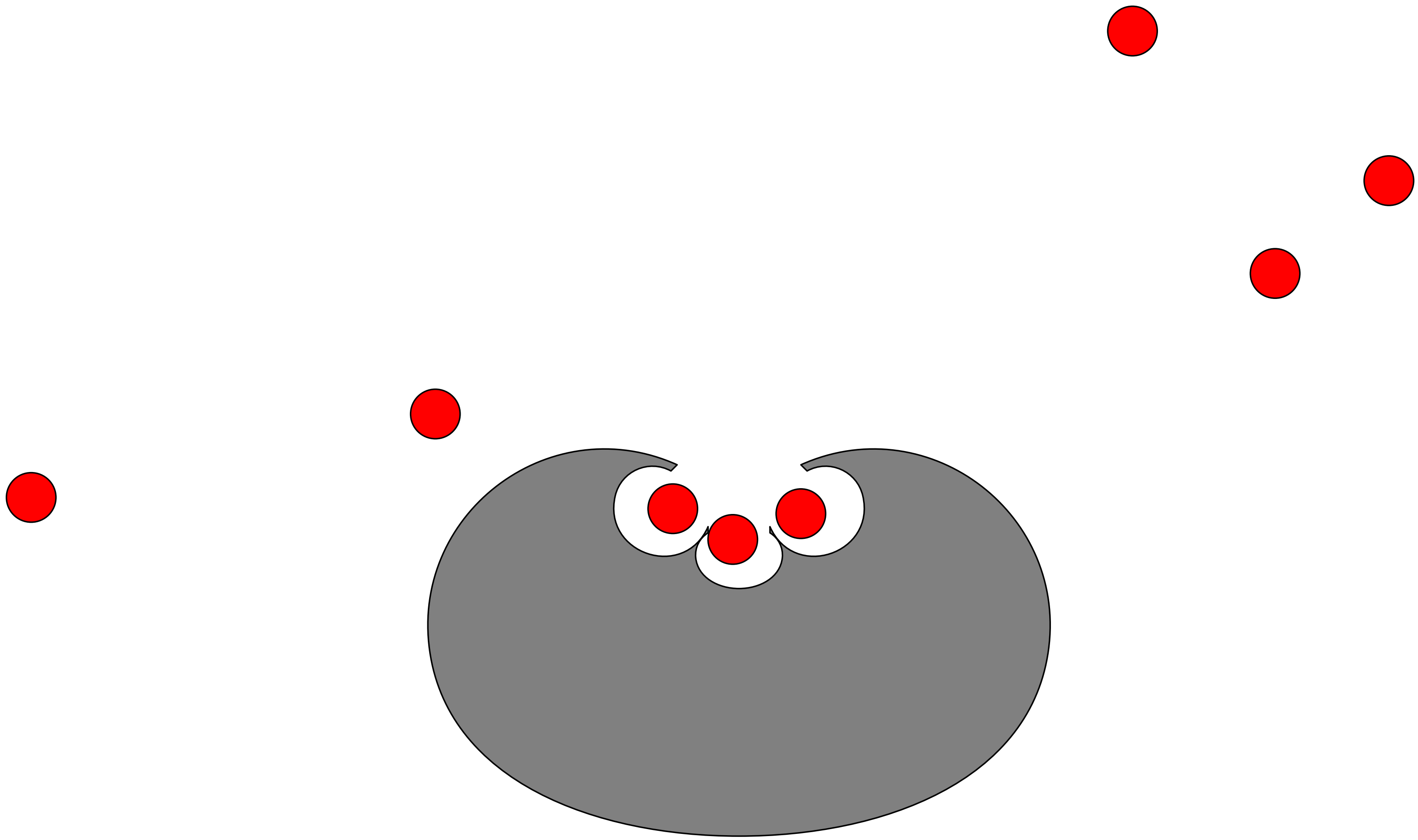

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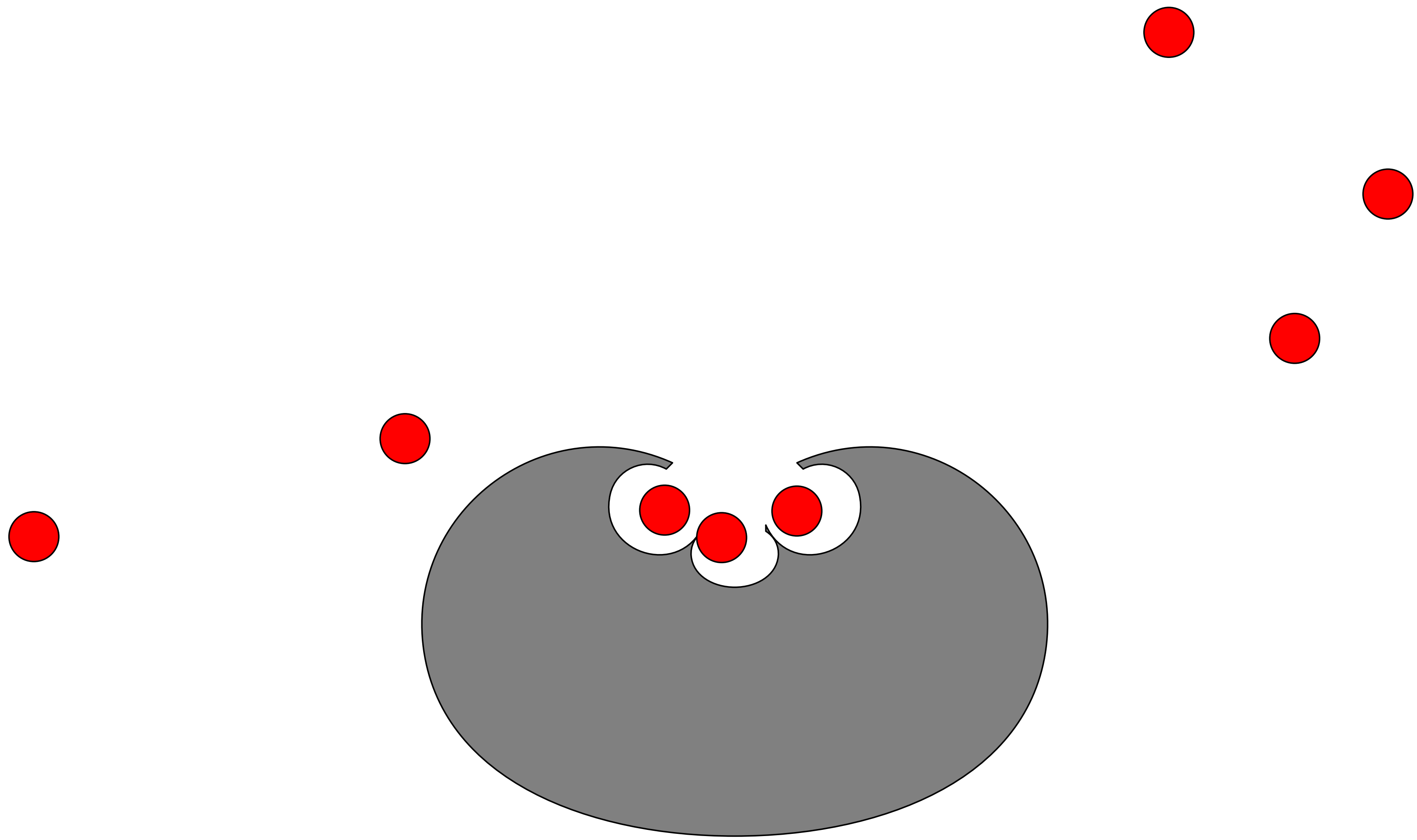

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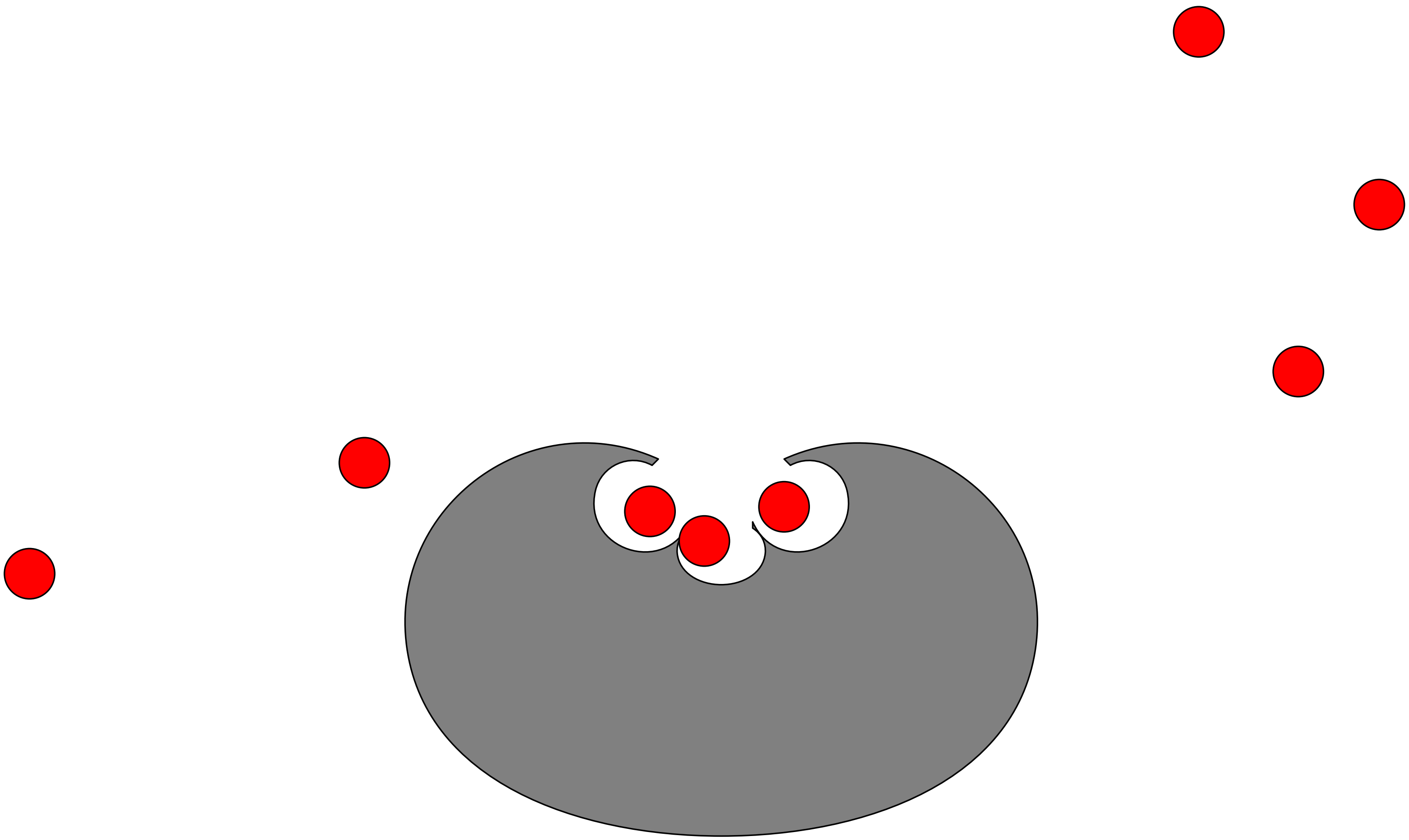

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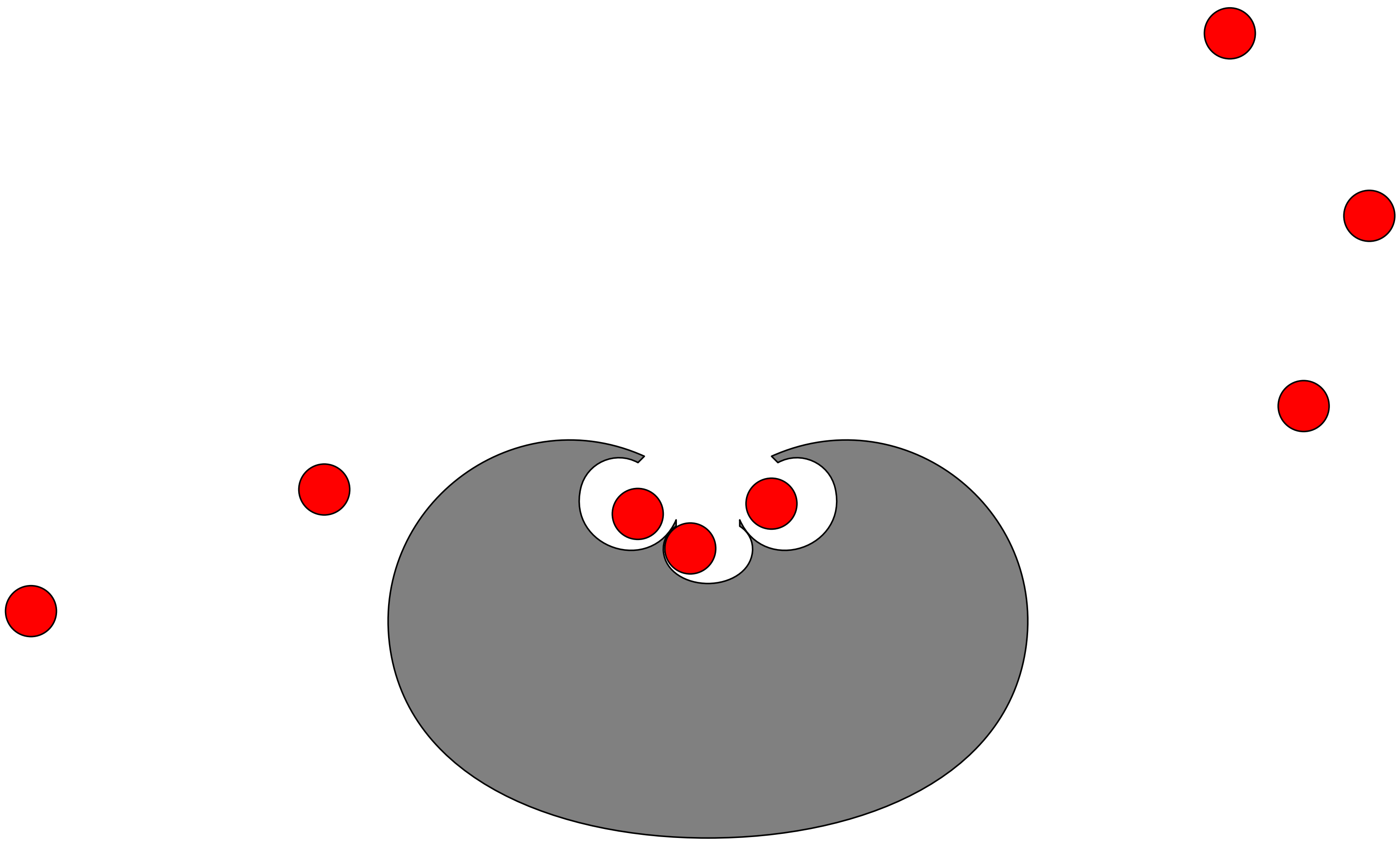

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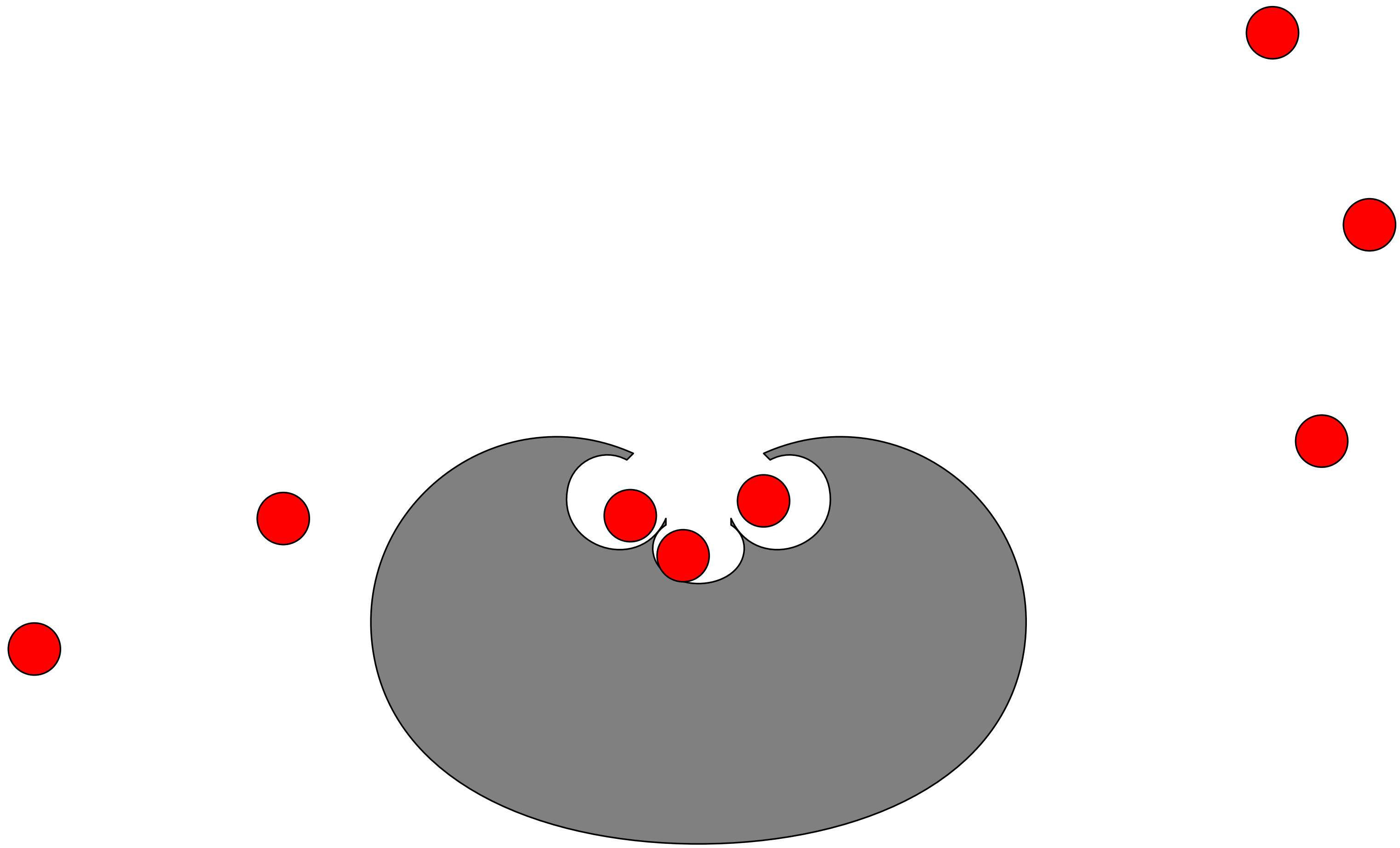

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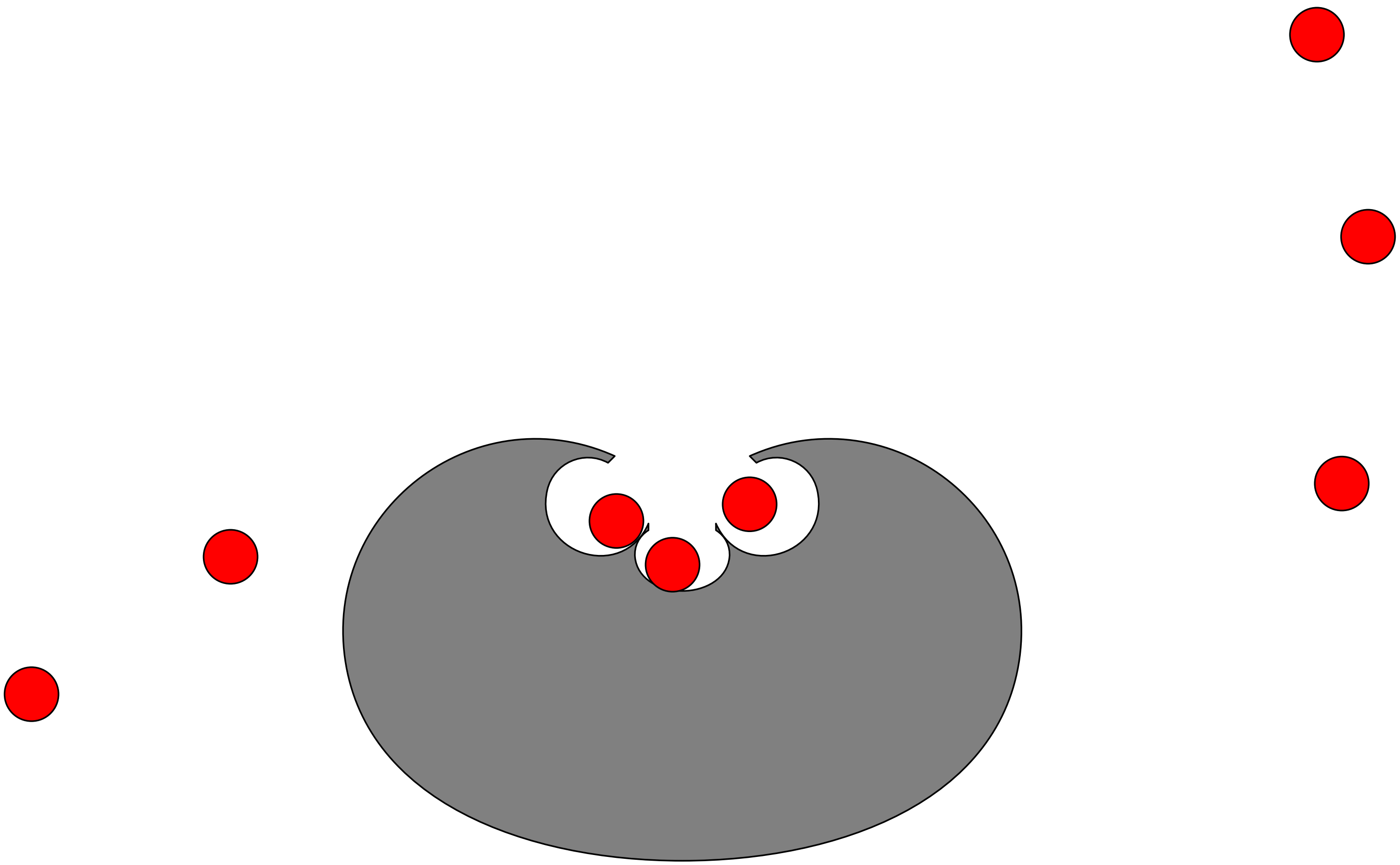

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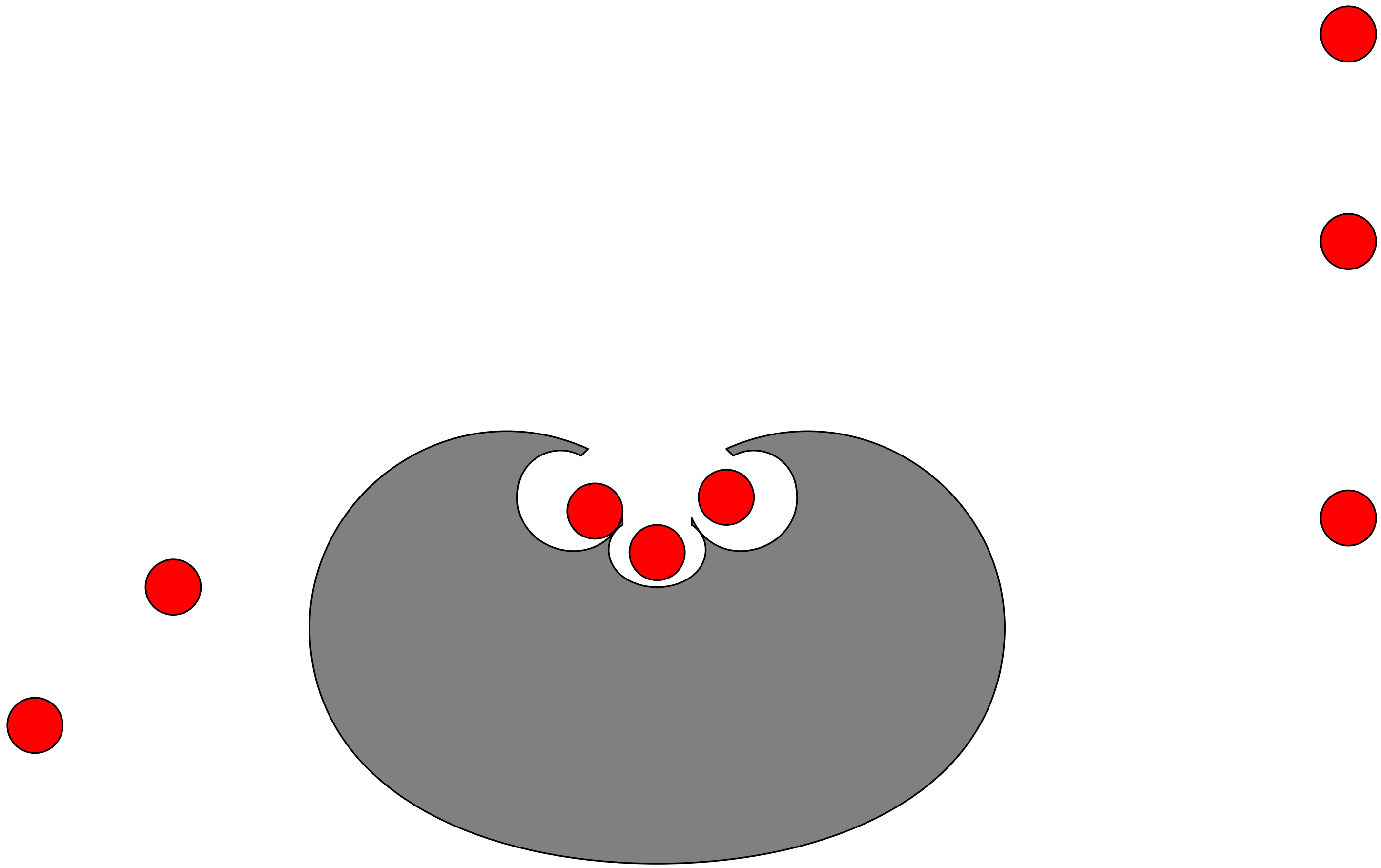

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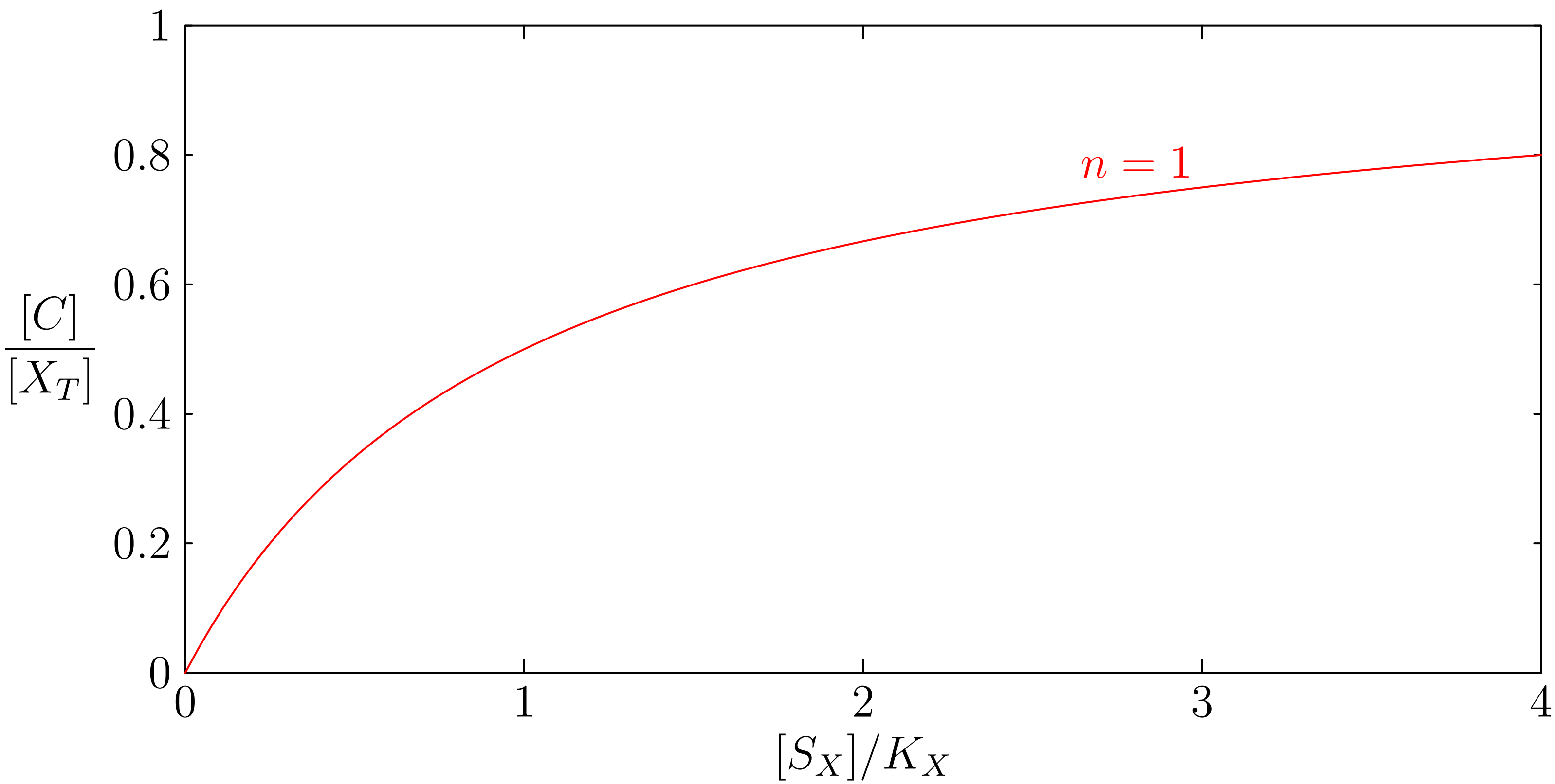

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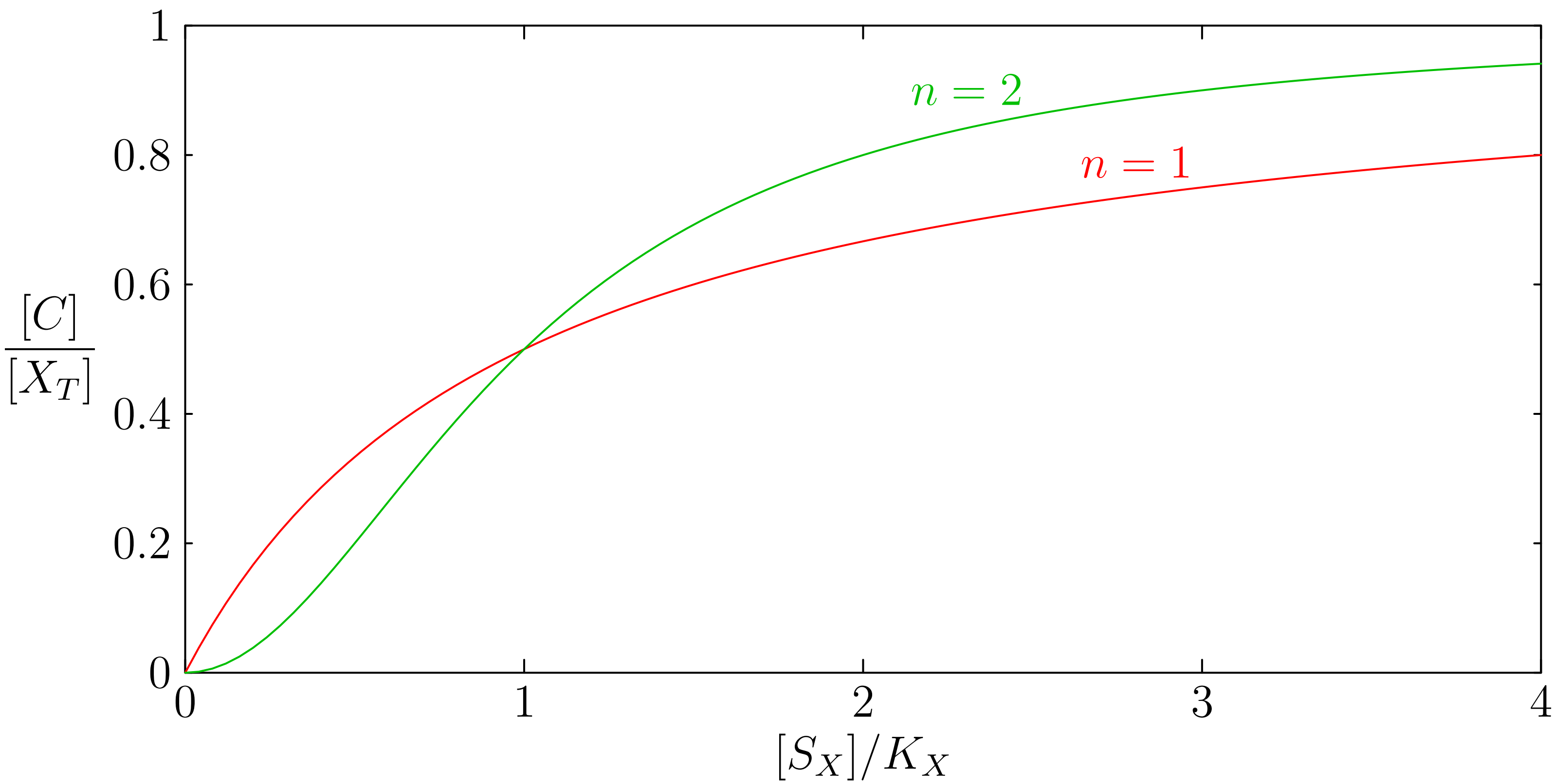

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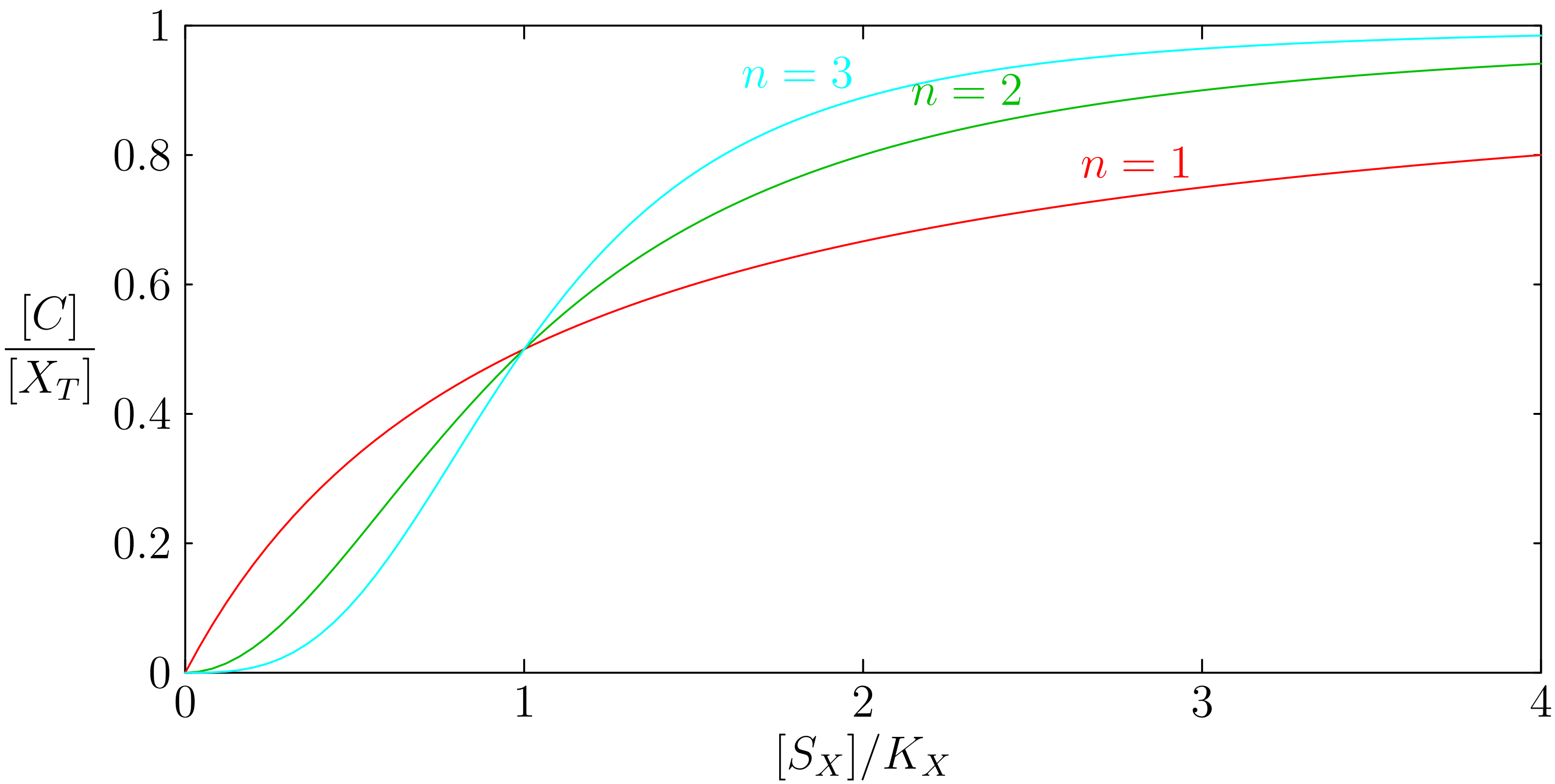

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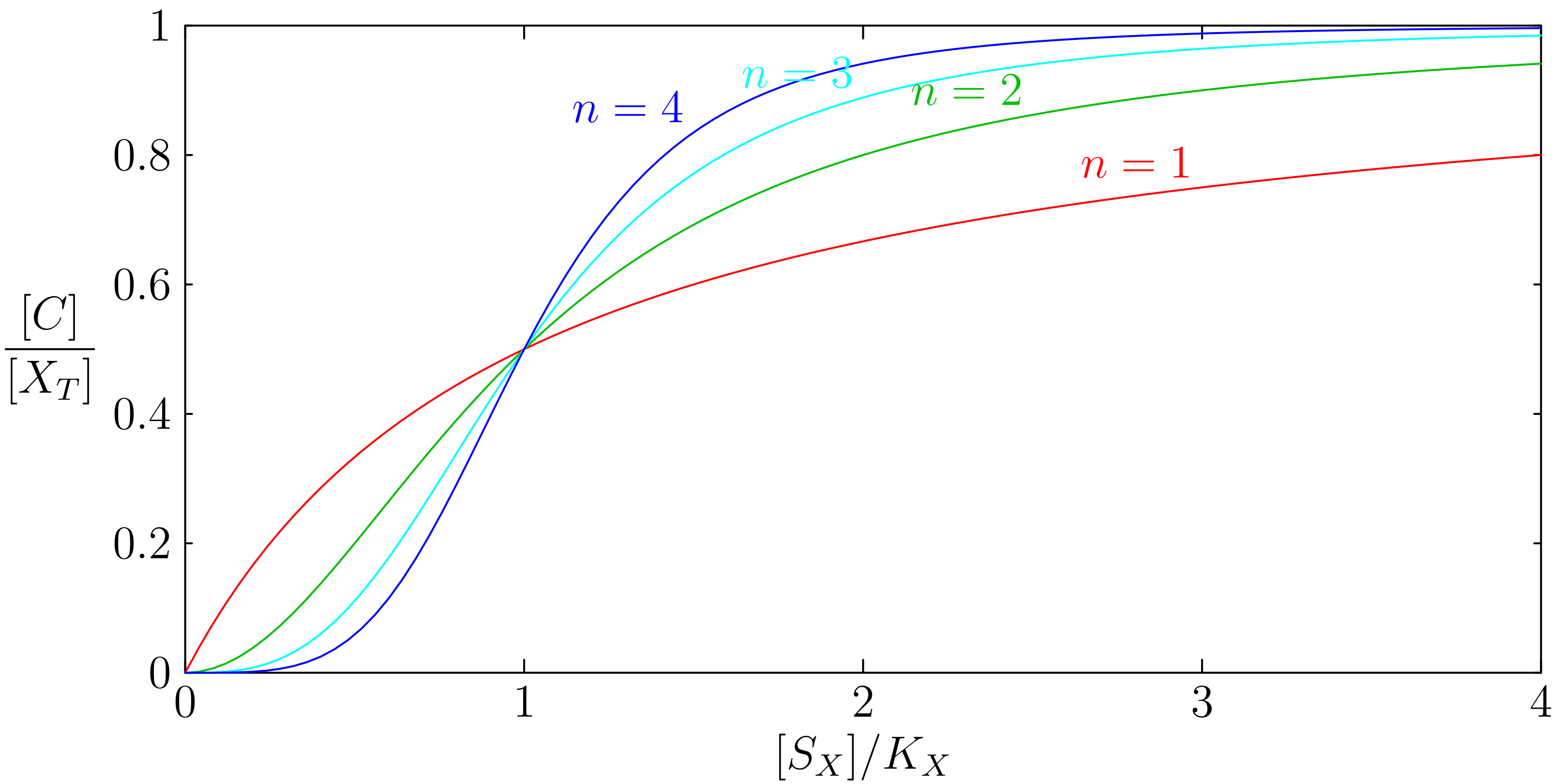

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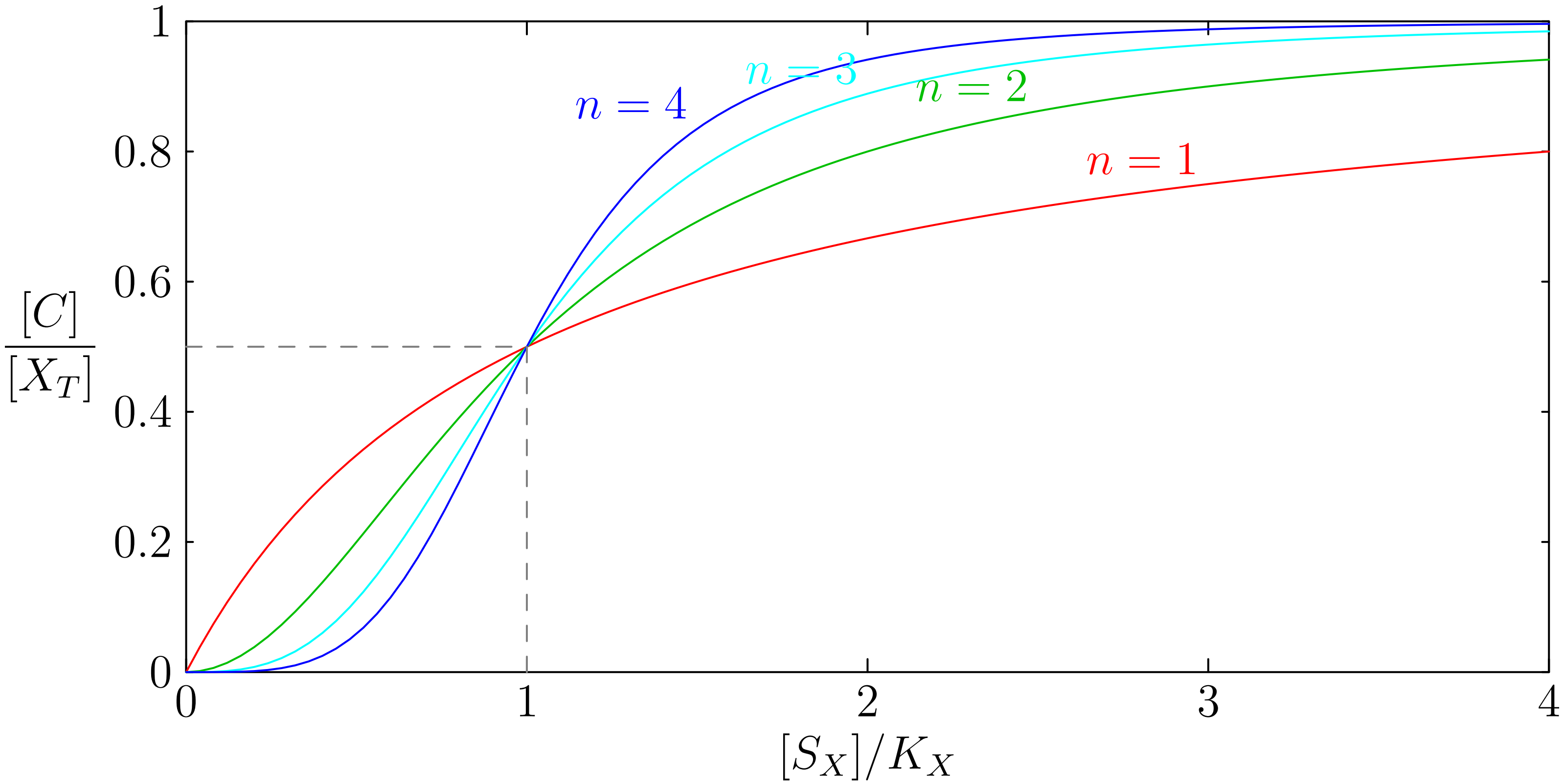

Activator and Repressor

- A protein that acts to increase the expression of a protein we call an **activator**

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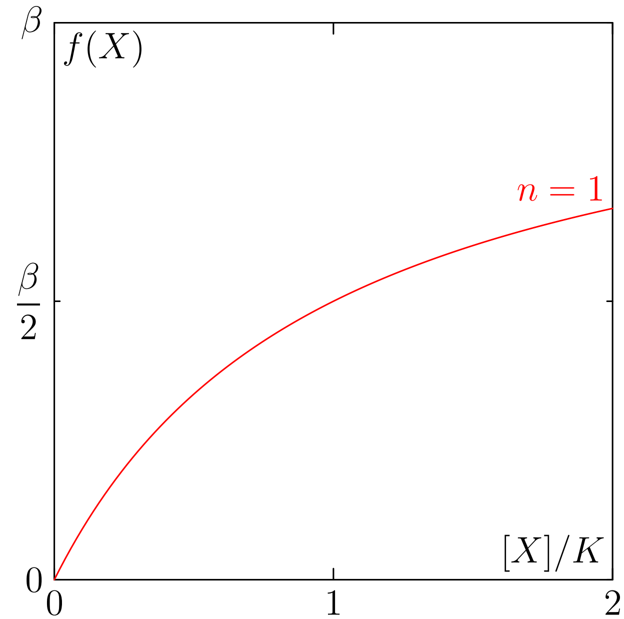

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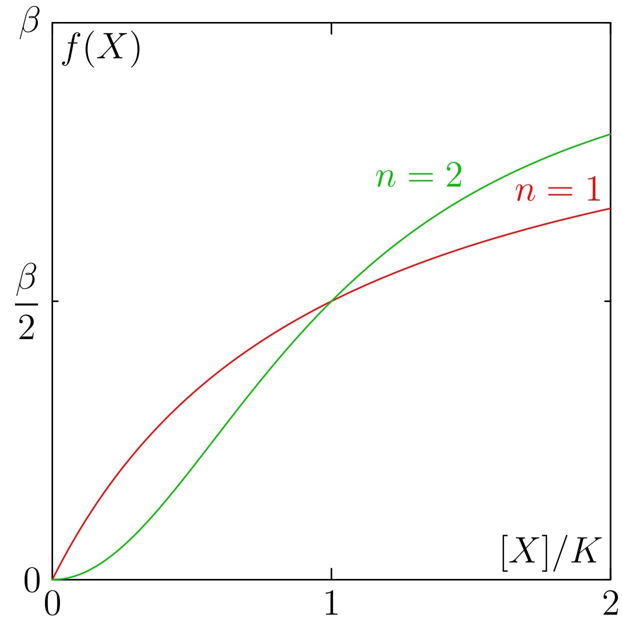

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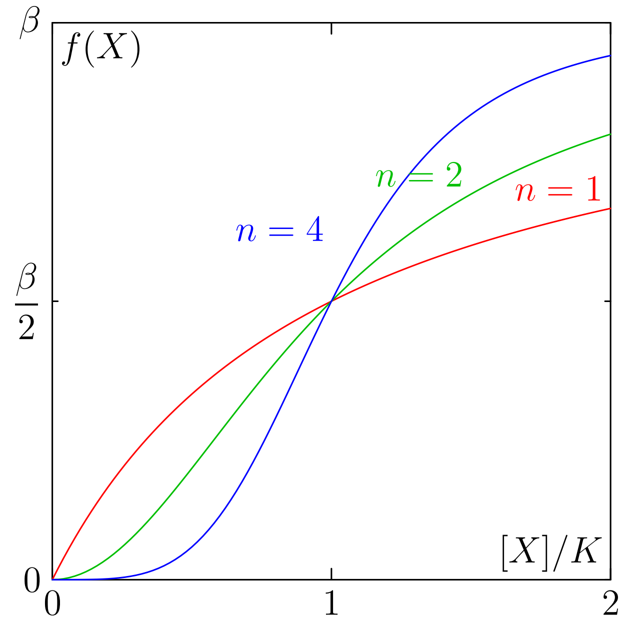

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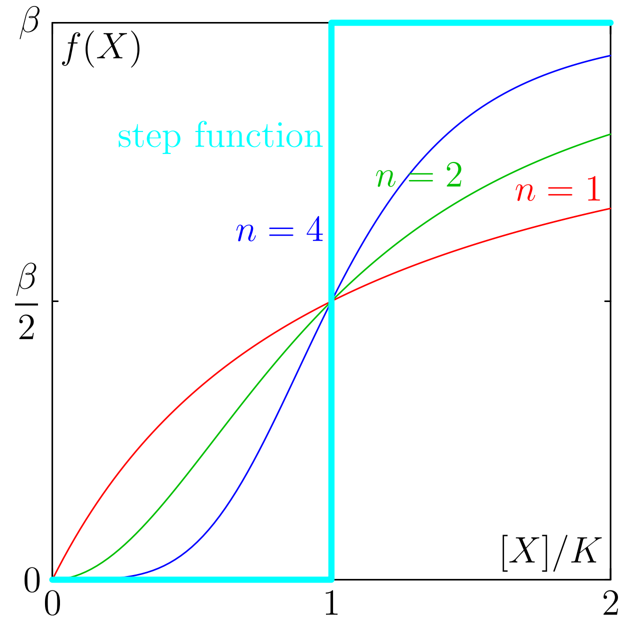

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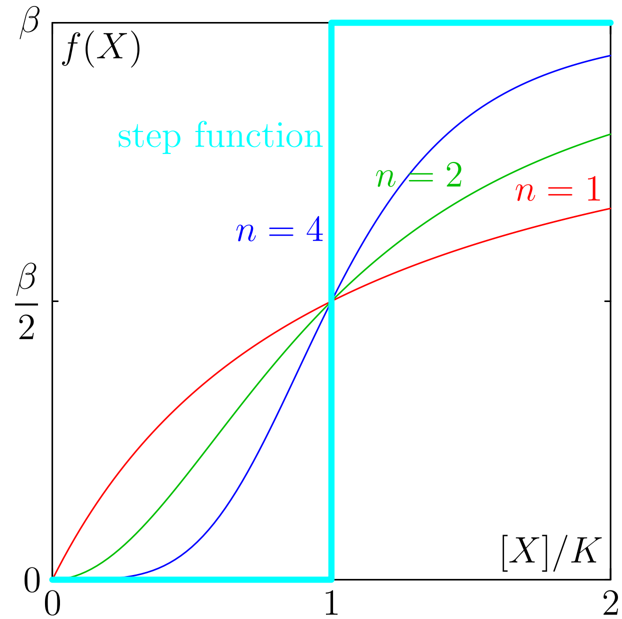

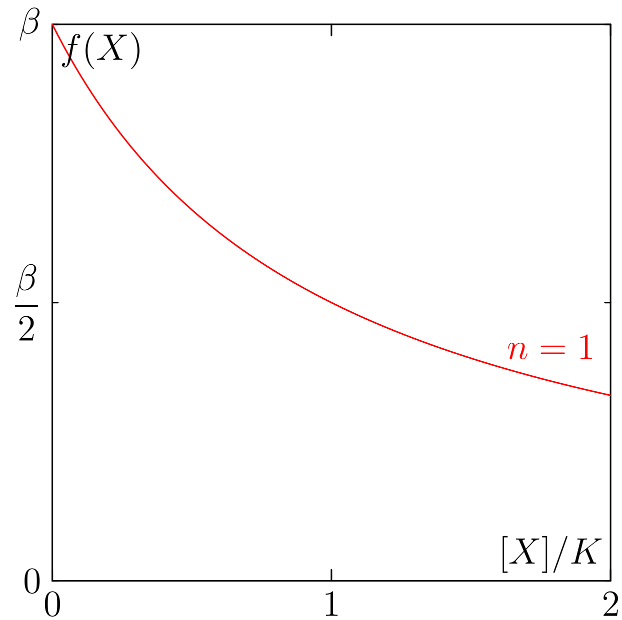

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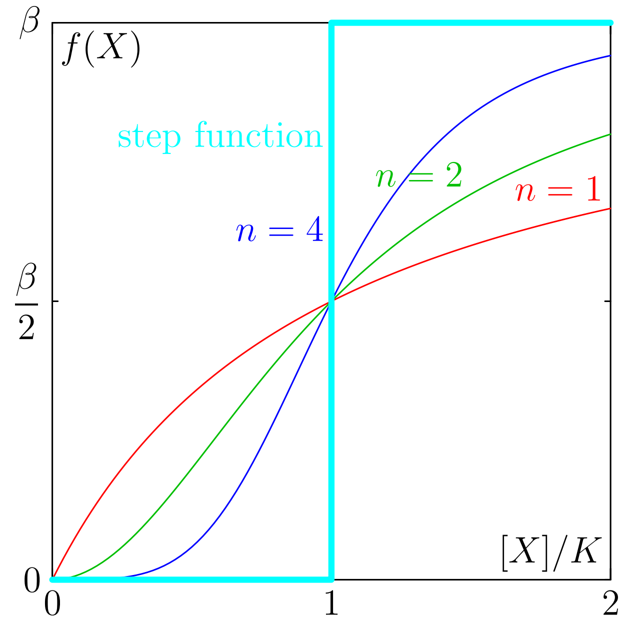

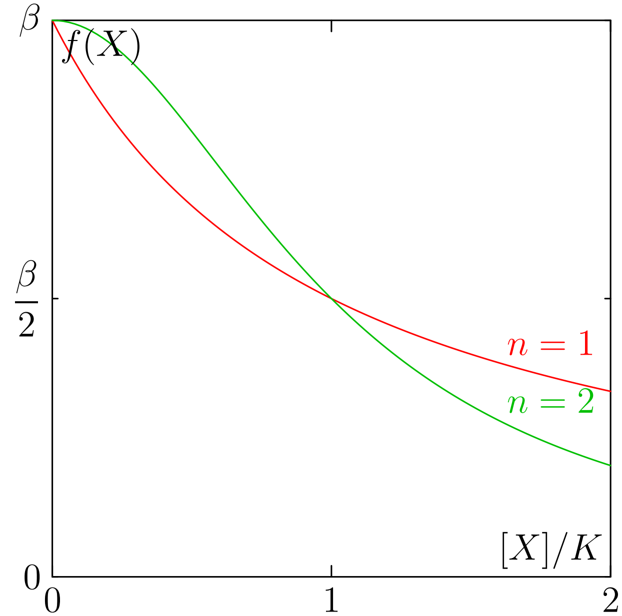

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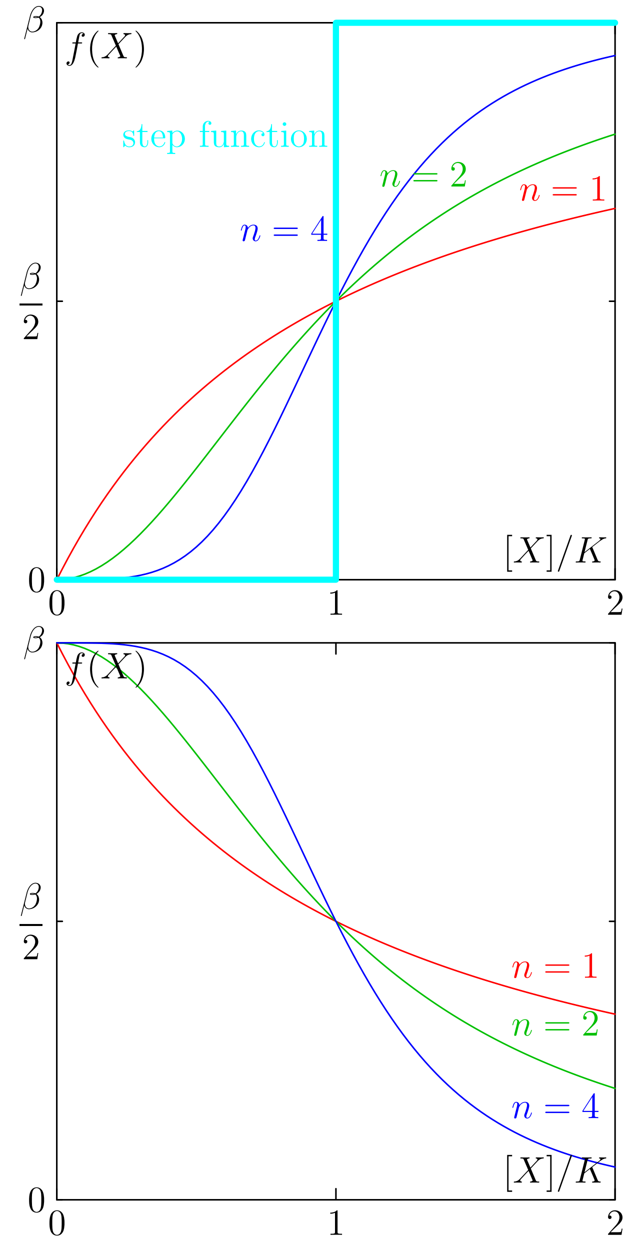

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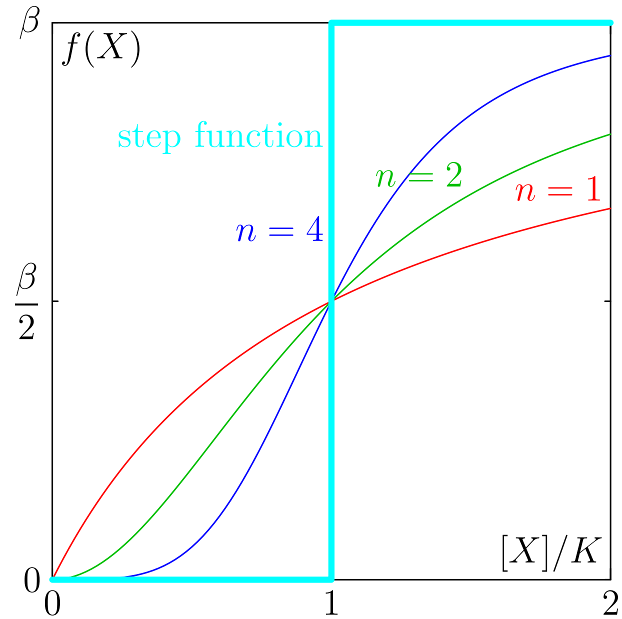
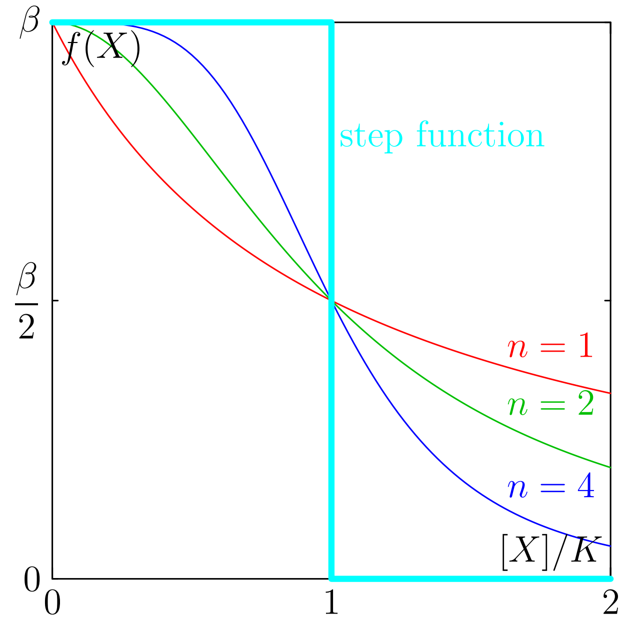
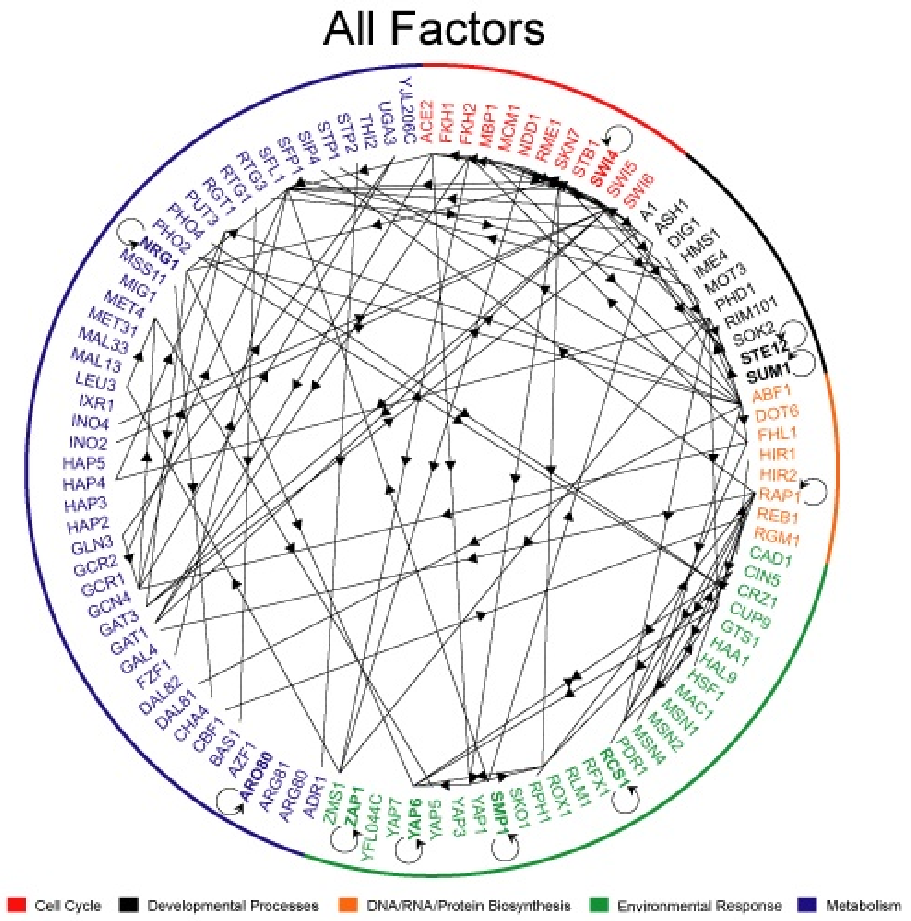




Outline

1. Gene Regulation
2. Chemical Equations
3. **Motifs**
4. Quiz



Understanding Gene Regulation Networks

- We can infer interactions between proteins as we described in a previous lecture

- Gene regulation networks tend to be rather complicated

- To make sense of them we wish to look for patterns, "**motifs**", that occur very often

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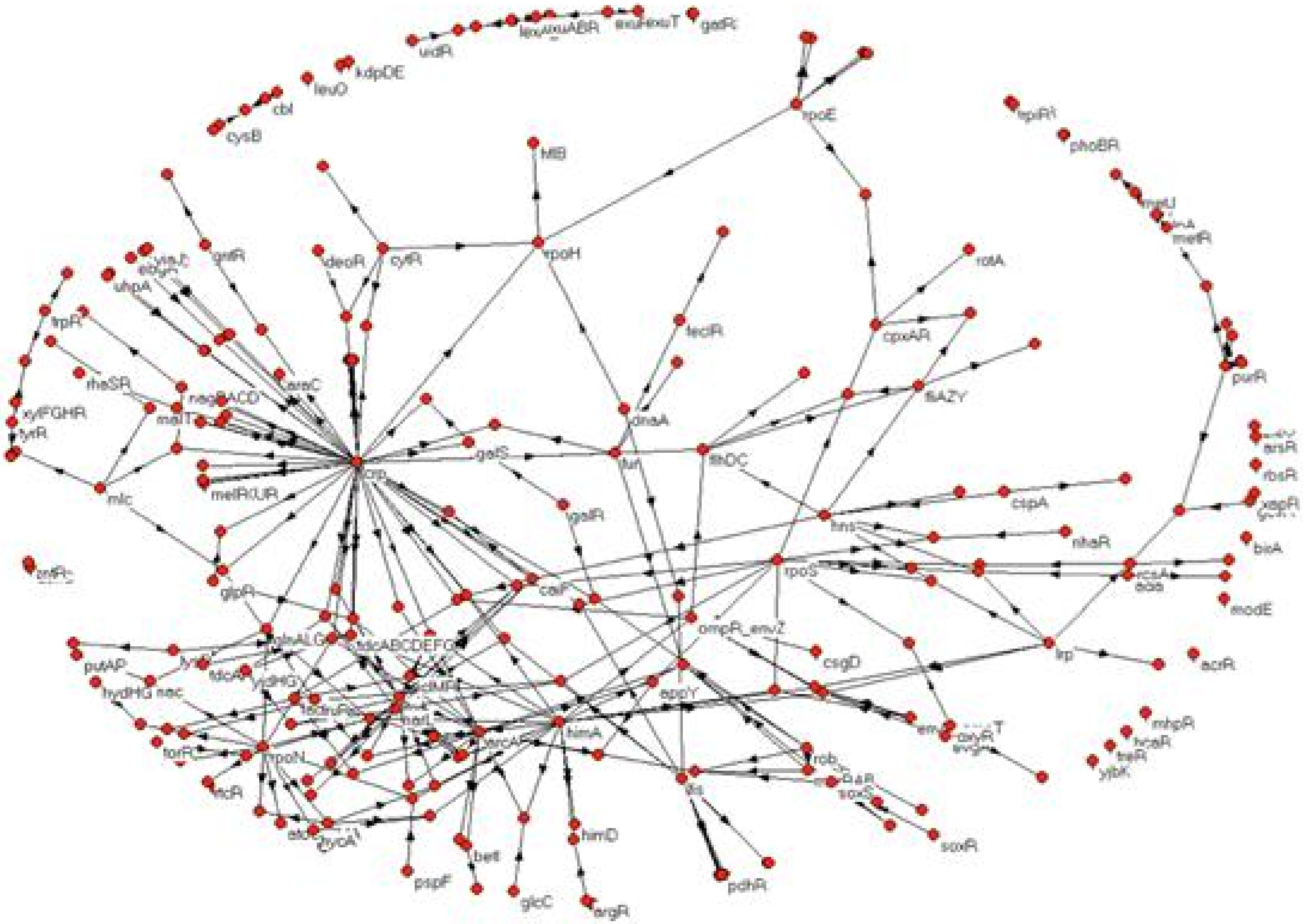
E-Coli GRN


rpoE
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crp
fur
flhDC
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hns
cpxAR
flhB
deoR
cytR
galR
araC
malT
mlc
glpR
fnr
arcA
himA
himD
rpoN
lrp
soxR
ompR_envZ
csgD
purR
modE
acrR

Auto-regulation

- In E-coli we have $N = 424$ nodes and $E = 519$ edges
- The total number of directed edges (including self-edges) is N^2
- The number of self edges is N so the probability of a randomly chosen edge being a self-edge is $p = 1/N$
- The probability that if we choose E edges k will be self-edges is

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- In fact we have $N_{\text{self}} = 40$

Auto-regulation

- In E-coli we have $N = 424$ nodes and $E = 519$ edges
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No Auto-Regulation

- To understand why auto-regulation is interesting let us first consider the case with no negative feedback
- We assume
$$\varnothing \xrightarrow{\hat{\beta}} \hat{X}$$
$$\hat{X} \xrightarrow{\hat{\alpha}} \varnothing$$
- So on average
$$\frac{\mathrm{d}[\hat{X}]}{\mathrm{d}t} = \hat{\beta} - \hat{\alpha}\,[\hat{X}]$$
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Solving Differential Equations

- The full dynamics is given by the solution to the differential equation and depend on the initial condition, $[\hat{X}(0)]$

- Most differential equations cannot be solved in closed form

- However, they are usually quite easy to solve numerically (though it is difficult to understand how the solution depend on the parameters)

- Our equation is linear and is easy to solve. For $[\hat{X}(0)] = 0$

$$\frac{\mathrm{d}[\hat{X}(t)]}{\mathrm{d}t} = \hat{\beta} - \hat{\alpha}\,[\hat{X}(t)] \qquad \Rightarrow \qquad [\hat{X}(t)] = \frac{\hat{\beta}}{\hat{\alpha}}\left(1 - \mathrm{e}^{-\hat{\alpha}\,t}\right)$$

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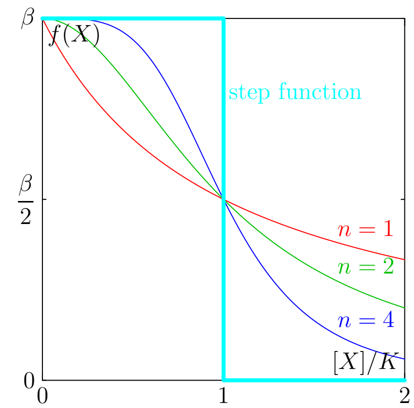
Auto-regulation

- We assume a negative feedback so that

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = f([X]) - \alpha\,[X]$$

- Where we have a Hill-type repressor

$$f([X]) = \frac{\beta}{1 + ([X]/K)^n} \approx \beta\,[\![[X] < K]\!]$$



- Starting from $[X(0)] = 0$ initially $[X(t)] = \beta\,t$

- When $[X(t)] > K$ then $f([X(t)]) = f(K+\epsilon) \approx 0$ and a steady state is reached

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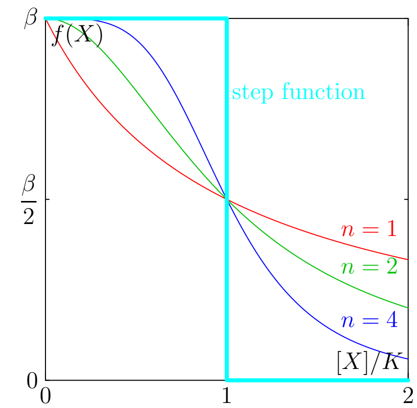

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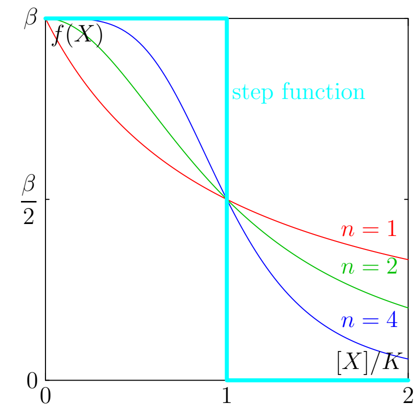

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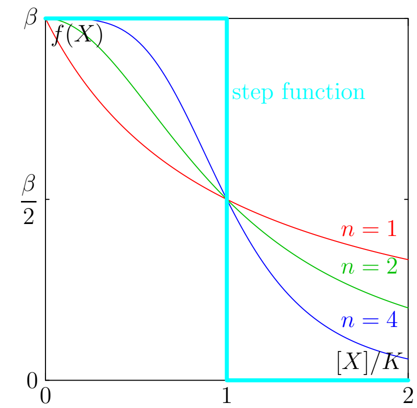

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Controlled Comparison

- To compare to having no feedback we want our two circuits to be as similar as possible
- Choose the same decay rate $\alpha = \hat{\alpha}$
- Choose $K = \hat{\beta}/\hat{\alpha}$ so that the steady state or equilibrium value of $X(t)$ is the same in both systems $[\hat{X}_{eq}] = [X_{eq}]$
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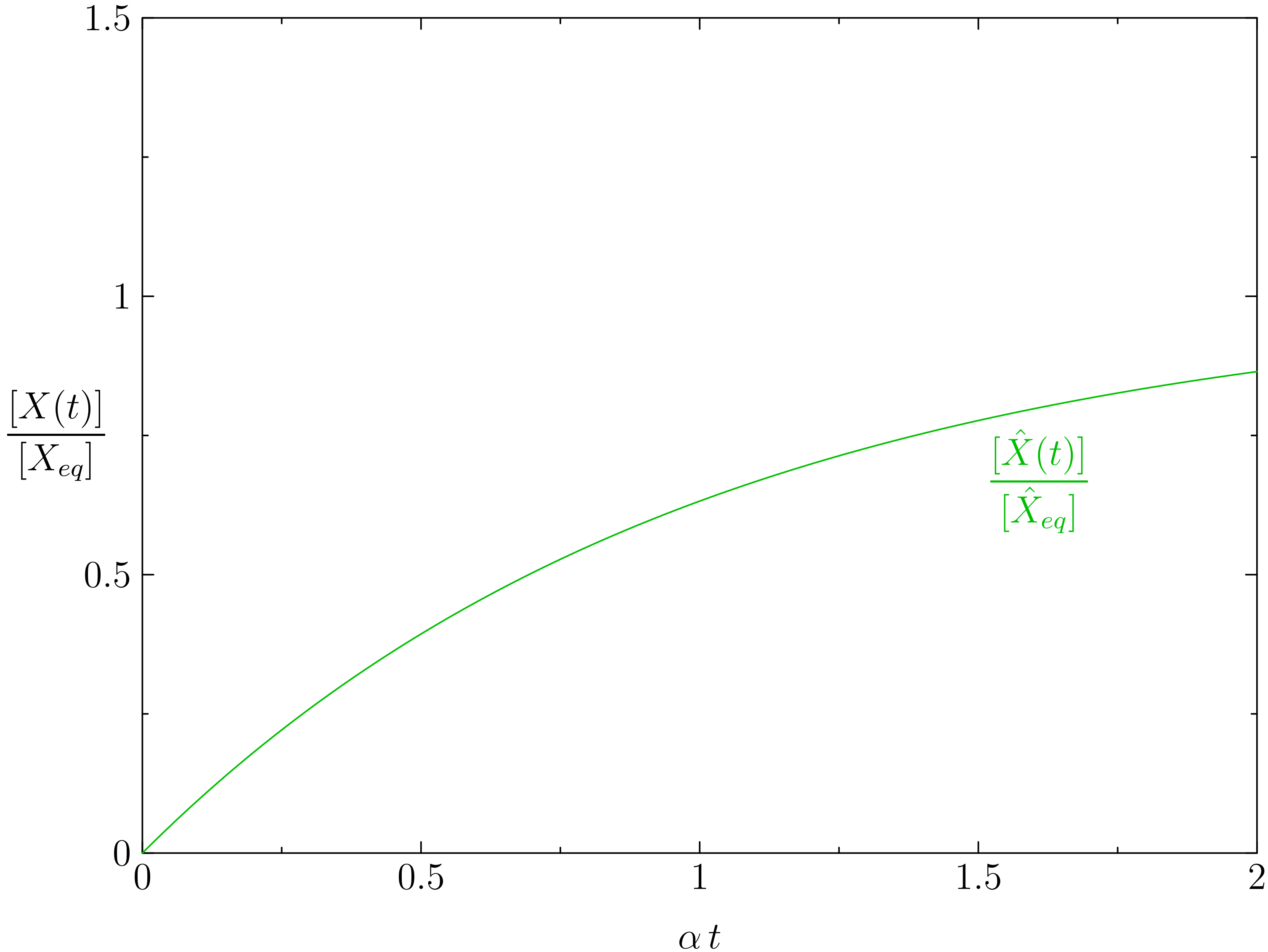

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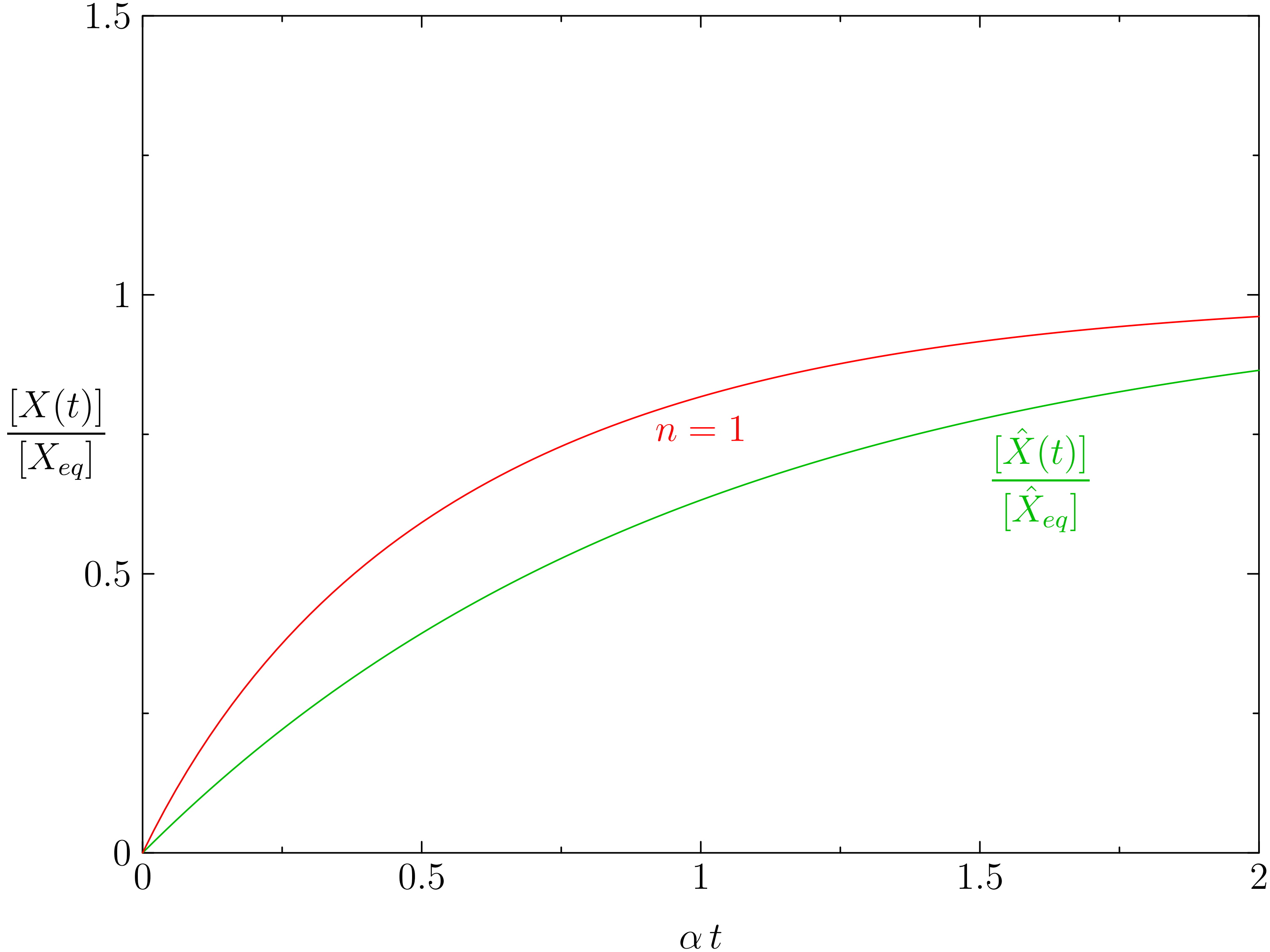

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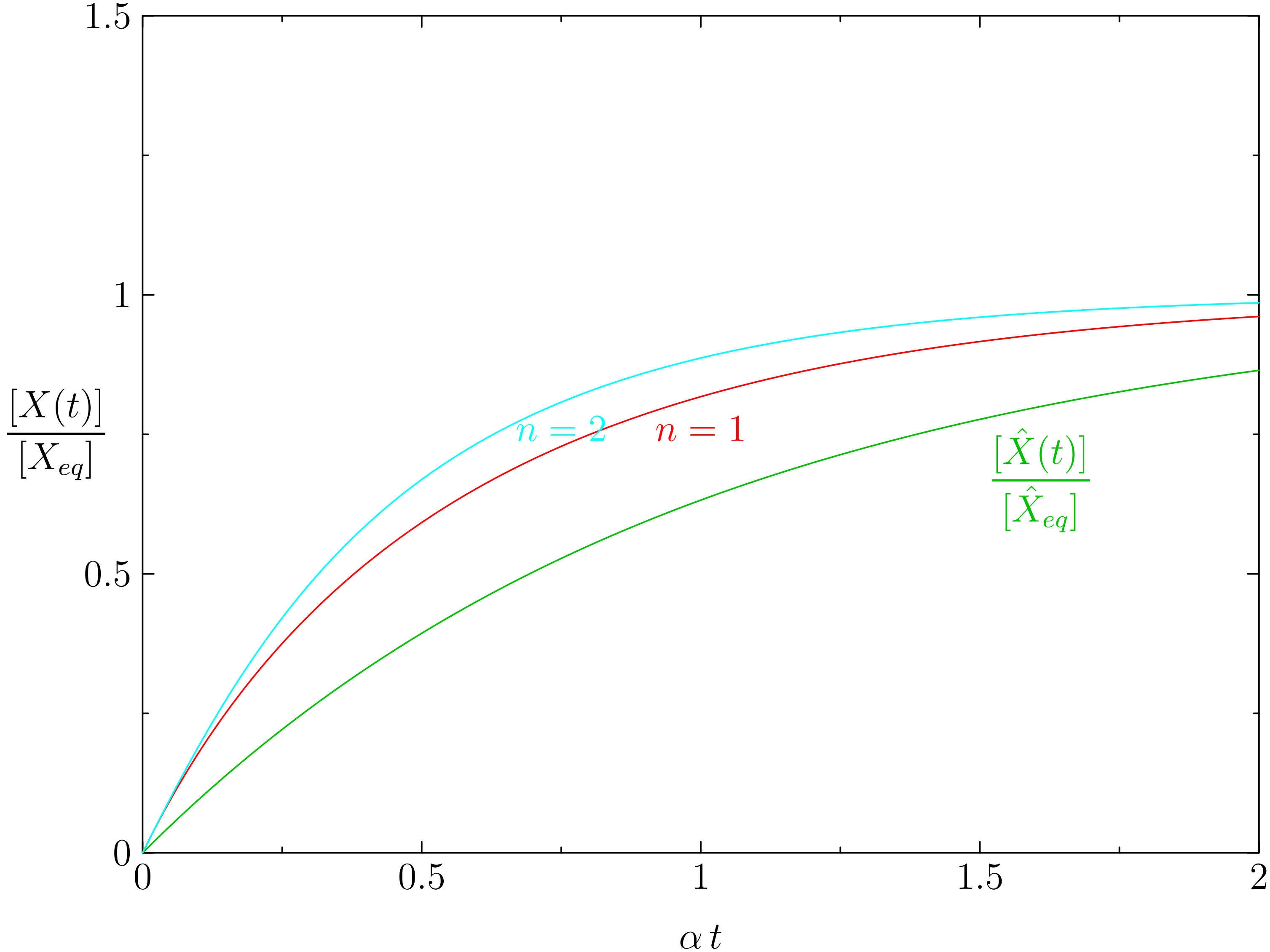

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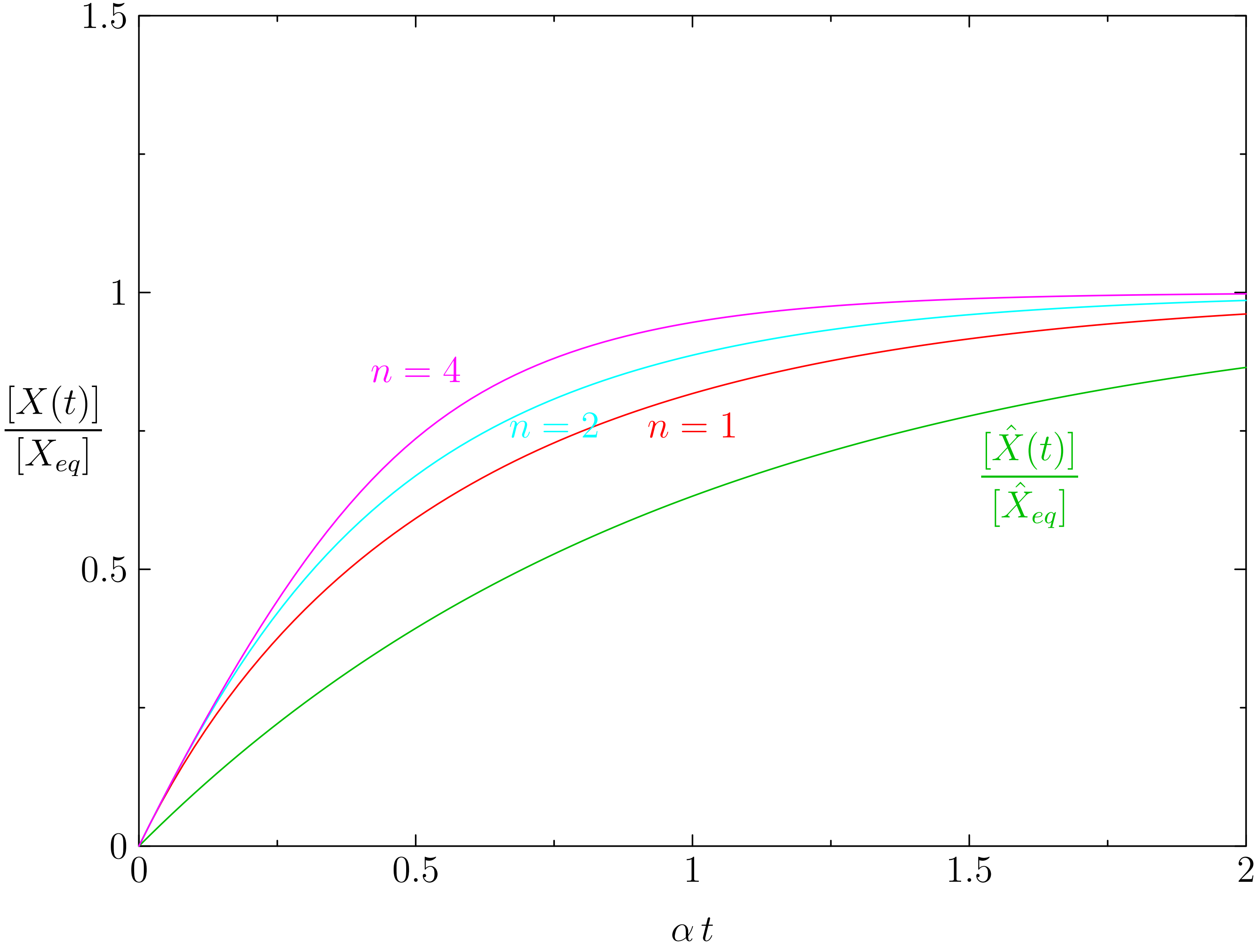

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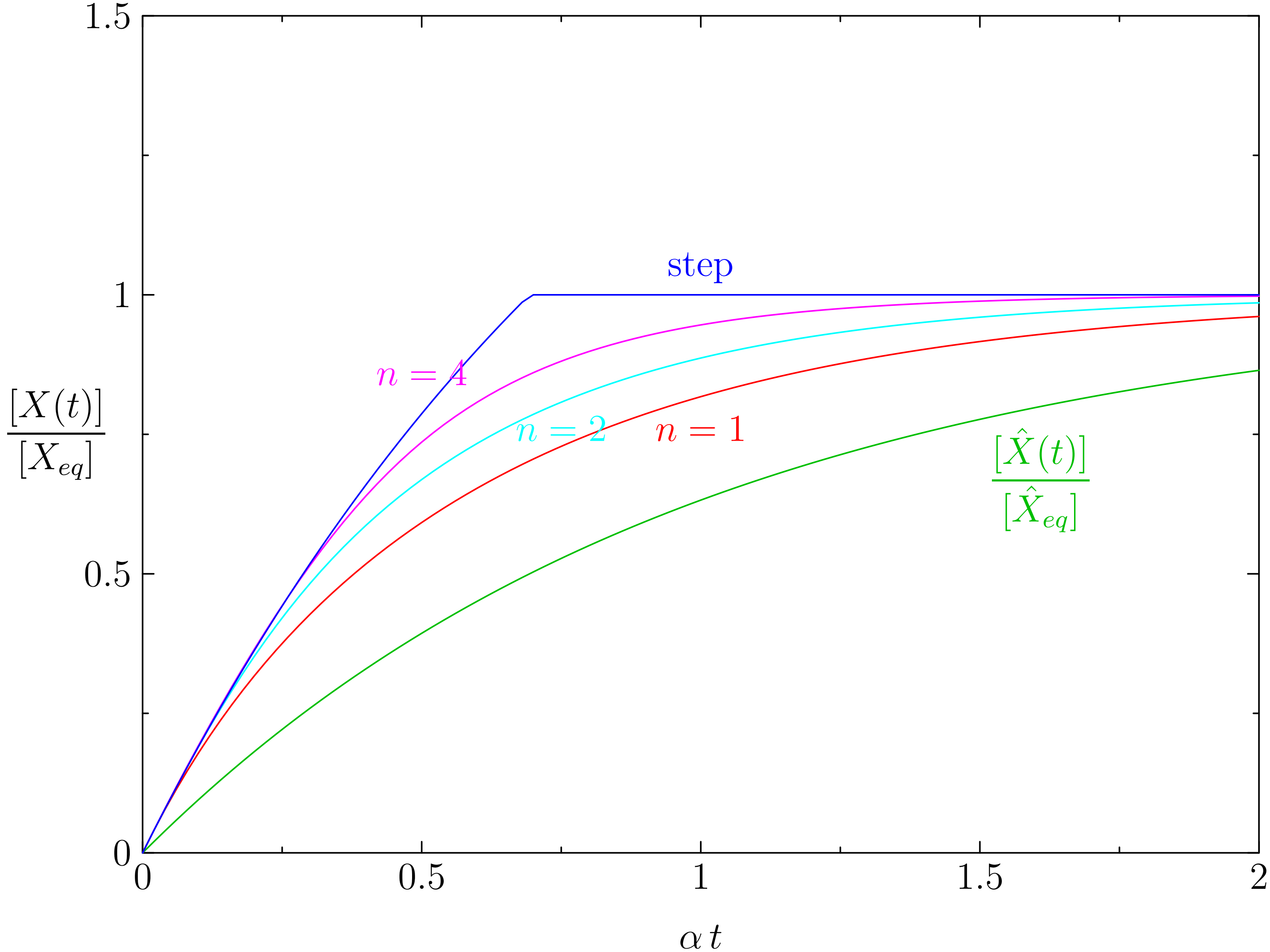

1.5
1
0.5
0
0
0.5
1
1.5
2
$\alpha\, t$
$\frac{[X(t)]}{[X_{eq}]}$
step
$n = 4$
$n = 2$
$n = 1$
$\frac{[\hat{X}(t)]}{[\hat{X}_{eq}]}$

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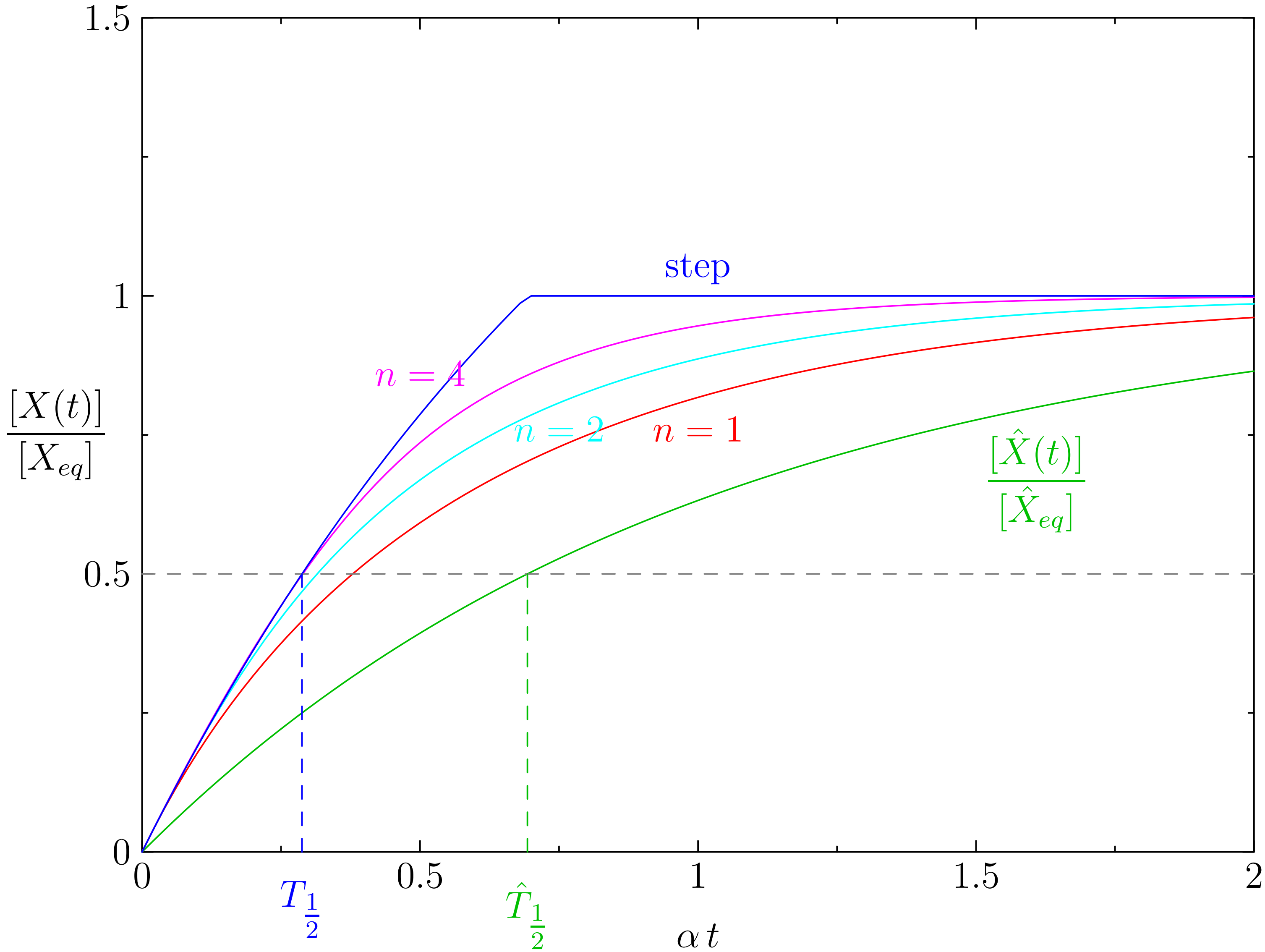

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Advantages of Auto-regulation

- Auto-regulation allows us to use a large production rate β for the same equilibrium, speeding up the equilibration time
- Note that we can independent evolve K (by mutations to the binding site of X in the promoter region) and β, by mutations in the binding site of RNA polymerase
- Another advantage is that β and α can strongly fluctuate during the life time of a cell caused by fluctuations in metabolic capacity
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- We can express the mean change in number of molecules through differential equations
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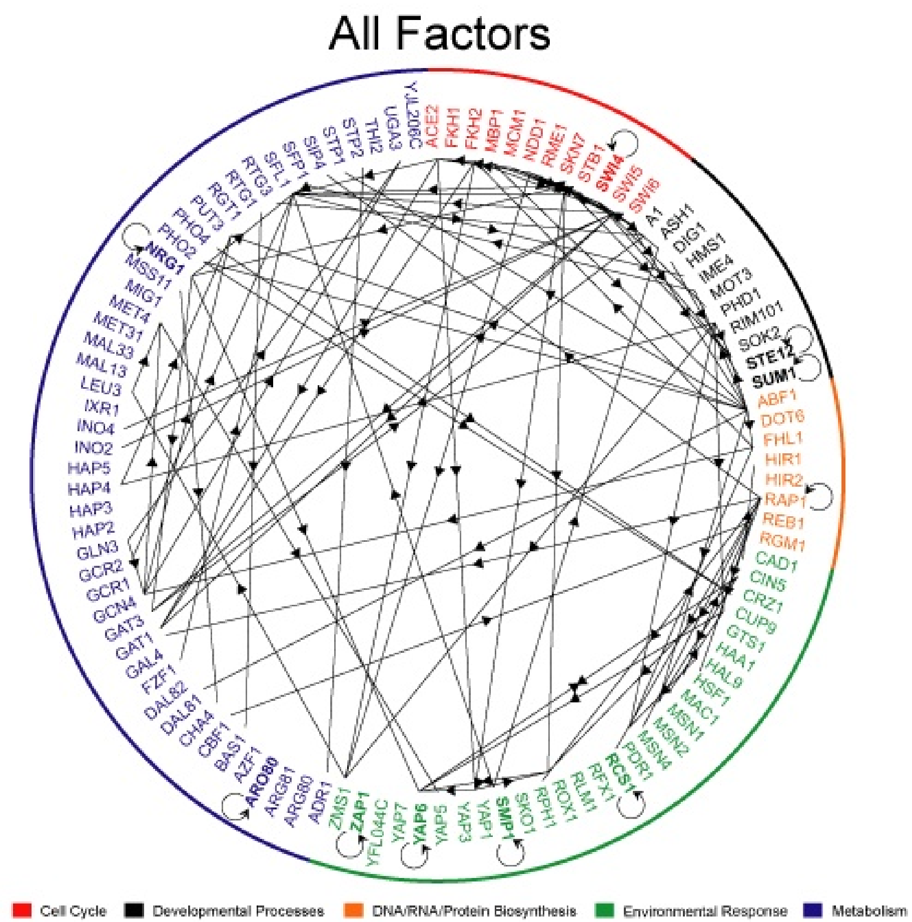
Outline

1. Gene Regulation
2. Chemical Equations
3. Motifs
4. **Quiz**



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2. A protein that creates RNA molecules
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- Answer: A protein that regulates transciption of RNA

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- For the "chemical equation"

$$X \xrightarrow{k} Y$$

what is the correct ODE for X?

1. $\frac{\mathrm{d}[X]}{\mathrm{d}t} = k\,[Y]$
2. $\frac{\mathrm{d}[X]}{\mathrm{d}t} = k\,[X]$
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$$4\,X + Y \xrightarrow{k} 2\,X + Z$$

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